Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = \{A^i\}_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients indexed by $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(A)_\sigma = \operatorname{tr}(A^{i_1} \cdots A^{i_N}).$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

Notation

$\mathbb{C}$, $\mathbb{N}$, $\mathbb{Z}$	complex numbers, natural numbers, integers
$M_D(\mathbb{C})$, $\mathrm{GL}_D(\mathbb{C})$	complex matrices and invertible complex matrices
$\operatorname{tr}(X)$, $\operatorname{span}_{\mathbb{C}}(S)$	trace and complex linear span
$A = \{A^i\}_{i=0}^{d-1}$	a translation-invariant MPS tensor
$w = (i_1, \dots, i_L)$, A^w	a word and the product $A^{i_1} \cdots A^{i_L}$
$\|V^{(N)}(A)\rangle$	the length-N matrix product vector
$V^{(N)}(A)_\sigma$	the coefficient of $\|V^{(N)}(A)\rangle$ at σ
$O_{AB}(N)$	the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$
$\mathcal{E}_A(X)$	the transfer map $\sum_i A^i X (A^i)^\dagger$
$A^{[L]}$	the length-L blocked tensor

In canonical form one writes

$$A^i = \bigoplus_{j=1}^{g} \mu_j A^i_j, \qquad B^i = \bigoplus_{k=1}^{h} \nu_k B^i_k,$$

where the displayed blocks are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N, then the bases of normal tensors have the same size. After relabelling the blocks of B, each matched pair satisfies

$$B^i_j = \zeta_j X_j A^i_j X_j^{-1},$$

for some invertible matrix X_j and nonzero scalar ζ_j, and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges and the weight comparison then combine into a single gauge conjugacy of the two weighted block-diagonal tensors. For a single injective block this reduces to the familiar conclusion $B^i = XA^iX^{-1}$ for all physical indices i.

Chapters 2–11 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 9 includes the invariant-projection splitting. Chapter 14 records the product-algebra block permutation argument. Chapter 10 supplies bases of normal tensors and the power-sum comparison. The after-blocking reductions in Section 9.6.2 give the common primitive block decompositions used immediately in Chapter 11, which records the conditional theorem statements.

The chapters after Chapter 11 collect applications, alternative formulations, and later tensor-network material. Chapter 12 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem. Chapter 13 develops the direct Fundamental Theorem for periodically repeated tensors in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 12.1 in Chapter 12). Chapter 14 gives the fixed-length algebraic Fundamental Theorem of [MGRPG$^+$18], and Chapter 15 records the PEPS analogs. Parent Hamiltonians, correlations, and examples are treated after the PEPS material. Channel representations and normal forms are then collected immediately before quantum dynamical semigroups and the operator-convexity material; entropy and matrix product density operators follow. These later topics use the Fundamental Theorem, or develop later finite-dimensional channel theory, rather than enter its proof.

Chapter 2

Matrix Product Vectors

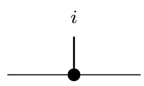
Fix a physical dimension d and a bond dimension D. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). Lean ✓ Stmt A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

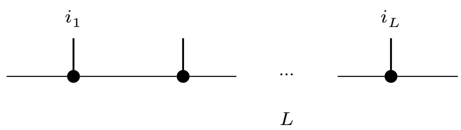


The black node denotes the tensor A, the horizontal legs are virtual, and the upper leg is the physical index i.

Definition 2.2 (Word evaluation). Lean ✓ Stmt Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \cdots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A, the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). Lean ✓ Stmt *For any words w_1, w_2, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.*

Proof. ✓ Proof Immediate from the definition of A^w. $\square$

Definition 2.4 (Matrix product vector). Lean ✓ Stmt The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

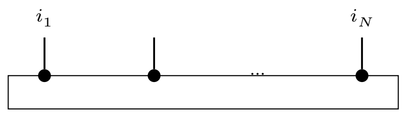
Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w, we also write

$$c_w(A) := \operatorname{tr}(A^w).$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

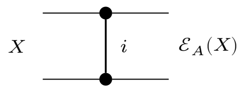


of N copies of the local tensor A, with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). Lean ✓ Stmt The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A, the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. ✓ Stmt The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Site-periodic tensor families

Definition 2.7 (Site-periodic tensor family). Lean ✓ Stmt For a fixed period m, a *site-periodic tensor family* is a family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D.

Definition 2.8 (Periodic same-state and gauge-equivalence relations). Lean Lean ✓ Stmt For site-periodic tensors A, B at fixed period m:

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

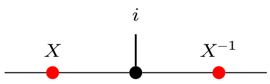
Lemma 2.9 (Equivalence structures for periodic relations). *Lean* *Lean* *✓ Stmt* *Both periodic relations are equivalence relations (reflexive, symmetric, transitive).*

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). Lean ✓ Stmt Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = XA^iX^{-1}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i, and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). Lean ✓ Stmt Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N.

Definition 2.12 (Same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions D_1 and D_2, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N.

Definition 2.13 (Positive-length same MPV - different bond dimensions). Lean ✓ Stmt For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Lemma 2.14 (Length-zero MPV coefficient). *Lean* *✓ Stmt* *For any MPS tensor A with bond dimension D, the length-zero MPV coefficient is D, the trace of the identity on the bond space.*

Lemma 2.15 (Positive-length equality with equal bond dimensions). *Lean* *✓ Stmt* *If two tensors have the same positive-length MPV family and the same bond dimension, then they have the same MPV family at every length.*

Remark 2.16. ✓ Stmt Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.17 (Proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate weakly proportional MPV families if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$.

Definition 2.18 (Nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every N there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem 2.10].

Definition 2.19 (Eventually nonzero proportional MPV). Lean ✓ Stmt Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N, there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma 10.4.

Lemma 2.20 (Elementary nonzero proportionality rules). Lean Lean Lean Lean ✓ Stmt *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar* 1.

Proof. ✓ Proof Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. ☐

Lemma 2.21 (Scaling of word evaluation). Lean ✓ Stmt *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w.

Proof. ✓ Proof Induction on w: the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ. ☐

Lemma 2.22 (Scaling of MPV components). Lean ✓ Stmt *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for every N and every configuration $\sigma = (i_1, \dots, i_N)$,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Apply Lemma 2.21 to the word $w = (i_1, \dots, i_N)$ of length N, then take the trace; the scalar ζ^N pulls out by linearity. ☐

Definition 2.23 (Gauge-phase equivalence). Lean Lean ✓ Stmt Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i,

$$B^i = \zeta X A^i X^{-1}.$$

Proof. ✓ Proof Gauge equivalence is the special case $\zeta = 1$. ☐

Remark 2.24. ✓ Stmt After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.25 (Gauge covariance of word evaluation). Lean ✓ Stmt *If $B^i = XA^iX^{-1}$ for all i, then for every word w,*

$$B^w = XA^wX^{-1}.$$

Proof. ✓ Proof Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

Theorem 2.26 (Gauge invariance of MPVs). Lean ✓ Stmt *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Let $B^i = XA^iX^{-1}$. By Lemma 2.25, $B^w = XA^wX^{-1}$ for every word w. Then $\mathrm{tr}(B^w) = \mathrm{tr}(XA^wX^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. □

Theorem 2.27 (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If, for every physical index i,*

$$B^i = \zeta XA^iX^{-1},$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

2.4 Injectivity and normality

Definition 2.28 (Injectivity). Lean ✓ Stmt A tensor A is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.29 (L-block injectivity). Lean ✓ Stmt A tensor A is *L-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Lemma 2.30 (One-site injectivity gives one-block injectivity). Lean ✓ Stmt *If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.*

Proof. ✓ Proof The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i. Thus the length-one word span is the same span as in one-site injectivity. □

Definition 2.31 (Normal tensor). Lean ✓ Stmt A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0-block injective.

Remark 2.32. ✓ Stmt The algebraic characterization of normality in Definition 2.31 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Lemma 8.68, Theorem 8.85, and Theorem 9.30. In particular, every subsequent use of "normal" here refers to this single notion; Definition 9.25 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.33 (Block-injective canonical form). Lean ✓ Stmt Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \ldots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \ldots, r\}$,
4. scaling factors $\mu_1, \ldots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [PGVWC07]; see also [CPGSV21] for the normal canonical form.

Remark 2.34. ✓ Stmt Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.28. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while Chapter 13 develops the separate periodic irreducible-form extension. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.35 (Block projection). Lean ✓ Stmt For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j-th summand and zero on all other summands.

Definition 2.36 (Block-diagonal matrix). Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.37 (Off-block characterization). *Lean* *✓ Stmt* *A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.*

Proof. ✓ Proof The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. □

Lemma 2.38 (Commutation with block projections). *Lean* *✓ Stmt* *If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j, then it is block diagonal.*

Proof. ✓ Proof Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.37 applies. □

Definition 2.39 (Block-diagonal tensor from blocks). Lean ✓ Stmt Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^{r} \mu_k A_k^i.$$

Lemma 2.40 (Word evaluation of a block-diagonal tensor). Lean ✓ Stmt *For a word w, the word evaluation of a block-diagonal tensor remains block diagonal:*

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

Proof. ✓ Proof Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k-th block. □

Theorem 2.41 (MPV decomposition). Lean ✓ Stmt *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^{r} \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^{r} \mu_k^N V^{(N)}(A_k)_\sigma.$$

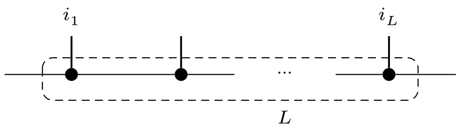
Proof. ✓ Proof By Lemma 2.40, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\operatorname{tr}(A^w) = \sum_k \mu_k^N \operatorname{tr}(A_k^w)$. □

2.6 Blocking

Definition 2.42 (Blocked tensor). Lean ✓ Stmt Given a tensor A and a blocking length L, the L-*blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} = A^{i_1} A^{i_2} \cdots A^{i_L}.$$

Blocking coarse-grains L neighbouring sites into one tensor: the inner virtual bonds are contracted, the L physical legs merge into a single composite index, and the two outer bonds remain as the bond of $A^{[L]}$.



Lemma 2.43 (Blocked word evaluation). Lean ✓ Stmt *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

Proof. ✓ Proof Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

Lemma 2.44 (Blocking preserves MPV coefficients). Lean ✓ Stmt *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.43). □

Theorem 2.45 (Blocking preserves same MPV). Lean ✓ Stmt *If A and B generate the same MPV family, then for any blocking length L, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. ✓ Proof By Lemma 2.44, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. □

2.7 MPV overlap

Definition 2.46 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.47 (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors A and B at system size N is

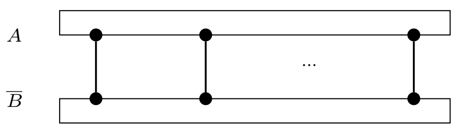
$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle := \sum_{\sigma\in\{0,\dots,d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.48 (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma\in\{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Diagrammatically, the overlap contracts the MPV ring of A against the conjugate ring of B along their shared physical indices, both virtual rings closed by the trace:



This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.49, it is the complex conjugate of the Hilbert-space inner product from Definition 2.47.

Lemma 2.49 (Overlap equals conjugate inner product). Lean ✓ Stmt $O_{AB}(N) = \overline{\langle V^{(N)}(A)|V^{(N)}(B)\rangle}$.

Proof. ✓ Proof The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. □

Lemma 2.50 (Overlap determined by positive-length MPV). Lean ✓ Stmt *If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ, then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N.*

Proof. ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [CPGSV21, Corollary IV.5]: there is an invertible X with $B^i = XA^iX^{-1}$ for every physical index i. The argument is purely algebraic. Equality of the matrix-product vectors gives $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w; the length-two and length-three traces produce a linear map T with $T(A^i) = B^i$ that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra $M_D(\mathbb{C})$ is a unital automorphism; and by Skolem–Noether every automorphism of $M_D(\mathbb{C})$ is conjugation by an invertible X. Evaluating at A^i closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 11.

3.1 The multiplicative linear extension

Theorem 3.1 (Nondegeneracy of the trace pairing). Lean ✓ Stmt *For $M \in M_D(\mathbb{C})$, $\operatorname{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. ✓ Proof Testing against the matrix unit E_{ji} gives $\operatorname{tr}(E_{ji}M) = M_{ij}$, so $\operatorname{tr}(MN) = 0$ for all N forces every entry of M to vanish. □

Lemma 3.2 (Same matrix-product vectors give trace agreement). Lean ✓ Stmt *If A and B generate the same matrix-product vectors, then $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$ for every word w.*

Proof. ✓ Proof The configuration $\sigma = w$ gives $V^{(|w|)}(A)_w = \operatorname{tr}(A^w)$. Equality of the matrix-product vectors gives $V^{(N)}(A) = V^{(N)}(B)$ for all N, so $\operatorname{tr}(A^w) = \operatorname{tr}(B^w)$. □

The trace conditions constrain every product of the matrices A^i. To bring the algebra structure of $M_D(\mathbb{C})$ to bear, record how a matrix pairs against the family $\{A^i\}$ under the trace.

Definition 3.3 (Trace pairing map). Lean ✓ Stmt For a tensor A, the trace pairing map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is

$$\Phi_A(M)_i = \operatorname{tr}(MA^i).$$

Theorem 3.4 (The trace pairing map is injective for injective tensors). Lean ✓ Stmt *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. ✓ Proof If $\Phi_A(M) = 0$, then $\mathrm{tr}(MA^i) = 0$ for every i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\mathrm{tr}(MN) = 0$ for all N, and $M = 0$ by Theorem 3.1. □

Lemma 3.5 (Injectivity transfers across a range inclusion). Lean ✓ Stmt *Let $\Phi, \Psi : V \to W$ be $\mathbb{C}$-linear maps with V finite-dimensional. If $\ker \Phi = \{0\}$ and $\operatorname{range} \Phi \subseteq \operatorname{range} \Psi$, then $\ker \Psi = \{0\}$.*

Proof. ✓ Proof By rank–nullity, $\ker \Phi = \{0\}$ gives $\dim \operatorname{range} \Phi = \dim V$, and the range inclusion gives

$$\dim V = \dim \operatorname{range} \Phi \leq \dim \operatorname{range} \Psi \leq \dim V,$$

so $\dim \operatorname{range} \Psi = \dim V$ and $\ker \Psi = \{0\}$. □

Theorem 3.6 (The linear extension exists and is unique). Lean ✓ Stmt *If A is injective and A, B generate the same matrix-product vectors, then there is a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ with $T(A^i) = B^i$ for all i.*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_A(A^i) = \Phi_B(B^i)$ for every i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\operatorname{range} \Phi_A \subseteq \operatorname{range} \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.4), Lemma 3.5 gives $\ker \Phi_B = \{0\}$. Let g be a left inverse of Φ_B and set $T = g \circ \Phi_A$; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

Theorem 3.7 (The linear extension is multiplicative). Lean ✓ Stmt *Under the hypotheses of Theorem 3.6, the map T satisfies $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$; since the $\{A^i\}$ span $M_D(\mathbb{C})$ this extends to $\Phi_B \circ T = \Phi_A$, so $\operatorname{range} \Phi_A = \operatorname{range}(\Phi_B \circ T) \subseteq \operatorname{range} \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.4), Lemma 3.5 makes Φ_B injective. Length-three trace agreement gives, for all i, j, k,

$$\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \mathrm{tr}(A^i A^j A^k) = \mathrm{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k,$$

so injectivity of Φ_B yields $T(A^i A^j) = B^i B^j$. Extend in two stages: for fixed j, the maps $M \mapsto T(MA^j)$ and $M \mapsto T(M)B^j$ agree on the spanning $\{A^i\}$, hence $T(MA^j) = T(M)B^j$ for all M; then for fixed M, the maps $N \mapsto T(MN)$ and $N \mapsto T(M)T(N)$ agree on the spanning $\{A^j\}$, hence $T(MN) = T(M)T(N)$ for all N. □

3.2 Inner automorphism and the single-block theorem

The rigidity of the full matrix algebra now finishes the proof. A nonzero multiplicative map on $M_D(\mathbb{C})$ is forced to be bijective (simplicity), unit-preserving (surjectivity), and conjugation by a single invertible matrix (Skolem–Noether). The next four results establish these steps and the nonvanishing of T, which the final theorem combines into the gauge X.

Theorem 3.8 (A nonzero multiplicative endomorphism of $M_D(\mathbb{C})$ is bijective). Lean ✓ Stmt *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra $M_D(\mathbb{C})$ is simple, so the kernel is $\{0\}$ or all of $M_D(\mathbb{C})$; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality. □

Lemma 3.9 (A surjective multiplicative map preserves the unit). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a surjective multiplicative $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$, so T is a $\mathbb{C}$-algebra homomorphism.*

Proof. ✓ Proof Choose X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}$. ☐

Lemma 3.10 (An injective tensor forces a nonzero partner). Lean ✓ Stmt *Let $D \geq 1$. If A is injective and A, B generate the same matrix-product vectors, then the B^i are not all zero.*

Proof. ✓ Proof If $B^i = 0$ for every i, then Lemma 3.2 on length-one words gives $\operatorname{tr}(A^i) = 0$ for all i. The $\{A^i\}$ span $M_D(\mathbb{C})$, so the trace functional vanishes identically, contradicting $\operatorname{tr}(\mathbb{1}) = D \neq 0$. ☐

Theorem 3.11 (Skolem–Noether). Lean ✓ Stmt *Every $\mathbb{C}$-algebra automorphism f of $M_D(\mathbb{C})$ is inner: there is $X \in \mathrm{GL}_D(\mathbb{C})$ with $f(M) = XMX^{-1}$ for all M.*

Proof. ✓ Proof Under the identification $M_D(\mathbb{C}) \cong \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$, every algebra automorphism of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$ is conjugation by an invertible linear map. Translating back gives the invertible X. ☐

Theorem 3.12 (Single-block Fundamental Theorem). Lean ✓ Stmt *Let A be an injective MPS tensor and B a tensor of the same bond dimension. If A and B generate the same matrix-product vectors, then there is $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = XA^iX^{-1}$ for all i.*

Proof. ✓ Proof Theorem 3.6 gives the unique linear T with $T(A^i) = B^i$, and Theorem 3.7 makes it multiplicative. It is nonzero: otherwise $B^i = 0$ for all i, against Lemma 3.10. By Theorem 3.8, T is bijective, and by Lemma 3.9 it fixes $\mathbb{1}$, so T is a $\mathbb{C}$-algebra automorphism. Theorem 3.11 gives $X \in \mathrm{GL}_D(\mathbb{C})$ with $T(M) = XMX^{-1}$; evaluating at $M = A^i$ gives $B^i = XA^iX^{-1}$. ☐

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 4.2. ✓ Stmt We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 4.3 (Trace-preserving map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\mathrm{tr}(E(X)) = \mathrm{tr}(X)$ for all X.

Definition 4.4 (Completely positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger.$$

By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all n.

Theorem 4.5 (CP maps are positive). *Lean* *✓ Stmt* *Every completely positive map is positive.*

Proof. ✓ Proof If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Theorem 4.6 (Channels are positive). *Lean* *✓ Stmt* *Every quantum channel is positive.*

Proof. ✓ Proof A quantum channel is completely positive by definition, so Theorem 4.5 applies. □

Theorem 4.7 (Positive maps preserve Hermiticity). *Lean* *✓ Stmt* *If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.*

Proof. ✓ Proof Decompose $X = X_+ - X_-$ into its positive and negative parts. Positivity makes $E(X_+)$ and $E(X_-)$ positive semidefinite, hence Hermitian, so their difference $E(X)$ is Hermitian. □

Definition 4.8 (Quantum channel). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.9 (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

Theorem 4.10 (Density matrices are compact). Lean ✓ Stmt *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

Proof. ✓ Proof The PSD cone is closed (preimage of the closed non-negative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\operatorname{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

Theorem 4.11 (Density matrices are convex). Lean ✓ Stmt *$\mathcal{D}_D$ is convex.*

Proof. ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. □

Theorem 4.12 (Density matrices are nonempty). Lean ✓ Stmt *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

Proof. ✓ Proof $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

Theorem 4.13 (Channels preserve density matrices). Lean ✓ Stmt *If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

Proof. ✓ Proof Complete positivity implies positivity (Theorem 4.5), so $E(\rho) \geq 0$; trace preservation gives $\operatorname{tr}(E(\rho)) = \operatorname{tr}(\rho) = 1$. □

4.2 Irreducibility

Definition 4.14 (Orthogonal projection). Lean ✓ Stmt A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 4.15 (Irreducible map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.16 (Injectivity implies irreducibility). Lean ✓ Stmt *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M, so $P = 0$ or $P = \mathbb{1}$. □

Theorem 4.17 (Transfer map is CP). Lean ✓ Stmt *The transfer map $\mathcal{E}_A(X) = \sum_i A^iX(A^i)^\dagger$ is completely positive.*

Proof. ✓ Proof It is already written in Kraus form, with Kraus operators $\{A^i\}$. □

Theorem 4.18 (Transfer map is a channel). Lean ✓ Stmt *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. ✓ Proof Complete positivity is the observation of Theorem 4.17. Trace preservation: $\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^i X (A^i)^\dagger) = \mathrm{tr}((\sum_i (A^i)^\dagger A^i) X) = \mathrm{tr}(X)$ by cyclicity and the normalization hypothesis. □

4.3 Cesàro fixed-point theorem

Definition 4.19 (Cesàro mean). Lean ✓ Stmt The *Cesàro mean* of a linear map E at order N, applied to a starting matrix ρ_0, is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.20 (Cesàro telescope). Lean ✓ Stmt $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. ✓ Proof Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^{N} E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. □

Theorem 4.21 (Hermitian fixed-point decomposition). Lean ✓ Stmt *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\mathrm{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Proposition 6.8].*

Proof. ✓ Proof Let Q be the support projection of P_+. Trace preservation and positivity give $\mathrm{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. □

Theorem 4.22 (Cesàro fixed-point theorem). Lean ✓ Stmt *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. ✓ Proof Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.12). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ by applying channel preservation to every iterate, adding positive semidefinite matrices, rescaling by the positive average factor, and using trace linearity. By compactness (Theorem 4.10), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 4.20) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

4.4 Fixed-point projection

Definition 4.23 (Fixed-point projection). Lean ✓ Stmt Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\mathrm{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.24 (Power decomposition). Lean ✓ Stmt *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. ✓ Proof P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. □

4.5 Peripheral spectrum and primitivity

Definition 4.25 (Peripheral spectrum). Lean ✓ Stmt The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T.

Definition 4.26 (Peripheral eigenvalues). Lean ✓ Stmt The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 4.27 (1 is a peripheral eigenvalue). Lean ✓ Stmt *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E.*

Proof. ✓ Proof ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Theorem 4.28 (Peripheral eigenvalues lie on the unit circle). Lean ✓ Stmt *Every peripheral eigenvalue λ satisfies $|\lambda| = 1$.*

Proof. ✓ Proof This is the defining condition in Definition 4.26. □

Theorem 4.29 (Finiteness of peripheral eigenvalues). Lean ✓ Stmt *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. ✓ Proof Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 4.30 (Power-stable peripheral eigenvalues are roots of unity). Lean ✓ Stmt *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. ✓ Proof Since μ^n is an eigenvalue for every n, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 4.31 (Peripheral powers imply root of unity). Lean ✓ Stmt *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is* closed under powers *($\mu \in \mathrm{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \mathrm{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. ✓ Proof The closure hypothesis provides $\mu^n \in \mathrm{peripheral}(E)$ for all n, which in particular means μ^n is an eigenvalue. Theorem 4.30 then gives $\mu^p = 1$ for some $p > 0$. □

4.5.1 Periodicity removal by powering

Remark 4.32. ✓ Stmt The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.33 (Common exponent for a finite set of roots of unity). Lean ✓ Stmt *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. ✓ Proof Take p to be the least common multiple of the exponents p_μ. □

Lemma 4.34 (Powering collapses peripheral eigenvalues). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

Proof. ✓ Proof The fixed point ρ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.35 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.31. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.5.2 The Kadison–Schwarz inequality and the multiplicative domain

Definition 4.36 (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \qquad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 4.37 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\left(\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right)$ and take the Schur complement of the lower-right block. □

Theorem 4.38 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then*

$$\mathrm{tr}(E(X)^\dagger E(X)) \le \mathrm{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation then identifies the traces of its two terms. ☐

Theorem 4.39 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \qquad R_i := XK_i^\dagger - K_i^\dagger E(X).$$

Proof. ✓ Proof Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. ☐

Theorem 4.40 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every i,*

$$XK_i^\dagger = K_i^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. ☐

Theorem 4.41 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. ☐

Theorem 4.42 (Left multiplicative identity from KS equality). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then for every $Y \in M_D(\mathbb{C})$,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. ✓ Proof The commutation relation of Theorem 4.40 lets one move $X^\dagger$ past each Kraus operator inside the sum. ☐

Definition 4.43 (Multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E.

Theorem 4.44 (Multiplicative domain is a $*$-subalgebra). Lean ✓ Stmt *For a unital Kraus map, the multiplicative domain is a $*$-subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. ☐

4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV16, Appendix A].

Lemma 4.45 (Peripheral closure via adjoint fixed point). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*
2. *E is irreducible.*

Then $\mathrm{peripheral}(E)$ *is closed under powers: if $\mu \in \mathrm{peripheral}(E)$ then $\mu^n \in \mathrm{peripheral}(E)$ for every $n \in \mathbb{N}$.*

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\mathrm{tr}(\rho G) = \mathrm{tr}(\rho E(X^\dagger X)) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(E^*(\rho) X^\dagger X) - \mathrm{tr}(\rho E(X)^\dagger E(X)) = \mathrm{tr}(\rho X^\dagger X) - |\mu|^2 \,\mathrm{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So X is invertible. By Theorem 4.40 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.42) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. □

Theorem 4.46 (Roots of unity via adjoint fixed point). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.*

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 4.45) with Theorem 4.31. □

Remark 4.47. ✓ Stmt Lemma 4.45 and Theorem 4.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicality.

Definition 4.48 (Channel period). Lean ✓ Stmt The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

Definition 4.49 (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.50 (Primitivity equals period one). Lean ✓ Stmt *A linear map with a nonzero fixed point is primitive if and only if its period is* 1.

Proof. ✓ Proof If $\mathrm{peripheral}(E) = \{1\}$, then $|\mathrm{peripheral}(E)| = 1$. Conversely, if $|\mathrm{peripheral}(E)| = 1$, then since $1 \in \mathrm{peripheral}(E)$ (Theorem 4.27), the only element is 1. □

Theorem 4.51 (Complementary transfer-map gap for primitive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\operatorname{tr}(\rho) \neq 0$. Let*

$$P(X) = \frac{\operatorname{tr}(X)}{\operatorname{tr}(\rho)}\rho$$

be the fixed-point projection associated to ρ. Assume that:

1. *E is primitive;*
2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*
3. *whenever $E(X) = X$ and $\operatorname{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\operatorname{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\operatorname{tr}(X) \neq 0$, trace preservation and $\operatorname{tr}(P(X)) = \operatorname{tr}(X)$ give

$$\operatorname{tr}(X) = \operatorname{tr}(E(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(P(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(X),$$

so $\nu \operatorname{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

4.6 Fixed-point algebra

Definition 4.52 (Kraus fixed points). Lean ✓ Stmt The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\operatorname{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 4.53 (Fixed points lie in the multiplicative domain). Lean ✓ Stmt *Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.*

Proof. ✓ Proof The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. □

Theorem 4.54 (Fixed points form a $*$-subalgebra). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\operatorname{Fix}(E)$ is a $*$-subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Theorem 6.12].*

Proof. ✓ Proof Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(XX^\dagger) = E(X)E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 4.42 yields $E(XY) = E(X)E(Y) = XY$. □

Remark 4.55. ✓ Stmt Theorem 4.53 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 4.56 (Projected elements lie in the multiplicative domain). Lean ✓ Stmt *Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 4.54, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under $\dagger$, so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

The vanishing of both Kadison–Schwarz gaps, again by the argument of Theorem 4.54, places $E(X)$ in the multiplicative domain. □

Theorem 4.57 (Choi–Effros left absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.58 (Choi–Effros right absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. ✓ Proof The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.59 (Choi–Effros symmetric absorption). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(XE(Y)).$$

Proof. ✓ Proof Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. □

Theorem 4.60 (Choi–Effros projected product). Lean ✓ Stmt *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. ✓ Proof The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

□

Definition 4.61 (Adjoint fixed points). Lean ✓ Stmt The *adjoint fixed-point set* $\mathrm{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 4.62 (Fixed points form a $*$-subalgebra in the Heisenberg picture). Lean Lean ✓ Stmt *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\mathrm{Fix}(E^*)$ is a $*$-subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Apply Theorem 4.54 to the adjoint family $\{K_i^\dagger\}$. □

Definition 4.63 (Kraus commutant). Lean Lean ✓ Stmt The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \forall i\}$. It is a $*$-subalgebra of $M_D(\mathbb{C})$.

Theorem 4.64 (Adjoint fixed points equal the Kraus commutant). Lean ✓ Stmt *Under the hypotheses of Theorem 4.62, the adjoint fixed-point $*$-subalgebra equals the Kraus commutant:*

$$\mathrm{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Theorem 6.13].

Proof. ✓ Proof ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any $*$-subalgebra inside $\mathrm{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \mathrm{Fix}(E^*)$ with $X^\dagger X \in \mathrm{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_i X$ and $XK_i^\dagger = K_i^\dagger X$. □

Theorem 4.65 (Kraus commutant inside the adjoint fixed points). Lean ✓ Stmt *Under the hypotheses above, the Kraus commutant is the largest $*$-subalgebra contained in the adjoint fixed-point set.*

Proof. ✓ Proof By Theorem 4.64, every $*$-subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a $*$-subalgebra. □

Chapter 5

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps, the multiplicative domain as a $*$-subalgebra on which the map restricts to a $*$-homomorphism, the extension to normal, subnormal, and commuting-dominant inputs for positive subunital maps, order and spectral-interval contractivity, and the transpose-trace map as a Schwarz map that is not completely positive. The peripheral-eigenvector and normal-block consequences feed the Fundamental Theorem of Chapter 11. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 21.

5.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 4.4) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a Kraus map (Theorem 4.17), so these results apply directly to transfer maps.

Definition 5.1 (Kraus map). Lean ✓ Stmt Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.4) if and only if it can be written in this form.

Definition 5.2 (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 5.3 (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$.

Definition 5.4 (Trace-preserving Kraus map). Lean ✓ Stmt A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 5.5 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof The 2×2 block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. □

Theorem 5.6 (Kadison–Schwarz for the adjoint channel). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. ✓ Proof The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 5.5 to $\{K_i^\dagger\}$. □

5.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 5.7 (k-positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is k-positive if $E \otimes \mathrm{id}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 5.8 (Two-positive map). Lean ✓ Stmt A linear map is *two-positive* if it is 2-positive.

Theorem 5.9 (Completely positive maps are k-positive). Lean ✓ Stmt *Every completely positive map is k-positive for all $k \geq 1$.*

Theorem 5.10 ((k+1)-positive implies k-positive). Lean ✓ Stmt *Positivity under amplification is monotone in k.*

Theorem 5.11 (Completely positive implies two-positive). Lean ✓ Stmt *Every completely positive map is two-positive.*

Theorem 5.12 (Two-positive implies positive). Lean ✓ Stmt *Every two-positive map is positive.*

Definition 5.13 (Unital linear map). Lean ✓ Stmt A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 5.14 (Kadison–Schwarz for unital two-positive maps). Lean ✓ Stmt *If E is unital and two-positive, then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

Proof. Form the 2×2 block matrix $\left(\begin{smallmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{smallmatrix}\right) \geq 0$. Apply $E \otimes \mathrm{id}_2$ (positive by 2-positivity) and read off the $(2,2)$-block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. □

Theorem 5.15 (Kraus-form Kadison–Schwarz as a special case). Lean ✓ Stmt *The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.*

5.1.2 Douglas-type factorization lemmas

Theorem 5.16 (Range inclusion implies right factorization). Lean ✓ Stmt *If* $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$, *then there exists* C *such that* $A = BC$.

Proof. In finite dimensions, $\operatorname{ran}(A) \subseteq \operatorname{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 5.17 (Vectorwise range criterion implies factorization). Lean ✓ Stmt *If* $Av \in \operatorname{ran}(B)$ *for every vector* v, *then there exists* C *with* $A = BC$.

Theorem 5.18 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then* $\operatorname{tr}(E(X)^\dagger E(X)) \leq \operatorname{tr}(X^\dagger X)$ *for all* $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof By Theorem 5.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\operatorname{tr}(E(X^\dagger X)) \geq \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. □

Lemma 5.19 (Trace pairing for Kraus map and adjoint map). Lean ✓ Stmt *For all* $X, Y \in M_D(\mathbb{C})$,

$$\operatorname{tr}(X\,E(Y)) = \operatorname{tr}(E^*(X)\,Y).$$

Lemma 5.20 (Positive-definite trace test). Lean ✓ Stmt *If* $A > 0$, $X \geq 0$, *and* $\operatorname{tr}(AX) = 0$, *then* $X = 0$.

Theorem 5.21 (Fixed-point peripheral eigenvectors satisfy KS equality). Lean ✓ Stmt *Let* E *be unital and let* E^* *be trace-preserving with a positive definite fixed point* $\rho > 0$. *If* $E(X) = \mu X$ *with* $|\mu| = 1$, *then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

5.2 Multiplicative domain

Theorem 5.22 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map* E, *the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. ✓ Proof Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 5.23 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let* E *be a unital Kraus map. If* $E(X^\dagger X) = E(X)^\dagger E(X)$ *for some* X, *then* $XK_i^\dagger = K_i^\dagger E(X)$ *for all* i.

Proof. ✓ Proof By Theorem 5.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 5.24 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let* E *be a Kraus map that is both unital and trace-preserving. If* $E(X) = \mu X$ *with* $|\mu| = 1$, *then the KS gap vanishes:* $E(X^\dagger X) = E(X)^\dagger E(X)$.

Proof. ✓ Proof The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 5.5. Its trace satisfies $\operatorname{tr}(G) = \operatorname{tr}(E(X^\dagger X)) - \operatorname{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\operatorname{tr}(E(X^\dagger X)) = \operatorname{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\operatorname{tr}(E(X)^\dagger E(X)) = |\mu|^2 \operatorname{tr}(X^\dagger X) = \operatorname{tr}(X^\dagger X)$. So $\operatorname{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

Theorem 5.25 (Left multiplicative identity from KS equality). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. ✓ Proof From Theorem 5.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$
\begin{aligned}
E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\
&= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\
&= E(X)^\dagger E(Y).
\end{aligned}
$$

□

5.2.1 Full multiplicative-domain structure

Definition 5.26 (Right and left multiplicative domains). Lean Lean ✓ Stmt For a Kraus map E, define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(XY) = E(X)E(Y)\},$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}),\ E(YX) = E(Y)E(X)\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Equations (5.11)–(5.12)].

Definition 5.27 (Full multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 5.28 (Multiplicative-domain characterization). Lean Lean ✓ Stmt *Let E be a unital Kraus map. Then*

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger,$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

Proof. ✓ Proof If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 5.25, and the right-domain implication follows by applying Theorem 5.25 to $X^\dagger$. □

Theorem 5.29 (One-sided multiplicative domains are subalgebras). Lean Lean ✓ Stmt *For a unital Kraus map E, both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Closure under addition follows from linearity of E. Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z, using Theorem 5.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E. □

Theorem 5.30 (Multiplicative domain is a ∗-subalgebra)**.** Lean ✓ Stmt *For a unital Kraus map E, the full multiplicative domain $\mathcal{A}(E)$ is a ∗-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].*

Proof. ✓ Proof If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 5.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 5.29, this gives a ∗-subalgebra. □

Theorem 5.31 (Restriction to multiplicative domain is a ∗-homomorphism)**.** Lean ✓ Stmt *If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \to M_D(\mathbb{C})$ is a ∗-algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].*

Proof. ✓ Proof Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 5.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. □

5.2.2 Schwarz inequality for normal operators

Theorem 5.32 (CP Schwarz inequality for normal operators)**.** Lean Lean ✓ Stmt *Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap*

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \le E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Proposition 5.1].

Proof. ✓ Proof The positivity statement is exactly Theorem 5.6. The second formulation is the Loewner-order reformulation of the first. □

Theorem 5.33 (Schwarz inequality for normal operators)**.** Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \le \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0.$$

This is [Wol12, Proposition 5.1].

Proof. ✓ Proof Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = Ue_ie_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown non-negative by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_iP_j)v_k\rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i)v_k\rangle\langle v_k, T(P_j)v_k\rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. □

Remark 5.34. The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative ∗-subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 5.35 (Schwarz inequality for subnormal operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Theorem 5.5].

Proof. ✓ Proof Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunital map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 5.33 to $\tilde{T}$ and N, then compress the resulting inequality back to $\mathbb{C}^D$. □

Theorem 5.36 (Schwarz inequality for commuting-dominant operators). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite* dominant *$B \geq 0$ with $A^\dagger A \leq B$ and B commuting with A ($BA = AB$), then*

$$T(A^\dagger)T(A) \leq T(B) \quad \text{and} \quad T(A)T(A^\dagger) \leq T(B).$$

This is [Wol12, Theorem 5.6].

Proof. ✓ Proof First take $B > 0$. Since B commutes with A and $A^\dagger A \leq B$, also $AA^\dagger \leq B$, so $D_R = (B - A^\dagger A)^{1/2}$ and $D_L = (B - AA^\dagger)^{1/2}$ are positive semidefinite. The block matrix $N = \begin{pmatrix} A & D_L \\ D_R & -A^\dagger \end{pmatrix}$ is normal, with $N^\dagger N = NN^\dagger = B \oplus B$. As in the subnormal case, extend T to a positive subunital map on $M_{2D}(\mathbb{C})$, apply the normal-operator Schwarz inequality (Theorem 5.33) to that extension and N, and compress back to $\mathbb{C}^D$. Since N is normal, both $T(A^\dagger)T(A) \leq T(B)$ and $T(A)T(A^\dagger) \leq T(B)$ follow. For general $B \geq 0$, apply this to $B_\varepsilon = B + \varepsilon\mathbb{1} > 0$ and let $\varepsilon \to 0$. □

Theorem 5.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs). Lean ✓ Stmt *Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A and satisfying $A^\dagger A \leq B$, then $E^*(A^\dagger)E^*(A) \leq E^*(B)$ and $E^*(A)E^*(A^\dagger) \leq E^*(B)$.*

Proof. ✓ Proof Apply Theorem 5.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). □

5.2.4 Positive maps preserve order and spectral intervals

Theorem 5.38 (Positive maps are monotone). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive and $A \leq B$, then $T(A) \leq T(B)$.*

Proof. ✓ Proof Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. □

Theorem 5.39 (Positive maps preserve adjoints). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. □

Theorem 5.40 (Order intervals are preserved). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$. If $a \le 0 \le b$ and $a\mathbb{1} \le A \le b\mathbb{1}$, then*

$$a\mathbb{1} \le T(A) \le b\mathbb{1}.$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. ✓ Proof By monotonicity, $T(a\mathbb{1}) \le T(A) \le T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \le 0$ and $T(\mathbb{1}) \le \mathbb{1}$, we have $aT(\mathbb{1}) \ge a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \le b\mathbb{1}$ since $b \ge 0$. Therefore $a\mathbb{1} \le T(A) \le b\mathbb{1}$. □

Theorem 5.41 (Spectral interval contractivity). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \le \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\mathrm{spec}(A) \subseteq [a, b]$ with $a \le 0 \le b$, then*

$$\mathrm{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. ✓ Proof For Hermitian matrices, the inclusion $\mathrm{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \le A \le b\mathbb{1}$. Apply Theorem 5.40 and translate the resulting inequalities back into spectral bounds. □

5.2.5 A positive Schwarz map that is not completely positive

Example 5.42 (Transpose-trace map on $M_2(\mathbb{C})$). Lean ✓ Stmt Consider the linear map $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\,\mathrm{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

Theorem 5.43 (The transpose-trace map is positive). Lean ✓ Stmt *The map from Example 5.42 is positive.*

Proof. ✓ Proof It is a non-negative linear combination of the transpose map and the map $A \mapsto \mathrm{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. □

Theorem 5.44 (The transpose-trace map satisfies the Schwarz inequality). Lean ✓ Stmt *For every $A \in M_2(\mathbb{C})$,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \ge 0,$$

where T is the map from Example 5.42.

Proof. ✓ Proof Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ, so it is enough to treat the case $\mathrm{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. □

Theorem 5.45 (The transpose-trace map is not completely positive). Lean ✓ Stmt *The map from Example 5.42 is not completely positive.*

Proof. ✓ Proof Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. □

Chapter 6

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf's spectral characterization of irreducibility. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

Theorem 6.1 (Brouwer fixed point on density matrices)**.** Lean Lean Lean Lean Lean ✓ Stmt *Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.*

Proof. ✓ Proof The proof starts from Brouwer's theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. ☐

6.1 Positive definiteness

Remark 6.2. ✓ Stmt Wolf's general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

Theorem 6.3 (PSD fixed point is positive definite)**.** Lean ✓ Stmt *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. ✓ Proof Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is non-negative; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i: the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M. Given any vector w, choose M with $Mv = w$. Then $\rho w = 0$ for every w, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Theorem 6.3(2)], the same conclusion is reached via the growth condition $(\mathbb{1} + \mathcal{E}_A)^{D-1}(\rho) > 0$ of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. ☐

Theorem 6.4 (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. ✓ Proof Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\mathcal{E}_A(QXQ)Q = \mathcal{E}_A(QXQ)$ for all X. By irreducibility of $\mathcal{E}_A$ (Definition 4.15), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2(1)]. ☐

Theorem 6.5 (Orthogonal trace condition). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\mathrm{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D-1$ such that*

$$\mathrm{tr}(B \cdot E^t(A)) > 0.$$

Proof. ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2(2)]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\mathrm{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\mathrm{tr}\left(B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\mathrm{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\mathrm{tr}(BA) = 0$). Hence at least one $k \in \{1, \ldots, D-1\}$ contributes a strictly positive term. This is [Wol12, Theorem 6.2(4)]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. ☐

6.2 Uniqueness

Theorem 6.6 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Both ρ and σ are positive definite by Theorem 6.3. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 6.3, τ must be zero, giving $\sigma = c_0\rho$. In [Wol12, Theorem 6.3, item 2] uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]; the same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition. ☐

Remark 6.7. ✓ Stmt The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 6.8 (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$, and if ρ is nonzero, then $\sigma = c\rho$ for some scalar c.*

Proof. ✓ Proof The irreducible positive-definiteness theorem (Theorem 6.4) upgrades every nonzero positive semidefinite fixed point to a positive definite one. The same minimal eigenvalue argument used in Theorem 6.6 then applies. ☐

Theorem 6.9 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E. Irreducibility of E passes to the adjoint transfer map $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot)K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.22) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \mathrm{tr}(\tau X) = \mathrm{tr}(\tau \cdot E(X)) = \mathrm{tr}(E^\dagger(\tau) \cdot X) = t \cdot \mathrm{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\mathrm{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. This is [Wol12, Theorem 6.3, item 3], the non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps; the proof follows Wolf's dual-map trace argument. □

6.3 Existence and the Perron–Frobenius theorem

Theorem 6.10 (Existence of PSD fixed point). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.18). The Cesàro fixed-point theorem (Theorem 4.22) then provides the fixed point; injectivity is not needed for this step. □

Theorem 6.11 (Quantum Perron–Frobenius). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. ✓ Proof Combine Theorems 6.10, 6.3, and 6.6. □

Theorem 6.12 (Quantum Perron–Frobenius for irreducible transfer maps). Lean ✓ Stmt *Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.*

Proof. ✓ Proof Existence is the channel fixed-point theorem (Theorem 6.10). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.8 gives uniqueness up to scalar multiple. □

6.4 Right- and left-canonical gauges

Theorem 6.13 (Right-canonical (unital) gauge). Lean ✓ Stmt *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization.*

Proof. ✓ Proof Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i\underbrace{SS^\dagger}_{\rho}(A^i)^\dagger(S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i\rho(A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

Theorem 6.14 (Left-canonical (trace-preserving) gauge). Lean ✓ Stmt *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.*

Proof. ✓ Proof Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger S S^{-1} = \mathbb{1}$. □

Theorem 6.15 (Square-root trace-preserving gauge from an adjoint fixed point). Lean ✓ Stmt *Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = \sigma$. Define*

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}.$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. ✓ Proof Since $\sigma^{1/2}$ is self-adjoint and $\sigma^{1/2}\sigma^{1/2} = \sigma$, each summand becomes $(\sigma^{1/2})^{-1}\,(A^i)^\dagger \sigma A^i\,\sigma^{-1/2}$. Summing over i and using the fixed-point equation leaves $(\sigma^{1/2})^{-1}\sigma\sigma^{-1/2} = \mathbb{1}$. □

Lemma 6.16 (Square-root trace-preserving gauge is a gauge equivalence). Lean ✓ Stmt *If σ is positive definite and $B^i = \sigma^{1/2}A^i\sigma^{-1/2}$, then A and B are gauge-equivalent.*

Proof. ✓ Proof The gauge matrix is $\sigma^{1/2}$, which is invertible because σ is positive definite. □

Remark 6.17. ✓ Stmt The right-canonical gauge of Theorem 6.13 and the left-canonical gauge of Theorem 6.14 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

6.5 Similarity preserves irreducibility

Lemma 6.18 (Similarity transform preserves irreducibility). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1}\, E(C\,X\,C^\dagger)\,(C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. ✓ Proof Suppose Q is an invariant projection for E': $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1}-Q)E'(QXQ)Q = 0$ for all X. Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E. Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

Lemma 6.19 (Similarity is an involution). Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1}\,E(C\,X\,C^\dagger)\,(C^\dagger)^{-1})\,C^\dagger = E(X).$$

Proof. ✓ Proof Direct computation: the inner conjugation $C \cdot (C^{-1}X(C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer conjugation by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma 6.20 (Irreducibility is invariant under similarity). Lean ✓ Stmt *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1} E(C\, X\, C^\dagger)\,(C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. ✓ Proof The forward direction follows from Lemma 6.18 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1}, which equals E by Lemma 6.19. The reverse direction is Lemma 6.18 itself (with $c = 1$). □

Theorem 6.21 (Scaling preserves irreducibility). Lean ✓ Stmt *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Proposition 6.6].*

Proof. ✓ Proof An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). □

6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

Theorem 6.22 (PSD eigenvector for positive maps). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ. Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 6.23 (Adjoint transfer map is nonzero). Lean ✓ Stmt *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. ✓ Proof If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

Theorem 6.24 (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]); the application to the adjoint transfer map follows [CPGSV16, Appendix A].

Proof. ✓ Proof By Theorem 6.23, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ is irreducible if and only if $\mathcal{E}_A^\dagger$ is irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ. Hence Theorem 6.22 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by

Theorem 6.21, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives σ_0 positive definite. ☐

Theorem 6.25 (TP-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix σ, and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the "spectral rescaling + TP gauge" step of [CPGSV16, Appendix A]. The similarity is generally non-unitary.

Proof. ✓ Proof By Theorem 6.24, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the square-root trace-preserving gauge construction (Theorem 6.15), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. Lemma 6.16 gives the required gauge equivalence. ☐

Theorem 6.26 (Unital-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix ρ, and a gauge-equivalent tensor B such that*

$$B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$$

is unital: $\sum_i B^i (B^i)^\dagger = \mathbb{1}$.

This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem 4, lines 765–770], stated for one irreducible nonzero block.

Proof. ✓ Proof Apply Theorem 6.24 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives $\mathcal{E}_A(\rho) = r\rho$ with ρ positive definite and $r > 0$. The unital gauge theorem then gives the displayed unital representative, and the same square-root similarity gives gauge equivalence to $r^{-1/2}A$. ☐

6.7 Exponential positivity for irreducible CP maps

Theorem 6.27 (Exponential truncation is positive definite). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2(3)].

Proof. ✓ Proof If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v. The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A, as established in the proof of Theorem 6.5) then forces $(\mathbb{1}+E)^{D-1}(A)$ to annihilate v, contradicting its positive definiteness. ☐

Theorem 6.28 (Exponential positivity for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2(3)].

Proof. ✓ Proof By Theorem 6.27, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. □

6.8 Ergodicity of irreducible channels

Theorem 6.29 (Unique density-matrix fixed point for an irreducible channel). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].*

Proof. ✓ Proof Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. □

Theorem 6.30 (Cesàro convergence for irreducible channels). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E. This is [Wol12, Corollary 6.3].

Proof. ✓ Proof Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.29, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. □

6.9 Spectral radius at the Perron eigenvalue

Theorem 6.31 (Spectral radius is the Perron eigenvalue). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r. This is [Wol12, Theorem 6.3(4)].*

Proof. ✓ Proof Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1}. The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map F satisfies $\rho(F) \leq 1$, since $\operatorname{tr}(F^n(X)) = \operatorname{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E. □

Theorem 6.32 (Real spectral-radius identity). Lean ✓ Stmt *Under the hypotheses of Theorem 6.31, the real-valued spectral radius of E is also equal to r.*

Proof. ✓ Proof This is the real-valued reformulation of Theorem 6.31. □

6.10 Spectral characterization of irreducibility

Definition 6.33 (Spectral properties). Lean ✓ Stmt A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number r, a positive definite right eigenvector ρ, and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ, and the spectral radius of E is equal to r.

Theorem 6.34 (Irreducibility gives the spectral properties). Lean ✓ Stmt *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.33.*

Proof. ✓ Proof Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.31. □

Theorem 6.35 (Spectral properties imply irreducibility). Lean ✓ Stmt *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.33, then it is irreducible.*

Proof. ✓ Proof Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.33. □

Theorem 6.36 (Irreducibility iff the spectral properties hold). Lean ✓ Stmt *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 6.33. This is [Wol12, Theorem 6.4].*

Proof. ✓ Proof Combine Theorems 6.34 and 6.35. □

Chapter 7

Transfer-Operator Gaps and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary transfer-map gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

This chapter concerns gaps in the spectrum of transfer operators, writing $\rho_{\mathrm{spec}}(\cdot)$ for the spectral radius: $\rho_{\mathrm{spec}}(F_{AB}) < 1$ for mixed transfer operators, and $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ after removing the fixed-point projection. The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian in Chapter 16.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

7.1 Mixed transfer operator

Definition 7.1 (Mixed transfer operator). Lean ✓ Stmt The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 7.2 (Rectangular mixed transfer operator). Lean ✓ Stmt For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\mathrm{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\mathrm{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Theorem 7.3 (Self-transfer). Lean ✓ Stmt *Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.*

Proof. ✓ Proof Substituting $B = A$ into the defining sum $F_{AB}(X) = \sum_i A^i X (B^i)^\dagger$ gives $\sum_i A^i X (A^i)^\dagger = \mathcal{E}_A(X)$. □

Lemma 7.4 (Iterated mixed transfer). Lean ✓ Stmt *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0,\dots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. ✓ Proof Induction on N: the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0,\dots,d-1\}^{N+1} \cong \{0,\dots,d-1\} \times \{0,\dots,d-1\}^N$. □

Lemma 7.5 (Matrix trace of iterated mixed transfer at the identity). Lean ✓ Stmt *For $N \geq 0$,*

$$\operatorname{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \operatorname{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not *be confused with the operator trace $\operatorname{Tr}(F_{AB}^N)$; its relation to the MPV overlap is proved in the next section.*

Proof. ✓ Proof Apply the matrix trace to each summand in Lemma 7.4 with $X = \mathbb{1}$: $\operatorname{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \operatorname{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_\sigma \operatorname{tr}(A^\sigma (B^\sigma)^\dagger)$. □

7.2 MPV overlap as transfer trace

Recall from Definition 2.48 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 7.6 (Trace expansion over matrix units). Lean ✓ Stmt *For any linear endomorphism T on $M_{D_1 \times D_2}(\mathbb{C})$, the operator trace can be expanded as*

$$\operatorname{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq},$$

where $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$ is the matrix with 1 in position (p,q) and 0 elsewhere.

Proof. ✓ Proof The operator trace $\operatorname{Tr}(T)$ equals the sum of $\operatorname{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p,q), and $\operatorname{tr}(M \cdot E_{qp}) = M_{pq}$. □

Lemma 7.7 (Overlap equals transfer trace). Lean ✓ Stmt *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \operatorname{Tr}(F_{AB}^N).$$

Proof. ✓ Proof Expand the operator trace using Lemma 7.6 as a sum over matrix units E_{pq}. Apply Lemma 7.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p,q)-entry, and reassemble. The resulting double sum over (p,q) and σ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. □

Lemma 7.8 (Rectangular overlap as transfer trace). Lean ✓ Stmt *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

Proof. ✓ Proof Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Lemma 7.7. □

Lemma 7.9 (Word trace as entrywise inner product). Lean ✓ Stmt *For a word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\mathrm{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

Proof. ✓ Proof Expand the matrix trace and the (j, j)-entry of $A^\sigma (B^\sigma)^\dagger$. □

7.3 Frobenius norm estimates

The transfer-operator gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

Definition 7.10 (Frobenius norm squared). Lean ✓ Stmt For a (possibly rectangular) matrix $X \in M_{m \times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 7.11 (Trace formula for Frobenius norm). Lean ✓ Stmt $\|X\|_F^2 = \mathrm{Re}\,\mathrm{tr}(X^\dagger X)$.

Proof. ✓ Proof Expand the matrix product and trace entry by entry. □

Definition 7.12 (Euclidean-space embedding). Lean ✓ Stmt The map $\mathrm{vec} : M_{m \times n}(\mathbb{C}) \to \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 7.13 (Norm of embedded matrix). Lean ✓ Stmt $\|\mathrm{vec}(X)\|^2 = \|X\|_F^2$.

Proof. ✓ Proof Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

7.4 Eigenvalue bound and transfer-operator gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

Theorem 7.14 (Eigenvalue bound). Lean ✓ Stmt *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \le 1$.*

Proof. ✓ Proof Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D \|X\|_{\mathrm{HS}}$ for all n, where C_D depends only on D. If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} = \|F_{AB}^n(X)\|_{\mathrm{HS}} \le C_D \|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \le 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 7.15 (Rectangular eigenvalue bound). Lean ✓ Stmt *Let A and B be normalized MPS tensors of bond dimensions $D_1 \ge 1$ and $D_2 \ge 1$. Every eigenvalue μ of F_{AB}^{rect} satisfies $|\mu| \le 1$.*

Proof. ✓ Proof The word-expansion and Frobenius-norm estimate used in Theorem 7.14 apply equally to rectangular matrices: if $F_{AB}^{\mathrm{rect}}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} \le D_1 \|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \le 1$. □

Lemma 7.16 (Spectral radius bound). Lean ✓ Stmt $\rho_{\mathrm{spec}}(F_{AB}) \le 1$ *for normalized tensors.*

Proof. ✓ Proof The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \le 1$ by Theorem 7.14. □

Theorem 7.17 (Spectral radius ≥ 1 implies gauge-phase equivalence). Lean ✓ Stmt *Let A, B be injective normalized MPS tensors. If $\rho_{\mathrm{spec}}(F_{AB}) \ge 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Since $\rho_{\mathrm{spec}}(F_{AB}) \le 1$ (Lemma 7.16) and $\rho_{\mathrm{spec}}(F_{AB}) \ge 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 6.11) and gauge each tensor separately to a unital Kraus family (Theorem 6.13): set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B. Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \left(\begin{smallmatrix} A'^i & 0 \\ 0 & B'^i \end{smallmatrix}\right)$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \left(\begin{smallmatrix} 0 & X' \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 5.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i.

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i. Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 5.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i. By Skolem–Noether (Theorem 3.11) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i, establishing gauge-phase equivalence (Definition 2.23).

The gauging is applied to the individual transfer maps $\mathcal{E}_A, \mathcal{E}_B$ separately; the mixed transfer map F_{AB} need not be simultaneously unital and trace-preserving. □

Lemma 7.18 (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt *Let A and B be normalized tensors, and let $X \neq 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has*

$$X'(B'^i)^\dagger = \mu (A'^i)^\dagger X', \qquad A'^i X' = \mu X' B'^i.$$

Proof. ✓ Proof Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. □

Lemma 7.19 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt *Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B', an invertible matrix X', and a scalar μ with $|\mu| = 1$ such that, for every physical index i,*

$$A'^i X' = \mu X' B'^i.$$

Then A and B are gauge-phase equivalent.

Proof. ✓ Proof Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B. □

Remark 7.20 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a *unital* Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention, $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the transfer-operator gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

Theorem 7.21 (Strict transfer-operator gap). Lean ✓ Stmt *Let A, B be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho_{\mathrm{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof $\rho_{\mathrm{spec}}(F_{AB}) \leq 1$ by Lemma 7.16. If $\rho_{\mathrm{spec}}(F_{AB}) = 1$, Theorem 7.17 gives gauge-phase equivalence, contradicting the hypothesis. □

7.5 Transfer-operator gap — rectangular case

Lemma 7.22 (Rectangular intertwining forces equal bond dimension). Lean ✓ Stmt *Let A and B be injective tensors of bond dimensions D_1 and D_2. If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i,*

$$X(B^i)^\dagger = \mu (A^i)^\dagger X, \qquad A^i X = \mu X B^i,$$

then $D_1 = D_2$.

Proof. ✓ Proof The intertwining relation shows that $\ker(X)$ is invariant under all generators of B. Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \leq D_2$. □

Theorem 7.23 (Rectangular transfer-operator gap). Lean ✓ Stmt *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2, with $D_1 \neq D_2$. Then $\rho_{\mathrm{spec}}(F_{AB}^{\mathrm{rect}}) < 1$.*

Proof. ✓ Proof The same word-expansion argument as in Theorem 7.15 gives $|\mu| \le 1$ for every eigenvalue of F_{AB}^{rect}. If $\rho_{\mathrm{spec}}(F_{AB}^{\mathrm{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A, B to unital families A', B' and transport X to a nonzero rectangular intertwiner X' with $A'^i X' = \mu X' B'^i$ for every i. This is the hypothesis of Lemma 7.22, so $D_1 = D_2$, contradicting the hypothesis. □

Theorem 7.24 (Rectangular overlap decay)**.** Lean ✓ Stmt *If $D_1 \ne D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof The transfer-operator gap $\rho_{\mathrm{spec}}(F_{AB}^{\mathrm{rect}}) < 1$ (Theorem 7.23) implies $(F_{AB}^{\mathrm{rect}})^N \to 0$, so the operator trace converges to 0. By Lemma 7.8, $O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N) \to 0$. □

7.6 MPV overlap decay

Theorem 7.25 (Transfer powers converge to zero)**.** Lean ✓ Stmt *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof The transfer-operator gap $\rho_{\mathrm{spec}}(F_{AB}) < 1$ (Theorem 7.21) implies $\|F_{AB}^n\|_{\mathrm{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every X. □

Theorem 7.26 (Overlap decay)**.** Lean ✓ Stmt *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof By Lemma 7.7, $O_{AB}(N) = \mathrm{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 7.6): $\mathrm{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 7.25, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. □

7.7 Block separation

The block-separation results below record the overlap asymptotics used in the canonical-form separation of Chapter 9.

Theorem 7.27 (Cross-correlation decay)**.** Lean ✓ Stmt *If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\mathrm{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof By Theorem 7.25, $F_{AB}^N(X) \to 0$ in operator norm, so $\mathrm{tr}(F_{AB}^N(X)) \to 0$. □

Theorem 7.28 (Self-correlation persists)**.** Lean ✓ Stmt *If ρ is a fixed point of $\mathcal{E}_A$, then $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$ for all $N \ge 0$. In particular this applies to the distinguished QPF fixed point from Theorem 6.11.*

Proof. ✓ Proof Since ρ is a fixed point, $\mathcal{E}_A^N(\rho) = \rho$ for every N, so $\mathrm{tr}(\mathcal{E}_A^N(\rho)) = \mathrm{tr}(\rho)$. □

7.8 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 7.29 (Gauge-equivalence from irreducible-TP spectral radius). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D. If $\rho_{\text{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Choose a modulus-one eigenvector as in Theorem 7.17. The proof follows the five-step structure of Theorem 7.17, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.11) and the right-canonical gauge (Theorem 6.13). The change is in Step 4: the Kadison–Schwarz equality gives $X'^{\dagger}X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of X'. Formally, Lemma 7.18 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Lemma 7.19 upgrades the intertwining to gauge-phase equivalence. □

Theorem 7.30 (Strict transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof Contrapositive of Theorem 7.29, combined with $\rho_{\text{spec}}(F_{AB}) \leq 1$ (Lemma 7.16). □

Theorem 7.31 (Transfer power decay (irreducible-TP)). Lean ✓ Stmt *Under the hypotheses of Theorem 7.30, $F^n_{AB}(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof By the Gelfand formula, $\rho_{\text{spec}}(F_{AB}) < 1$ implies $\|F^n_{AB}\|_{\text{op}} \to 0$. □

Theorem 7.32 (Overlap decay (irreducible-TP)). Lean ✓ Stmt *Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine the transfer-operator gap (Theorem 7.30) with the overlap–trace identity (Lemma 7.7). □

Theorem 7.33 (Rectangular transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F^{\text{rect}}_{AB}) < 1$.*

Proof. ✓ Proof Assume for contradiction that $\rho_{\text{spec}}(F^{\text{rect}}_{AB}) = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. Gauge A, B separately to unital families A', B' and transport X to $X' \neq 0$. The matrix $X'X'^{\dagger}$ ($D_1 \times D_1$) is a nonzero positive-semidefinite fixed point of $\mathcal{E}_{A'}$, so by Theorem 6.4 it is positive definite, hence invertible; thus X' has full row rank and $D_1 \leq D_2$. Symmetrically $X'^{\dagger}X'$ ($D_2 \times D_2$) is positive definite, giving full column rank and $D_2 \leq D_1$. Hence $D_1 = D_2$, contradicting the hypothesis. □

Theorem 7.34 (Rectangular overlap decay (irreducible-TP)). Lean ✓ Stmt *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. ✓ Proof Combine Theorem 7.33 with Lemma 7.8. □

7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation E_σ is the special case where the fixed-point $*$-subalgebra is the scalar algebra $\mathbb{C} \cdot \mathbb{1}$ (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

Definition 7.35 (Conditional expectation). Lean ✓ Stmt Given a $*$-subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S, and satisfies $P(X) = X$ for all $X \in S$.

Definition 7.36 (Scalar conditional expectation). Lean ✓ Stmt For $\sigma \geq 0$ with $\mathrm{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}.$$

Lemma 7.37 (Evaluation formula). Lean ✓ Stmt $E_\sigma(X) = \dfrac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}$.

Lemma 7.38 (Unitality). Lean ✓ Stmt $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Lemma 7.39 (Idempotence). Lean ✓ Stmt *$E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X.*

Lemma 7.40 (Range is scalar). Lean ✓ Stmt *For every X, there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c\,\mathbb{1}$.*

Lemma 7.41 (Fixes scalar operators). Lean ✓ Stmt *For $c \in \mathbb{C}$, $E_\sigma(c\,\mathbb{1}) = c\,\mathbb{1}$.*

Lemma 7.42 (Absorbs the adjoint map). Lean ✓ Stmt *If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.*

Lemma 7.43 (Adjoint map absorbs scalar projection). Lean ✓ Stmt *If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.*

Lemma 7.44 (Image lies in adjoint fixed points). Lean ✓ Stmt *For all X, $E_\sigma(X)$ is fixed by T^*.*

Theorem 7.45 (Scalar map is a conditional expectation). Lean ✓ Stmt *Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

Lemma 7.46 (Complete positivity of the scalar conditional expectation). Lean ✓ Stmt *For any positive semidefinite σ, the scalar conditional expectation E_σ is a completely positive map.*

Proof. ✓ Proof Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i|\, S$ indexed by $(j,i) \in \{1, \dots, D\}^2$. Using $|j\rangle\langle i|\, M\, |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$ and summing first over i and then over j,

$$\sum_{j,i} K_{j,i}\, X\, K_{j,i}^\dagger = \sum_j |j\rangle\langle j|\, \mathrm{tr}(SXS) = \mathrm{tr}(\sigma X)\mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\mathrm{tr}(\sigma))^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the non-negative real number $(\mathrm{tr}(\sigma))^{-1}$, and therefore completely positive. □

Lemma 7.47 (σ-trace preservation of the scalar conditional expectation). Lean ✓ Stmt *If $\mathrm{tr}(\sigma) \neq 0$, then for every X*

$$\mathrm{tr}(\sigma\, E_\sigma(X)) = \mathrm{tr}(\sigma X).$$

Proof. ✓ Proof Direct computation: $\mathrm{tr}(\sigma \cdot \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)} \mathbb{1}) = \frac{\mathrm{tr}(\sigma X)}{\mathrm{tr}(\sigma)}\, \mathrm{tr}(\sigma) = \mathrm{tr}(\sigma X)$. □

7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

Lemma 7.48 (Lower-triangular Kraus condition implies corner invariance). Lean ✓ Stmt *Let A be an MPS tensor and let P be an orthogonal projection. If, for every physical index i,*

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P\mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P-corner. □

Lemma 7.49 (Support projection invariance). Lean ✓ Stmt *Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ. Then*

$$PE(PXP)P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i. Apply Lemma 7.48. □

Definition 7.50 (Stationary state). Lean ✓ Stmt For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\mathrm{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.11).

Definition 7.51 (Stationary support). Lean ✓ Stmt The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 7.50).

Theorem 7.52 (Stationary support is full). Lean ✓ Stmt *For an irreducible channel E, the stationary support equals the identity:* $\mathrm{supp}(\rho_\infty) = \mathbb{1}$.

Proof. ✓ Proof The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Lemma 7.49). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. □

Remark 7.53 (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.

7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$-structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a $\mathbb{C}$-algebra and $*$-structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 7.54 (Support projection annihilates the kernel). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \operatorname{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho\, v = 0$, then $P\, v = 0$.*

Proof. ✓ Proof Diagonalize $\rho = U\, \operatorname{diag}(\lambda)\, U^\dagger$ with $\lambda_j \geq 0$. Writing $w := U^\dagger v$, the hypothesis $\rho v = 0$ gives $\lambda_j\, w_j = 0$ for every j; hence $w_j = 0$ whenever $\lambda_j > 0$. The support projection acts as $\operatorname{diag}(\chi_{\lambda_j > 0})$ in the eigenbasis, so $P\, v = U\, \operatorname{diag}(\chi_{\lambda_j>0})\, w = 0$. □

Definition 7.55 (Corner submodule / idempotent-corner identification). Lean ✓ Stmt For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{P\, X\, P \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$-linear spaces by the identity on underlying matrices.

Remark 7.56 ($\mathbb{C}$-algebra and $*$-structure on the corner). Lean Lean Lean Lean ✓ Stmt The corner algebra associated to an idempotent already carries a ring structure with unit P. In the matrix case it also has $\mathbb{C}$-module and $\mathbb{C}$-algebra structures and, under the additional assumption $P^\dagger = P$, star, star ring, and star module structures over $\mathbb{C}$.

Theorem 7.57 (Faithful compression onto the support). Lean ✓ Stmt *Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \operatorname{supp}(\rho)$, and let $V : \mathbb{C}^D \to \mathbb{C}^k$ be a compression isometry satisfying $V\, V^\dagger = \mathbb{1}_k$ and $V^\dagger\, V = P$. Then the compression*

$$V\, \rho\, V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

Proof. ✓ Proof Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$, set $u := V^\dagger w$, and note that $VV^\dagger = \mathbb{1}_k$ forces $u \neq 0$, while $V^\dagger V = P$ forces $Pu = u$. The quadratic form computes as $w^\dagger (V \rho V^\dagger) w = u^\dagger \rho u$, which is non-negative by positive semidefiniteness. If it vanishes, then $\rho u = 0$, and by Lemma 7.54 also $Pu = 0$, contradicting $Pu = u \neq 0$. □

7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

Theorem 7.58 (Fixed-point algebra is semisimple). Lean ✓ Stmt *The adjoint-fixed-point $*$-subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. ✓ Proof The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. Following [Wol12, Theorem 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^* x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so $x^* x$ is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^* x\|^{2^k} = \|(x^* x)^{2^k}\| = 0$, hence $x^* x = 0$, and the C*-identity gives $x = 0$. □

Theorem 7.59 (Abstract Wedderburn–Artin decomposition). Lean ✓ Stmt *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$-subalgebra is $\mathbb{C}$-algebra isomorphic to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C}).$$

Proof. ✓ Proof Combine semisimplicity (Theorem 7.58) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. □

Theorem 7.60 (Fixed-point algebra decomposition with multiplicities). Lean ✓ Stmt *There exist integers n, block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$-algebra isomorphism from the adjoint-fixed-point $*$-subalgebra to*

$$\prod_{k=1}^{n} M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^{n} d_k m_k \leq D.$$

Proof. ✓ Proof Apply the abstract Wedderburn decomposition (Theorem 7.59). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. □

Remark 7.61 (Block form of the fixed-point algebra). Wolf's block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_{k} M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

Theorem 7.62 (Weighted fixed points form a $*$-subalgebra). Lean ✓ Stmt *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $*$-subalgebra of $M_D(\mathbb{C})$.

Proof. ✓ Proof Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.54 to B. The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. □

7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

Theorem 7.63 (Exponential positivity condition). Lean Lean ✓ Stmt *Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A, then E is irreducible (full equivalence* Lean *).*

Theorem 7.64 (Kraus-span equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

Theorem 7.65 (Combined primitivity equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 7.66 (Pairwise primitivity equivalences). Lean Lean Lean Lean ✓ Stmt These are the pairwise equivalences in [Wol12, Theorem 6.8].

Theorem 7.67 (Scalar fixed-point algebra case). Lean ✓ Stmt *In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\mathrm{tr}(\rho X)}{\mathrm{tr}(\rho)}\mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$-subalgebra.*

7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

Definition 7.68 (Block-permutation structure). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι, a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \mathrm{Sym}(\iota)$ (not necessarily a single cycle—in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k)\,T(X), \quad T(X P_k) = T(X)\,T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 7.69 (Multi-cycle block-permutation structure). Lean ✓ Stmt A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ consists of a finite cycle index ι, a per-cycle period $m : \iota \to \mathbb{N}_{>0}$, a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections. For every $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$, the per-cycle cyclic action is

$$T(P_{c,k+1}) = P_{c,k}.$$

The multiplicative-domain factorisations are $T(P_{c,k}\,X) = T(P_{c,k})\,T(X)$ and $T(X\,P_{c,k}) = T(X)\,T(P_{c,k})$.

Lemma 7.70 (Per-cycle corner preservation). Lean ✓ Stmt *For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C})\, P_{c,k}$.*

Proof. ✓ Proof By induction on n, the n-th iterate T^n sends $P_{c,k+n}\,X\,P_{c,k+n}$ to $P_{c,k}\,T^n(X)\,P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k+m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. □

Lemma 7.71 (Common-period corner preservation). Lean ✓ Stmt *If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C})\, P_{c,k}$ of the decomposition.*

Proof. ✓ Proof Write $N = m(c)\, q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Lemma 7.70. □

Definition 7.72 (Flattening to a single-index block-permutation structure). Lean ✓ Stmt A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c\in\iota}\{0, \ldots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.76.

Lemma 7.73 (Peripheral eigenvectors are invertible). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. ✓ Proof The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. □

Lemma 7.74 (Peripheral eigenvalues: product closure). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E, then so is $\mu\nu$.*

Proof. ✓ Proof Take eigenvectors X, Y for μ, ν. The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 7.73), $YX \neq 0$ and $|\mu\nu| = 1$. □

Theorem 7.75 (Peripheral eigenvalues: cyclic structure). Lean ✓ Stmt *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is* not *required). Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \ldots, \gamma^{m-1}\}$.*

Proof. ✓ Proof The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. □

Theorem 7.76 (Peripheral eigenvalues form a cyclic group). Lean ✓ Stmt *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m-th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \ldots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Apply Theorem 7.75 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \operatorname{tr}(P_0) = D$ with $\operatorname{tr}(P_0) \in \mathbb{N}$. □

Theorem 7.77 (Period divides bond dimension). Lean ✓ Stmt *Let E be as in Theorem 7.76, and let γ be a primitive m-th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. ✓ Proof Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 7.76. □

7.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 7.78 (Corner preservation). Lean ✓ Stmt For an orthogonal projection P and linear map E, the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X.

Definition 7.79 (Corner subspace). Lean ✓ Stmt The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 7.80 (Restricted corner map). Lean ✓ Stmt The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 7.81 (Irreducible restriction on a corner). Lean ✓ Stmt Irreducibility of the corner-restricted map.

Definition 7.82 (Rank of an orthogonal projection). Lean ✓ Stmt The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Lemma 7.83 (Compression isometry for an orthogonal projection). Lean ✓ Stmt *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n. The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P.*

Lemma 7.84 (Rank equals trace for an orthogonal projection). Lean ✓ Stmt *The rank of an orthogonal projection P equals its trace: $n = \operatorname{tr} P$.*

Lemma 7.85 (Peripheral unitary eigenvector). Lean ✓ Stmt *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Lemma 7.86 (Powers on a peripheral-unitary orbit). Lean ✓ Stmt *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n.*

Lemma 7.87 (Normalized peripheral unitary). Lean ✓ Stmt *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Lemma 7.88 (Cyclic projections from a peripheral unitary). Lean ✓ Stmt *A primitive m-th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 7.89 (Cyclic decomposition for irreducible Schwarz maps). Lean ✓ Stmt *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Lemma 7.90 (Power maps preserve each cyclic sector). Lean ✓ Stmt *Appropriate powers of the channel preserve each sector corner.*

Lemma 7.91 (Irreducibility of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is irreducible.*

Lemma 7.92 (Primitivity of restricted sector dynamics). Lean ✓ Stmt *Each sector restriction is primitive.*

Lemma 7.93 (Corner invariance at the period). Lean ✓ Stmt *The channel raised to the period preserves every cyclic corner.*

7.14.2 Group structure of the peripheral spectrum

Lemma 7.94 (Peripheral product closure). Lean ✓ Stmt *The product of two peripheral eigenvalues is peripheral.*

Lemma 7.95 (Peripheral inverse closure). Lean ✓ Stmt *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 7.96 (Peripheral cyclic group with period divisibility). Lean ✓ Stmt *The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.*

Theorem 7.97 (Peripheral eigenvalue multiplicity one). Lean ✓ Stmt *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. ✓ Proof Given a peripheral unitary eigenvector U for γ, the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ-eigenvector X. By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot \mathbb{1}$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. □

7.15 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where T_ϕ is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses.

Theorem 7.98 (Complementary transfer-map gap implies trace convergence to 1). Lean ✓ Stmt *Let E be trace-preserving with fixed point ρ ($\mathrm{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.23). If $\rho_{\mathrm{spec}}(E - P) < 1$, then $\mathrm{Tr}(E^n) \to 1$ as $n \to \infty$.*

Proof. ✓ Proof By the power decomposition (Theorem 4.24), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\mathrm{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\mathrm{Tr}(E^n) \to \mathrm{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$, we have $\mathrm{Tr}(P) = \frac{\mathrm{tr}(\rho)}{\mathrm{tr}(\rho)} = 1$. □

Theorem 7.99 (Self-overlap convergence from a complementary transfer-map gap). Lean ✓ Stmt *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

Proof. ✓ Proof By Lemma 7.7 (with $A = B$), $O_{AA}(N) = \mathrm{Tr}(\mathcal{E}_A^N)$. By Theorem 7.98, $\mathrm{Tr}(\mathcal{E}_A^N) \to 1$. □

Lemma 7.100 (Norm of self-overlap convergence). Lean ✓ Stmt *If A is an MPS tensor and $O_{AA}(N) \to 1$ in $\mathbb{C}$ as $N \to \infty$, then $\|O_{AA}(N)\| \to 1$ as $N \to \infty$.*

Proof. ✓ Proof Follows from the continuity of the norm: $O_{AA}(N) \to 1$ implies $\|O_{AA}(N)\| \to \|1\| = 1$. □

Chapter 8

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D, the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

8.1 Cumulative span

Definition 8.1 (Word span). Lean ✓ Stmt The *word span* at length n is the subspace

$$S_n(A) = \mathrm{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

Definition 8.2 (Cumulative span). Lean ✓ Stmt The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 8.3 (Monotonicity of cumulative span). Lean ✓ Stmt *$T_n(A) \le T_{n+1}(A)$ for all n.*

Proof. ✓ Proof T_{n+1} includes all words of length $\le n+1$, which contains all words of length $\le n$. □

Lemma 8.4 (Dimension bound). Lean ✓ Stmt *$\dim T_n(A) \le D^2$ for all n.*

Proof. ✓ Proof $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. □

Theorem 8.5 (Cumulative span stabilisation). Lean ✓ Stmt *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \ge n$.*

Proof. ✓ Proof If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2}, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. □

Theorem 8.6 (Dimension growth). Lean ✓ Stmt *If $T_n(A) \ne T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. ✓ Proof $T_n \le T_{n+1}$ (Lemma 8.3) and $T_n \ne T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. □

Lemma 8.7 (Word span characterises block injectivity). Lean ✓ Stmt *$S_L(A) = M_D(\mathbb{C})$ if and only if A is L-block injective.*

Proof. ✓ Proof Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. □

Lemma 8.8 (Zero-length word span). Lean ✓ Stmt *If $D \geq 2$, then $S_0(A) \neq M_D(\mathbb{C})$.*

Proof. ✓ Proof The zero-length products consist only of the identity, so $S_0(A) = \mathrm{span}_{\mathbb{C}}\{\mathbb{1}\}$ has dimension 1. Since $\dim M_D(\mathbb{C}) = D^2 \geq 4$, this span cannot be all of $M_D(\mathbb{C})$. □

Corollary 8.9 (0-block injectivity). Lean ✓ Stmt *If $D \geq 1$ and A is 0-block injective, then $D = 1$.*

Proof. ✓ Proof By Lemma 8.7, 0-block injectivity gives $S_0(A) = M_D(\mathbb{C})$. Lemma 8.8 excludes $D \geq 2$, hence the nonzero bond dimension must be 1. □

Lemma 8.10 (Two-sided span from one nonzero matrix). Lean ✓ Stmt *If $C \in M_D(\mathbb{C})$ is nonzero, then*

$$\mathrm{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}).$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

Proof. ✓ Proof Choose a nonzero entry C_{pq}. Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the displayed span contains the standard matrix basis. □

Lemma 8.11 (Exact words around a nonzero middle matrix). Lean ✓ Stmt *If A is L-block injective and $X \neq 0$, then*

$$\mathrm{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. ✓ Proof By L-block injectivity, the length-L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the preceding lemma to products with R and S chosen among those word products. □

8.2 Fitting decomposition

Definition 8.12 (Fitting decomposition). Lean ✓ Stmt A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V,
4. the generalized eigenspaces are linearly independent.

Theorem 8.13 (Fitting decomposition exists). Lean ✓ Stmt *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. ✓ Proof Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 8.14 (Nilpotent power bound). Lean ✓ Stmt *If f is nilpotent on a space of dimension n, then $f^n = 0$.*

Proof. ✓ Proof A nilpotent endomorphism on an n-dimensional space has nilpotency index $\leq n$. □

8.3 Nonzero trace product

Theorem 8.15 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). Lean ✓ Stmt *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 8.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0, hence $T_{L_0} = M_D(\mathbb{C})$). □

Theorem 8.16 (Cumulative span reaches $M_D(\mathbb{C})$). Lean ✓ Stmt *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Take n from Theorem 8.15; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

Theorem 8.17 (Nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. ✓ Proof By Theorem 8.16, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\operatorname{tr}(\mathbb{1}) = D \neq 0$). □

Theorem 8.18 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). Lean ✓ Stmt *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \operatorname{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. ✓ Proof The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. □

Theorem 8.19 (Sharp nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2 - \operatorname{kr}(A) + 1$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. ✓ Proof By Theorem 8.18, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. □

Theorem 8.20 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 2$, then there exists a word w of* positive *length $1 \leq |w| \leq D^2 - \operatorname{kr}(A) + 1$ such that $\operatorname{tr}(A^w) \neq 0$. This strengthens Theorem 8.19 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \mathrm{span}\{A^w : 1 \le |w| \le n\}$. For $D \ge 2$, the span $S_0(A) = \mathrm{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\ge d' \ge 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i, so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \ge 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\mathrm{tr}(\mathbb{1}) = D \ge 2 \neq 0$). □

8.4 Eigenvector extraction

Theorem 8.21 (Nonzero trace implies eigenvector)**.** Lean ✓ Stmt *If $M \in M_D(\mathbb{C})$ has $\mathrm{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

Proof. ✓ Proof The trace is the sum of the eigenvalues of M, counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue μ. Choosing a nonzero vector $\varphi \in \ker(M - \mu\mathbb{1})$ gives $M\varphi = \mu\varphi$. □

Theorem 8.22 (Eigenvalue extraction)**.** Lean ✓ Stmt *If A is normal, then there exist a word w_0 with $|w_0| \le D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. ✓ Proof By Theorem 8.17, there is a word w_0 with $\mathrm{tr}(A^{w_0}) \neq 0$ and $|w_0| \le D^2$. By Theorem 8.21, A^{w_0} has a nonzero eigenvector. □

8.5 Eigenvector spreading

Definition 8.23 (Cumulative vector span)**.** Lean ✓ Stmt Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w\varphi : |w| \le n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 8.24 (Vector span stabilisation)**.** Lean ✓ Stmt *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \ge n$.*

Proof. ✓ Proof Same argument as Theorem 8.5: left-multiplication by A^i cannot escape the stabilised span. □

Lemma 8.25 (Vector span from matrix span)**.** Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N. Applying such matrices to φ recovers all of $\mathbb{C}^D$. □

Theorem 8.26 (Eigenvector spreading)**.** Lean ✓ Stmt *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. ✓ Proof The vector span K_n is monotone and $\dim K_n \le D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 8.24). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 8.16, so Lemma 8.25 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. □

8.6 Proof of the cumulative Wielandt bound

Theorem 8.27 (Wielandt chain). Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:*

1. $T_{D^2}(A) = M_D(\mathbb{C})$ *(cumulative span stabilises);*
2. *there exists a word w_0 with $|w_0| \leq D^2$ and $\operatorname{tr}(A^{w_0}) \neq 0$;*
3. *there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\,\varphi$;*
4. *for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Items (1)–(3) follow from Theorems 8.16 and 8.22. Item (4) combines these with the eigenvector spreading bound (Theorem 8.26). □

Theorem 8.28 (Wielandt bound). Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2. This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, stated below as a separate statement.*

Proof. ✓ Proof Immediate from Theorem 8.16. □

Remark 8.29 (Explicit blocking-length estimates). ✓ Stmt Theorem 8.28 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.42 below. The left-canonical normal blocked-injectivity estimate D^4 is proved below in Theorem 8.49. The sharper bi-canonical blocking estimate $3D^5$ from [SPGWC10] is not proved here.

Definition 8.30 (One-step augmentation). Lean ✓ Stmt Let A be an MPS tensor with Kraus family $\{A^i\}_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\widetilde{A}$ with physical dimension $d+1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A:

$$\widetilde{A}^0 = X, \qquad \widetilde{A}^{i+1} = A^i.$$

Lemma 8.31 (Kraus operators lie in the one-step span). Lean ✓ Stmt *Every Kraus operator A^i belongs to the one-step span $S_1(A)$.*

Proof. ✓ Proof This is the length-one word with letter i. □

Lemma 8.32 (Redundant one-step augmentation). Lean Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. Then*

$$S_n(\widetilde{A}) = S_n(A)$$

for every $n \geq 0$.

Proof. ✓ Proof The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. □

Lemma 8.33 (Normality and Kraus rank under one-step augmentation). Lean Lean ✓ Stmt *Let $X \in S_1(A)$, and let $\widetilde{A}$ be the one-step augmentation of A by X. If A is normal, then $\widetilde{A}$ is normal, and*

$$\operatorname{kr}(\widetilde{A}) = \operatorname{kr}(A).$$

Proof. ✓ Proof Since all exact word spans are unchanged by Lemma 8.32, eventual full word span for A gives eventual full word span for $\widetilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. ☐

Theorem 8.34 (Wielandt case (2), invertible generator). Lean Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension D, and suppose that some Kraus operator A^{i_0} is invertible. Then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \le D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since $\dim S_1(A) = \mathrm{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \mathrm{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. ☐

Theorem 8.35 (Wielandt case (2), one-step invertible element). Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \ge 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains an invertible matrix X, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}).$$

In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator, obtaining $\widetilde{A}$. Primitivity with a positive-definite fixed point and normalization give normality of A, hence normality of $\widetilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\widetilde{A}$ is invertible, so Theorem 8.34 gives $S_{D^2-\mathrm{kr}(\widetilde{A})+1}(\widetilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A. ☐

Corollary 8.36 (Wielandt case (2), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.35,*

$$\iota(A) \le D^2 - \mathrm{kr}(A) + 1,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$.

Proof. ✓ Proof The displayed word-span equality gives an admissible value in the defining minimum for $\iota(A)$. ☐

Corollary 8.37 (Wielandt case (2), invertible Kraus operator). Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \ge 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is invertible, then*

$$S_{D^2-\mathrm{kr}(A)+1}(A) = M_D(\mathbb{C}), \qquad \iota(A) \le D^2 - \mathrm{kr}(A) + 1.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.31. ☐

Lemma 8.38 (Rank-one extraction from a non-invertible eigenvector). Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \ge 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exists a vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$,*

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

Proof. ✓ Proof Let r be the nilpotent index of left multiplication by A^{i_0}. The dimension-growth argument gives a level n_0 for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since $A^{i_0}\varphi = \mu\varphi$ and $\mu \neq 0$, iterating gives $(A^{i_0})^r\varphi = \mu^r\varphi$, hence

$$\varphi = \mu^{-r}(A^{i_0})^r\varphi \in \operatorname{range}((A^{i_0})^r).$$

Hence, for every $\psi \in \mathbb{C}^D$,

$$|\varphi\rangle\langle\psi| \in S_{r+n_0}(A).$$

The nilpotent-index estimate gives

$$r + n_0 \leq D^2 - D + 1.$$

Since

$$A^{i_0}|\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu|\varphi\rangle\langle\psi|$$

and $\mu \neq 0$, the equality

$$|\varphi\rangle\langle\psi| = \mu^{-1}A^{i_0}|\varphi\rangle\langle\psi|$$

implies

$$|\varphi\rangle\langle\psi| \in S_k(A) \implies |\varphi\rangle\langle\psi| \in S_{k+1}(A)$$

because left multiplication by A^{i_0} maps $S_k(A)$ into $S_{k+1}(A)$. Iterating from $k = r+n_0$ and using $r + n_0 \leq D^2 - D + 1$ gives

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

□

Theorem 8.39 (Wielandt case (3), one-step non-invertible element)**.** Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then*

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as A. The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D - 1$, and Lemma 8.38 gives the rank-one operators $|\varphi\rangle\langle\psi|$ in length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A. □

Corollary 8.40 (Wielandt case (3), index bound)**.** Lean ✓ Stmt *Under the hypotheses of Theorem 8.39,*

$$\iota(A) \leq D^2.$$

Proof. ✓ Proof The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. □

Corollary 8.41 (Wielandt case (3), non-invertible Kraus operator). Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ, then*

$$S_{D^2}(A) = M_D(\mathbb{C}), \qquad \iota(A) \leq D^2.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.31. □

Theorem 8.42 (Wielandt's inequality — general bound). Lean ✓ Stmt *Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive with a positive-definite fixed point. Then*

$$\iota(A) \leq (D^2 - \mathrm{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. ✓ Proof

1. $D = 1$*:* In this case $\iota(A) = 0$, so the bound holds trivially.

2. $D \geq 2$*:* By Theorem 8.20, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\mathrm{tr}(A^w) \neq 0$, where $d' = \mathrm{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 8.21). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 8.53), and the encoding of w as a single block index gives $B^{i_0} = A^w$.

   (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

   In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \mathrm{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$.

□

Remark 8.43 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.59: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 8.44 (Fixed-length vector span). Lean ✓ Stmt The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \mathrm{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 8.45 (Word span products). Lean ✓ Stmt *$S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m+n$.*

Proof. ✓ Proof Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$ with $|w_1w_2| = m + n$. □

Lemma 8.46 (Eigenvector padding). Lean ✓ Stmt *Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then for all n,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0. The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n. The reverse inclusion $H_n \leq K_n$ is immediate. □

Theorem 8.47 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt *Assume:*

1. *$H_n(A, \varphi) = \mathbb{C}^D$ (length-n word products applied to φ span all of $\mathbb{C}^D$), and*
2. *for each standard basis vector e_j, the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.*

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij}. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 8.45. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. □

Lemma 8.48 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Each length-n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-nL word in the original alphabet $\{0, \dots, d-1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. □

Theorem 8.49 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \leq D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [CPGSV16, Section II, line 332] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [SPGWC10, Theorem 1]. The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.

Proof. ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since A is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound $(D^2 - \operatorname{rank} S_1(A) + 1)D^2$ is at most D^4. The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.48. □

Corollary 8.50 (Uniform injective blocking for separated normal-canonical families). Lean Lean ✓ Stmt *Let $(A_x)_{x\in X}$ be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$ is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.*

Here the family is a basis of normal-canonical representatives in the sense of Definition 10.9; the blocks are distinct and the weight moduli are non-increasing but need not be strictly ordered. It assumes that the representative family has already been selected; it does not reconstruct the full CPSV sector decomposition with repeated copies inside one sector. The restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

Proof. ✓ Proof Apply Theorem 8.49 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. □

Remark 8.51 (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.47 is the second step. Theorem 8.58 supplies the rank-one hypothesis in blocked form, and Theorem 8.59 then gives a blocked, fixed-length word-span saturation result.

Remark 8.52 (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.28 gives cumulative saturation $T_{D^2}(A) = M_D(\mathbb{C})$. This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.59, supplies a length N for which $S_N(A) = M_D(\mathbb{C})$. This is the statement that matches the injectivity of Γ_N and can be transported through blocking by Lemma 8.48. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

Theorem 8.53 (Blocking preserves normality). Lean ✓ Stmt *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. ✓ Proof Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 8.45. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 8.54, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. □

Lemma 8.54 (Chunking: word span embeds into blocked word span). Lean ✓ Stmt *$S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L.*

Proof. ✓ Proof Every length-nL word in the original alphabet can be split into n consecutive chunks of length L, each of which is a single blocked Kraus operator. By induction on n, using the word span product lemma (Lemma 8.45). □

Theorem 8.55 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L, then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0.*

Proof. ✓ Proof By definition, $(A^{[L]})^{j_0} = A^{w_0}$. □

Theorem 8.56 (Blocked fixed-length matrix spanning). Lean ✓ Stmt *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m.*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.
Remark. *The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above gives only the existential conclusion.*

Proof. ✓ Proof Apply Theorem 8.47 to the blocked tensor $A^{[L]}$, which is normal by Theorem 8.53. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 8.55. Transfer back to the original tensor via Lemma 8.48. □

Theorem 8.57 (Blocked fixed-length matrix spanning from word eigenvectors). Lean ✓ Stmt *Suppose A is normal and $L > 0$. If a length-L word has a nonzero eigenvector and a length-L word has a nonzero transpose eigenvector, then there exists N such that $S_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof The rank-one extraction theorem supplies the blocked rank-one element required by Theorem 8.56; apply the blocked assembly theorem. □

Theorem 8.58 (Rank-one extraction for blocked tensor). Lean ✓ Stmt *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0, nonzero vectors φ, ψ, nonzero scalars μ, ν, and $m \in \mathbb{N}$ such that:*

1. *$A^{\sigma_0}\varphi = \mu\varphi$,*
2. *$(A^{\tau_0})^T\psi = \nu\psi$,*
3. *$\varphi\psi^T \in S_m(A^{[N_0]})$.*

Proof. ✓ Proof Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\mathrm{tr}(B^{j_0}) \neq 0$. By Theorem 8.21, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0, a nonzero vector ψ, and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 8.53. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). ☐

Theorem 8.59 (Fixed-length word-span saturation via blocking)**.** Lean ✓ Stmt *If A is a normal MPS tensor with bond dimension D, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. ✓ Proof By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 8.58 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 8.56 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. ☐

Remark 8.60 (Blocking argument for fixed-length spanning). ✓ Stmt Theorem 8.59 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.58) and then applying the fixed-length matrix-spanning theorem (Theorem 8.56). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

8.9 Primitive MPS tensors

Definition 8.61 (Primitive MPS tensor)**.** Lean ✓ Stmt Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ, then $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires $\rho > 0$; see Remark 8.67. The stronger positive-definite hypothesis is imposed separately in Lemma 8.68 and Theorem 8.78.

Theorem 8.62 (Primitive overlap convergence)**.** Lean ✓ Stmt *If A is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

Proof. ✓ Proof Immediate from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.99). ☐

8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.61: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1.

Lemma 8.63 (Primitive MPS tensors have nonzero word products)**.** Lean ✓ Stmt *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length* exactly *n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w\rho(A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. ☐

Lemma 8.64 (Iterated transfer does not vanish). Lean ✓ Stmt *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Lemma 8.63, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. □

8.10.1 From primitivity to normality

The primitivity condition of Definition 8.61 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

Lemma 8.65 (Fixed-point uniqueness from a complementary transfer-map gap). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ, then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ: $\sigma = \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Then $\operatorname{tr}(\sigma') = 0$ and $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $\mathcal{E}_A - P_\rho$, contradicting the complementary transfer-map gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. □

Lemma 8.66 (Complement powers tend to zero). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. □

Remark 8.67 (Primitivity: transfer-map gap vs. positive definiteness). ✓ Stmt The primitivity condition used in Section 8.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ, and the complementary transfer-map gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the transfer-map gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \operatorname{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Lemma 8.68 gives irreducibility. Together with the Burnside argument below (Section 8.10.3) and Theorem 8.84, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \operatorname{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 8.79. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

8.10.2 Primitive implies irreducible

Lemma 8.68 (Primitive tensors with positive-definite fixed point are irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV16, Section 2.3]).*

Proof. ✓ Proof By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^i P = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).

2. *Convergence:* By Lemma 8.66, $\mathcal{E}_A^n(\sigma) \to \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho = 0$.

3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\mathrm{tr}(P\rho P) > 0$, so $\frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.

4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

8.10.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.16) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 8.69 (Algebra span). Lean ✓ Stmt For an MPS tensor A, let $\mathrm{alg}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}_{i=0}^{d-1}$:

$$\mathrm{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

Definition 8.70 (Invariant submodule). Lean ✓ Stmt A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i.

Definition 8.71 (Irreducible action). Lean ✓ Stmt The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.72 (Irreducible tensor implies irreducible action). *Lean* *✓ Stmt* *If A is irreducible as a tensor (Definition 9.16), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.71).*

Proof. ✓ Proof Let $W \leq \mathbb{C}^D$ be an A-invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i. Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. □

Theorem 8.73 (Burnside's theorem for Kraus algebras). *Lean* *✓ Stmt* *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. ✓ Proof Transport $\mathrm{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. □

Remark 8.74. ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 8.75 (Irreducible action implies eventual cumulative spanning). Lean ✓ Stmt *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.73, $\mathrm{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \cdots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. □

8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect primitivity in Definition 8.61, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Lemma 8.76 (Primitive tensors with positive-definite fixed point have irreducible transfer map). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Lemma 8.65): every PSD fixed point is proportional to ρ, so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. ✓ Proof By Lemma 8.65, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. □

Lemma 8.77 (Primitive tensors with positive-definite fixed point are strongly irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.*

Proof. ✓ Proof Combine Lemma 8.76 (irreducibility) with the peripheral-spectrum primitivity from the transfer-map gap hypothesis (Theorem 8.62). □

Theorem 8.78 (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ, then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

This is the key connection between primitivity in Definition 8.61, formulated via a complementary transfer-map gap, and normality; it removes the need for the aperiodicity-based argument.

Proof. ✓ Proof By Lemma 8.77, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.65 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). □

8.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

Remark 8.79 (Aperiodicity condition). ✓ Stmt We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}_{i=0}^{d-1}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply

$\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.65) without passing through aperiodicity.

Remark 8.80 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\mathrm{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n, and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \mathrm{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 8.81 (Word span monotonicity from aperiodicity). Lean ✓ Stmt *If* $\mathbb{1} \in S_1(A)$, *then* $S_n(A) \leq S_{n+1}(A)$ *for all* n.

Proof. ✓ Proof For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 8.45. □

Theorem 8.82 (Cumulative span equals word span under aperiodicity). Lean ✓ Stmt *If* $\mathbb{1} \in S_1(A)$, *then* $T_n(A) = S_n(A)$ *for all* n.

Proof. ✓ Proof By Theorem 8.81, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term S_n. □

Theorem 8.83 (Cumulative spanning plus aperiodicity implies normality). Lean ✓ Stmt *If* $T_N(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$, *then* A *is normal.*

Proof. ✓ Proof By Theorem 8.82, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Theorem 8.84 (Algebra span plus aperiodicity implies normality). Lean ✓ Stmt *If* $\mathrm{alg}(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$, *then* A *is normal.*

Proof. ✓ Proof Since $\mathrm{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.83. □

Theorem 8.85 (Irreducible tensor plus aperiodicity implies normality). Lean ✓ Stmt *If* A *is an irreducible tensor,* $D > 0$, *and* $\mathbb{1} \in S_1(A)$, *then* A *is normal. This assembles the full chain:*

$$\textit{irreducible tensor} \xrightarrow{8.72} \textit{irreducible action} \xrightarrow{\textit{Burnside}} \mathrm{alg}(A) = M_D(\mathbb{C}) \xrightarrow{8.84} \textit{normal}.$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. The Burnside step is supplied by Theorems 8.72 and 8.73.

Proof. ✓ Proof By Theorem 8.72, A acts irreducibly on $\mathbb{C}^D$. By Theorem 8.73, $\mathrm{alg}(A) = M_D(\mathbb{C})$. By Theorem 8.84 with the aperiodicity hypothesis, A is normal. □

Remark 8.86 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in Section 8.11 are needed only for the block-injective argument through Definition 2.31, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV16]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

Theorem 8.87 (TP-primitive irreducible tensor is injective after blocking)**.** Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible tensor, and primitive transfer map. Then some positive blocking $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof If $D = 1$, trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans $M_1(\mathbb{C})$; hence A is one-site injective, so Lemma 2.30 gives the positive witness $L = 1$. Suppose now that $D \geq 2$. The normality theorem (Theorem 9.46) gives a block-injectivity witness L. If $L = 0$, Corollary 8.9 would force $D = 1$, a contradiction. Thus $L > 0$, and the blocked-chain equivalence identifies this L-block injectivity with one-site injectivity of the blocked tensor $A^{[L]}$. □

Remark 8.88. See Theorem 8.53 for the formalized statement that blocking preserves normality.

Chapter 9

Canonical Form Reduction

This is the first of three chapters that together prove the Fundamental Theorem of matrix product states. Here an arbitrary translation-invariant MPS tensor is reduced to a weighted family of primitive blocks; Chapter 10 constructs the basis of normal tensors carried by those blocks and compares their coefficients, and Chapter 11 proves the theorem itself. The reduction follows [PGVWC07], [CPGSV16], [CPGSV21], and [Wol12, Chapter 6]. Its goal is to expose the primitive blocks inside the tensor. The transfer operator of an arbitrary tensor can be reducible and can have a period greater than one; the reduction splits it along its invariant subspaces and brings the period of each block down to one, so that each block is primitive, with peripheral spectrum $\{1\}$. The blocks are returned in a fixed gauge, and the sections below carry out the reduction.

Two normalizations appear. The unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ is used in [PGVWC07, Theorem 4]; the trace-preserving (left-canonical) orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, whose transfer map is the Frobenius adjoint of $\mathcal{E}_A$.

Theorem 9.1 (Translation-invariant canonical form)**.** Lean ✓ Stmt *Let A be a translation-invariant MPS representation of bond dimension D. There is a positive scalar s and a finite family of blocks A_k, possibly empty, such that the globally rescaled tensor $s^{-1}A$ has the same MPV coefficients on every positive-length ring as*

$$\bigoplus_k \mu_k A_k^i.$$

The weights are positive and satisfy $1 \geq |\mu_k|$; if the family is nonempty, then some weight has norm 1. For each block,

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \qquad \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k,$$

where Λ_k is diagonal, positive, and full-rank. Each block has positive bond dimension. Moreover the only fixed points of $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ are the scalar multiples of $\mathbb{1}$: for every X,

$$\mathcal{E}_{A_k}(X) = X \Longrightarrow \exists\, c \in \mathbb{C}, X = c\,\mathbb{1}.$$

The total bond dimension of the decomposition is at most D.

This is the positive-length form of [PGVWC07, Theorem 4], with the global spectral-radius normalization step made explicit.

Remark 9.2 (Source proof order). The proof follows [PGVWC07, Theorem 4] in order: normalize the spectral radius and use a full-rank positive fixed point to obtain the unital gauge; derive the invariant support projection from a singular positive fixed point by the positivity contradiction; split the finite-ring trace into the supported and orthogonal summands; iterate, using a non-scalar fixed point to split further until each block has only scalar fixed points; and finally repeat the fixed-point argument for the dual map and diagonalize the resulting full-rank positive fixed point by a unitary gauge. The primitive-block results below are used only after their hypotheses have been obtained from this argument.

Remark 9.3 (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV16, Section 2.3] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking by some period $p \geq 1$, the direct sum of an all-zero summand and a weighted family of normal tensors $\bigoplus_k \mu_k B_k$ with $|\mu_k| \leq 1$ and at least one $|\mu_k| = 1$. The closest proved results here are Theorem 9.47 and Theorem 10.40; the source upper normalization $|\mu_k| \leq 1$ with a unit-modulus witness is the one remaining input, recorded in Remark 10.41.

9.1 Normal tensors and canonical forms

This section records the CPSV definitions of *normal tensor* and *canonical form* from [CPGSV16]. The basis of normal tensors is stated in Chapter 10. Write

$$\sigma_\partial(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\}$$

for the peripheral spectrum of a transfer map $\mathcal{E}$, where $r(\mathcal{E})$ is its spectral radius. The stronger predicates used later (Definition 9.25 and the canonical-form predicate of Chapter 10) add left-canonical normalization and spectral primitivity. The weight moduli are taken non-increasing, not strictly decreasing; equal-modulus and repeated-copy sectors are retained and compared by the coefficient argument of Chapter 10.

Definition 9.4 (CPSV normal tensor (NT)). Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is a *normal tensor* if, after the spectral-radius normalization of [CPGSV16], (i) A admits no nontrivial invariant orthogonal projection and (ii) the associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has peripheral spectrum $\sigma_\partial(\mathcal{E}_A) = \{1\}$.

Remark 9.5. Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum $\{1\}$ while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

The CPSV *canonical form* (CF) of a tensor A is the normal-block decomposition

$$A^i \cong \bigoplus_{k=1}^{r} \mu_k A_k^i$$

with weights normalized by $|\mu_k| \leq 1$ and with at least one $|\mu_k| = 1$; see [CPGSV16, Section 2.3]. In the statements below the structural part is supplied by the stronger normal canonical form of Definition 9.25, and the global weight normalization is invoked as a separate source hypothesis where needed.

Remark 9.6 (Relation to the strong predicates). A tensor A which is both irreducible (Definition 9.16) and has primitive transfer map is normal in the sense of Definition 9.4. The reverse implications from the stronger predicates of Definition 9.25 and of Chapter 10 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

9.1.1 Normalization conventions

Remark 9.7 (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar ζ rescales by $|\zeta|^2$; unit-modulus phase rescaling preserves $\sum_i (A^i)^\dagger A^i = \mathbb{1}$; and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$ with $\eta_k := \mu_k/|\mu_k|$.

Left-canonical normalization is required before the blocks can be compared. Without it the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k,\ \mathcal{V}(A_k) = \mathcal{V}(B_k)$ already fails for two scalar blocks: for $\mu = (1, 2)$, $A = (1, 3/2)$, $B = (3, 1/2)$ the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 10, which requires no ordering of the weight moduli.

9.2 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of $\mathcal{E}_A$ that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

Theorem 9.8 (Unitary conjugation preserves the MPV family). Lean ✓ Stmt *If U is a unitary matrix, then A and $U^\dagger AU$ generate the same MPV family.*

Proof. ✓ Proof By cyclicity of the trace, $\mathrm{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \mathrm{tr}(A^{i_1} \cdots A^{i_N})$. □

Theorem 9.9 (Invariant projection gives a two-block decomposition). Lean ✓ Stmt *Suppose P is an orthogonal projection satisfying, for every physical index i,*

$$(\mathbb{1} - P)A^i P = 0.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ Proof Diagonalize P by a unitary U. Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 9.8. In this basis the relation $(\mathbb{1} - P)A^i P = 0$ makes each letter upper block triangular. Write

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}$$

for the conjugated letters in this split basis. Since $B^{i_1} \cdots B^{i_N}$ is again upper block triangular, its trace satisfies

$$\mathrm{tr}(B^{i_1} \cdots B^{i_N}) = \mathrm{tr}(A_1^{i_1} \cdots A_1^{i_N}) + \mathrm{tr}(A_2^{i_1} \cdots A_2^{i_N}).$$

Hence the block-diagonal tensor $A_1 \oplus A_2$ has the same MPV coefficients as the conjugated tensor. □

Theorem 9.10 (Nontrivial invariant projection gives strict two-block reduction). Lean ✓ Stmt *Under the hypotheses of Theorem 9.9, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2, such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.*

Proof. ✓ Proof Apply Theorem 9.9. In the diagonal form of P, the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. □

Definition 9.11 (Support projection). Lean ✓ Stmt For a positive semidefinite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ. Via the spectral decomposition $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$,

$$P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger,$$

where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 9.12 (Invariant projection predicate). Lean ✓ Stmt An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^i P = 0$ for all i. A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.16).

Theorem 9.13 (PSD fixed point gives invariant projection). Lean ✓ Stmt *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be its support projection. Then $(\mathbb{1} - P)A^i P = 0$ for all i: the range of P is invariant under every A^i.*

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, one gets $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i. Since ρ is supported on the range of P, this forces $(\mathbb{1} - P)A^i P = 0$.

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1]. □

Theorem 9.14 (Two-block decomposition). Lean ✓ Stmt *If $\mathcal{E}_A$ has a nonzero PSD fixed point ρ that is not positive definite, then there exist MPS tensors A_1, A_2 of smaller bond dimensions with $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.*

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not positive definite). Theorem 9.13 gives $(\mathbb{1} - P)A^i P = 0$, so a unitary change of basis block-diagonalizing P puts A^i in upper-triangular block form with diagonal blocks B^i_{11}, B^i_{22}. Theorem 9.8 preserves the MPV under the conjugation, and the trace of an upper-triangular product equals the sum of the diagonal-block traces, so deleting the off-diagonal blocks preserves all MPV coefficients. □

Theorem 9.15 (Non-scalar fixed point gives a further split). Lean Lean ✓ Stmt *Suppose A is unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If X is a Hermitian fixed point of $\mathcal{E}_A$ which is not a scalar multiple of $\mathbb{1}$, then there is a nonzero PSD fixed point ρ which is not positive definite, and consequently A admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than D. This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem 4].*

Proof. ✓ Proof Let $\lambda_{\max}$ be the largest eigenvalue of X. Then $\rho := \lambda_{\max}\mathbb{1} - X$ is positive semidefinite, singular, and nonzero (since X is not scalar), and unitality with the fixed-point equation for X gives $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 9.14. The non-scalar hypothesis is needed because scalar multiples of $\mathbb{1}$ are fixed by every unital map and yield no nontrivial support projection. □

9.2.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting on bond dimension decomposes any tensor into irreducible blocks; the remaining lemmas record the spectral facts about irreducible unital blocks and the algebra-spanning input used later.

Definition 9.16 (Irreducible tensor). Lean ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 9.17 (Irreducible block decomposition). Lean ✓ Stmt *Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ with each A_k irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.*

Proof. ✓ Proof By strong induction on D. If A is irreducible, take $r = 1$. Otherwise a nontrivial invariant projection P gives the upper-triangular splitting of Theorem 9.14; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting. $\square$

Theorem 9.18 (Scalar fixed points in an irreducible unital block). Lean ✓ Stmt *If A is irreducible and unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and $\mathcal{E}_A(X) = X$, then X is a scalar multiple of $\mathbb{1}$. This is the final fixed-point assertion in the proof of [PGVWC07, Theorem 4].*

Proof. ✓ Proof Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar. $\square$

Theorem 9.19 (Unital gauge from a full-rank Perron–Frobenius eigenvector). Lean Lean Lean ✓ Stmt *Let ρ be positive definite with $\mathcal{E}_A(\rho) = r\rho$ for some $r > 0$. For $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$,*

$$\sum_i B^i(B^i)^\dagger = \mathbb{1}.$$

If $r = 1$, then the unscaled gauge $B^i := \rho^{-1/2}A^i\rho^{1/2}$ is unital and generates the same MPV family as A. This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem 4].

Proof. ✓ Proof The positive square root of ρ is invertible. Substituting $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ into $\sum_i B^i(B^i)^\dagger$ and using $\mathcal{E}_A(\rho) = r\rho$ gives the identity. For $r = 1$ the change from A to B is a similarity, so cyclicity of the trace gives equality of all MPV coefficients. $\square$

Theorem 9.20 (Unitary diagonal positive fixed point in TP gauge). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1}, \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. ✓ Proof The TP hypothesis makes $\mathcal{E}_A$ a channel with a nonzero PSD fixed point. Irreducibility makes $\mathcal{E}_A$ an irreducible CP map, and Theorem 6.4 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization. $\square$

The passage from irreducible tensors to full algebra spanning uses Burnside's theorem: an irreducible subalgebra of $M_D(\mathbb{C})$ is all of $M_D(\mathbb{C})$. It is developed alongside the Wielandt material in Chapter 8; see Theorem 8.75.

Lemma 9.21 (The algebra span reaches a finite cumulative span). Lean ✓ Stmt *If the generated unital algebra satisfies* $\mathrm{alg}(A) = M_D(\mathbb{C})$*, then there exists* N *such that* $T_N(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Every element of $\mathrm{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n\geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \cdots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. □

Lemma 9.22 (Irreducible action gives an irreducible tensor). Lean ✓ Stmt *If the letters of* A *act irreducibly on* $\mathbb{C}^D$*, then* A *has no nontrivial invariant orthogonal projection.*

Proof. ✓ Proof If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A. Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. □

9.3 Normal canonical form

This section fixes the two canonical-form predicates used downstream: the block-injective form of Definition 9.23 and the spectral *normal* canonical form of Definition 9.25. It then records how the self-overlap clause and gauge-phase equivalence follow from primitivity, and closes with the periodicity-removal step that produces primitive transfer maps.

Recall from Definition 2.48 and Lemma 7.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma\in\{0,\ldots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \mathrm{Tr}(F_{AB}^N).$$

This notation enters the self-overlap clause of the canonical-form predicate below.

Definition 9.23 (Canonical form conditions). Lean ✓ Stmt A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \to 1$ for each k.

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7(3)]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the transfer-operator gap proof.

Remark 9.24 (Normalization convention). [PGVWC07, Theorem 4] uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$; here the blocks are in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 6.14).

Definition 9.25 (Normal canonical form conditions). Lean ✓ Stmt Scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfy the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each $\mathcal{E}_{A_k}$ is primitive (equivalently, $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$),
4. $|\mu_1| \geq |\mu_2| \geq \cdots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here "normal" is the spectral sense of [CPGSV16], which by [SPGWC10, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.31.

Theorem 9.26 (Self-overlap is derived in normal canonical form). Lean ✓ Stmt *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \to 1$ for every block k.*

Proof. ✓ Proof Fix a block k. Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_{A_k}$ with fixed-point projector P; primitivity gives the complementary gap $\rho_{\mathrm{spec}}(\mathcal{E}_{A_k} - P) < 1$ (Theorem 4.51). Theorem 7.99 then yields $O_{A_k A_k}(N) \to 1$. □

Remark 9.27. See Theorem 7.29 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

Lemma 9.28 (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt *Let A and B have the same bond dimension. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then the mixed transfer map has spectral radius at least 1.*

Proof. ✓ Proof If the mixed transfer spectral radius were < 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace; via $\mathrm{Tr}(F_{AB}^N) = \langle V^{(N)}(A)|V^{(N)}(B)\rangle$ the overlap would tend to 0, contradicting the hypothesis. □

Theorem 9.29 (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt *Let A and B be irreducible MPS tensors of the same bond dimension with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $\|\langle V^{(N)}(A)|V^{(N)}(B)\rangle\| \to 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof By Lemma 9.28 the mixed transfer spectral radius is at least 1, and Theorem 7.29 then gives gauge-phase equivalence. □

9.3.1 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have peripheral spectrum larger than $\{1\}$, namely the roots of unity coming from its period. Blocking by the least common multiple of those periods raises the transfer map to a power whose peripheral spectrum is $\{1\}$, removing the periodicity. In [CPGSV21] this is the passage from irreducibility to primitivity by blocking to period one.

Theorem 9.30 (Blocking yields a peripheral-spectrum primitive transfer map). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$.*

Proof. ✓ Proof Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, which is unital with irreducible Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint of K, so Theorem 4.46 shows every peripheral eigenvalue is a root of unity. Since $\sigma_\partial(\mathcal{E}_A)$ is finite (Theorem 4.29), Lemma 4.33 gives a common exponent p with $\mu^p = 1$ for all $\mu \in \sigma_\partial(\mathcal{E}_A)$, and Lemma 4.34 gives $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$. Blocking by p takes the pth power of the transfer map, so $\mathcal{E}_{A[p]}$ is primitive. □

Remark 9.31. Theorem 9.30 is the periodicity-removal-by-blocking step of [CPGSV16, Appendix A] for an irreducible block already in TP gauge.

9.4 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits the PGVWC07 unital orientation together with a diagonal positive-definite dual fixed point (Theorem 9.32, the Perron–Frobenius gauge of Chapter 6, and Theorem 9.18); the full PGVWC07 canonical form is Theorem 9.1.

Theorem 9.32 (Left-canonical normalization for irreducible TP blocks)**.** Lean ✓ Stmt *Let A be an irreducible MPS tensor in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*
2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*
3. *Λ is diagonal and positive definite,*
4. *A and B generate the same MPV family.*

This is the left-canonical normalization step (Canonical Form II) of [CPGSV16, Appendix A]. The unitary conjugation is a second step inside TP gauge, performed only after TP normalization.

Proof. ✓ Proof Theorem 9.20 provides the unitary and the diagonal PD fixed point, and unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.8). □

Theorem 9.33 (Dual diagonalization for irreducible unital blocks)**.** Lean ✓ Stmt *Let A be an irreducible MPS tensor in unital gauge $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}, \qquad \sum_i (B^i)^\dagger \Lambda B^i = \Lambda,$$

and A, B generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem 4].

Proof. ✓ Proof Apply Theorem 9.32 to the conjugate-transposed family $i \mapsto (A^i)^\dagger$: unitality of A is trace preservation for this family, whose transfer map is the dual of $\mathcal{E}_A$. Translating back gives the displayed equations. □

Lemma 9.34 (Common scalar rescaling of block weights)**.** Lean ✓ Stmt *Multiplying every block weight in a block-diagonal MPS tensor by a scalar c multiplies every length-N MPV coefficient by c^N.*

Proof. ✓ Proof Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor c^N out of the sum. □

Theorem 9.35 (Normal canonical form from a primitive block decomposition)**.** Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_k \mu_k A_k$, and that each A_k is irreducible, satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, and has primitive transfer map, all $\mu_k \neq 0$, and all bond dimensions are positive. Then there exist a blocking length $p > 0$, weights (ν_k), and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.*

Proof. ✓ Proof The given blocks are already primitive, so take $p = 1$. Reorder the blocks by non-increasing weight modulus, with ties allowed, and set $\nu_k := \mu_k$ and $B_k := A_k$ in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.25 and generates the same MPV family as A. □

Remark 9.36 (Scope of Theorem 9.35)*.* Theorem 9.35 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV16, Section 2.3]; that primitive-block reduction, including the zero-tail count, is Theorem 9.47 in Section 9.6.2.

9.5 Zero-block separation and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. An all-zero block contributes $V^{(N)} = 0$ for every $N \geq 1$, so it adds nothing to the positive-length MPV family; at $N = 0$ the MPV coefficient is the bond dimension. The all-zero summand therefore enters the output not as a tensor but as a single count D_0 in the length-zero identity $D = D_0 + \sum_k D_k$, and all positive-length content is an equality of MPV families. This is the positive-length convention used throughout the reduction: comparisons of weighted block sums are stated for $N \geq 1$, with the zero block carried only by D_0. After separating the zero summand, each nonzero block is placed in the trace-preserving gauge, the furthest unconditional reduction available before periodicity removal.

Theorem 9.37 (Zero-block separation)**.** Lean ✓ Stmt *Every MPS tensor A admits a block decomposition into a zero-block bond dimension D_0 and a finite family of nonzero blocks $(B_k)_{k=1}^r$, each irreducible, with at least one nonzero Kraus operator, and of positive bond dimension, recorded by two facts. The positive-length equality of MPV families: for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r B_k)_\sigma.$$

The length-zero dimension identity

$$D = D_0 + \sum_{k=1}^{r} D_k,$$

where D_0 is the total bond dimension of the all-zero blocks.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 9.17) and partition the blocks by whether each has a nonzero Kraus operator; D_0 accumulates the bond dimensions of the all-zero blocks. For every all-zero irreducible block C_j and every $N \geq 1$, $V^{(N)}(C_j)_\sigma = 0$, so

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r B_k)_\sigma.$$

At $N = 0$ the MPV coefficient is the bond dimension, giving $D = D_0 + \sum_{k=1}^{r} D_k$. □

Theorem 9.38 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ Stmt *Suppose A generates the same MPV family as $\bigoplus_{k=1}^{r} B_k$, where each B_k is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights $(\mu_k)_{k=1}^{r}$ and blocks $(C_k)_{k=1}^{r}$ such that, for every N and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^{r} \mu_k C_k)_\sigma = \sum_{k=1}^{r} \mu_k^N V^{(N)}(C_k)_\sigma,$$

each C_k irreducible and left-canonical, $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$, with positive bond dimension. This is the blockwise form used in the arbitrary-tensor trace-preserving gauge below.

Proof. ✓ Proof The Perron–Frobenius TP-gauge construction is applied to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights μ_k. □

9.5.1 Trace-preserving gauge for arbitrary tensors

Theorem 9.39 (TP-gauge reduction for arbitrary tensors with zero-block separation). Lean ✓ Stmt *For any MPS tensor A, there exist a zero-block bond dimension D_0 and nontrivial blocks $(B_k)_{k=1}^{r}$ with nonzero weights $(\mu_k)_{k=1}^{r}$, each block irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), and of positive bond dimension, such that for every $N \geq 1$ and every σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^{r} \mu_k B_k)_\sigma,$$

and the zero-block contribution is recorded by the length-zero identity

$$D = D_0 + \sum_{k=1}^{r} D_k.$$

Proof. ✓ Proof Theorem 9.37 supplies D_0 and the nonzero irreducible blocks $(C_k)_{k=1}^{r}$. For every $N \geq 1$ and every σ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^{r} C_k)_\sigma.$$

The blockwise TP-gauge theorem then replaces each C_k by a left-canonical block B_k with weight μ_k. Again for every $N \geq 1$ and every σ,

$$V^{(N)}(\textstyle\bigoplus_{k=1}^{r} C_k)_\sigma = V^{(N)}(\bigoplus_{k=1}^{r} \mu_k B_k)_\sigma.$$

The length-zero identity $D = D_0 + \sum_{k=1}^{r} D_k$ is preserved. □

9.6 Blocking, period removal, and primitive blocks

Blocking by an integer p carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. The first consequence is the existence of a common blocking period under which every transfer map in a finite family becomes primitive; the period-removal and primitive-block reductions then proceed from that common period.

Theorem 9.40 (Common blocking period for a block family). Lean ✓ Stmt *Let $(B_k)_{k=1}^{r}$ be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period p_k making its transfer map primitive. Then $p = \mathrm{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k.*

Proof. ✓ Proof Each p_k divides p, so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ carries over to $\mathcal{E}_{B_k^{[p]}}$ via the divisibility identity for transfer-map powers. □

The following stability statements are used when a primitive irreducible block is blocked again for a common-refinement or injectivity length. This later blocking is distinct from the period-removal length that produced the sector block.

Theorem 9.41 (Blocking preserves irreducibility for primitive irreducible blocks)**.** Lean ✓ Stmt *Let A be left-canonical, irreducible, primitive, of positive bond dimension. For every $P > 0$, the blocked tensor $A^{[P]}$ is irreducible: there is no orthogonal projection Q with $Q \neq 0$, $Q \neq \mathbb{1}$ such that, for every P-blocked index α,*

$$(\mathbb{1} - Q)(A^{[P]})^{\alpha} Q = 0.$$

Proof. ✓ Proof The blocked transfer map is the corresponding positive power of the original one:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X).$$

Hence a positive-definite fixed point of $\mathcal{E}_A$ remains fixed by $\mathcal{E}_{A^{[P]}}$. Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence its transfer map is irreducible, which is the irreducibility criterion for $A^{[P]}$. □

Theorem 9.42 (Extra blocking of primitive irreducible sector blocks)**.** Lean ✓ Stmt *Let A be trace-preserving, primitive, and tensor-irreducible. For every $k > 0$,*

$$\sum_{\alpha} ((A^{[k]})^{\alpha})^{\dagger} (A^{[k]})^{\alpha} = \mathbb{1}, \qquad \sigma_{\partial}(\mathcal{E}_{A^{[k]}}) = \{1\},$$

and $A^{[k]}$ is tensor-irreducible. This k is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. ✓ Proof Trace preservation follows from blocking a trace-preserving tensor, primitivity from a positive power of a primitive transfer map, and irreducibility from Theorem 9.41. □

9.6.1 Period removal by cyclic sectors

In the proof of [CPGSV17a] the required step is periodicity removal by blocking: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the result has no nontrivial p-periodic vectors [CPGSV16, Section 2.3]. The cyclic-sector decomposition comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6].

Theorem 9.43 (Period removal by blocking)**.** Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period-removing length $m \geq 1$ such that $A^{[m]}$ admits:*

1. *left-canonical sector tensors $(C_k)_{k=1}^m$ of nonzero bond dimension,*
2. *a unit-weight direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$,*
3. *orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,*
4. *commutation relations $P_k A_i^{[m]} = A_i^{[m]} P_k$ for every blocked letter i,*
5. *the sector trace formula $f_{C_k}(\sigma) = \operatorname{tr}(P_k A_{\sigma_1}^{[m]} \cdots A_{\sigma_N}^{[m]})$,*

6. *linear isomorphisms onto the compression corners* $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,
7. *transfer-intertwining identities* $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$,
8. *multiplicativity and adjoint compatibility* $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$ *and* $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$.

This is the cyclic-sector part of the period-removal step of [CPGSV16, Section 2.3]; the projections P_k *come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).*

Proof. ✓ Proof The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these with the adjoint transfer map of $A^{[m]}$. Apply the block decomposition from adjoint-fixed projections. □

Theorem 9.44 (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let* A *be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum* $\{\gamma^j : 0 \le j < m\}$ *for a primitive* m*th root of unity* γ*, with blocked cyclic-sector decomposition* $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$*, and linear isomorphisms* $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$*. Assume explicitly that the* P_u *are orthogonal projections summing to* $\mathbb{1}$ *and satisfying* $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$*, that each sector has nonzero bond dimension, and that the compression maps satisfy*

$$\varphi_u(\mathcal{E}_{C_u^\dagger}(X)) = \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \qquad \varphi_u(XY) = \varphi_u(X)\varphi_u(Y), \qquad \varphi_u(X^\dagger) = \varphi_u(X)^\dagger.$$

Here $P_u A^{[m]}$ *denotes the blocked tensor with letters* $P_u(A^{[m]})^\alpha$*. Then every sector tensor satisfies* $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$ *and is tensor-irreducible.*

Proof. ✓ Proof Theorem 9.62 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u. The compression equivalences φ_u intertwine the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \le j < m\} = \{1\},$$

because γ is a primitive mth root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across φ_u; then use Theorem 9.54 to pass from the adjoint blocked map to the primal transfer map of C_u. □

Theorem 9.45 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let* A *be a left-canonical irreducible tensor. Then there is a period* $m \ge 1$ *and a unit-weight decomposition of* $A^{[m]}$ *into sector blocks* $(C_k)_{k=1}^m$*, each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of* $A^{[m]}$ *already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.61).*

Proof. ✓ Proof Left-canonicality makes $K_i := (A^i)^\dagger$ unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.43 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 9.44 to the sector blocks. □

9.6.2 Reduction to primitive blocks

This subsection completes the reduction after period removal. A TP-primitive-irreducible block is normal, and an arbitrary tensor becomes, after a positive blocking length, a weighted family of left-canonical primitive blocks with the zero-block dimension recorded separately. For two tensors with the same MPV family, the cyclic sectors can be expressed over one common blocked alphabet, which is the form used by the Fundamental Theorem.

Theorem 9.46 (TP-primitive irreducible tensor is normal). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. ✓ Proof Irreducibility upgrades the PSD fixed point to positive definite (Theorem 6.4), and Theorem 8.78 then gives normality. □

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of primitive irreducible blocks.

Theorem 9.47 (Arbitrary tensor to primitive block decomposition). Lean ✓ Stmt *For any MPS tensor A, there exist $p \geq 1$, a zero-tail bond dimension D_0, nonzero weights $(\mu_k)_{k=1}^r$, and blocks $(B_k)_{k=1}^r$ such that, for every blocked σ of positive length N,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The zero-block contribution is recorded separately by the length-zero identity $D_0 + \sum_{k=1}^r D_k = D$. Each block is left-canonical and primitive,

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}, \qquad \sigma_\partial(\mathcal{E}_{B_k}) = \{1\},$$

and each bond dimension is positive.

Proof. ✓ Proof Theorem 9.39 supplies the zero-block dimension, the left-canonical nontrivial blocks, the positive-length equality, and the identity $D = D_0 + \sum_k D_k$. Theorem 9.40 provides a common blocking period p making all transfer maps primitive, Lemma 9.60 keeps the blocked weights nonzero, and Lemma 9.58 carries the positive-length equality through blocking. □

Theorem 9.48 (Cyclic sectors after the zero-block split). Lean ✓ Stmt *If A and B generate the same MPV family, then after the zero-block split and the trace-preserving gauge there are weighted nonzero-block tensors $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ with, for every $N \geq 1$ and σ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma,$$

and the two nonzero parts agree:

$$V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

Each nonzero block has primitive irreducible cyclic sectors: for each j, there is $m_j \geq 1$ with a unit-weight sector decomposition $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$, and similarly for B_k.

Proof. ✓ Proof Apply Theorem 9.39 separately to A and B. Its positive-length conclusion gives, for $N \geq 1$,

$$V^{(N)}(A)_\sigma = V^{(N)}(\textstyle\bigoplus_j \mu_j^A A_j)_\sigma, \qquad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

Chaining these through $V(A) = V(B)$ identifies the two nonzero parts. Finally apply Theorem 9.45 to every trace-preserving irreducible nonzero-weight block. □

Theorem 9.49 (Common blocking length for cyclic sectors). Lean ✓ Stmt *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length p. For every $N \geq 1$ and every blocked σ, each blocked tensor equals its powered nonzero part,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma,$$

and the two powered nonzero parts agree,

$$V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma = V^{(N)}\big(\textstyle\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\big)_\sigma.$$

Proof. ✓ Proof Start from Theorem 9.48. With p_A, p_B the least common multiples of the period-removal lengths on the two sides, use $p = \mathrm{lcm}(p_A, p_B)$. The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet; for every $N \geq 1$,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\big(\textstyle\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\big)_\sigma.$$

The same applies on the B side, and the two-sided blocking lemma gives equality of the two powered nonzero parts. □

Theorem 9.50 (Unconditional common primitive block decompositions). Lean ✓ Stmt *Let A and B have the same MPV family. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

Proof. ✓ Proof Apply Theorem 9.49 to choose the common blocking length and the two common cyclic-sector families. For either one of these families, write G_k for its cyclic-sector blocks, and write (ν_x, C_x) for the weights and blocks of the flattened common-sector family from Lemma 9.68. The common-alphabet nonzero-part identity gives, for every length N and blocked word σ,

$$V^{(N)}(\textstyle\bigoplus_k \mu_k^p G_k^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_x \nu_x C_x)_\sigma,$$

Applying this on the two sides of the comparison yields a positive length p and weighted primitive block families such that, for every $N \geq 1$ and blocked word σ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_x \mu_x^A A_x)_\sigma, \qquad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\textstyle\bigoplus_y \mu_y^B B_y)_\sigma,$$

with the two weighted nonzero parts equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families. □

A weighted block sum with distinct sectors is the simplest sector decomposition: one block per sector, each of multiplicity one. Equal-modulus and repeated-copy sectors are not single scalar powers, already for $A = C \oplus (-C)$ with $V^{(N)}(A)_\sigma = (1 + (-1)^N)\, V^{(N)}(C)_\sigma$; they are compared by the basis-of-normal-tensors construction of Chapter 10.

Definition 9.51 (Single-copy sector decomposition). Lean ✓ Stmt Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *single-copy sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1, and sector weight μ_k. As a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\textstyle\bigoplus_k \mu_k B_k)_\sigma.$$

Lemma 9.52 (Single-copy sector decomposition preserves the represented family). Lean ✓ Stmt *The single-copy sector decomposition of* $(\mu_k, B_k)_k$ *represents the same MPS family as* $\bigoplus_k \mu_k B_k$.

Proof. ✓ Proof Expanding the sector decomposition gives $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$, the finite block-sum expression for $\bigoplus_k \mu_k B_k$. □

9.7 Blocking and weighted direct sums

Blocking by a length p commutes with weighted direct sums up to the pth power of each weight: for nonzero weights (μ_k) and blocks (B_k),

$$\left(\bigoplus_k \mu_k B_k\right)^{[p]} \sim \bigoplus_k \mu_k^p\, B_k^{[p]},$$

where $\sim$ is equality of MPV families. When two block families are reblocked to a common length $p = m_k e_k$, the p-blocked alphabet is identified with the e_k-fold tensor power of the m_k-blocked alphabet, so each sector reblocked through its own period agrees with the directly p-blocked block.

Lemma 9.53 (Canonical form from peripheral primitivity). Not ready *Let* $(\mu_k, A_k)_{k=1}^r$ *be nonzero weights and injective left-canonical blocks with positive bond dimensions, non-increasing moduli* $|\mu_1| \geq \cdots \geq |\mu_r| > 0$, *and* $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$. *Then* $(\mu_k, A_k)_{k=1}^r$ *satisfies the canonical-form conditions of Definition 9.23, and* $O_{A_k A_k}(N) \to 1$ *for every* k.

Lemma 9.54 (Adjoint primitivity equivalence). Lean ✓ Stmt *Let* $\mathcal{E}$ *be a finite-dimensional completely positive map. Then* $\mathcal{E}$ *is primitive if and only if its Frobenius adjoint* $\mathcal{E}^\dagger$ *is primitive; equivalently,* $\sigma_\partial(\mathcal{E}) = \{1\}$ *iff* $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$, *since the peripheral spectrum of* $\mathcal{E}^\dagger$ *is the complex conjugate of that of* $\mathcal{E}$.

Lemma 9.55 (Primitivity under blocking by a multiple). Lean ✓ Stmt *Let* A *have positive bond dimension and let* $p_0, p \geq 1$ *with* $p_0 \mid p$. *If* $\mathcal{E}_{A^{[p_0]}}$ *is primitive, then* $\mathcal{E}_{A^{[p]}}$ *is primitive.*

Lemma 9.56 (Blocking the weighted block sum). Lean ✓ Stmt *For nonzero weights* $(\mu_k)_{k=1}^r$, *blocks* $(B_k)_{k=1}^r$, *and any* $p \geq 1$,

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p\, B_k^{[p]},$$

where $\sim$ *denotes generation of the same MPV family at every length.*

Lemma 9.57 (Positive-length equality survives blocking). Lean ✓ Stmt *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

Lemma 9.58 (Positive-length blocking of a weighted block sum). Lean ✓ Stmt *If* A *and* $\bigoplus_k \mu_k B_k$ *have the same MPV coefficients at every positive length, then after any positive blocking length* p, $A^{[p]}$ *and* $\bigoplus_k \mu_k^p B_k^{[p]}$ *have the same MPV coefficients at every positive length.*

Lemma 9.59 (Positive-length equality of powered block sums). Lean ✓ Stmt *If* $\bigoplus_j \mu_j^A A_j$ *and* $\bigoplus_k \mu_k^B B_k$ *have the same MPV coefficients at every positive length, then after any positive common blocking length* p, $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$ *and* $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$ *have the same MPV coefficients at every positive length.*

Lemma 9.60 (Nonzero weights survive blocking). Lean ✓ Stmt *If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

Lemma 9.61 (Common reblocking of cyclic-sector families). Lean ✓ Stmt *Given a finite family $(B_k)_{k=1}^r$, each with primitive irreducible cyclic sectors, there is a common blocking length p such that each $B_k^{[p]}$ admits a unit-weight direct-sum decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

Lemma 9.62 (Sector irreducibility for the adjoint blocked map). Lean ✓ Stmt *Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of its cyclic-sector decomposition, and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map with $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.*

Lemma 9.63 (Iterated blocking at the transfer-map level). Lean ✓ Stmt *For any MPS tensor A and $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Definition 9.64 (Primitive irreducible cyclic-sector decomposition). Lean ✓ Stmt *A has primitive irreducible cyclic sectors* of period $m \geq 1$ if there is a family $(C_k)_{k=1}^m$ with $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ (unit weights), each C_k left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.

Lemma 9.65 (Common reblocking at a prescribed multiple). Lean ✓ Stmt *With hypotheses as in Lemma 9.61, and given any common multiple p of the per-block period-removal lengths (m_k), the common reblocking can be performed at the prescribed length p.*

Lemma 9.66 (Structural properties of the common reblocked sectors). Lean ✓ Stmt *Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.61 is trace-preserving ($\sum_i (B^i)^\dagger B^i = \mathbb{1}$), has primitive transfer map ($\sigma_\partial(\mathcal{E}_B) = \{1\}$), is tensor-irreducible, and has positive bond dimension.*

Lemma 9.67 (Common-alphabet flattening). Lean ✓ Stmt *With $p = m_k e_k$ for each k, the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with $B_k^{[p]}$ in MPV family, after the canonical identification of the p-blocked alphabet $d^{[p]}$ with the e_k-fold tensor power of the m_k-blocked alphabet $d^{[m_k]}$.*

Lemma 9.68 (Reindexed nonzero-part identity). Lean ✓ Stmt *The powered nonzero-part decomposition $\bigoplus_k \mu_k^p B_k^{[p]}$ carries over under the common-alphabet identification of Lemma 9.67: the common-alphabet sectors have the same nonzero-part decomposition as the direct p-blocked tensors.*

Lemma 9.69 (Direct blocking in the common alphabet). Lean ✓ Stmt *For every original block B_k, direct blocking at the common length $p = m_k e_k$ agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write $\widetilde{B}_k$ for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

9.8 Scope of the canonical-form reduction

Remark 9.70 (Scope relative to [PGVWC07], Section III). The reduction follows [CPGSV17a] and [CPGSV21], using site-independent tensors, transfer-map primitivity, and CPM fixed-point normalization. The periodic decomposition theorem of [PGVWC07, Theorem 5] contains an explicit state-vector decomposition. Here we use only the blocked cyclic-sector consequence, Theorem 9.43; the explicit decomposition of the state vector into p-periodic summands lies outside this chapter.

Chapter 10

Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor A is a finite family of normal blocks $\{A_j\}$ whose MPV states $\{|V^{(N)}(A_j)\rangle\}$ span $|V^{(N)}(A)\rangle$ at every length and are eventually linearly independent. Two payoffs follow. First, equal MPV families force a bijective gauge-phase matching of the basis sectors $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$. Second, the sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ recover the per-copy weight multisets by Newton–Girard. The references are [PGVWC07], [CPGSV16], and [CPGSV21].

10.1 Bases of normal tensors

Definition 10.1 (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \ldots, A_g)$ with bond dimensions $(D_1, \ldots, D_g)$ such that each A_j is normal (Definition 2.31); at each system size N the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$; and for sufficiently large N those block MPVs are linearly independent. This is [CPGSV21, Definition IV.2]. In the canonical-form characterization of [CPGSV16, Section 2.3], a BNT is a minimal family of representatives for $\widetilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$ among the normal blocks.

Definition 10.2 (Basis of normal tensors, coefficient form). Lean ✓ Stmt The coefficient form of a basis of normal tensors: each block A_j is a CPSV normal tensor (Definition 9.4), the MPV of A is given at every length by explicit coefficients $c_j^{(N)} \in \mathbb{C}$ with $|V^{(N)}(A)\rangle = \sum_{j=1}^{g} c_j^{(N)} |V^{(N)}(A_j)\rangle$, and the block MPVs $\{|V^{(N)}(A_j)\rangle\}$ are eventually linearly independent.

The eventual linear-independence clause is produced from the asymptotic overlaps: diagonal overlaps tending to one and off-diagonal overlaps tending to zero make the Gram matrix converge to the identity.

Lemma 10.3 (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ Stmt *Let I be a finite index set and let $(A_i)_{i \in I}$ be a finite family of tensors such that $O_{A_i A_i}(N) \to 1$ for each i, and $O_{A_i A_j}(N) \to 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. □

Lemma 10.4 (Eventual linear independence from overlap orthonormality). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g}$ be a finite family of tensors such that $O_{A_jA_j}(N) \to 1$ for each j and $O_{A_iA_j}(N) \to 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent for all sufficiently large N.*

Proof. ✓ Proof Apply Lemma 10.3 with $I = \{1, \dots, g\}$. □

The same conclusion is often more convenient with an explicit threshold rather than the eventual quantifier.

Theorem 10.5 (Existential BNT independence from overlap limits). Lean ✓ Stmt *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ Proof This is Lemma 10.4, restated with an explicit threshold N_0. □

Eventual linear independence is also what converts an equality of two finite linear combinations into equality of their coefficients; this is the mechanism behind the coefficient identities of Chapter 11.

Lemma 10.6 (Eventual coefficient extraction from linear independence). Lean ✓ Stmt *If a finite family of vectors is linearly independent for all sufficiently large N, and two finite linear combinations of that family are equal for all sufficiently large N, then the corresponding coefficients are equal for all sufficiently large N.*

Proof. ✓ Proof Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. □

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 10.6.

Lemma 10.7 (Eventual linear independence for two families). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_jB_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N.

Proof. ✓ Proof Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. □

We now isolate the separation condition that supplies the off-diagonal decay and combine it with the canonical-form hypotheses.

Definition 10.8 (No gauge-phase-equivalent distinct blocks). Lean ✓ Stmt A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 10.9 (Normal canonical form with BNT separation). Lean ✓ Stmt A family is in *normal canonical form with BNT separation* if it satisfies Definition 9.25 and distinct blocks of the same bond dimension are not gauge-phase equivalent. The weight moduli are non-increasing but need not be strictly decreasing, so equal-modulus blocks are allowed; the grouping into a

BNT is governed by gauge-phase equivalence, not by distinctness of moduli. Repeated equal-modulus copies of one basis tensor are not separate blocks here; they appear as the weights $\mu_{j,q}$ in $c_N^{(j)} = \sum_q \mu_{j,q}^N$ in the sector decomposition below.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

Theorem 10.10 (Cross-overlap decay from explicit BNT hypotheses)**.** Lean ✓ Stmt *Let $(A_j)_{j=1}^r$ be a family in which each A_j is injective and left-canonical, $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$, and distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \to 0$ for every pair $j \neq k$.*

Proof. ✓ Proof If the bond dimensions differ, Theorem 7.24 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.26 gives the decay. □

Lemma 10.11 (BNT from explicit blockwise hypotheses)**.** Lean ✓ Stmt *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume each A_k is injective and left-canonical, $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, that $O_{A_k A_k}(N) \to 1$ for each k, and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 10.10 supplies the off-diagonal decay. Therefore Lemma 10.4 gives the eventual linear independence required in Definition 10.1. □

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

Remark 10.12 (Comparing the two normality conditions). The spectral normality of Definition 9.25 and the algebraic normality of Definition 2.31 agree: the Wielandt theory of Chapter 8 supplies the implication from primitivity to eventual block injectivity used here.

Lemma 10.13 (Restricted normal-CF-BNT hypotheses give a BNT)**.** Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.*

Proof. ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 10.1. □

10.2 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace at every length, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

Lemma 10.14 (Invertible change of basis). Lean ✓ Stmt *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \mathrm{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^{g} U_{ij} v_i$$

for every j.

Proof. ✓ Proof Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^{g}$; these coefficients form the columns of U. If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. □

Lemma 10.15 (Eventual change of basis from equal spans). Lean ✓ Stmt *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N, then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large N. At such an N, the span equality lets us apply Lemma 10.14. □

The change-of-basis matrix is now used to match the blocks. We first treat equal block counts, then remove that restriction.

Theorem 10.16 (Permutation rigidity for equal block counts). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g}$ and $(B_j)_{j=1}^{g}$ be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N, then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.*

Proof. ✓ Proof For large N, Lemma 10.15 gives an invertible matrix U_N relating the two MPV families. Fix j and write $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$. With $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$ and $b_j(N) = (O_{A_i B_j}(N))_i$, this gives

$$G_A(N)\overline{u_j(N)} = b_j(N), \qquad O_{B_j B_j}(N) = u_j(N)^\top G_A(N)\overline{u_j(N)}.$$

If all mixed overlaps $O_{A_i B_j}(N)$ tended to 0, then $b_j(N) \to 0$; since $G_A(N) \to I$, one obtains $\overline{u_j(N)} \to 0$ and hence $u_j(N) \to 0$. Thus $O_{B_j B_j}(N) \to 0$, contradicting $O_{B_j B_j}(N) \to 1$. Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 7.24 and 7.26 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B-family makes f injective, hence bijective, and its inverse is the required permutation. □

Lemma 10.17 (Permutation rigidity with asymptotically orthonormal overlaps). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors of positive bond dimension satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N,*

$$\mathrm{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \le j \le g_A\} = \mathrm{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \le k \le g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j.

Proof. ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large N. At such an N, the common span has the same dimension when computed from the A-family or the B-family, so $g_A = g_B$. After identifying the two index sets, Theorem 10.16 yields the permutation conclusion. □

The two preceding results obtain the block matching from the asymptotic orthonormality of the overlaps. The same matching is obtained by the exact sector comparison of Section 10.6, which pairs the sectors of two canonical forms through a bijection fixed by their bond dimensions and gauge-phase equivalences. The proof of the Fundamental Theorem in Chapter 11 proceeds through the sector comparison.

The same overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths; this is an input to the exact sector matching later in the chapter.

Theorem 10.18 (Non-equivalent irreducible TP blocks are eventually non-proportional)**.** Lean Lean ✓ Stmt *Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \geq N_0$ such that $|V^{(N)}(A)\rangle$ is not proportional to $|V^{(N)}(B)\rangle$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.*

Proof. ✓ Proof Suppose that for all sufficiently large N one had $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \qquad O_{AB}(N) = c_N O_{BB}(N).$$

Since $O_{AA}(N) \to 1$ and $O_{BB}(N) \to 1$, the first identity gives $|c_N| \to 1$, and hence $|O_{AB}(N)| \to 1$. The irreducible trace-preserving overlap decay theorem gives $O_{AB}(N) \to 0$ when $A \nsim_{\text{gp}} B$, a contradiction. □

10.3 Newton–Girard identities and power-sum recovery

The coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

Theorem 10.19 (Newton–Girard trace recursion)**.** Lean Lean ✓ Stmt *If $A, B \in M_n(\mathbb{C})$ satisfy $\operatorname{tr}(A^k) = \operatorname{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.*

Proof. ✓ Proof Let e_m denote the mth elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$m e_m = \sum_{i=1}^{m} (-1)^{i-1} e_{m-i} \operatorname{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\operatorname{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. □

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient $\sum_q \mu_{j,q}^N$ then equals $\sum_q (\nu_{j,q} e^{i\phi_j})^N$, and Newton–Girard recovers the weight multiset with multiplicity.

Theorem 10.20 (Finite-range equal power sums imply equal multisets). Lean Lean Lean Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n-1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \le k \le n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Two variants relax the equal-cardinality assumption. If $\alpha : \{0, \dots, m-1\} \to \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \to \mathbb{C}$ have no zero entries and their power sums agree for $1 \le k \le \max(m, n)$, then $m = n$ and the multisets are equal. Without a nonzero hypothesis, the list form $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \le N \le \max(x_a, x_b)$ gives equality of the nonzero multisets; if it also holds at $N = 0$ then $x_a = x_b$ and the full lists agree.

Proof. ✓ Proof Apply the finite-range Newton–Girard theorem to $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$ and extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$; the nonzero hypothesis makes the padded-zero counts $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without that hypothesis, delete the zero entries first (unchanged for positive N); the equality at $N = 0$ gives $x_a = x_b$. □

Theorem 10.21 (Equal power sums imply equal multisets). Lean ✓ Stmt *Let $\alpha, \beta : \{0, \dots, n-1\} \to \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \ge 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. ✓ Proof Construct diagonal matrices $\operatorname{diag}(\alpha)$ and $\operatorname{diag}(\beta)$. The power-sum hypothesis gives $\operatorname{tr}(\operatorname{diag}(\alpha)^k) = \operatorname{tr}(\operatorname{diag}(\beta)^k)$. By Theorem 10.19, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

10.4 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate of [CPGSV16, Section 2], defined on a sector decomposition with raw sector weights $\mu_{j,q}$ and coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$; no equal-modulus factorization is assumed.

10.4.1 Sector decompositions and phase matching

The sector decomposition and the MPV phase-equivalence predicates introduced here feed the BNT canonical-form predicate below.

Definition 10.22 (Sector weights and sector decomposition). Lean Lean ✓ Stmt A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$, nonzero weights $\mu_{j,q}$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights.

Lemma 10.23 (Sector decomposition formula). Lean ✓ Stmt *For a sector decomposition S with associated tensor A_{tot}, the MPV satisfies*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

Definition 10.24 (MPV phase equivalence for nonzero-weight blocks). Lean ✓ Stmt Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every system size N and every physical word,

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

The two bond dimensions are allowed to differ.

Definition 10.25 (Common MPV phase cover). Lean ✓ Stmt For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

Definition 10.26 (MPV phase classes). Lean Lean Lean ✓ Stmt For a finite family of blocks, put $j \sim k$ when there is a nonzero scalar factor ζ such that

$$|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$$

for every length N. The associated tuple enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and proves that distinct representatives are not MPV-phase equivalent.

Definition 10.27 (BNT canonical form). Lean ✓ Stmt Given a sector decomposition P, the BNT canonical-form predicate records the minimal BNT hypotheses of [CPGSV16, CPGSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, Section 2.3] canonical-form normalization holds,

$$\forall (j,q),\ |\mu_{j,q}| \le 1, \qquad \exists\, (j_*, q_*) : |\mu_{j_*,q_*}| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^{g} \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV16, Section 2.3] and [CPGSV21, Definition IV.2]. The canonical-form normalization admits every example of the source paper (in particular $C \oplus D$, $C \oplus (-C)$, $C \oplus (1/2)C$, and $C \oplus (-C) \oplus (1/2)C$) and is not derivable from the per-block normality hypotheses alone.

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

Lemma 10.28 (Raw two-layer sector coefficient identity). Lean ✓ Stmt *For every sector decomposition P, length N, and basis index j,*

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV16, Section 2.3].

Proof. ✓ Proof Immediate from the definition of the sector coefficient. □

Lemma 10.29 (Cross-overlap decay between distinct basis blocks)**.** Lean ✓ Stmt *If P is a BNT canonical form and $j \neq k$, then $O_{P_j P_k}(N) \to 0$.*

Proof. ✓ Proof Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. □

Lemma 10.30 (Combined-family eventual linear independence)**.** Lean ✓ Stmt *Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N. This is [CPGSV16, Appendix A].

Proof. ✓ Proof Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 10.29), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 10.7). □

Lemma 10.31 (Unit-modulus weight witness from the canonical-form normalization)**.** Lean ✓ Stmt *Under the BNT canonical form, the [CPGSV16, Section 2.3] unit-modulus existential provides indices (j_*, q_*) with $|\mu_{j_*,q_*}| = 1$.*

Proof. ✓ Proof The canonical-form normalization includes a unit-modulus weight $|\mu_{j_*,q_*}| = 1$. □

Example 10.32 ($C \oplus (1/2)C$, unequal-modulus admissibility). Lean ✓ Stmt The single-sector two-copy decomposition with raw weights $(1, 1/2)$ is a BNT canonical form, so unequal-modulus copies are admissible: $|1/2| \leq 1$ and the unit witness is the first copy.

10.4.2 Prepared-block construction of BNT canonical forms

The construction below separates the after-blocking preparation of suitable blocks from the normalized-weight BNT construction: the prepared blocks are trace-preserving, primitive, irreducible, and injective; normalized weights then quotient gauge-phase-equivalent blocks into sectors to produce the canonical form of Definition 10.27.

Lemma 10.33 (Common injective blocking for primitive irreducible families)**.** Lean ✓ Stmt *Let $(B_k)_{k=1}^{r}$ be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single $L \geq 1$ such that each $B_k^{[L]}$ is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

Proof. ✓ Proof For each block, Theorem 8.87 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem 9.42 preserves the other hypotheses. □

Theorem 10.34 (Prepared BNT blocks after blocking)**.** Lean ✓ Stmt *Let A be an MPS tensor. Then there is a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the p-blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*

3. $A^{[p]}$ *and* $\bigoplus_k \mu_k B_k$ *generate the same MPV family at every positive length.*

No weight-modulus normalization is asserted at this stage.

Proof. ✓ Proof Apply the common primitive irreducible block decomposition to (A, A), giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Lemma 10.33 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one. □

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

Theorem 10.35 (BNT from normal blocks with normalized weights)**.** Lean ✓ Stmt *Let* $r \in \mathbb{N}$*, and for each* $k = 1, \dots, r$ *let* $D_k \in \mathbb{N}$*,* $\mu_k \in \mathbb{C}$*, and let* B_k *be an MPS tensor of physical dimension* d *and bond dimension* D_k*. Assume:*

1. ***positive bond dimensions:*** $D_k > 0$ *for all* k*,*
2. ***TP normalization:*** $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$*,*
3. ***primitivity:*** $\mathcal{E}_{B_k}$ *is primitive for each* k*,*
4. ***irreducibility:*** *each* B_k *is irreducible,*
5. ***nonzero weights:*** $\mu_k \neq 0$ *for all* k*,*
6. ***bounded weights:*** $|\mu_k| \leq 1$ *for all* k*,*
7. ***global unit witness:*** $|\mu_k| = 1$ *for some* k*.*

Then there exists a sector decomposition P *such that* P *and* $\bigoplus_{k=1}^r \mu_k B_k$ *generate the same MPV family at every length, and* P *satisfies the basis-of-normal-tensors conditions (Definition 10.2) together with the multiplicity, span, and weight-bound conditions of Definition 10.27.*

Proof. ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives B_j, $B_{j'}$ $(j \neq j')$ the cross-overlap decays,

$$\lim_{N\to\infty} O_{B_j B_{j'}}(N) = 0, \qquad \lim_{N\to\infty} O_{B_j B_j}(N) = 1,$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions. □

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs X and Y of positive bond dimension have MPV families related by $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ with $\zeta \neq 0$. The normalized self-overlaps give $|O_{XX}(N)| \to 1$. The phase relation gives the cross-overlap identity

$$O_{YX}(N) = \zeta^N O_{XX}(N), \qquad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \to 1$. If their bond dimensions differed, the rectangular overlap-decay theorem would give

$$O_{YX}(N) \to 0,$$

a contradiction.

Lemma 10.36 (Total dimension after grouping by BNT representatives). Lean ✓ Stmt *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

Proof. ✓ Proof The collapsed decomposition counts one representative dimension for each assigned copy. Let $e(j, q)$ denote the original CF summand assigned to the qth copy of representative P_j. Since $(j, q) \mapsto e(j, q)$ reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^{g} \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^{g} \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^{r} D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension. □

Lemma 10.37 (Prepared phase-equivalent NTs have the same bond dimension). Lean ✓ Stmt *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then X and Y have the same bond dimension.*

Proof. ✓ Proof Choose $\zeta \neq 0$ with $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ for every length N. The normalized self-overlaps give $|O_{XX}(N)| \to 1$, and the phase relation gives

$$O_{YX}(N) = \zeta^N O_{XX}(N), \qquad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \to 1$. If the two bond dimensions differed, the rectangular overlap-decay theorem would force

$$O_{YX}(N) \to 0,$$

contradicting the preceding limit. □

Lemma 10.38 (Prepared BNT representative classes preserve bond dimension). Lean ✓ Stmt *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.*

Proof. ✓ Proof Apply Lemma 10.37 to each representative and each assigned summand. □

Lemma 10.39 (Prepared collapsed BNT total dimension). Lean ✓ Stmt *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

Proof. ✓ Proof Every assigned summand shares the bond dimension of its representative, by the equal-dimension fact above. The displayed regrouping identity in Lemma 10.36 then identifies the collapsed sum over pairs (j, q) with $\sum_{k=1}^{r} D_k$, the original direct-sum dimension. □

Combining the after-blocking preparation with the normalized-weight construction gives the arbitrary-input statement, conditional on the weight normalization.

Theorem 10.40 (Conditional BNT form after blocking for an arbitrary tensor). Lean Lean ✓ Stmt *Let A be an MPS tensor. Then there exists a blocking length $p \geq 1$, nonzero weights μ_k, and blocks B_k over the blocked alphabet, such that:*

1. every block has positive bond dimension;

2. every block is TP-normalized, primitive, irreducible, and injective;

3. $A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.

Moreover, under $|\mu_k| \le 1$ for every k and $|\mu_k| = 1$ for at least one k, there exists a sector decomposition P over the blocked alphabet in the BNT canonical form of Definition 10.27 such that $A^{[p]}$ and P generate the same MPV family at every positive length; the identity upgrades to all lengths once the total bond dimension of P equals that of $A^{[p]}$.

Proof. ✓ Proof The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV16, Section 2.3] makes; Theorem 10.35 then produces the sector decomposition, following the BNT selection of [CPGSV16, Appendix A]. At length zero the MPV coefficient is the bond dimension, so equal total bond dimensions supply the missing length-zero coefficient. The remaining rescaling step is recorded in Remark 10.41. □

Remark 10.41 (Arbitrary-input reduction). The reduction is conditional on the two weight-normalization hypotheses of Theorem 10.40.

10.5 Coefficient comparison and copy-weight recovery

The exact matching of Section 10.6 produces, on each matched pair, a coefficient identity. This section turns that identity into equality of the per-copy weight multisets [CPGSV16, Appendix A]. No analytic non-decay estimate enters: the fixed-length linear independence isolates each sector coefficient exactly.

10.5.1 Matched-sector weight multiset equality

Given a matched pair $(j_0, \tilde{k}_0)$ and a nonzero scalar ζ with $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$ past a threshold N_0, this section records two conclusions: the multiset equality of the per-copy weights from power-sum recovery (Theorem 10.21), and its upgrade to an explicit per-copy permutation. Together they give the $\mu_{j,q} = \nu_{j,q} e^{i\varphi_j}$ recovery of [CPGSV16, Appendix A]. The coefficient identity is supplied, in the equal-MPV case, by Lemma 11.3 of Chapter 11; in the proportional case it is the remaining hypothesis of the conditional proportional global gauge (Theorem 11.10).

Lemma 10.42 (Matched-sector weight multiset equality). Lean ✓ Stmt *Let P, Q be sector decompositions with weights $\mu_{j,q}$ and $\nu_{k,q'}$, and fix indices j_0, $\tilde{k}_0$, and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$. Suppose that for every $N > N_0$,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q).$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$, and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix A] read on a single matched sector.

Proof. ✓ Proof Expand both sides of the coefficient identity into per-copy power sums $\sum_q \mu_{j_0,q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 10.20), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. $\square$

Lemma 10.43 (Matched-sector weight-permutation extractor). Lean ✓ Stmt *Under the hypotheses of Lemma 10.42, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q,*

$$\nu_{\tilde{k}_0,\tau(q)} = \zeta^{-1} \cdot \mu_{j_0,q}.$$

This is the per-copy realisation of [CPGSV16, Appendix A]: the multiset equality from Lemma 10.42 is upgraded to a concrete indexing between the matched P- and Q-copies.

Proof. ✓ Proof Invoke Lemma 10.42 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. $\square$

10.6 Strong existential matching by exact linear independence

The exact matching reads [CPGSV16, Appendix A] as a statement at fixed length: equal MPV families force each basis sector of one form to pair with a basis sector of the other. The non-vanishing of the sector coefficients isolates the matched sector exactly, with no limit of a single-sector projection.

Lemma 10.44 (Sector coefficient is not eventually zero). Lean ✓ Stmt *For a BNT canonical form P and any sector j, the sector coefficient $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$ is not eventually zero in N. Only nonvanishing of the copy weights is used, not any modulus condition.*

Proof. ✓ Proof If $c_{P,j}^{(N)} = 0$ for all $N \geq M$, the signed geometric-sequence vanishing lemma forces $\sum_q \mu_{j,q}^k = 0$ at every exponent k, in particular $\sum_q 1 = r_j = 0$ at $k = 0$, contradicting $r_j > 0$. Repeated ratios are handled by the same extrapolation, so distinct weights are not needed. $\square$

Theorem 10.45 (Exact single-sector matching). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family (at all positive lengths). For every sector j of P there exist a sector k of Q with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap. No per-sector unit-modulus hypothesis is required.*

Proof. ✓ Proof Suppose j decays against every sector of Q, and let T be the set of P-sectors with this property. For $j' \notin T$ the overlap dichotomy turns a non-decaying overlap with some $Q_{k(j')}$ into a gauge-phase equivalence, hence a scalar $\omega_{j'} \in \mathbb{C}^\times$ with $V^{(N)}(P_{j'}) = \omega_{j'}^N V^{(N)}(Q_{k(j')})$, so $V^{(N)}(P_{j'})$ lies in the span of $\{V^{(N)}(Q_k)\}_k$. The family $\{P_{j'} : j' \in T\} \cup \{Q_k\}_k$ then has all pairwise cross-overlaps decaying, hence is eventually linearly independent (Lemma 10.30). Rewriting the common-MPV identity over this family,

$$\sum_{j' \in T} c_N^{(j')} V^{(N)}(P_{j'}) = \sum_k \Big(c_N^{(k)} - \sum_{j' \notin T,\, k(j')=k} c_N^{(j')} \omega_{j'}^N \Big) V^{(N)}(Q_k),$$

exact coefficient extraction forces $c_N^{(j)} = 0$ at each fixed large N, contradicting Lemma 10.44. Hence a matched sector k exists; the gauge-phase equivalence is recovered from the non-decaying overlap by the irreducible trace-preserving overlap dichotomy. ☐

Theorem 10.46 (Strong existential matching, full-basis form). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Then for every sector k of Q there exist a sector j of P with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

Proof. ✓ Proof Apply Theorem 10.45 with the roles of P and Q swapped and the same-MPV hypothesis flipped by symmetry of gauge-phase equivalence and overlap convergence. ☐

10.6.1 Bijective matching by symmetry

Applying the exact matching once more with P and Q swapped gives the reverse direction; the basis-distinctness clause forces injectivity both ways, so $g_a = g_b$ and the source relabelling $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$ holds on the full basis [CPGSV16, Appendix A].

Theorem 10.47 (Bijective matching by symmetry). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Then there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

such that for every $k \in J(Q)$ the bond dimensions match, the basis tensors $P_{\beta(k)}$ and Q_k are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.

Proof. ✓ Proof Theorem 10.45 applied to (Q, P) gives a matching map $m : J(Q) \to J(P)$, and applied to (P, Q) a matching map $J(P) \to J(Q)$.

Injectivity of m. If $m(k_1) = m(k_2) = j$, the two witnesses give a common centre P_j gauge-phase equivalent, after identifying bond dimensions, to both Q_{k_1} and Q_{k_2}; transitivity yields a gauge-phase equivalence $Q_{k_1} \sim Q_{k_2}$ under $\dim_Q(k_1) = \dim_Q(k_2)$. The basis-distinctness clause on Q then forces $k_1 = k_2$; the reverse map is injective by symmetry.

Bijection. The two injections give equal finite sector counts, so m is a bijection $\beta : J(Q) \simeq J(P)$. ☐

Chapter 11

Proof of the Fundamental Theorem

For BNT canonical forms P and Q, write the weighted direct sums over normal-tensor sectors $P^i = \bigoplus_j \mu_j P^i_j$ and $Q^i = \bigoplus_k \nu_k Q^i_k$. The proportional theorem gives a bijection β of the normal-tensor sectors with unit-modulus phases ζ_k, invertible matrices X_k, and $Q^i_k = \zeta_k X_k P^i_{\beta(k)} X_k^{-1}$; the equal-MPV corollary collapses the per-block phases into a single X with $Q^i_{\mathrm{tot}} = X P^i_{\mathrm{tot}} X^{-1}$. Given the sector matching of Chapter 10, extract the per-sector coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q)$ from eventual linear independence, recover the copy weights by finite power-sum comparison, and combine the block gauges into $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.

Throughout, "eventually" abbreviates "for all sufficiently large N": there is N_0 such that the stated relation holds for every $N > N_0$. The comparison is made at a fixed such N, where BNT linear independence detects a vanishing coefficient even when $|\mu| < 1$ lets it decay.

11.1 Coefficient identity, copy weights, and the block-diagonal gauge

The bijective matching theorem of Chapter 10 (Theorem 10.47) gives the sector phases ζ_k and block gauges X_k. The coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188] then turns the equality of total MPV families into the per-sector coefficient identity $c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q)$. Neither step uses a per-sector unit-modulus hypothesis.

Lemma 11.1 (Unit phase from matched normalized BNT blocks). Lean ✓ Stmt *Let P and Q be BNT canonical forms. If, for every length N and word σ, a sector P_j and a sector Q_k have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma$$

then $|\zeta| = 1$.

Proof. ✓ Proof The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The displayed MPV identity gives, for every N,

$$O_{Q_k Q_k}(N) = |\zeta|^{2N}\, O_{P_j P_j}(N).$$

Since $O_{Q_k Q_k}(N) \to 1$ and $O_{P_j P_j}(N) \to 1$, taking limits forces $|\zeta|^{2N} \to 1$ and hence $|\zeta| = 1$. $\square$

Theorem 11.2 (Proportional sector matching with unit phases). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$, scalars $\zeta_k \in \mathbb{C}$, and block gauges X_k such that $|\zeta_k| = 1$ and, for every sector k and physical index i,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

Equivalently, for every length N and word σ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

This is [CPGSV16, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].

Proof. ✓ Proof Apply the proportional sector-matching theorem to obtain the bijection and dimension-compatible gauge-phase equivalences. Choosing the gauges and scalars gives $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, and therefore

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N).$$

The normalized self-overlap limits give $O_{Q_k Q_k}(N) \to 1$ and $O_{P_{\beta(k)} P_{\beta(k)}}(N) \to 1$, so $|\zeta_k|^{2N} \to 1$ and hence $|\zeta_k| = 1$. The MPV scalar-power identity follows from the same gauge equation: $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. ☐

Lemma 11.3 (Full-basis coefficient identity from matched sectors). Lean ✓ Stmt *Let P and Q be BNT canonical forms with the same MPV family, matched by a bijection $\beta : J(Q) \simeq J(P)$ with bond-dimension equalities and a gauge-phase equivalence between Q_k and $P_{\beta(k)}$ for every k. Then each sector carries a unit-modulus phase ζ_k, and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

This is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].

Proof. ✓ Proof For each matched sector, the gauge-phase equivalence gives $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. Lemma 11.1 gives $|\zeta_k| = 1$. Applying Lemma 11.4 to the same phases gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N. ☐

Lemma 11.4 (Full-basis coefficient identity from matched phases). Lean ✓ Stmt *Let P be a BNT canonical form, let Q be a sector decomposition, and assume P_{tot} and Q_{tot} have the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

and scalars $\zeta_k \in \mathbb{C}$ such that, for every $k \in J(Q)$, every length N, and every spin configuration σ of length N,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N \, V^{(N)}(P_{\beta(k)})_\sigma.$$

Then for every $k \in J(Q)$ there is N_0 such that, for all $N > N_0$,

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q).$$

This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.

Proof. ✓ Proof For every N, expand the equality of total MPVs as

$$\sum_{j\in J(P)} c_N^{(j)}(P)|V^{(N)}(P_j)\rangle\rangle = \sum_{k\in J(Q)} c_N^{(k)}(Q)|V^{(N)}(Q_k)\rangle\rangle.$$

The phase relation gives $|V^{(N)}(Q_k)\rangle\rangle = \zeta_k^N|V^{(N)}(P_{\beta(k)})\rangle\rangle$, so reindexing the Q-sum by β onto the P-basis yields

$$\sum_{j\in J(P)} c_N^{(j)}(P)|V^{(N)}(P_j)\rangle\rangle = \sum_{j\in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q)|V^{(N)}(P_j)\rangle\rangle.$$

For all sufficiently large N the P-basis MPV family $\{|V^{(N)}(P_j)\rangle\rangle\}_{j\in J(P)}$ is linearly independent, by the BNT eventual linear independence of P. Lemma 10.6 then extracts the coefficient equalities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ from this eventual linear independence, for all sufficiently large N. □

The coefficient identity is now converted, sector by sector, into a matching of the individual copy weights.

Definition 11.5 (Copy-weight matching for matched BNT sectors)**.** Lean ✓ Stmt Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, with sector phases ζ_k. A copy-weight matching consists of copy permutations τ_k such that, for all sectors k and copies q,

$$\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}.$$

This is the copy-level conclusion of [CPGSV16, Appendix MPV proof, line 1188].

Lemma 11.6 (Copy-weight matching from matched-sector coefficient identities)**.** *Lean* *✓ Stmt* *Let the sectors of P and Q be matched by $\beta : J(Q) \simeq J(P)$, with nonzero sector phases ζ_k. If the matched-sector coefficient identities*

$$c_N^{(\beta(k))}(P) = \zeta_k^N\, c_N^{(k)}(Q)$$

hold eventually, then the phases ζ_k determine a copy-weight matching $\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}$. This is, sector by sector, [CPGSV16, Appendix MPV proof, line 1188].

Proof. ✓ Proof Eventually the coefficient identity in sector k reads

$$\sum_r \mu_{\beta(k),r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k\nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 10.43) applies sector by sector, producing a copy permutation τ_k with $\mu_{\beta(k),\tau_k(q)} = \zeta_k\nu_{k,q}$; since $\zeta_k \neq 0$ this is $\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}$. □

The matched block gauges and copy weights now combine into a single block-diagonal gauge.

Theorem 11.7 (Global gauge from matched sectors and copy weights)**.** *Lean* *✓ Stmt* *Suppose that the sectors of two sector decompositions P and Q are matched by a bijection $\beta : J(Q) \simeq J(P)$, with bond-dimension equalities, copy permutations τ_k, nonzero phases ζ_k, and block gauges X_k. If, for every sector k, copy q, and physical index i,*

$$Q_k^i = \zeta_k\, X_k P_{\beta(k)}^i X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)},$$

then, in the flattened Q-coordinates,

$$Q_{\text{flat}}^i = X\, P_{\text{matched}}^i\, X^{-1}, \qquad X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

Proof. ✓ Proof For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k),\tau_k(q)})(\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k),\tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge X_k. Lemma 14.48 combines these block conjugacies into the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. □

Theorem 11.8 (Global gauge from a copy-weight matching)**.** Lean ✓ Stmt *Suppose that the sectors of P and Q are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

If the same phases also give a copy-weight matching, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is only a reformulation of Theorem 11.7.

Proof. ✓ Proof The copy-weight matching supplies $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$. Together with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, these are exactly the two displayed hypotheses of Theorem 11.7. □

We can now state the proportional global gauge, first from a copy-weight matching and then from the coefficient identities that produce one.

Theorem 11.9 (Conditional proportional global gauge from a copy-weight matching)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives β, the bond-dimension equalities, the unit phases ζ_k, and the block gauges X_k. If the remaining proportional coefficient comparison supplies a copy-weight matching for these same phases, then the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The proportional sector-matching theorem supplies β, X_k, and unit phases ζ_k satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $|\zeta_k| = 1$, each ζ_k is nonzero. A copy-weight matching gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$, so Theorem 11.8 gives the direct-sum gauge. □

Theorem 11.10 (Conditional proportional global gauge from coefficient identities)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

for every sector k and physical index i. If these same phases satisfy the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N \, c_N^{(k)}(Q)$$

for all sufficiently large N, then they determine a copy-weight matching and hence the flattened Q-tensor is obtained from the matched-coordinate P-tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

Proof. ✓ Proof The sector-matching theorem gives β, X_k, and $|\zeta_k| = 1$ with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $\zeta_k \neq 0$, the coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ produce $\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}$ by Lemma 11.6. Theorem 11.9 then applies to this copy-weight matching. □

In the equal-MPV case all ingredients are unconditional.

Lemma 11.11 (Equal-MPV matched copy-weight witnesses)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k, and block gauges X_k such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector,

$$\nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}.$$

Proof. ✓ Proof The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k)\rangle = |\zeta_k|^{2N}\langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)})\rangle$$

together with the BNT normalization limits on both sides forces $|\zeta_k| = 1$. The matched MPV phase identities also give the eventual coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Lemma 11.6 applies the finite power-sum comparison in each sector and gives the copy-weight identities. □

Theorem 11.12 (Full-basis global gauge in matched coordinates)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then there exist:*

- *a bijection $\beta : J(Q) \simeq J(P)$,*
- *bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$,*
- *copy-number equalities $r_{\beta(k)}(P) = r_k(Q)$ and copy permutations τ_k,*
- *unit-modulus phases ζ_k with $|\zeta_k| = 1$ and block gauges X_k,*

such that, for every sector k, copy q, and physical index i,

$$Q_k^i = \zeta_k\, X_k P_{\beta(k)}^i X_k^{-1}, \qquad \nu_{k,q} = \zeta_k^{-1}\mu_{\beta(k),\tau_k(q)}.$$

Moreover, if

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$$

is the flattened block-diagonal gauge, then, for every physical index i, the flattened Q-coordinates satisfy

$$Q_{\text{flat}}^i = X\, P_{\text{matched}}^i\, X^{-1}.$$

This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].

Proof. ✓ Proof Lemma 11.11 gives β, the bond-dimension identifications, the phases ζ_k, the block gauges X_k, and copy permutations τ_k satisfying

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}.$$

Theorem 11.8 combines the block conjugacies into the flattened direct-sum conjugation by $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. $\square$

Theorem 11.13 (Literal equal-MPV gauge after sector-coordinate permutation)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors have the same MPV family. Then the total bond dimensions agree; write their common value as D. There is*

$$Y \in \mathrm{GL}_D(\mathbb{C})$$

such that, for every physical index i,

$$Q^i_{\mathrm{tot}} = Y\, P^i_{\mathrm{tot}}\, Y^{-1},$$

where P^i_{tot} and Q^i_{tot} are both regarded as matrices in $M_D(\mathbb{C})$ using the total bond-dimension equality. This is the BNT form of [CPGSV16, Corollary II.2 and Appendix MPV proof, lines 1189–1192].

Proof. ✓ Proof Apply Theorem 11.12 to obtain the flattened gauge after the sectors of P have been reindexed by the bijection between basis blocks of P and Q. The sector-coordinate permutation ρ, induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed P-side direct sum with the total tensor P_{tot}, transported along the total bond-dimension equality. Composing the permutation gauge for ρ with the sector-permuted gauge gives the displayed single matrix Y. $\square$

11.2 The equal and proportional cases

Theorem 11.14 (Multi-block fundamental theorem for normal tensors)**.** Lean ✓ Stmt *Let P and Q be BNT canonical forms (Definition 10.27), with bases of normal tensors $(P_j)_{j=1}^{g_a}$ and $(Q_k)_{k=1}^{g_b}$. If the total tensors P_{tot}, Q_{tot} generate eventually nonzero proportional MPV families — for all sufficiently large N there is a nonzero scalar c_N with $V^{(N)}(P_{\mathrm{tot}}) = c_N V^{(N)}(Q_{\mathrm{tot}})$ — then $g_a = g_b$. After relabelling the basis of normal tensors of P by a bijection $\beta : J(Q) \simeq J(P)$, there are scalars $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$ and invertible matrices Y_k such that, for every k and every physical index i,*

$$Q^i_k = \zeta_k\, Y_k\, P^i_{\beta(k)}\, Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension ($\dim P_{\beta(k)} = \dim Q_k$) and the same gauge-phase class.

Proof. ✓ Proof The proportional sector-matching Theorem 11.2, applied to the BNT canonical forms P and Q, supplies the bijection β, the bond-dimension equalities, the unit phases ζ_k, and the block gauges Y_k with $Q^i_k = \zeta_k Y_k P^i_{\beta(k)} Y_k^{-1}$; the matching needs no per-sector unit-modulus hypothesis. $\square$

This is [CPGSV16, Theorem 2.10]; see also [CPGSV21, Theorem IV.4 and Corollary IV.5]. The statement is given on the BNT canonical-form surface (Definition 10.27), which is the paper's "A and B in canonical form" with chosen bases of normal tensors; the reduction of an *arbitrary* proportional-MPV pair to a common primitive block decomposition is Theorem 9.49 of Chapter 9.

Theorem 11.15 (Equal BNT canonical forms are globally conjugate). Lean ✓ Stmt *Let P and Q be BNT canonical forms (Definition 10.27) whose total tensors P_{tot}, Q_{tot} generate the same MPV family for every system size N. Then their total bond dimensions coincide; writing the common value as D, there is one invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that, for every physical index i,*

$$Q^i_{\text{tot}} = X\, P^i_{\text{tot}}\, X^{-1}.$$

This is [CPGSV16, Corollary 2.11] on the BNT canonical-form surface; the canonical form of [CPGSV16, Section 2.3], a direct sum of normal tensors with scalar weights, is a BNT canonical form. The reduction of an arbitrary *pair of same-MPV tensors to a common BNT canonical form is Chapter 9.*

Proof. ✓ Proof This is Theorem 11.13. The BNT comparison of Chapter 10 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. The exact linear-independence matching needs no per-sector unit-modulus hypothesis, and the BNT canonical form needs no one-site injectivity. ☐

This is also [CPGSV21, Corollary IV.5].

Chapter 12

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS+08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and virtual intertwiners enter the string-order criteria.

12.1 Physical Symmetries Give Virtual Gauges

Definition 12.1 (Physical-index rotation). Lean ✓ Stmt Given a matrix $M \in M_d(\mathbb{C})$ and an MPS tensor A, the *physical-index rotation* is the tensor $(A_M)^i := \sum_j M_{ij} A^j$. When the matrix is a unitary u, the rotation is written A_u.

Corollary 12.2 (Periodic symmetry yields gauge equivalence up to a root of unity). Lean ✓ Stmt *Suppose that for any two tensors in irreducible form II with the same MPV family there is $m > 0$ making them gauge equivalent up to an m-th root of unity (Theorem 13.24). Let A be in irreducible form II and let M be a matrix such that $B := A_M$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there is $m > 0$ such that A and B are gauge equivalent up to an m-th root of unity.*

Proof. ✓ Proof Apply the assumed equivalence (Theorem 13.24) to A and B, which are in irreducible form II with $\mathcal{V}(A) = \mathcal{V}(B)$. □

12.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 12.3 (Twisted tensor). Lean ✓ Stmt Given a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is defined, for each $i \in \{0, \ldots, d-1\}$, by

$$\widetilde{A}^i_g := \sum_{j=0}^{d-1} U(g)_{ij} A^j.$$

Definition 12.4 (On-site symmetry). Lean ✓ Stmt A tensor A is *on-site symmetric* under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ for every $g \in G$.

Lemma 12.5 (Functoriality of twisting). Lean ✓ Stmt *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by gh is the same as first twisting by h and then by g.

Proof. ✓ Proof Expanding both sides using $U(gh) = U(g)\,U(h)$, then swapping the order of summation. $\square$

Lemma 12.6 (Identity twist). Lean ✓ Stmt *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

Proof. ✓ Proof Since $U(1) = \mathbb{1}_d$, each component reduces to $\sum_j \delta_{ij}\, A^j = A^i$. $\square$

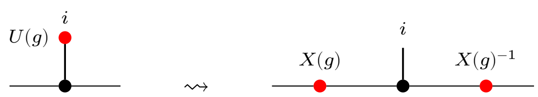
Lemma 12.7 (Symmetric twists are gauge equivalent). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to A.*

Proof. ✓ Proof On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. $\square$

Remark 12.8 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies A and $\widetilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 12.7. Gauge uniqueness and the twist composition law then give Theorem 12.15. The string-order results later in this chapter use this virtual representation as an input.

Theorem 12.9 (Virtual symmetry equation). Lean ✓ Stmt *If A is injective and on-site symmetric under U, then for every $g \in G$ there exist an invertible matrix $X(g) \in \mathrm{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\widetilde{A}^i_g = \varphi(g)\, X(g)\, A^i\, X(g)^{-1}$ for all i.*

Proof. ✓ Proof The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 12.7 to obtain $X(g)$, then set $\varphi(g) = 1$. $\square$

Lemma 12.10 (Gauge uniqueness up to scalar). Lean ✓ Stmt *Let A be an injective tensor. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\,A^i\,X^{-1} = Y\,A^i\,Y^{-1}$ for all i, then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof The matrix $Z = Y^{-1}X$ commutes with every A^i. Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 14.12, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 12.11 (Gauge uniqueness from equality of MPV families). Lean ✓ Stmt *Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both satisfy $B^i = X\,A^i\,X^{-1} = Y\,A^i\,Y^{-1}$ for all i, then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. ✓ Proof This is exactly Lemma 12.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. □

Definition 12.12 (Scalar 2-cochain)**.** Lean ✓ Stmt A *scalar 2-cochain* on a group G is a function $\omega : G \times G \to \mathbb{C}^{\times}$.

Definition 12.13 (Projective representation)**.** Lean ✓ Stmt Given a scalar 2-cochain ω, a *projective representation* of G on $\mathbb{C}^D$ is a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\,\rho(h) = \omega(g, h)\,\rho(gh).$$

Lemma 12.14 (Associativity forces the cocycle condition)**.** Lean ✓ Stmt *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:*

$$\omega(g, h)\,\omega(gh, k) = \omega(g, hk)\,\omega(h, k).$$

Proof. ✓ Proof Expanding $\rho(g)\,(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. □

Theorem 12.15 (Virtual representation theorem for injective MPS)**.** Lean ✓ Stmt *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. Then there exist:*

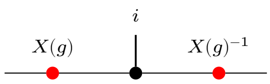
1. *a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^{\times}$, and*
2. *a map $\rho : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,*

$$\rho(g)\,\rho(h) = \omega(g, h)\,\rho(gh).$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}.$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof Since A is injective and on-site symmetric, each twisted tensor $\widetilde{A}_g$ is gauge equivalent to A (Lemma 12.7). Choose $X(g) \in \mathrm{GL}_D(\mathbb{C})$ with $\widetilde{A}^i_g = X(g)\,A^i\,X(g)^{-1}$ for all i.

By functoriality (Lemma 12.5), the product $X(h)\,X(g)$ also intertwines A with $\widetilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 12.10) gives a nonzero scalar $\omega'(h, g)$ with $X(h)\,X(g) = \omega'(h, g)\,X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g)\,\rho(h) = \omega(g, h)\,\rho(gh)$ where $\omega(g, h) := \omega'(h^{-1}, g^{-1})$. □

Corollary 12.16 (2-cocycle identity for the factor system)**.** Lean ✓ Stmt *If $D > 0$, the factor system ω from Theorem 12.15 satisfies the 2-cocycle identity for all $g, h, k \in G$:*

$$\omega(g, h)\,\omega(gh, k) = \omega(g, hk)\,\omega(h, k).$$

Proof. ✓ Proof Theorem 12.15 gives ρ and ω. By Lemma 12.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. □

12.3 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 12.17 (Coboundary). Lean ✓ Stmt A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi\colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}.$$

Definition 12.18 (Cohomologous cocycles). Lean ✓ Stmt Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi\colon G \to \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g,h).$$

Theorem 12.19 (Cohomology relation is an equivalence relation). *Lean* *✓ Stmt* *The relation "cohomologous to" is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.*

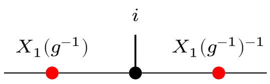
Proof. ✓ Proof Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1}. Transitivity: compose the witnesses φ and ψ pointwise. $\square$

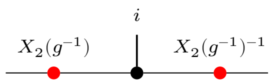
Lemma 12.20 (Coboundary iff cohomologous to the trivial cocycle). *Lean* *✓ Stmt* *A cocycle ω is a coboundary if and only if $\omega \sim \mathbb{1}$, where $\mathbb{1}(g,h) = 1$ for all g, h.*

Proof. ✓ Proof Both directions follow by absorbing or introducing the factor $\mathbb{1}(g,h) = 1$ via $x \cdot 1 = x$. $\square$

Theorem 12.21 (Gauge independence of the cocycle class). *Lean* *✓ Stmt* *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U, then $\omega_1 \sim \omega_2$.*

Proof. ✓ Proof For each g, gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g)\, X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges



and


describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g,h) = f(g)\, f(h)\, f(gh)^{-1}\omega_1(g,h)$. $\square$

Definition 12.22 (Multiplicative 2-cocycle condition). Lean ✓ Stmt A scalar 2-cochain $\omega\colon G \times G \to \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g,h)\omega(gh,k) = \omega(g,hk)\omega(h,k).$$

Definition 12.23 (Second cohomology quotient). Lean ✓ Stmt The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 12.24 (Projective equivalence). Lean ✓ Stmt Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 12.25 (Projective equivalence and cohomologous cocycles). Lean ✓ Stmt *Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2. Then*

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. ✓ Proof This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. □

12.3.1 Non-triviality via the commutator phase

For an abelian symmetry group the cohomology class of a factor system is detected by an anti-symmetric pairing, which gives a computable test for non-triviality.

Definition 12.26 (Commutator phase). Lean ✓ Stmt The *commutator phase* of a 2-cochain ω at a pair g, h is $\beta_\omega(g,h) := \omega(g,h)\,\omega(h,g)^{-1}$. It is defined for any $\mathbb{C}^\times$-valued 2-cochain, the cocycle condition is not needed.

Theorem 12.27 (The commutator phase is a class invariant). Lean ✓ Stmt *If $\omega_1 \sim \omega_2$ and $gh = hg$, then $\beta_{\omega_1}(g,h) = \beta_{\omega_2}(g,h)$.*

Proof. ✓ Proof Write $\omega_1(g,h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g,h)$ for a coboundary witness φ. Substituting into the ratio and using $gh = hg$ and the commutativity of $\mathbb{C}^\times$,

$$\beta_{\omega_1}(g,h) = \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\,\omega_2(g,h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\,\omega_2(h,g)} = \frac{\omega_2(g,h)}{\omega_2(h,g)} = \beta_{\omega_2}(g,h),$$

the φ-factors cancelling since $\varphi(gh) = \varphi(hg)$. □

Definition 12.28 (Non-trivial cohomology class). Lean ✓ Stmt A factor system ω has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

Theorem 12.29 (Commutator-phase non-triviality test). Lean ✓ Stmt *If a commuting pair g, h has $\beta_\omega(g,h) \neq 1$, then ω has a non-trivial class.*

Proof. ✓ Proof The trivial cocycle has commutator phase 1 at every pair, so by Theorem 12.27 any cocycle cohomologous to it has $\beta = 1$ at every commuting pair. Contrapositively, $\beta_\omega(g,h) \neq 1$ for one commuting pair forces ω to be non-cohomologous to the trivial cocycle. □

12.4 Permutation-Based Physical Symmetry

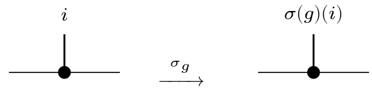
The linear-combination twist $\widetilde{A}^i_g = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

Definition 12.30 (Permutation symmetry datum). Lean ✓ Stmt An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \to M_d(\mathbb{C})$ and a physical-index action $\sigma : G \to \mathrm{End}(\{0, \dots, d-1\})$.

Definition 12.31 (Permutation-twisted tensor). Lean ✓ Stmt Given an on-site symmetry datum (u, σ), the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



Lemma 12.32 (Identity permutation twist). Lean ✓ Stmt *If $\sigma(1) = \mathrm{id}$, then $(A^{\sigma_1})^i = A^i$ for all i.*

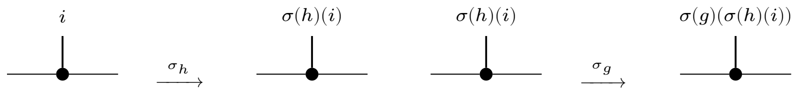
Proof. ✓ Proof Immediate from $\sigma(1)(i) = i$ for all i. □

Composition order. The linear-combination twist (Lemma 12.5) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g. The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 12.33 (Composition of permutation twists). Lean ✓ Stmt *If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then*

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. ✓ Proof Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

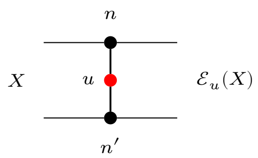
12.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS+08] detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 12.34 (Twisted transfer map). Lean ✓ Stmt Given an MPS tensor A and a physical-index matrix u, the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n'|u|n\rangle A_n \, X \, A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

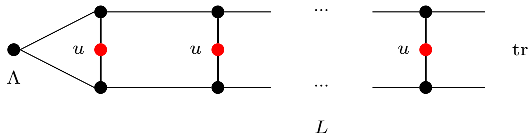


The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 12.35 (String order parameter). Lean ✓ Stmt The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \mathrm{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length-L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 12.36 (Boundary string order parameter). Lean ✓ Stmt For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \mathrm{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$.

Definition 12.37 (Local symmetry). Lean ✓ Stmt An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger.$$

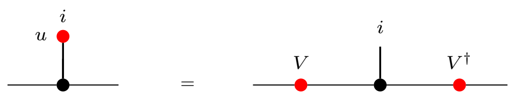
The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V.

Definition 12.38 (String order). Lean ✓ Stmt String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \le \|R_L(u; X, Y)\|$ for all L, where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Thus some boundary-refined string-order sequence does not decay to zero.

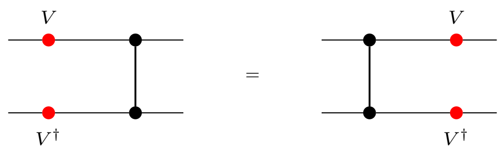
Lemma 12.39 (String order prevents decay to zero). Lean ✓ Stmt *If A has string order for u with respect to a boundary state Λ, then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \to \infty$.*

Proof. ✓ Proof A uniform lower bound $c \le \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. □

Definition 12.40 (Local physical intertwining). Lean ✓ Stmt The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 12.41 (Transfer-map covariance). Lean ✓ Stmt Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:

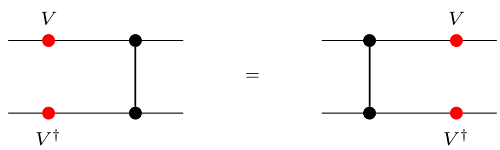


Definition 12.42 (Commutation in the doubled transfer picture). Lean ✓ Stmt In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V\,\mathcal{E}_1(X)\,V^\dagger = \mathcal{E}_1(VXV^\dagger).$$

Theorem 12.43 (Transfer-map covariance and doubled commutation are equivalent). Lean ✓ Stmt *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V.*

Proof. ✓ Proof Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides. ☐

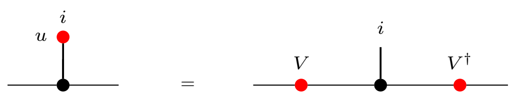
Lemma 12.44 (Unitary Kraus mixing). Lean ✓ Stmt *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel:*

$$\sum_i \left(\sum_j u_{ij} A_j\right) Y \left(\sum_j u_{ij} A_j\right)^\dagger = \sum_i A_i\, Y\, A_i^\dagger.$$

Proof. ✓ Proof This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 19.20. ☐

Theorem 12.45 (Local physical intertwining implies transfer-map covariance). Lean ✓ Stmt *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

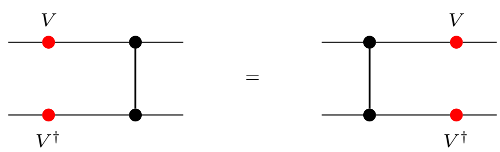
Proof. ✓ Proof Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u-boxes disappear by the unitary Kraus-mixing identity. ☐

Theorem 12.46 (Transfer-map covariance implies local physical intertwining). Lean ✓ Stmt *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. ✓ Proof Set $B^i := V\,A^i\,V^\dagger$. The transfer-map covariance relation



implies that the two Kraus families B and A define the same transfer map. Kraus freedom for equal channels converts $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Definition 12.47 (Twisted companion tensor). Lean ✓ Stmt For an MPS tensor A and a physical-index matrix u, define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}}\, A^{n'}.$$

Lemma 12.48 (Twisted transfer as a mixed transfer operator). Lean ✓ Stmt *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u):*

$$\mathcal{E}_u = F_{A,B_u}.$$

Proof. ✓ Proof Expand both definitions and rearrange the double sum. $\square$

Theorem 12.49 (Twisted-transfer eigenvalues have modulus at most one). Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. ✓ Proof By Lemma 12.48, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ. $\square$

Theorem 12.50 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). Lean ✓ Stmt *Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If*

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u.

Proof. ✓ Proof By Lemma 12.48, the twisted transfer map is the mixed transfer operator F_{A,B_u}. Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge the eigenvalue equation becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.29 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for A and B_u. $\square$

Lemma 12.51 (Spectral radius < 1 forces boundary-string decay). Lean ✓ Stmt *If $\rho(\mathcal{E}_u) < 1$, then, as $L \to \infty$, for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has*

$$R_L(u; X, Y) \longrightarrow 0.$$

Proof. ✓ Proof In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \to 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \operatorname{tr}(\Lambda X Z)$ gives the stated scalar convergence. $\square$

Theorem 12.52 (String order gives gauge-phase equivalence with the companion tensor). Lean ✓ Stmt *Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ, then A is gauge-phase equivalent to the companion tensor B_u.*

Proof. ✓ Proof If $\rho(\mathcal{E}_u) < 1$, Lemma 12.51 forces every boundary string-order sequence to decay to zero, contradicting Definition 12.38. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 12.49 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 12.50 then gives the gauge-phase equivalence. □

Theorem 12.53 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u, then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu \, V A^i V^\dagger.$$

Proof. ✓ Proof Start from an invertible intertwiner X between A and B_u. The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V, and the original intertwining relation becomes the phased covariance equation above, with $|\mu| = 1$. □

Theorem 12.54 (Virtual unitary gives a twisted-transfer eigenmatrix). Lean ✓ Stmt *Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu \, V A^i V^\dagger$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. ✓ Proof Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. □

Theorem 12.55 (Boundary-state invariance of the virtual unitary). Lean ✓ Stmt *Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\mathrm{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i,*

$$\sum_j u_{ij} A^j = \mu \, V A^i V^\dagger,$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. ✓ Proof The covariance relation shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. □

Lemma 12.56 (Local symmetry implies string order). Lean ✓ Stmt *Assume $\mathrm{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ, then string order exists for u.*

Proof. ✓ Proof Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 12.54, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \operatorname{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

Theorem 12.57 (Local symmetry iff a unitary peripheral eigenmatrix exists). Lean ✓ Stmt *Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\operatorname{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that*

$$\mathcal{E}_u(V) = \mu V.$$

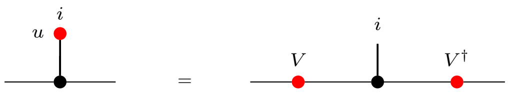
By Theorem 12.49, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ Proof The forward direction is Theorem 12.54. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 12.50 gives gauge-phase equivalence with the companion tensor; then Theorem 12.53 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 12.55 gives the boundary-state invariance needed for local symmetry. $\square$

Theorem 12.58 (String order $\Leftrightarrow$ local symmetry). Lean ✓ Stmt *For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\operatorname{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.*

Proof. ✓ Proof If string order exists, Theorem 12.52 gives a gauge-phase equivalence with the companion tensor. Theorem 12.53 extracts a virtual unitary and phase from that equivalence, and Theorem 12.55 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 12.56 turns local symmetry into a non-decaying boundary string-order sequence. $\square$

Theorem 12.59 (Virtual unitary from string order). Lean ✓ Stmt *For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ, then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i. Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*



Proof. ✓ Proof String order first gives gauge-phase equivalence with the companion tensor by Theorem 12.52. Theorem 12.53 then normalizes the resulting intertwiner to a unitary and produces the phase μ. $\square$

12.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 12.60 (Same SPT phase). Lean ✓ Stmt Injective tensors A and B, both symmetric under an on-site representation $U : G \to \mathrm{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 12.61 (String order holds for injective symmetric MPS). Lean ✓ Stmt *Let A be an injective tensor, symmetric under a unitary representation U. Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\mathrm{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g.*

Proof. ✓ Proof The virtual representation theorem produces $\rho(g^{-1}) \in \mathrm{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 12.50 gives gauge-phase equivalence with the companion tensor, and Theorem 12.53 extracts a virtual unitary and phase. By Theorem 12.55, this unitary preserves the boundary state, so Lemma 12.56 gives string order. □

Theorem 12.62 (String order is an SPT invariant). Lean ✓ Stmt *If A and B are injective, both on-site symmetric under a unitary representation U, and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\mathrm{tr}(\Lambda_A) = 1$, $\mathrm{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B. In particular, since Definition 12.60 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

Proof. ✓ Proof Both directions follow from Theorem 12.61, which shows that string order holds universally for injective symmetric tensors with canonical normalisations. □

Chapter 13

Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 13.24) following [DlCCSPG17], and gives its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form II to a virtual Z-gauge, and the equivalence between p-refinability of the matrix-product-vector family and p-divisibility of the transfer map.

13.1 Periodic irreducible form and physical-index rotation

Definition 13.1 (Periodic tensor). Lean ✓ Stmt An irreducible MPS tensor A of bond dimension D is *m-periodic* (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m-th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 4.49).

Definition 13.2 (Irreducible form). Lean ✓ Stmt A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j \, A_j^i,$$

where each block A_j is an m_j-periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DlCCSPG17, Section II]. It extends the normal canonical form (Definition 9.25), which requires the stronger condition that every block is 1-periodic.

Remark 13.3 (Relation to normal canonical form). Not ready Every tensor in normal canonical form (Definition 9.25) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

Physical-index rotation was introduced in Definition 12.1 (Chapter 12). The following compatibility theorems are proved in the present chapter.

Theorem 13.4 (Transfer map preserved by physical rotation). *Lean ✓ Stmt If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.*

Proof. ✓ Proof This is the unitary freedom of Kraus representations from [Wol12, Theorem 2.1(4)]. □

Theorem 13.5 (Left-canonical preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.*

Proof. ✓ Proof Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. □

Theorem 13.6 (Irreducibility preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.*

Proof. ✓ Proof Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1-P)\, A_k\, P = 0$ for all k via the orthogonality of u, contradicting irreducibility of A. □

Theorem 13.7 (Periodicity preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is periodic with period m, then A_u is periodic with the same period.*

Proof. ✓ Proof Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. □

Theorem 13.8 (Equality of MPV families preserved by physical rotation). Lean ✓ Stmt *If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.*

Proof. ✓ Proof Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 13.9 (Physical rotation distributes over block-diagonal composition). Lean ✓ Stmt *The rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.*

Proof. ✓ Proof Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. □

Theorem 13.10 (Irreducible form preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is in irreducible form, then A_u is in irreducible form with the same block structure and rotated blocks.*

Proof. ✓ Proof Each block remains periodic by Theorem 13.7, equality of MPV families transfers via Theorem 13.8, and block-diagonal compatibility follows from Theorem 13.9. □

13.1.1 The Z-gauge degree of freedom

Definition 13.11 (Entrywise periodic Z-gauge ratio). Lean ✓ Stmt Given complex weights μ, ν, define the entrywise Z-gauge ratio by

$$z(\mu, \nu) := \mu/\nu.$$

Lemma 13.12 (Matched powers imply root-of-unity ratio). Lean ✓ Stmt *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

Lemma 13.13 (Ratio rescales denominator back to numerator). Lean ✓ Stmt *If $\nu \neq 0$, then*

$$z(\mu, \nu)\nu = \mu.$$

Definition 13.14 (Diagonal periodic Z-gauge matrix). Lean ✓ Stmt For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \operatorname{diag}(z(\mu_i, \nu_i))_i.$$

Lemma 13.15 (Diagonal Z-gauge has finite order under matched powers). Lean ✓ Stmt *If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i, then*

$$Z^m = \mathbb{1}.$$

Lemma 13.16 (Diagonal Z-gauge transports diagonal weights). Lean ✓ Stmt *Under $\nu_i \neq 0$ for all i,*

$$Z \operatorname{diag}(\nu) = \operatorname{diag}(\mu).$$

Definition 13.17 (Z-gauge equivalence). Lean ✓ Stmt Let A be an m-periodic irreducible tensor of bond dimension D. A *Z-gauge transformation* of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i, and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m-th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m, because the factor $\operatorname{tr}(Z^N) = \operatorname{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod m$. Two tensors A, B are *Z-gauge equivalent* if there exists an invertible Y and a Z-gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i.

Remark 13.18 (Z-gauge is trivial for period-1 blocks). Lean ✓ Stmt When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z-gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z-gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

Theorem 13.19 (Blocking a $\mathbb{Z}_m$-gauge gives an ordinary gauge). Lean ✓ Stmt *If A and B are $\mathbb{Z}_m$-gauge equivalent, then their m-blocked tensors $A^{[m]}$ and $B^{[m]}$ are gauge equivalent in the ordinary sense.*

Proof. ✓ Proof Write the $\mathbb{Z}_m$-gauge relation as $ZA^i = YB^iY^{-1}$ with $Z^m = \mathbb{1}$ and $[A^i, Z] = 0$. For a blocked letter, commute one factor of Z past each physical letter in the associated length-m word. The total prefactor becomes $Z^m = \mathbb{1}$, so the entire blocked word is related by the same ordinary gauge Y. □

Corollary 13.20 (Blocking a $\mathbb{Z}_m$-gauge preserves the MPV family). Lean ✓ Stmt *If A and B are $\mathbb{Z}_m$-gauge equivalent, then the blocked tensors $A^{[m]}$ and $B^{[m]}$ have the same MPV family.*

Proof. ✓ Proof Apply Theorem 13.19 and then the gauge invariance of MPV families. □

13.2 Periodic Fundamental Theorem

The theorems in Chapter 11 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. [DlCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

13.2.1 Common-period blocking lemmas

Definition 13.21 (LCM period for a finite block family). Lean ✓ Stmt Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 13.22 (Common-period blocked family). Lean ✓ Stmt Given a finite family of tensors (B_i) with designated periods (p_i), block each B_i to the shared period $\mathrm{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 13.23 (Common-period blocking preserves transfer-map primitivity). *Lean* *✓ Stmt* *If each tensor B_i has a primitive transfer map after blocking by its own positive period p_i, then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.*

Proof. ✓ Proof Each p_i divides $\mathrm{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. □

13.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 ("equal case") of [DlCCSPG17]. The statements recorded here are conditional components of that theorem: the final comparison still assumes periodic-overlap and power-equality hypotheses, so the literature theorem has not yet been obtained here as a single unconditional result.

Theorem 13.24 (Periodic Fundamental Theorem). *Not ready* *Lean* *Lean* *Lean* *Lean* *Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j\in J}$ and $\{B_k\}_{k\in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).*

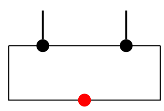
Then $|J| = |K|$ and there is a bijection $\pi : J \to K$ such that A_j and $B_{\pi(j)}$ are m_j-periodic for the same period m_j. Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i,*

such that, for every physical index i,

$$Z_j\, A_j^i = Y_j\, B_{\pi(j)}^i\, Y_j^{-1}$$

For each periodic block, the extra freedom is encoded by



Z_j

rather than by an ordinary gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i, with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. *The proof is conditional on the periodic-overlap dichotomy [DlCCSPG17, Proposition 3.3], which is left here as a separate hypothesis.*

13.2.3 Periodic overlap dichotomy

Definition 13.25 (Cyclic sector decomposition). Lean ✓ Stmt A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m, each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \operatorname{tr}(P_k (A^{(m)})^\sigma)$ for every word σ, and each block carries a $*$-algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k. This is the inverse cyclic indexing of the off-diagonal convention in [DlCCSPG17, Appendix A]. There the same adjoint transfer map is written E^*, and the source projectors satisfy $E^*(P_u) = P_{u+1}$ together with $A^i = \sum_u P_u A^i P_{u+1}$. Here P_k corresponds to the source sector $u = -k \pmod m$, which turns the source shift into the displayed adjoint-transfer relation.

Theorem 13.26 (Self-overlap along period multiples). Lean ✓ Stmt *If A is m-periodic, then $\lim_{N\to\infty} O_{AA}(mN) = m$.*

Theorem 13.27 (Different periods imply asymptotic orthogonality). Lean ✓ Stmt *If A and B have periods $m_A \neq m_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$.*

Proof. ✓ Proof If $D_A \neq D_B$, use Theorem 13.33. Assume $D_A = D_B$ and $A \sim_{\mathrm{gp}} B$, say $B^i = \zeta X A^i X^{-1}$ with $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \neq 0$. The transfer maps satisfy

$$\mathcal{E}_B(Y) = \zeta\bar{\zeta} X\, \mathcal{E}_A(X^{-1} Y X^{-*})\, X^*.$$

With $\mathcal{E}_A(\rho_A) = \rho_A$, put $\sigma = X\rho_A X^*$. Then $\mathcal{E}_B(\sigma) = |\zeta|^2\sigma$, while $\mathcal{E}_B(\rho_B) = \rho_B$ for a nonzero positive fixed point ρ_B. Perron–Frobenius eigenvalue uniqueness [Wol12, Theorem 6.3] gives $|\zeta|^2 = 1$. Hence $\mathcal{E}_B$ is conjugate to $\mathcal{E}_A$, so $\sigma_{\mathrm{per}}(\mathcal{E}_A) = \sigma_{\mathrm{per}}(\mathcal{E}_B)$. Since

$$\sigma_{\mathrm{per}}(\mathcal{E}_A) = \{e^{2\pi i r/m_A} : 0 \le r < m_A\}, \qquad \sigma_{\mathrm{per}}(\mathcal{E}_B) = \{e^{2\pi i r/m_B} : 0 \le r < m_B\},$$

equality of the two finite sets gives $m_A = m_B$, a contradiction. Thus $A \nsim_{\mathrm{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N\to\infty} O_{AB}(N) = 0$. $\square$

Lemma 13.28 (Sector-block normality for periodic tensors). Lean ✓ Stmt *For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.*

Theorem 13.29 (No sector match implies asymptotic orthogonality). Lean ✓ Stmt *Suppose A and B have the same total bond dimension and the same period m. Let A_u and B_v be period-blocked sector tensors of virtual dimensions d_u and e_v, indexed by $u, v \in \mathbb{Z}_m$, such that*

$$\bar{A} \sim \bigoplus_{u\in\mathbb{Z}_m} A_u, \qquad \bar{B} \sim \bigoplus_{v\in\mathbb{Z}_m} B_v$$

with unit weights, and the two decompositions are cyclic sector decompositions. Assume each sector is left-canonical,

$$\sum_\alpha (A_u^\alpha)^\dagger A_u^\alpha = I_{d_u}, \qquad \sum_\alpha (B_v^\alpha)^\dagger B_v^\alpha = I_{e_v},$$

Assume $\forall u,\ d_u \neq 0$ and $\forall v,\ e_v \neq 0$, and whenever $d_u = e_v$, the corresponding sector tensors are not gauge-phase equivalent after identifying their virtual spaces:

$$d_u = e_v \implies A_u \nsim_{\mathrm{gp}} B_v.$$

Then $\lim_{N\to\infty} O_{AB}(N) = 0$.

Proof. ✓ Proof Write $\bar{A} \sim \bigoplus_{u\in\mathbb{Z}_m} A_u$ and $\bar{B} \sim \bigoplus_{v\in\mathbb{Z}_m} B_v$. For every N,

$$O_{\bar{A}\bar{B}}(N) = \sum_{u\in\mathbb{Z}_m}\sum_{v\in\mathbb{Z}_m} O_{A_uB_v}(N).$$

For each sector pair (u, v),

$$d_u = e_v, \quad A_u \nsim_{\mathrm{gp}} B_v \quad \Longrightarrow \quad \lim_{N\to\infty} O_{A_uB_v}(N) = 0,$$

and

$$d_u \neq e_v \quad \Longrightarrow \quad \lim_{N\to\infty} O_{A_uB_v}(N) = 0.$$

The finite double sum therefore gives

$$\lim_{N\to\infty} O_{\bar{A}\bar{B}}(N) = \sum_{u\in\mathbb{Z}_m}\sum_{v\in\mathbb{Z}_m} \lim_{N\to\infty} O_{A_uB_v}(N) = 0.$$

If $A \sim_{\mathrm{gp}} B$, then $\bar{A} \sim_{\mathrm{gp}} \bar{B}$, so for some $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \neq 0$, $(\bar{B})^\alpha = \zeta X(\bar{A})^\alpha X^{-1}$. Thus $V^{(N)}(\bar{B})_\sigma = \zeta^N V^{(N)}(\bar{A})_\sigma$, and

$$O_{\bar{A}\bar{B}}(N) = \bar{\zeta}^N O_{\bar{A}\bar{A}}(N), \qquad O_{\bar{B}\bar{B}}(N) = |\zeta|^{2N} O_{\bar{A}\bar{A}}(N).$$

Since $\lim_{N\to\infty} O_{\bar{A}\bar{A}}(N) = \lim_{N\to\infty} O_{\bar{B}\bar{B}}(N) = m > 0$,

$$1 = \lim_{N\to\infty} \frac{O_{\bar{B}\bar{B}}(N)}{O_{\bar{A}\bar{A}}(N)} = \lim_{N\to\infty} |\zeta|^{2N},$$

hence $|\zeta| = 1$. Therefore

$$\lim_{N\to\infty} |O_{\bar{A}\bar{B}}(N)| = \lim_{N\to\infty} O_{\bar{A}\bar{A}}(N) = m \neq 0,$$

contradicting $\lim_{N\to\infty} O_{\bar{A}\bar{B}}(N) = 0$. Thus $A \nsim_{\mathrm{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N\to\infty} O_{AB}(N) = 0$. □

Lemma 13.30 (Sector-match propagation under translation). Lean Not ready *Let A and B be left-canonical tensors with period m, and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Suppose $d_{u_0} = e_{v_0}$, $d_{u_0} \neq 0$, and, after identifying the virtual spaces, $A_{u_0} \sim_{\mathrm{gp}} B_{v_0}$. Then for every $l \in \mathbb{Z}_m$ there is an equality $d_{u_0+l} = e_{v_0+l}$ and, after this identification,*

$$A_{u_0+l} \sim_{\mathrm{gp}} B_{v_0+l}.$$

Lemma 13.31 (Per-site proportionality from blocked sector match). Lean Not ready *Let A and B be left-canonical tensors with period m, and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Assume $d_u \neq 0$ and A_u is normal for every u. Fix $q \in \mathbb{Z}_m$ and assume that for every u there are $d_u = e_{u+q}$, $\zeta_u \neq 0$, and $X_u \in \mathrm{GL}_{d_u}(\mathbb{C})$ such that $B^\alpha_{u+q} = \zeta_u X_u A^\alpha_u X_u^{-1}$ for every blocked physical index α. The source obtains a phase-only product identity after absorbing these sector gauges into the B-sectors: with Q_v the cyclic sector projectors of B, it writes*

$$\widetilde{B}^i_v = U_v Q_v B^i Q_{v+1} U^\dagger_{v+1}.$$

The contraction input is then, for every physical block $(i_1, \dots, i_m)$,

$$A_u^{i_1} A_{u+1}^{i_2} \cdots A_{u+m-1}^{i_m} = e^{i\lambda_{u+q}} \widetilde{B}_{u+q}^{i_1} \widetilde{B}_{u+q+1}^{i_2} \cdots \widetilde{B}_{u+q+m-1}^{i_m}.$$

Then there are a phase ξ and an invertible matrix U such that $A^i = e^{i\xi} U B^i U^{-1}$ for every physical index i.

Theorem 13.32 (Sector match yields repeated-block equivalence). Lean Not ready *Let A and B be periodic tensors with the same period m. Let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical, and assume every sector A_u has nonzero virtual dimension. If there are sectors u_0, v_0 with $d_{u_0} = e_{v_0}$ and $A_{u_0} \sim_{\mathrm{gp}} B_{v_0}$ after identifying their virtual spaces, then A and B are equivalent up to repeated blocks.*

Theorem 13.33 (Different bond dimensions imply asymptotic orthogonality). Lean ✓ Stmt *If two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \to 0$ as $N \to \infty$.*

Theorem 13.34 (Periodic overlap dichotomy). Lean Not ready *Let A and B be periodic tensors. Then at least one of the following alternatives holds:*

1. *$\langle V^{(N)}(A) | V^{(N)}(B) \rangle \to 0$ as $N \to \infty$;*
2. *A and B are equivalent up to repeated blocks.*

Proof. Not ready If $m_A \neq m_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$ by Theorem 13.27. Assume $m_A = m_B = m$. If $D_A \neq D_B$, then $\lim_{N\to\infty} O_{AB}(N) = 0$ by Theorem 13.33. Assume $D_A = D_B$.

Choose period-blocked sector decompositions

$$\bar{A} \sim \bigoplus_{u \in \mathbb{Z}_m} A_u, \qquad \bar{B} \sim \bigoplus_{v \in \mathbb{Z}_m} B_v.$$

If no pair (u, v) satisfies

$$d_u = e_v, \qquad A_u \sim_{\mathrm{gp}} B_v,$$

then Theorem 13.29 gives $\lim_{N\to\infty} O_{AB}(N) = 0$.

Otherwise there are $u_0, v_0 \in \mathbb{Z}_m$ such that

$$d_{u_0} = e_{v_0}, \qquad A_{u_0} \sim_{\mathrm{gp}} B_{v_0}.$$

Theorem 13.32 then gives repeated-block equivalence between A and B. □

Theorem 13.35 (Eventual linear independence of a periodic basis). Lean Not ready *For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p, there is N_0 such that for every $N \geq N_0$ the vectors $\{|V^{(pN)}(A_j)\rangle\}_j$ are linearly independent.*

Remark 13.36 (Comparison with the blocking approach). Not ready Theorem 13.24 and the equal-MPV corollary of Section 11.2 both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Section 11.2, following [CPGSV16]): for a common multiple p of the periods,

$$A^{[p]} \sim \bigoplus_a C_a, \qquad B^{[p]} \sim \bigoplus_b D_b,$$

with all C_a, D_b normal. The Fundamental Theorem gives, after a permutation of the normal summands,

$$D_{\pi(a)}^{\alpha} = X_a C_a^{\alpha} X_a^{-1}.$$

This is a gauge statement for $A^{[p]}$ and $B^{[p]}$; the cyclic phases have been absorbed into the period-p words.

2. *Periodic approach* (Theorem 13.24, following [DlCCSPG17]): keep the cyclic sectors

$$A^i = \sum_{u \in \mathbb{Z}_m} P_u A^i P_{u+1}, \qquad B^i = \sum_{v \in \mathbb{Z}_m} Q_v B^i Q_{v+1}.$$

Sector matching gives phases ϕ_v, a shift q, and unitaries U_v such that

$$P_u A^i P_{u+1} = e^{i\xi} e^{i\phi_{u+q}} U_{u+q} Q_{u+q} B^i Q_{u+q+1} U_{u+q+1}^{-1} e^{-i\phi_{u+q+1}}.$$

In the equal-MPV theorem the proportional phase is absorbed, so one obtains $ZA^i = UB^iU^{-1}$ with $Z^m = \mathbb{1}$; the trace only sees powers of the corresponding m-th roots of unity.

Chapters 9 and 11 follow only the blocking argument: remove periodicity by blocking, then apply the Fundamental Theorem. The periodic extension treats periodic blocks directly.

Theorem 13.37 (Peripheral proportional case from exact MPV equality)**.** Lean Not ready *Let A and B be periodic tensors with the same matrix product vectors. Then their bond dimensions agree, and after identifying the bond spaces, A and B are equivalent up to repeated blocks.*

Proof. Not ready By Theorem 13.34, either the overlap $\langle V^{(N)}(A)|V^{(N)}(B)\rangle$ tends to 0 or the bond dimensions agree and A and B are equivalent up to repeated blocks. In the decay case, equality of the MPV families gives, for every N,

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle = \langle V^{(N)}(A)|V^{(N)}(A)\rangle.$$

Along multiples of the period m_a this would imply

$$0 = \lim_{N\to\infty} \langle V^{(m_aN)}(A)|V^{(m_aN)}(B)\rangle = \lim_{N\to\infty} \langle V^{(m_aN)}(A)|V^{(m_aN)}(A)\rangle = m_a,$$

contradicting $m_a > 0$. □

Theorem 13.38 (Peripheral proportional case from phase rescaling)**.** Lean Not ready *Assume that whenever two periodic tensors have proportional MPV families, one can rescale one tensor by a unit-modulus phase so that the MPV families agree exactly. Then proportional periodic MPV families already force the tensors to agree up to repeated blocks.*

Proof. Not ready Apply the phase-rescaling hypothesis to replace A by a periodic tensor A' with the same MPV family as B. Theorem 13.37 then yields repeated-block equivalence between A' and B. Since A' differs from A only by a unit-modulus phase, A and A' are themselves equivalent up to repeated blocks, and transitivity gives the claim. □

Remark 13.39 (Proportional periodic FT)*.* Not ready There is also a proportional periodic Fundamental Theorem (Theorem 3.4, "proportional case", in [DlCCSPG17]). The former formulation of this theorem assumed externally supplied coefficient arrays with nonzero limits as hypotheses. That is not the hypothesis set of the source theorem. A faithful statement must instead derive the nondecaying cross-overlap witnesses from the proportional MPV hypothesis and the periodic BNT structure.

13.3 Physical symmetries on periodic MPS

Theorem 13.40 (Physical symmetry yields a virtual Z-gauge). Lean ✓ Stmt *Let A be an MPS tensor in irreducible form II and let $U : G \to \mathrm{GL}_d(\mathbb{C})$ be a representation of a group G on the physical leg, acting unitarily ($U(g)\,U(g)^\dagger = \mathbb{1}$). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 13.24) as an explicit hypothesis on the pair (d, D), expressed by the hypothesis* Lean *$d\,D$. If A is on-site symmetric under U — that is, the twisted tensor $\widetilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$-gauge equivalence between A and $\widetilde{A}_g$.*

Concretely there are matrices Z_g and Y_g, with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying, for every physical index i,

$$Z_g\,A^i = Y_g\,\widetilde{A}_g^i\,Y_g^{-1}$$

Proof. ✓ Proof The twisted tensor $\widetilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition 12.1. By Theorem 13.10 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 13.24) to obtain the asserted $\mathbb{Z}_{m_g}$-gauge equivalence. □

Definition 13.41 (Periodic projective-rigidity hypothesis). Lean ✓ Stmt Let A be an MPS tensor with physical dimension d and bond dimension D, and assume it is on-site symmetric under a group action $U : G \to \mathrm{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \to \mathrm{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \to \mathbb{C}^\times$ such that

1. every Y_g realises the intertwining from [DlCCSPG17, Corollary 4.1] with the twisted tensor $\widetilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h\,Y_g = u(g, h)\,Y_{gh}$.

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §12.2); in the genuinely periodic setting the bond-space commutant of A is generated by the Z_g matrices, so coherence of the factor system u is an independent analytic assumption.

Theorem 13.42 (Projective-representation upgrade). Lean ✓ Stmt *Under the hypotheses of Corollary 13.40 together with the periodic projective-rigidity hypothesis (Definition 13.41), the Y_g family defines a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \to \mathbb{C}^\times$ and virtual gauges $X : G \to \mathrm{GL}_D(\mathbb{C})$ satisfying*

$$X(g)\,X(h) = \omega(g, h)\,X(gh),$$

while for each g the matrix $X(g^{-1})$ still realises the $\mathbb{Z}_{m_g}$-intertwining from [DlCCSPG17, Corollary 4.1] between A and the twisted tensor $\widetilde{A}_g$.

Proof. ✓ Proof Unpack the rigidity hypothesis to extract the family (Y, u) together with the intertwiners from [DlCCSPG17, Corollary 4.1]. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g)\,X(h) = \omega(g, h)\,X(gh)$ is then exactly the (h^{-1}, g^{-1})-instance of the factor-system identity, using $(gh)^{-1} = h^{-1}g^{-1}$. The intertwining relation between A and $\widetilde{A}_g$ transports unchanged. □

Theorem 13.43 (Cocycle identity for the upgraded factor system). Lean ✓ Stmt *If the bond dimension D is positive, the factor system ω produced by Theorem 13.42 satisfies the multiplicative 2-cocycle identity $\omega(g, h)\,\omega(gh, k) = \omega(g, hk)\,\omega(h, k)$.*

Proof. ✓ Proof Associativity of matrix multiplication applied to $X(g)\,X(h)\,X(k)$, combined with the projective multiplication law, gives

$$\omega(g,h)\,\omega(gh,k)\,X((gh)k) = \omega(g,hk)\,\omega(h,k)\,X(g(hk)).$$

Since $(gh)k = g(hk)$, both sides are scalar multiples of the same matrix $X((gh)k)$. The matrix lies in $\mathrm{GL}_D(\mathbb{C})$ and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires $D > 0$ so that the underlying index set is inhabited. Cancelling then yields $\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k)$ in $\mathbb{C}^\times$, hence in Units($\mathbb{C}$). □

Remark 13.44 (SPT classification). The cohomology class of ω in $H^2(G, U(1))$ classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not established here; only the underlying 2-cocycle identity of Theorem 13.43 is available.

Theorem 13.45 (Projective representation and cocycle from symmetry). Lean ✓ Stmt *Theorems 13.42 and 13.43 combine into a single statement: from the periodic projective-rigidity hypothesis (Definition 13.41), one obtains a projective representation of G on the bond space whose factor system is a genuine 2-cocycle whenever $D > 0$.*

13.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DlCCSPG17].

Definition 13.46 (p-divisible CP map). Lean ✓ Stmt A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is *p-divisible* if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 13.47 (p-refinable MPS tensor). Lean ✓ Stmt An MPS tensor B with physical dimension d and bond dimension D is *p-refinable* if there exist another such tensor A and an isometry $W : \mathbb{C}^d \to (\mathbb{C}^d)^{\otimes p}$ (encoded as a matrix $W \in M_{d^p\times d}(\mathbb{C})$ with $W^\dagger W = \mathbb{1}$) such that the p-blocked tensor $A^{[p]}$ matches the W-image of B at the level of MPV coefficients:

$$V^{(N)}(A^{[p]})_\tau = \sum_{\sigma\in\{0,\dots,d-1\}^N} \left(\prod_k W_{\tau(k),\sigma(k)}\right) V^{(N)}(B)_\sigma$$

for every $N \in \mathbb{N}$ and every $\tau \in \{0,\dots,d^p-1\}^N$.

In Dirac notation, this encodes $|V_{pN}(A)\rangle = W^{\otimes N}\,|V_N(B)\rangle$ for all N.

Theorem 13.48 (Pullback of a refinement witness). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. If B is p-refinable, then there exist a tensor A, an isometry $W : \mathbb{C}^d \to (\mathbb{C}^d)^{\otimes p}$, and the pullback tensor*

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha,i} B^i \qquad (\alpha \in \{0,\dots,d^p-1\})$$

such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. ✓ Proof Choose (A, W) from the refinement data. The pullback construction gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W. The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of B_j. Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of C. □

Definition 13.49 (Blocked equal-case Z-gauge extraction hypothesis). Lean ✓ Stmt Fix (d, D, p). This hypothesis says that whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$-gauge equivalence between C and $A^{[p]}$.

Definition 13.50 (Blocked-to-root reconstruction hypothesis from Z-gauge). Lean ✓ Stmt Fix (d, D, p). This hypothesis says that if $m \geq 1$ and a blocked tensor C in irreducible form II is $\mathbb{Z}_m$-gauge equivalent to $A^{[p]}$, then there exists a tensor A' with $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ and $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Definition 13.51 (Blocked equal-case root hypothesis). Lean ✓ Stmt Fix (d, D, p). This states the combined root-existence statement: whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, one can replace A by a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Theorem 13.52 (Blocked equal-case root from Z-gauge extraction). Lean ✓ Stmt *If the blocked equal-case Z-gauge extraction hypothesis and the blocked-to-root reconstruction hypothesis both hold, then the combined blocked equal-case root hypothesis holds.*

Proof. ✓ Proof Apply the Z-gauge extraction hypothesis to obtain a blocked $\mathbb{Z}_m$-gauge witness between C and $A^{[p]}$, and then apply the reconstruction hypothesis to that witness. □

Theorem 13.53 (Reduction to the blocked equal-case root hypothesis). Lean ✓ Stmt *Assume the blocked equal-case root hypothesis. Then every p-refinable tensor B in irreducible form II admits a left-canonical root A' with $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$.*

Proof. ✓ Proof Apply Theorem 13.48 to the refinement witness of B. This produces a blocked tensor C in irreducible form II with $\mathcal{E}_C = \mathcal{E}_B$ and the same MPV family as $A^{[p]}$. The blocked equal-case root hypothesis then gives a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$. Combining the two transfer-map identities yields $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$. □

Theorem 13.54 (Forward direction from the blocked equal-case root hypothesis). Lean ✓ Stmt *Assume the blocked equal-case root hypothesis. Then for every tensor B in irreducible form II, p-refinability of B implies p-divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof Combine Theorem 13.53 with the witness-level forward implication: once $\mathcal{E}_B$ is identified with $\mathcal{E}_{A'^{[p]}}$ for a left-canonical A', the blocking identity gives $\mathcal{E}_B = (\mathcal{E}_{A'})^p$. □

The following conditional statements isolate the refinability–divisibility equivalence from [DlCCSPG17, Theorem 4.1].

Theorem 13.55 (p-refinability iff transfer-map p-divisibility). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the two root hypotheses used in the one-way statements: the forward left-canonical root hypothesis for refinability ⇒ divisibility and the reverse transfer-map root hypothesis for divisibility ⇒ refinability. Then B is p-refinable iff its transfer map $\mathcal{E}_B$ is p-divisible.*

Proof. ✓ Proof Bundle the forward implication (Theorem 13.56) with the reverse implication (Theorem 13.59) into a single biconditional. The forward direction assumes the left-canonical root hypothesis; the reverse direction assumes only the transfer-map root hypothesis. Together they give the full equivalence under those hypotheses. □

Theorem 13.56 (p-refinability implies transfer-map p-divisibility)**.** Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. Assume a left-canonical root hypothesis: every refinement witness gives a tensor A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. Then p-refinability of B implies p-divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof The left-canonical root hypothesis supplies a witness A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (left-canonical) and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. The left-canonical property makes $\mathcal{E}_A$ a CPTP channel, and the blocking identity $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$ then identifies $\mathcal{E}_B$ with the p-th power of the channel $\mathcal{E}_A$. □

Definition 13.57 (Blocked-to-root channel-root hypothesis from Z-gauge)**.** Lean ✓ Stmt Fix (d, D, p). This sharpened reconstruction hypothesis asks for a CPTP map $\mathcal{E}'$ such that $\mathcal{E}_C = (\mathcal{E}')^p$ and the Choi/Kraus rank of $\mathcal{E}'$ is at most d, whenever C is related to $A^{[p]}$ by a blocked $\mathbb{Z}_m$-gauge witness.

Theorem 13.58 (Channel-root criterion for blocked-to-root reconstruction)**.** Lean ✓ Stmt *If the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.*

Proof. ✓ Proof Choose a tensor A' realizing the bounded Kraus family of the root channel. Because the channel is trace-preserving, the Kraus operators of A' satisfy $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$. The identity $\mathcal{E}_C = (\mathcal{E}_{A'})^p$ is then equivalent to $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ by the blocking formula. □

Theorem 13.59 (Transfer-map p-divisibility implies p-refinability)**.** Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the reverse transfer-map root hypothesis: there is a tensor A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is p-divisible. Then p-divisibility of $\mathcal{E}_B$ implies p-refinability of B.*

Proof. ✓ Proof From the reverse transfer-map root hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so [Wol12, Theorem 2.1(4)] supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding the coefficient of $A^{[p]}$ at the word τ by linearity of matrix multiplication and of the trace, the contributions reorganise into the V-weighted sum over length-N words of B required by Definition 13.47, exhibiting V as the isometry W in the statement of p-refinability. □

Remark 13.60 (Continuum-limit application)*.* Not ready The original motivation for Theorem 13.55 is a continuum-limit construction in the spirit of [DlCCSPG17]: refinability along arbitrarily large p is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

Chapter 14

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG+18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly inside $M_D(\mathbb{C})$. Two lemmas carry the whole fixed-length argument: the virtual bond gauge (Lemma 14.13) and the proportionality lemma (Lemma 14.14). From them follow the main chain theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains. A second, parallel development treats block-diagonal tensors through a product-algebra automorphism and its block-permutation decomposition, leading to the direct-sum gauge-equivalence statements and the equal-case block matching with differing bond dimensions.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are not pursued here.

14.1 Non-translation-invariant periodic chains

The algebraic proof of [MGRPG+18] keeps a fixed periodic chain length and lets the local tensors vary from site to site. The necessary objects—the coefficient of a configuration, the same-state relation, sitewise injectivity, and cyclic gauge equivalence—are recorded first.

Definition 14.1 (Periodic non-TI chain data)**.** Lean Lean Lean Lean Lean ✓ Stmt Fix a chain length n. A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \qquad k = 0, \dots, n-1,$$

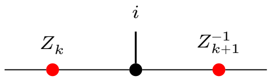
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \operatorname{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ. The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i, with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 14.2 (Cyclic gauge equivalence is an equivalence relation). Lean Lean Lean ✓ Stmt *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. ✓ Proof Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. □

14.2 One-sided inverse and physical realisation of virtual insertions

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \to M_D(\mathbb{C})$ is surjective and admits a right inverse. The resulting decomposition map expresses any matrix as a linear combination of the spanning family $\{A^i\}$, and its multiplicative refinement re-expresses a virtual insertion on the bond as a physical operation on the local index.

Lemma 14.3 (Existence of a right inverse). Lean ✓ Stmt *If A is injective, then there exists a $\mathbb{C}$-linear map $\Psi : M_D(\mathbb{C}) \to \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \mathrm{id}_{M_D(\mathbb{C})}$.*

Proof. ✓ Proof Surjectivity of Φ_A yields a linear section by choice. □

Definition 14.4 (Decomposition map). Lean ✓ Stmt Let A be an injective MPS tensor. The *decomposition map* $\Psi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A.

Lemma 14.5 (Decomposition property). Lean ✓ Stmt *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i \, A^i.$$

Proof. ✓ Proof Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 14.6 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \mathrm{id}$, so the choice does not affect the conclusions.

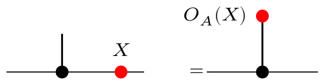
14.2.1 Physical realisation of virtual insertions

For an injective tensor A, the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices. Its multiplicativity (Theorem 14.9) turns the bond algebra into an algebra acting on the physical leg, and is the source of the algebra homomorphism used in the virtual bond gauge.

Definition 14.7 (Physical realisation map). Lean ✓ Stmt Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:



The map $X \mapsto O_A(X)$ is $\mathbb{C}$-linear.

Lemma 14.8 (Defining property of the physical realisation map). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and every physical index i,*

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Proof. ✓ Proof By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \mathrm{id}$. □

Theorem 14.9 (Physical realisation is multiplicative). Lean ✓ Stmt *Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,*

$$O_A(XY) = O_A(X)\, O_A(Y).$$

Proof. ✓ Proof Applying Lemma 14.8 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A:

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij}\, \Psi_A(A^j Y).$$

Taking the kth component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij}\, O_A(Y)_{jk} = (O_A(X)\, O_A(Y))_{ik}.$$

Since this holds for all i and k, the matrices are equal. □

Remark 14.10 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 14.9 is already the general injective statement: for the fixed choice of Ψ_A in Definition 14.4, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

14.3 The virtual bond gauge and the proportionality lemma

The two structural lemmas of [MGRPG+18] carry the entire fixed-length argument. The virtual bond gauge (Lemma 14.13) turns equality of the combined MPV families into the virtual-bond conjugacy $X \mapsto Z^{-1}XZ$ by a fixed invertible matrix Z; the proportionality lemma (Lemma 14.14) shows that two tensor pairs agreeing under all virtual insertions are proportional. Both rest on two elementary facts: injectivity is preserved by blocking, and the centre of $M_D(\mathbb{C})$ consists of scalars.

14.3.1 Two preliminary facts

Lemma 14.11 (Blocking preserves injectivity). Lean ✓ Stmt *Let $A_0, \ldots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \cdots \otimes A_{m-1})^{(i_0,\ldots,i_{m-1})} := A_0^{i_0} A_1^{i_1} \cdots A_{m-1}^{i_{m-1}}$ is again injective.*

Proof. ✓ Proof The span of the blocked matrices contains all products $A_0^{i_0} \cdots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. □

Lemma 14.12 (Centre of $M_D(\mathbb{C})$). Lean ✓ Stmt *If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}$.*

Proof. ✓ Proof Standard: Z commutes with every matrix unit E_{ij}, which forces all off-diagonal entries to vanish and all diagonal entries to agree. □

14.3.2 Same state forces a virtual bond gauge

The 3-site virtual-bond statement is used later in the proof of the PEPS fundamental theorem. Its hypothesis is the all-length MPV equality of the combined tensors, and its conclusion is the conjugacy of the middle-bond insertion data.

Lemma 14.13 (Virtual bond gauge – 3-site case). Lean ✓ Stmt *Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \mathrm{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:*

$$\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Z^{-1} X Z\, B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2.

In [MGRPG+18, Lemma 1], the statement is given for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is not pursued here.

Proof. ✓ Proof The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\mathrm{tr}(A_0^{i_0} X\, A_1^{i_1} A_2^{i_2}) = \mathrm{tr}(B_0^{i_0} Y\, B_1^{i_1} B_2^{i_2})$ for some Y depending on X. Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 14.9. Setting $T(X) = Y$ defines a $\mathbb{C}$-linear multiplicative map, hence a nonzero $\mathbb{C}$-algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 3.8. Skolem–Noether (Theorem 3.11) gives $Z \in \mathrm{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1}XZ$. □

14.3.3 Aligned gauges force proportionality

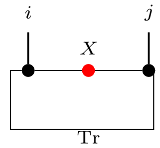
After applying Lemma 14.13 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG+18] shows that this agreement forces the local tensors to be proportional.

Lemma 14.14 (Aligned gauges force proportionality – Lemma 2 of [MGRPG+18]). Lean ✓ Stmt
Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D. Suppose that for all $X \in M_D(\mathbb{C})$ (on the internal bond), all $Y \in M_D(\mathbb{C})$ (on the external bond), and all physical indices i, j,

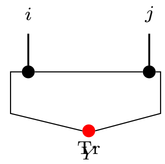
$$\mathrm{tr}(A_1^i\, X\, A_2^j) = \mathrm{tr}(B_1^i\, X\, B_2^j) \tag{14.1}$$

$$\mathrm{tr}(A_2^j\, Y\, A_1^i) = \mathrm{tr}(B_2^j\, Y\, B_1^i) \tag{14.2}$$

Diagrammatically, inserting X on the bond between the two sites (Eq. 14.1):



while the insertion Y on the outer trace bond (Eq. 14.2) is:



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

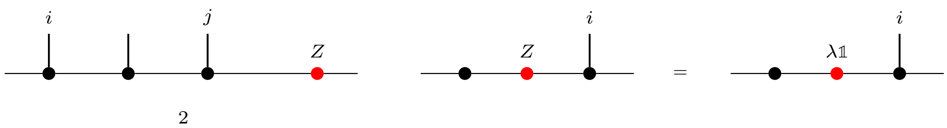
$$A_1^i = \lambda\, B_1^i \qquad \textit{and} \qquad A_2^j = \lambda^{-1}\, B_2^j$$

for all physical indices i, j.

Proof. ✓ Proof By cyclicity and non-degeneracy of the trace pairing, the conditions in Equations (14.1) and (14.2) give, for all physical indices i, j,

$$A_2^j A_1^i = B_2^j B_1^i,$$
$$A_1^i A_2^j = B_1^i B_2^j.$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2}, and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 14.12, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. □

14.4 The Fundamental Theorem for injective chains

The chain theorem combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. Its hypothesis is all-length MPV equality of the combined tensors; the fixed-length same-state form is obtained from a separate same-state reduction hypothesis, treated below.

Theorem 14.15 (Fundamental Theorem for injective chains). Lean ✓ Stmt *Let* $\mathbf{A} = (A_0, \dots, A_{n-1})$ *and* $\mathbf{B} = (B_0, \dots, B_{n-1})$ *be two* n*-site chains with common bond dimension* D*, and assume that* $\mathbf{A}$ *is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then* $\mathbf{A}$ *and* $\mathbf{B}$ *are cyclically gauge equivalent.*

Proof. ✓ Proof Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the combined tensor by Lemma 14.11. Apply the single-block Fundamental Theorem (Theorem 3.12) to the two combined tensors. This yields one invertible matrix X such that every combined matrix of $\mathbf{B}$ equals the corresponding combined matrix of $\mathbf{A}$ conjugated by X. Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i, so the constant choice $Z_k = X$ is a cyclic gauge. □

Lemma 14.16 (Uniform site rescaling rescales the combined tensor). Lean ✓ Stmt *Let* $\mathbf{A} = (A_0, \dots, A_{n-1})$ *be an* n*-site chain and let* $\zeta \in \mathbb{C}$*. If* $\zeta\mathbf{A}$ *denotes the chain obtained by replacing each site tensor* A_k *with* ζA_k*, and if* $\widetilde{\mathbf{A}}$ *denotes the combined tensor of* $\mathbf{A}$*, then*

$$\widetilde{\zeta\mathbf{A}} = \zeta\,\widetilde{\mathbf{A}}.$$

Proof. ✓ Proof Combining the site and physical indices does not change which local matrix is read at the component (k, i), so both sides evaluate to ζA_k^i there. □

Theorem 14.17 (Injective-chain Fundamental Theorem up to a global phase). Lean ✓ Stmt *Let* $\mathbf{A} = (A_0, \dots, A_{n-1})$ *and* $\mathbf{B} = (B_0, \dots, B_{n-1})$ *be two* n*-site chains with common bond dimension* D*, and assume that* $\mathbf{A}$ *is injective. Suppose that the combined tensors* $\widetilde{\mathbf{A}}$ *and* $\widetilde{\mathbf{B}}$ *are gauge-phase equivalent. Then there exist invertible matrices* $Z_0, \dots, Z_{n-1} \in \mathrm{GL}_D(\mathbb{C})$ *and a nonzero scalar* $\zeta \in \mathbb{C}^\times$ *such that*

$$B_k^i = \zeta\, Z_k A_k^i Z_{k+1}^{-1}$$

for all sites k *and physical indices* i*, where* $k+1$ *is interpreted cyclically modulo* n*.*

Proof. ✓ Proof If the combined tensor of $\mathbf{B}$ differs from that of $\mathbf{A}$ by a gauge and a nonzero scalar ζ, rescale each site of $\mathbf{B}$ by ζ^{-1}. Lemma 14.16 removes this common scalar from the combined tensor, so Theorem 14.15 applies to the rescaled chain. Multiplying the resulting sitewise relations by ζ gives the displayed conclusion. □

14.4.1 From fixed-length chain states to combined-tensor MPV agreement

The chain theorem assumes equality of the *combined-tensor* MPV families. Converting the fixed-length same-state hypothesis into this form is isolated as a separate hypothesis, which the source proof establishes by the virtual bond gauge and proportionality lemmas above.

Definition 14.18 (Same-state reduction hypothesis). Lean ✓ Stmt A *same-state reduction hypothesis* for dimensions (d, D) is the assertion that whenever $\mathbf{A}$ is an injective n-site chain with $n \geq 3$ and $\mathbf{A}, \mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Lemma 14.19 (Same state gives combined-tensor MPV agreement). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be n-site chains with common bond dimension D, with $n \geq 3$, and assume that* $\mathbf{A}$ *is injective. If* $\mathbf{A}$ *and* $\mathbf{B}$ *generate the same chain state, then the combined tensors of* $\mathbf{A}$ *and* $\mathbf{B}$ *generate the same MPV family.*

Proof. ✓ Proof This is exactly the defining property of the same-state reduction hypothesis. □

Theorem 14.20 (Fundamental Theorem for injective chains from same state). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be n-site chains with common bond dimension D, with $n \geq 3$, and assume that* $\mathbf{A}$ *is injective. If* $\mathbf{A}$ *and* $\mathbf{B}$ *generate the same chain state, then* $\mathbf{A}$ *and* $\mathbf{B}$ *are cyclically gauge equivalent.*

Proof. ✓ Proof Apply Lemma 14.19 to convert the fixed-length chain-state hypothesis into equality of the combined-tensor MPV families, then invoke Theorem 14.15. □

Remark 14.21 (Relationship to the translation-invariant all-sizes theorem). Theorem 14.15 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 3.12) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = XA^iX^{-1}$. The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

14.4.2 Finite-length MPV agreement

The single-block theorem can be strengthened by weakening its hypothesis from all-length to finite-length MPV agreement: for an injective tensor, agreement from any threshold N_0 onward already implies full agreement.

Definition 14.22 (Finite-length MPV agreement). Lean ✓ Stmt Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$, meaning that for every $N \geq N_0$ and $\sigma \in [d]^N$,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$$

Theorem 14.23 (Finite-length agreement implies full agreement). Lean ✓ Stmt *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.*

Proof. ✓ Proof Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length-N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length-N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\operatorname{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\operatorname{tr}(w_A)$ and $\operatorname{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 14.24 (Strengthened single-block FT (finite-length)). Lean ✓ Stmt *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. ✓ Proof Apply Theorem 14.23 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 3.12). □

14.5 Corollaries: translation-invariant, non-uniform, and blocked chains

The chain theorem specializes to three settings: constant translation-invariant chains, non-uniform chains whose state happens to be translation-invariant, and constant chains of blocked tensors.

14.5.1 Translation-invariant chains

Theorem 14.25 (Single gauge for translation-invariant chains). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D, and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that*

$$B^i = X\, A^i\, X^{-1}$$

for every physical index i.

Proof. ✓ Proof Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the combined tensor by Lemma 14.11, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. □

Corollary 14.26 (Translation-invariant collapse to one gauge). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D, and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \mathrm{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that*

$$B^i = \lambda\, Z^{-1}\, A^i\, Z$$

for every physical index i.

Proof. ✓ Proof Apply Theorem 14.25 to obtain $B^i = XA^iX^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$ and the displayed identity is exactly the same gauge relation rewritten in the form used by the corollary. □

14.5.2 Non-translation-invariant descriptions of translation-invariant states

Corollary 14.27 (Translation-invariant states from nonuniform chains). Not ready *Assume a same-state reduction hypothesis for (d, D). Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-translation-invariant chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a* *translation-invariant description with the same bond dimension: there exists an injective tensor B such that the constant chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication,* not *conjugation), where Z_0 is extracted from the cyclic shift.*

Proof. Apply Theorem 14.20 to $\mathbf{A}$ and its cyclic shift. This gives invertible gauges $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$. Iterating these identities yields $A_k^i = L_k^{-1} A_0^i R_k$, where the products L_k, R_k satisfy $R_k L_{k+1}^{-1} = Z_0^{-1}$. Substituting into the chain trace shows that the same state is generated by the constant tensor $B^i := A_0^i Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. ☐

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 12.

14.5.3 Blocked chains

Lemma 14.28 (Block injectivity and blocked-tensor injectivity). Lean ✓ Stmt *A tensor A is L-block injective if and only if its L-blocked tensor $A^{[L]}$ is injective.*

Proof. ✓ Proof The blocked physical alphabet is just a reindexing of words of length L. Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent. ☐

Definition 14.29 (Constant blocked chain). Lean ✓ Stmt For an MPS tensor A, a blocking length L, and a chain length n, the *blocked chain* is the constant n-site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 14.30 (Blocked chains are injective). Lean ✓ Stmt *If A is L-block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.*

Proof. ✓ Proof By Lemma 14.28, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. ☐

Theorem 14.31 (Fundamental Theorem for blocked chains). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D, let L be a blocking length, and let n be a chain length. Assume that A is L-block injective and that the combined tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.*

Proof. ✓ Proof By Lemma 14.30, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 14.15 to the two constant blocked chains. ☐

Remark 14.32. The theorem above is the blocked-chain result established here. Recovering site-wise gauges for the original unblocked tensors gives the normal-MPS corollary of [MGRPG+18]; that step is outside this blocked-chain statement.

14.6 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 14.15) handles non-translation-invariant periodic chains. A parallel development addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition. Throughout this section, assume $D_k \geq 1$ for every block k.

14.6.1 Block permutation algebra

Definition 14.33 (Block ideal). Lean ✓ Stmt Let $\{R_k\}_{k=0}^{r-1}$ be simple rings. The k-th *block ideal* in $\prod_{j=0}^{r-1} R_j$ is

$$I_k := \{x \in \prod_{j=0}^{r-1} R_j : x_j = 0 \text{ for } j \neq k\}.$$

Theorem 14.34 (Each block ideal is an atom). Lean ✓ Stmt *Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=0}^{r-1} R_j$.*

Proof. ✓ Proof Under the order isomorphism between two-sided ideals of $\prod_{j=0}^{r-1} R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k-th coordinate and $\bot$ elsewhere. This tuple is an atom, hence so is I_k. □

Theorem 14.35 (Ring isomorphisms permute block ideals). Lean ✓ Stmt *Let $\{R_k\}_{k=0}^{r-1}$ be simple rings. Every ring isomorphism $T : \prod_k R_k \to \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k-th block ideal to the $\pi(k)$-th block ideal.*

Proof. ✓ Proof By Theorem 14.34, each block ideal is an atom in the lattice of two-sided ideals of $\prod_k R_k$. A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation π. □

Theorem 14.36 (Dimension preservation). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. If T is an algebra automorphism of $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k.*

Proof. ✓ Proof The restriction of T to the k-th block ideal induces a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ by comparing dimensions as $\mathbb{C}$-vector spaces. □

Theorem 14.37 (Per-block component maps are bijections). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. Let T be a ring automorphism of $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$, and let π be the induced permutation from Theorem 14.35. For each block index k, the map*

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ is bijective.

Proof. ✓ Proof Theorem 14.35 forces a single-support element in the k-th block ideal to remain supported on the $\pi(k)$-th block after applying T. Injectivity follows from injectivity of T. For surjectivity, apply the same argument to T^{-1} and the inverse permutation. □

Theorem 14.38 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). Lean ✓ Stmt *Assume $D_k \geq 1$ for all k. Every $\mathbb{C}$-algebra automorphism T of $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$ decomposes as per-block inner automorphisms followed by a block permutation π: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_0, \dots, M_{r-1})$ and every block index k, $T(M)_{\pi(k)} = X_k M_k X_k^{-1}$, where $T(M)_j$ denotes the j-th block component of $T(M)$.*

Proof. ✓ Proof Theorem 14.35 gives the permutation π. Theorem 14.36 identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k, Theorem 14.37 gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to

$M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the elements supported on one factor of T; after reindexing by the dimension equality, this is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem 3.11) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, Section IV.A] for the MPS context. □

Remark 14.39 (Ring level versus algebra level). Theorem 14.35 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem 14.37 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 14.36 and 14.38 require an upgrade to $\mathbb{C}$-algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$-vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$-algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

14.6.2 Per-block linear extension and the product automorphism

Per-block equality of MPV families produces a unique linear map on each matrix block; assembling these maps componentwise yields an algebra automorphism of the product, to which the block-permutation decomposition applies.

Definition 14.40 (Per-block linear extension). Lean ✓ Stmt Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$-linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i.

Lemma 14.41 (Per-block multiplicativity). *Lean* *✓ Stmt* *For each block k, the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)\,T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.*

Proof. ✓ Proof Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. □

Lemma 14.42 (Per-block bijectivity). *Lean* *✓ Stmt* *Each T_k is bijective.*

Proof. ✓ Proof A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. □

Definition 14.43 (Product algebra automorphism). Lean ✓ Stmt The *product algebra automorphism* is the $\mathbb{C}$-algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the componentwise map on the product algebra obtained by applying the corresponding extension to each summand.

Theorem 14.44 (Block-permutation decomposition). Lean ✓ Stmt *The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$(T(M))_{\sigma(k)} = X_k\, M\, X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k. Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. ✓ Proof Apply Theorem 14.38 to the product-algebra automorphism of Definition 14.43. The theorem writes the nonzero input block as $\pi(k)$ in output component k; for the statement above we set $\sigma := \pi^{-1}$, so an element supported in block k lands in block $\sigma(k)$. □

14.6.3 Nondegeneracy of the product trace pairing

Lemma 14.45 (Nondegeneracy of the product trace pairing). Lean ✓ Stmt *Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \mathrm{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.*

Proof. ✓ Proof Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\mathrm{tr}(M_k N_k) = 0$ for all N_k, so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. □

Lemma 14.46 (Injectivity of the product Gram map). Lean ✓ Stmt *Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$ is injective.*

Proof. ✓ Proof Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \mathrm{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). □

14.7 Gauge equivalence for direct sums

The first lemmas in this section are elementary facts about block-diagonal tensors: they assemble already-known block gauges, and they record the easy consequences of applying the injective single-block theorem to each block separately. They do not derive the block matching appearing in [CPGSV16, Theorem 2.10]. In the source proof, that matching is obtained from BNT overlap, independence, and coefficient comparison; the corresponding global gauge assembly is Theorem 11.12.

Definition 14.47 (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$, first form the dependent block-diagonal element $\bigoplus_k X_k$ and then reindex the dependent block-coordinate index to the standard index $\{0, \dots, \sum_k D_k - 1\}$ of the assembled tensor. The resulting element of $\mathrm{GL}_{\sum_k D_k}(\mathbb{C})$ is the global gauge used in the conjugation formula for $\bigoplus_k \mu_k A_k$.

Lemma 14.48 (Explicit direct-sum conjugation by the flattened gauge). Lean Lean ✓ Stmt *If each block satisfies $B_k^i = X_k A_k^i X_k^{-1}$, then the weighted direct sums satisfy, for every physical index i,*

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1},$$

where X is the flattened block-diagonal gauge of Definition 14.47. Consequently the two assembled tensors are gauge equivalent with this explicit witness.

Proof. ✓ Proof The block-diagonal matrix product is computed blockwise. The scalar weight μ_k commutes with the per-block conjugation, so each diagonal block of the right-hand side is $\mu_k B_k^i$. Reindexing from the dependent block coordinates to the standard bond index preserves multiplication and inverses. The gauge equivalence follows directly from the conjugation identity. □

Lemma 14.49 (Gauge equivalence of direct sums from per-block gauge equivalence)**.** Lean ✓ Stmt *If each pair (A_k, B_k) is gauge equivalent, then the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof Choose per-block gauge matrices via the hypothesis; then apply Lemma 14.48. □

Lemma 14.50 (Gauge equivalence of direct sums with phase-weight absorption)**.** Lean ✓ Stmt *Let A_k, B_k be block families, and let $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ and scalars $\zeta_k \in \mathbb{C}$ satisfy $B_k^i = \zeta_k \cdot X_k A_k^i X_k^{-1}$ for every block k and physical index i. If the weights satisfy $\mu_A(k) = \mu_B(k)\zeta_k$, then the weighted block-diagonal tensors $\bigoplus_k \mu_A(k) A_k$ and $\bigoplus_k \mu_B(k) B_k$ are gauge equivalent.*

Proof. ✓ Proof Per-block, the weight identity $\mu_A(k) = \mu_B(k)\zeta_k$ rewrites the scaled relation as $\mu_B(k)\, B_k^i = X_k\, (\mu_A(k)\, A_k^i)\, X_k^{-1}$. Each weighted block then satisfies an ordinary conjugation, and the block-diagonal conjugation formula assembles the per-block X_k into the flattened global gauge $\bigoplus_k X_k$ relating $\bigoplus_k \mu_A(k) A_k$ to $\bigoplus_k \mu_B(k) B_k$. □

14.7.1 Same-index direct-sum corollaries from the single-block theorem

Remark 14.51. See Definition 2.39 for the formalized block-diagonal tensor construction. The results in this subsection assume that the block correspondence has already been fixed; they do not prove [CPGSV16, Theorem 2.10]. They are therefore retained here as same-index assembly corollaries, not among the BNT preparation steps used in the source proof or in the after-blocking canonical-form reduction.

Lemma 14.52 (Per-block application of the single-block theorem)**.** Lean ✓ Stmt *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. ✓ Proof Apply the single-block Fundamental Theorem (Theorem 3.12) to each block independently. □

Remark 14.53 (Per-block restatement, not the multi-block paper theorem). This is a block-sum reformulation of the per-block injective single-block FT: the block pairing (A_k, B_k) is assumed already given and the blocks already matched. It is not the multi-block Fundamental Theorem of [CPGSV16, Theorem 2.10, lines 349–352], which derives the block matching and permutation from only the global same-MPV hypothesis.

Lemma 14.54 (Per-block same MPV implies global same MPV)**.** Lean ✓ Stmt *If A_k and B_k generate the same MPV family for each block k, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. ✓ Proof By the MPV decomposition (Theorem 2.41), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. □

Remark 14.55 (Direct-sum congruence). This is a direct-sum congruence statement: per-block MPV equality lifts to global MPV equality via the block-diagonal decomposition formula.

Lemma 14.56 (Gauge equivalence for same-index direct sums). Lean ✓ Stmt *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof Apply the per-block single-block Fundamental Theorem (Lemma 14.52) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. □

Remark 14.57 (Block-diagonal gauge assembly, not the FT block-matching theorem). This combines per-block gauges into a block-diagonal gauge for the global assembled tensor, given prematched same-index blocks. It is not the FT block-matching theorem: it does not derive the block pairing or permutation from a global same-MPV hypothesis; the block correspondence is assumed in advance.

Lemma 14.58 (Gauge equivalence of direct sums from per-block MPV equality). Lean ✓ Stmt *Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions D_k, each A_k injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k. Then:*

1. *Per-block gauge equivalence: $B_k^i = X_k\, A_k^i\, X_k^{-1}$ for some $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$, for every k and i.*
2. *Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Explicitly, for $X = \bigoplus_k X_k$,

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1}.$$

Proof. ✓ Proof Per-block gauge equivalence follows from applying the single-block Fundamental Theorem to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. With $X = \bigoplus_k X_k$, the displayed identity follows by multiplying block-diagonal matrices block by block. □

Lemma 14.59 (Explicit per-block gauges). Lean ✓ Stmt *Under the hypotheses of Lemma 14.58, there exist invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k\, A_k^i\, X_k^{-1}$$

for every block k and every physical index i.

Proof. ✓ Proof Extract the gauge matrix from the per-block part of Lemma 14.58 for each block. □

Lemma 14.60 (Product-algebra decomposition of per-block MPV equality). Lean ✓ Stmt *Assume that every block dimension is nonzero. Under the hypotheses of Lemma 14.58, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation σ of the blocks, compatible block-dimension equalities, and invertible matrices X_k such that, after reindexing the target block $\sigma(k)$ back to block k, the action on the kth matrix block is $M \mapsto X_k M X_k^{-1}$.*

Proof. ✓ Proof Apply Theorem 14.44 to the product-algebra automorphism built from the per-block linear extensions. □

14.7.2 MPV invariance under reindexing

Lemma 14.61 (MPV invariance under block permutation). Lean ✓ Stmt *Permuting the block index set and transporting the weights correspondingly does not change the* $\mathrm{SameMPV}_2$ *family of the weighted block-diagonal tensor.*

Proof. ✓ Proof The sum of per-block MPV contributions is invariant under rearrangement of the summands. □

Lemma 14.62 (MPV invariance under dimension cast). Lean ✓ Stmt *If two block-dimension families agree pointwise, the corresponding weighted block-diagonal tensors generate the same* $\mathrm{SameMPV}_2$ *family after transporting the blocks across the dimension equalities.*

Proof. ✓ Proof Substituting the equality of dimensions makes the two tensor families identical. □

14.7.3 Converse: gauge equivalence implies same MPV

Lemma 14.63 (Gauge equivalence of direct sums implies same MPV). Lean ✓ Stmt *If the weighted block-diagonal tensors* $\bigoplus_k \mu_k A_k$ *and* $\bigoplus_k \mu_k B_k$ *are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Gauge equivalence preserves all traces of word evaluations, hence all MPV coefficients. □

14.7.4 Canonical-form tensor as a weighted direct sum

Lemma 14.64 (Canonical-form tensor agrees with block sum). Lean ✓ Stmt *For a canonical-form record* C*, the assembled tensor* T_C *is definitionally equal to the weighted block-diagonal tensor* $\bigoplus_k \mu_k^C A_k^C$*, where* μ_k^C *are the block weights and* A_k^C *are the block tensors stored in* C*.*

Proof. ✓ Proof The equality holds by definition; T_C is defined as the weighted block-diagonal tensor formed from the weights and block tensors of C. □

Theorem 14.65 (Fundamental theorem for canonical forms with the same block structure). Lean ✓ Stmt *Let* C *be a canonical-form record with block weights* μ_k^C *and block tensors* A_k^C*. Let* $\{B_k\}$ *be a family of MPS tensors on the same block dimensions, and assume that each* A_k^C *is injective and generates the same MPV family as* B_k*. Then* T_C *and* $\bigoplus_k \mu_k^C B_k$ *are gauge equivalent.*

Proof. ✓ Proof Rewrite T_C as a weighted block-diagonal tensor via Lemma 14.64, then apply the multi-block global gauge-equivalence lemma (Lemma 14.56). □

Lemma 14.66 (Per-block equality of MPV families and per-block gauge equivalence). Lean ✓ Stmt *Let* $\{A_k\}$ *be a family of injective block tensors. Then* $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ *for all* k *if and only if* A_k *and* B_k *are gauge equivalent for all* k*.*

Proof. ✓ Proof The forward direction applies the single-block Fundamental Theorem to each block. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. □

14.7.5 Equal-case block matching with differing bond dimensions

The same-index lemmas above assume a fixed block correspondence. The equal-case theorem on the BNT canonical form instead derives the sector matching from the same-MPV hypothesis alone, allowing different sector counts and bond dimensions before matching.

Lemma 14.67 (Equal-case block matching with differing bond dimensions). Lean ✓ Stmt *Let P and Q be two BNT canonical forms (Definition 10.27), each admitting a unit-modulus copy weight in every basis sector, possibly with different sector counts and different bond dimensions before matching. If their block-diagonal tensors generate the same MPV family (Definition 2.12), then the basis sectors are bijectively matched by a permutation β, each matched basis pair is gauge-phase equivalent (Definition 2.23), the matched copy multiplicities agree, and the copy weights agree up to a unit-modulus phase ζ and a copy-index reindexing. This is the sector-data form of the equal-case theorem [CPGSV16, Corollary 2.11, lines 354–361 and appendix 1172–1192]; the corresponding global gauge corollary is recorded in Chapter 11.*

Proof. ✓ Proof On the BNT canonical form, the proof proceeds in three steps: bijective block matching from non-decaying cross-overlaps, coefficient identity via the global-gauge phase ζ, and multiset comparison of the scalar power sums recovering matched copy multiplicities and weights up to ζ^{-1}. □

14.8 Translation-invariant reduction and global symmetry

The single-gauge corollary for translation-invariant chains specializes to a reduction statement: equal states force one tensor to be a gauge of the other and inherit injectivity. Composing two gauges of the same tensor then yields the scalar law underlying the global-symmetry consequences.

Theorem 14.68 (Translation-invariant reduction corollary). Lean ✓ Stmt *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ with $B^i = X\, A^i\, X^{-1}$ for all i.*
2. *B is injective.*

Proof. ✓ Proof Theorem 14.25 gives a single gauge matrix X with $B^i = XA^iX^{-1}$ for all i. Since gauge equivalence preserves injectivity, B is injective as well. □

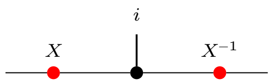
Theorem 14.69 (Translation-invariant reduction from same state). Lean ✓ Stmt *Assume a same-state reduction hypothesis for (d, D). Let A and B be MPS tensors of bond dimension D, and let $n \geq 3$. If the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same chain state, with A injective, then the same conclusion as in Theorem 14.68 holds: B is gauge equivalent to A, and B is injective.*

Proof. ✓ Proof The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their combined-tensor MPV families. Then Theorem 14.68 applies. □

14.8.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 12.15) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 14.70 (Gauge ratio commutes with tensor matrices). Lean ✓ Stmt *Diagrammatically, the hypothesis is that the two local gauges*



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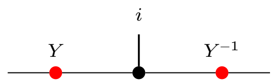


produce the same target tensor B^i from the same source tensor A^i. If $X, Y \in \mathrm{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i.

Proof. ✓ Proof Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. ☐

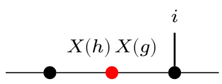
Lemma 14.71 (Gauge ratio is scalar for injective tensors). Lean ✓ Stmt *Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

Proof. ✓ Proof The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 14.70, $Y^{-1}X$ commutes with all A^i. Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 14.12, it is a scalar matrix. ☐

Theorem 14.72 (Projective multiplication law for gauge matrices). Lean ✓ Stmt *Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \to \mathrm{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \mathrm{GL}_D(\mathbb{C})$ satisfy $\widetilde{A}^i_g = X(g)\, A^i\, X(g)^{-1}$ for all i. Then for all $g, h \in G$, there exists $c(g,h) \in \mathbb{C}$ such that*

$$X(h)\, X(g) = c(g,h)\, X(g \cdot h).$$

In fact $c(g,h) \neq 0$ since both sides are invertible, so $c(g,h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



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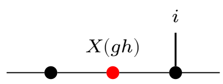


and both sides conjugate A^i to the same twisted tensor $\widetilde{A}^i_{gh}$.

Proof. ✓ Proof By the composition law $\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g$ (Lemma 12.5), both $X(h)\, X(g)$ and $X(gh)$ conjugate A to $\widetilde{A}_{gh}$. Lemma 14.71 then gives the scalar factor. ☐

14.9 Closing remarks

Remark 14.73. Appendix A of [MGRPG+18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L + 1$ by a more refined blocking argument. The injective PEPS theorem of Chapter 15 and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 14.13 and Lemma 14.14); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 15

Fundamental Theorems for Injective and Normal PEPS

The following definitions and results extend the Fundamental Theorem to injective and normal PEPS, following Theorems 2 and 3 of [MGRPG⁺18].

Definition 15.1 (Graph edge for PEPS). Lean ✓ Stmt For a finite simple graph G, an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v. The ordering is only used to avoid double counting.

Definition 15.2 (Incident edges at a vertex). Lean ✓ Stmt For a vertex v, an *incident edge* is an edge of G having v as one of its endpoints.

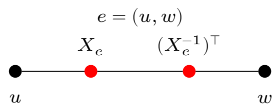
Definition 15.3 (PEPS tensor on a graph). Lean ✓ Stmt A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e, together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical index lies in $\{0, \dots, d-1\}$.

Definition 15.4 (Virtual configuration). Lean ✓ Stmt A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e.

Definition 15.5 (PEPS state coefficient). Lean ✓ Stmt For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

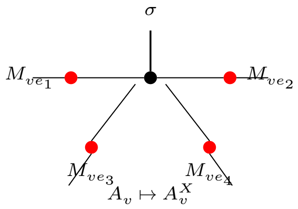
Definition 15.6 (Equality of PEPS states). Lean ✓ Stmt Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

Definition 15.7 (Edge gauge at a vertex). Lean ✓ Stmt For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \mathrm{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$. Diagrammatically, the ordered edge carries opposite endpoint actions:



Definition 15.8 (Gauge vertex action). Lean ✓ Stmt The gauged tensor at vertex v is A_v^X. For each incident edge ie, write M_{ie} for the matrix specified by Definition 15.7, applied at the vertex v

to the gauge on the underlying edge of ie. Then $A_v^X(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))A_v(\eta',\sigma)$. In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



Definition 15.9 (Apply gauge). Lean ✓ Stmt The PEPS tensor obtained by applying edge-gauge matrices to A, with vertex tensors A_v^X.

Definition 15.10 (Absorb edge gauges). Lean ✓ Stmt Given a tensor family B and an oriented gauge matrix X_e on every edge, the absorbed tensor family is $\widetilde{B} = B^X$; at each vertex, its local tensor is obtained by inserting the oriented endpoint matrices on the incident virtual legs.

Theorem 15.11 (Absorbed gauges keep the bond spaces). Lean ✓ Stmt *The absorbed tensor family has $D_{\widetilde{B}}(e) = D_B(e)$ for every edge e.*

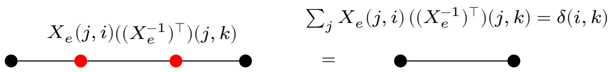
Proof. ✓ Proof Since $\widetilde{B} = B^X$ is defined with the same bond-dimension function as B, one has $D_{\widetilde{B}}(e) = D_{B^X}(e) = D_B(e)$ for every edge e. □

Theorem 15.12 (Local formula for absorbed gauges). Lean ✓ Stmt *If M_{ie} is the oriented endpoint matrix attached to an incident edge ie at v, then $\widetilde{B}_v(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))B_v(\eta',\sigma)$.*

Proof. ✓ Proof Expanding $\widetilde{B} = B^X$ at a vertex gives $\widetilde{B}_v(\eta,\sigma) = (B^X)_v(\eta,\sigma) = \sum_{\eta'}(\prod_{ie} M_{ie}(\eta(ie),\eta'(ie)))B_v(\eta',\sigma)$. □

Definition 15.13 (Gauge equivalence for PEPS). Lean ✓ Stmt Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 15.14 (Gauge invariance of PEPS state). Lean ✓ Stmt *Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:*



Theorem 15.15 (Gauge equivalence implies same state). Lean ✓ Stmt *Gauge equivalence implies that the two PEPS tensors represent the same state.*

Definition 15.16 (Local tensor evaluation). Lean ✓ Stmt The local tensor evaluated at vertex v with virtual-index weighting f, computing $\sum_\eta(\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta,\sigma)$. This is multilinear in the components of f, not linear in the full tuple.

Definition 15.17 (Vertex injectivity). Lean ✓ Stmt A PEPS tensor A is *vertex-injective* if, at every vertex v, the family of physical vectors

$$(A_v(\eta,\cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the injectivity formulation of [MGRPG+18, Section 3], where the local tensor is interpreted as a map from virtual configurations to the physical space.

Definition 15.18 (Local tensor map). Lean ✓ Stmt For a fixed vertex v, the family of local physical vectors $\eta \mapsto A_v(\eta, \cdot)$ extends by linearity to a map from the coefficient space on local virtual configurations at v into $\mathbb{C}^d$.

Definition 15.19 (Local left inverse). Lean ✓ Stmt If A is vertex-injective, one may choose a linear map Ψ_v from $\mathbb{C}^d$ to the coefficient space on local virtual configurations at v such that $\Psi_v \circ \Phi_v = \mathbb{1}$.

Definition 15.20 (Physical realization of a local virtual operator). Lean ✓ Stmt For an endomorphism T of the coefficient space on local virtual configurations at v, define the local physical operator

$$O_T = \Phi_v \circ T \circ \Psi_v.$$

Theorem 15.21 (Local virtual operations on the image). Lean ✓ Stmt *For every coefficient function c on the local virtual configurations at v,*

$$O_T(\Phi_v(c)) = \Phi_v(Tc).$$

Proof. ✓ Proof This is immediate from $\Psi_v \circ \Phi_v = \mathbb{1}$. ☐

Theorem 15.22 (Injectivity of local physical realization). Lean ✓ Stmt *If two virtual operations have the same canonical physical realization, then they agree.*

Proof. ✓ Proof Evaluate both physical realizations on vectors in the image of the local tensor map and use vertex injectivity. ☐

Definition 15.23 (Local projector). Lean ✓ Stmt The local projector is the physical realization of the identity virtual operator:

$$P_v = \Phi_v \circ \Psi_v.$$

Definition 15.24 (Virtual pullback of a local physical operator). Lean ✓ Stmt For a physical operator O on the local physical space at v, define its virtual pullback by

$$\Psi_v \circ O \circ \Phi_v.$$

This is the local injectivity step used in the physical-to-virtual direction of the insertion-algebra argument in [MGRPG+18, Section 3].

Theorem 15.25 (Virtual pullback realizes the projected physical action). Lean ✓ Stmt *For every coefficient function c, applying the local tensor map after the virtual pullback gives the projection of $O(\Phi_v(c))$ onto the image of Φ_v.*

Proof. ✓ Proof Expand the definitions. The chosen left inverse gives $\Psi_v \Phi_v = \mathbb{1}$, and the remaining map is the local projector. ☐

Theorem 15.26 (Virtual pullback for image-preserving physical actions). Lean ✓ Stmt *If O sends every vector in the image of Φ_v back into that image, then its virtual pullback realizes exactly the action of O on $\Phi_v(c)$.*

Proof. ✓ Proof Use the preceding theorem and the image-preservation hypothesis. □

Theorem 15.27 (Recovery of a virtual operation from a physical realization). Lean ✓ Stmt *If a physical operator O realizes a virtual operation T on every vector in the image of Φ_v, then pulling back O recovers T.*

Proof. ✓ Proof After applying the local tensor map, both virtual operations have the same image. Vertex injectivity makes the local tensor map injective, so the virtual operations agree. □

Theorem 15.28 (Extensionality of physical-to-virtual pullback). Lean ✓ Stmt *Two physical endpoint operations have the same virtual pullback exactly when their projected actions agree on every vector in the image of the local tensor map:*

$$T_O = T_{O'} \iff P_v O \Phi_v(c) = P_v O' \Phi_v(c) \quad \text{for all } c.$$

Proof. ✓ Proof Apply the local tensor map to the two virtual pullbacks and use the pullback specification. The converse follows from vertex injectivity; the forward direction is obtained by rewriting the two virtual pullbacks. □

Theorem 15.29 (Pullback of the canonical physical realization). Lean ✓ Stmt *Pulling back the canonical physical realization of a virtual operation recovers the original virtual operation.*

Proof. ✓ Proof The canonical physical realization agrees with the virtual operation on the image of the local tensor map. Apply the preceding recovery theorem. □

Theorem 15.30 (Physical realization of a pulled-back physical operator). Lean ✓ Stmt *Let $P_v = \Phi_v \circ \Psi_v$ be the local projector, and let $T_O = \Psi_v \circ O \circ \Phi_v$ be the virtual pullback of a physical operator O. Then*

$$\Phi_v \circ T_O \circ \Psi_v = P_v \circ O \circ P_v.$$

Proof. ✓ Proof Expand the definitions of the physical realization, the virtual pullback, and the local projector. □

Theorem 15.31 (Projected local recovery). Lean ✓ Stmt *Suppose that the compressed physical action $P_v O P_v$ agrees with the canonical physical realization of a virtual operation T. Then the virtual pullback of O is T.*

Proof. ✓ Proof The physical realization of the virtual pullback of O is $P_v O P_v$. By hypothesis this is the physical realization of T. Injectivity of the canonical physical realization gives equality of the virtual operations. □

Theorem 15.32 (Projected local recovery equivalence). Lean ✓ Stmt *The compressed physical action $P_v O P_v$ is the canonical physical realization of a virtual operation T if and only if the virtual pullback of O is T.*

Proof. ✓ Proof One implication follows by realizing both virtual operations physically and using the formula for the physical realization of a pulled-back operator. The converse is the preceding recovery theorem. □

Definition 15.33 (Matrix action on one incident virtual edge). Lean ✓ Stmt A matrix on one edge incident to v acts on the coefficient space of local virtual configurations by applying the matrix to that edge coordinate and summing over the input edge index while keeping the residual virtual configuration fixed.

Theorem 15.34 (One-edge matrix action on a basis configuration). Lean ✓ Stmt *If the coefficient function is the unit basis function at one local virtual configuration, and the matrix acts on the distinguished incident edge, the result is the sum over the new distinguished edge index with the residual local boundary held fixed.*

Proof. ✓ Proof Evaluate both sides on a local virtual configuration and split into the two cases according to whether the residual boundary agrees with the fixed residual boundary. □

Theorem 15.35 (Local tensor form of a one-edge basis action). Lean ✓ Stmt *Applying the local tensor map after a one-edge matrix action on a basis virtual configuration gives the corresponding linear combination of local tensor components.*

Proof. ✓ Proof Use the preceding coefficient identity and linearity of the local tensor map. □

Theorem 15.36 (Physical realization of one-edge virtual matrix action). Lean ✓ Stmt *If A is vertex-injective, then a matrix acting on one edge incident to v is realized by a physical operator on the physical space at v.*

Proof. ✓ Proof Apply the physical realization of local virtual operations to the one-edge matrix action. □

Definition 15.37 (Split at one incident edge). Lean ✓ Stmt Choosing one incident edge at v identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

Definition 15.38 (Middle region of an edge). Lean ✓ Stmt For an edge $e = (u, v)$, the middle region is the complement $V \setminus \{u, v\}$.

Theorem 15.39 (Edge-centered vertex partition). Lean ✓ Stmt *For an edge $e = (u, v)$, the three regions $\{u\}$, $V \setminus \{u, v\}$, and $\{v\}$ cover the whole vertex set.*

Proof. ✓ Proof Every vertex is either u, v, or different from both. □

Theorem 15.40 (Nonempty middle region from a cardinal bound). Lean ✓ Stmt *If $2 < |V|$, then for every edge $e = (u, v)$ the middle region $V \setminus \{u, v\}$ is nonempty.*

Proof. ✓ Proof The middle region has cardinality $|V| - 2$. Hence $2 < |V|$ gives $|V| - 2 > 0$. □

Theorem 15.41 (Product splitting at an edge). Lean ✓ Stmt *For any edge $e = (u, v)$ and any function f on the vertex set with values in a commutative monoid M,*

$$\prod_{x \in V} f(x) = f(u) \left(\prod_{x \in V \setminus \{u,v\}} f(x) \right) f(v).$$

Proof. ✓ Proof Isolate the two distinguished endpoints in the product over V and group the remaining factors into the middle region. □

Theorem 15.42 (PEPS coefficient splitting at an edge). Lean ✓ Stmt *For any edge $e = (u, v)$, each summand in the PEPS coefficient factors into the tensor contribution at u, the product over the middle region $V \setminus \{u, v\}$, and the tensor contribution at v.*

Proof. ✓ Proof Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand. □

Definition 15.43 (Edge-centered boundary data). Lean ✓ Stmt For an edge $e = (u, v)$, the boundary data consist of the virtual index on e, the remaining virtual indices incident to u, and the remaining virtual indices incident to v. These are the indices over which the endpoint tensors couple to the blocked middle region.

Definition 15.44 (Boundary data read from a virtual configuration). Lean ✓ Stmt A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

Definition 15.45 (Boundary compatibility at an edge). Lean ✓ Stmt A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

Definition 15.46 (Middle configurations over fixed boundary data). Lean ✓ Stmt For fixed boundary data β, the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with β.

Definition 15.47 (Virtual configurations fibred by edge boundary data). Lean ✓ Stmt A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

Definition 15.48 (Left endpoint local configuration from boundary data). Lean ✓ Stmt The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

Definition 15.49 (Right endpoint local configuration from boundary data). Lean ✓ Stmt The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

Theorem 15.50 (Left endpoint reconstruction for compatible boundary data). *Lean* *✓ Stmt* *If a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.*

Theorem 15.51 (Right endpoint reconstruction for compatible boundary data). *Lean* *✓ Stmt* *If a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.*

Definition 15.52 (Boundary data with an inserted edge matrix). Lean ✓ Stmt For an edge $e = (u, v)$, inserted-edge boundary data consist of two independent indices on the bond e, together with the residual virtual indices incident to u and to v. These are the boundary indices used when a matrix is inserted on the distinguished bond.

Definition 15.53 (Diagonal inserted-edge boundary data). Lean ✓ Stmt Ordinary edge-centered boundary data determine inserted-edge boundary data by using the same bond index at both endpoints. This is the diagonal boundary datum selected by the identity matrix in the edge-insertion coefficient.

Definition 15.54 (Right-endpoint inserted-boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data together with an independent right endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the left bond index and the two residual endpoint boundaries.

Definition 15.55 (Left-endpoint inserted-boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data together with an independent left endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the right bond index and the two residual endpoint boundaries.

Definition 15.56 (Diagonal inserted-boundary configurations). Lean ✓ Stmt The diagonal inserted-boundary configurations are those inserted-edge boundary data for which the two distinguished bond indices agree. This is the support of the identity matrix inside the inserted-boundary sum.

Definition 15.57 (Diagonal boundary reindexing). Lean ✓ Stmt Ordinary edge-boundary data are equivalent to diagonal inserted-boundary data. This is the finite reindexing map used after the identity matrix has removed all off-diagonal inserted-boundary summands.

Definition 15.58 (Open middle tensor at an edge). Lean ✓ Stmt Fixing the residual endpoint data but leaving the bond e out of the middle region, the open middle tensor is the contraction over all virtual indices away from e that are compatible with those endpoint residual data.

Definition 15.59 (Middle physical configurations at an edge). Lean Lean ✓ Stmt For an edge $e = (u, v)$, the middle physical configurations are the physical indices on the vertices in $V \setminus \{u, v\}$. A global physical configuration restricts to such a middle configuration.

Definition 15.60 (Middle tensor with middle physical index). Lean Lean ✓ Stmt The ordinary and open middle contractions may be written using only the middle physical configuration, rather than a physical configuration on all vertices. This is the middle tensor in the three-site chain obtained after the edge blocking in arXiv:1804.04964, Section 3.

Theorem 15.61 (Middle restriction of the blocked tensor). *Lean* *Lean* *✓ Stmt* *The middle contractions written with a global physical configuration are the corresponding middle-indexed contractions evaluated on the restricted middle physical configuration.*

Proof. ✓ Proof Rewrite the product over $V \setminus \{u, v\}$ as the product over the corresponding finite set of middle vertices. □

Definition 15.62 (Restricting middle configurations away from the edge). Lean ✓ Stmt An ordinary middle configuration for the edge-blocked contraction restricts to a virtual configuration on all edges except the distinguished edge. This is the first reindexing map needed to compare the identity insertion with the ordinary edge-blocked middle tensor.

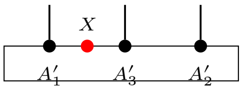
Definition 15.63 (Restoring the distinguished edge index). Lean ✓ Stmt Conversely, a compatible open middle configuration becomes an ordinary middle configuration after the fixed edge-boundary index is restored on the distinguished edge. Together with the restriction map, this gives the finite reindexing underlying the identity-insertion specialization.

Definition 15.64 (Open-middle reindexing equivalence). Lean ✓ Stmt For fixed edge-boundary data, ordinary middle configurations are equivalent to compatible open-middle configurations. This is the finite reindexing used when the inserted matrix is the identity.

Theorem 15.65 (Middle-indexed open tensor reindexing). *Lean* *✓ Stmt* *The ordinary middle tensor and the open middle tensor agree after restoring the fixed distinguished-edge index and reindexing the finite sum, with the physical index already restricted to the middle block.*

Proof. ✓ Proof Reindex the ordinary middle configurations by the equivalence with compatible open-middle configurations, then compare each local tensor entry. □

Definition 15.66 (Edge insertion coefficient). Lean ✓ Stmt For an edge $e = (u, v)$ and a matrix X acting on the bond space of e, the edge insertion coefficient is the contraction obtained by inserting X between the endpoint tensors and the open middle tensor:



Theorem 15.67 (Middle tensor reindexing for an opened edge). Lean ✓ Stmt *Restoring the fixed distinguished-edge index identifies the ordinary blocked middle tensor with the open middle tensor. This is the finite reindexing used before the identity matrix collapses the two inserted bond indices to the diagonal.*

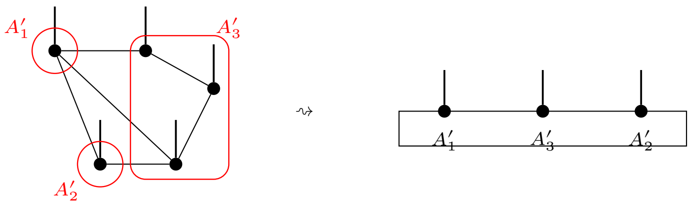
Theorem 15.68 (Diagonal summand of the identity insertion). Lean ✓ Stmt *On the diagonal boundary datum selected by the identity matrix, the inserted-edge summand is the ordinary edge-blocked summand. Thus the remaining identity-insertion step is purely a finite summation statement: the off-diagonal inserted boundary data vanish and the diagonal data reindex the ordinary edge-boundary sum.*

Theorem 15.69 (Off-diagonal summands of the identity insertion). Lean ✓ Stmt *If the two inserted bond indices are distinct, then the corresponding summand in the identity insertion is zero. This is the pointwise off-diagonal vanishing used before the inserted-boundary sum is restricted to its diagonal part.*

Theorem 15.70 (Identity insertion recovers the edge-blocked coefficient). Lean ✓ Stmt *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the ordinary edge-blocked coefficient.*

Proof. ✓ Proof Split the inserted-boundary sum into the diagonal part, where the two inserted bond indices agree, and the off-diagonal part. The off-diagonal summands vanish for the identity matrix, while the diagonal summands reindex through the equivalence with ordinary edge-boundary data. □

Definition 15.71 (Blocked middle tensor at an edge). Lean ✓ Stmt Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region $V \setminus \{u, v\}$. The graph-local contraction is read as a three-site chain:



Definition 15.72 (Edge-blocked PEPS coefficient). Lean ✓ Stmt The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

Theorem 15.73 (Edge-blocked coefficient equals the PEPS coefficient). Lean ✓ Stmt *For every edge e, the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.*

Proof. ✓ Proof Write $\mathrm{Coeff}_A(\sigma)$ for the PEPS state coefficient and $C_A(e; \sigma)$ for the edge-blocked coefficient, and put $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. The equivalence $\omega \leftrightarrow (\beta, \omega_{M_e})$ between global virtual configurations and an edge boundary configuration with a compatible middle configuration rewrites the PEPS coefficient as

$$\sum_{\omega} A_u(\omega|_u, \sigma_u) \left(\prod_{x \in M_e} A_x(\omega|_x, \sigma_x) \right) A_v(\omega|_v, \sigma_v) = \sum_{\beta} A_u(\beta_u, \sigma_u) C_{A, M_e}(\beta, \sigma|_{M_e}) A_v(\beta_v, \sigma_v).$$

Under the equivalence, $\omega|_u = \beta_u$ and $\omega|_v = \beta_v$; the inner sum over ω_{M_e} is exactly $C_{A,M_e}(\beta, \sigma|_{M_e})$. Thus $C_A(e;\sigma) = \mathrm{Coeff}_A(\sigma)$. □

Theorem 15.74 (Identity insertion equals the PEPS coefficient). *Lean* *✓ Stmt* *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the original PEPS state coefficient.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. With $\mathbb{1}$ inserted on e, the inserted-boundary sum is

$$C_A(e, \mathbb{1}; \sigma) = \sum_{a,b,\lambda,\rho} \delta_{a,b} A_u(a, \lambda; \sigma_u) C^{\mathrm{open}}_{A,M_e}(\lambda, \rho; \sigma|_{M_e}) A_v(b, \rho; \sigma_v).$$

Here a, b are the two copies of the distinguished bond index, while λ, ρ are the remaining virtual configurations at u and v. The factor $\delta_{a,b}$ leaves only the diagonal $a = b$, and the open-middle reindexing gives $C^{\mathrm{open}}_{A,M_e}(\lambda, \rho; \sigma|_{M_e}) = C_{A,M_e}(a, \lambda, \rho; \sigma|_{M_e})$. Hence $C_A(e, \mathbb{1}; \sigma) = C_A(e; \sigma) = \mathrm{Coeff}_A(\sigma)$. □

Theorem 15.75 (Same state gives the same edge-blocked coefficients). *Lean* *✓ Stmt* *If two PEPS tensors have the same state, then their edge-blocked coefficients agree at every edge and every physical configuration.*

Proof. ✓ Proof For every physical configuration σ,

$$C_A(e; \sigma) = \mathrm{Coeff}_A(\sigma) = \mathrm{Coeff}_B(\sigma) = C_B(e; \sigma).$$

□

Theorem 15.76 (Same state gives the same identity insertions). *Lean* *✓ Stmt* *If two PEPS tensors have the same state, then their edge-centered identity insertions agree at every edge and every physical configuration.*

Proof. ✓ Proof For every physical configuration σ,

$$C_A(e, \mathbb{1}; \sigma) = \mathrm{Coeff}_A(\sigma) = \mathrm{Coeff}_B(\sigma) = C_B(e, \mathbb{1}; \sigma).$$

□

Definition 15.77 (Middle tensor family at an edge). Lean Lean ✓ Stmt For the edge-blocked three-site chain at $e = (u, v)$, the middle tensor is the family obtained by fixing the two residual endpoint boundaries and evaluating the open middle contraction on the middle physical index. Its injectivity is the middle-block part of the assertion following the edge-blocking diagram.

Definition 15.78 (Edge-blocked three-site injectivity). Lean ✓ Stmt For a chosen edge $e = (u, v)$, the edge-blocked three-site chain is injective when the left endpoint tensor map, the middle tensor family, and the right endpoint tensor map are all injective.

Definition 15.79 (Singleton blocked region). Lean Lean Lean Lean ✓ Stmt For a singleton block $\{v\}$, the blocked physical index is the original physical index and the blocked tensor is

$$T_{A,\{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Its injectivity is $\mathrm{inj}(T_{A,\{v\}})$. The comparison with a region-injectivity predicate κ records the implication

$$\mathrm{inj}(T_{A,\{v\}}) \Longrightarrow \kappa(\{v\}).$$

Theorem 15.80 (Vertex injectivity gives singleton-block injectivity). Lean ✓ Stmt *If A is vertex-injective, then every singleton blocked tensor is injective. For every vertex v,*

$$\mathrm{inj}(T_{A,\{v\}}).$$

Proof. ✓ Proof For $R = \{v\}$, the contraction defining $T_{A,R}$ has one factor. Hence

$$T_{A,\{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Thus $\mathrm{inj}(A_v)$ is exactly $\mathrm{inj}(T_{A,\{v\}})$. Vertex injectivity gives $\forall v, \mathrm{inj}(A_v)$, and therefore $\forall v, \mathrm{inj}(T_{A,\{v\}})$. For a fixed v, this is the displayed assertion. $\square$

Theorem 15.81 (Singleton comparison gives the region base case). Lean Lean ✓ Stmt *Assume that a region-injectivity predicate κ satisfies, for every vertex v,*

$$\mathrm{inj}(T_{A,\{v\}}) \Longrightarrow \kappa(\{v\}).$$

Then, for every vertex v,

$$A \text{ vertex-injective} \Longrightarrow \kappa(\{v\}).$$

Proof. ✓ Proof For each v, the preceding theorem gives A vertex-injective $\Longrightarrow \mathrm{inj}(T_{A,\{v\}})$. The comparison hypothesis gives $\mathrm{inj}(T_{A,\{v\}}) \Longrightarrow \kappa(\{v\})$. Their composition is the claimed implication. $\square$

Theorem 15.82 (Finite regions from singleton regions and union closure). Lean ✓ Stmt *Assume that κ satisfies singleton comparison and union closure: $\mathrm{inj}(T_{A,\{v\}}) \Longrightarrow \kappa(\{v\})$ for every vertex v, and $\kappa(R) \wedge \kappa(S) \Longrightarrow \kappa(R \cup S)$. If A is vertex-injective, then every nonempty finite region R satisfies $\kappa(R)$.*

Proof. ✓ Proof For $R \neq \emptyset$, write $R = \bigcup_{v \in R}\{v\}$. The preceding theorem turns singleton comparison and vertex injectivity into $\kappa(\{v\})$ for every $v \in R$. Applying the finite-union consequence of union closure to these singleton regions gives $\kappa(R)$. $\square$

Definition 15.83 (Region injectivity supplies the edge-middle tensor). Lean ✓ Stmt A comparison from finite-region injectivity to the edge-middle tensor records the following assertion: if the middle vertex region $V \setminus \{u, v\}$ is injective after blocking, then the middle tensor in the edge-blocked three-site chain is injective. This isolates the finite-region contraction claim in [MGRPG+18, Section 3].

Theorem 15.84 (Endpoint injectivity in the edge-blocked chain). Lean ✓ Stmt *If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective.*

Proof. ✓ Proof If $e = (u, v)$, vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, which are the left and right endpoint tensor maps of the edge-blocked chain. $\square$

Theorem 15.85 (Endpoint injectivity at all edges). Lean ✓ Stmt *If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective at every edge.*

Proof. ✓ Proof For each $e = (u, v)$, Theorem 15.84 gives $\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,v})$. $\square$

Theorem 15.86 (Reducing edge-blocked injectivity to the middle block). Lean ✓ Stmt *For a vertex-injective PEPS tensor, injectivity of the edge-blocked middle tensor gives the full injectivity statement for the edge-blocked three-site chain.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The endpoint theorem supplies $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, while the hypothesis is $\mathrm{inj}(T_{A,M_e})$. Hence

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 15.87 (All edge-blocked chains from middle-tensor injectivity). Lean ✓ Stmt *If every edge-middle tensor is injective, then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.*

Proof. ✓ Proof For each $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. Vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$, while the hypothesis gives $\mathrm{inj}(T_{A,M_e})$. Hence

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v})$$

is the injectivity assertion for the edge-blocked three-site chain at e. □

Theorem 15.88 (Region injectivity gives edge-blocked injectivity). Lean ✓ Stmt *Suppose the middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose this region-injectivity assertion has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at the edge $e = (u, v)$.*

Proof. ✓ Proof Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The comparison gives $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. Vertex injectivity supplies $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$. The preceding reduction then gives

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 15.89 (Middle tensors from all middle regions). Lean ✓ Stmt *Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then every edge-middle tensor family is injective.*

Proof. ✓ Proof For each edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The hypotheses give $\kappa(M_e)$. The comparison implication $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$ yields $\mathrm{inj}(T_{A,M_e})$ for every edge. □

Theorem 15.90 (All edge-blocked chains from middle-region injectivity). Lean ✓ Stmt *Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.*

Proof. ✓ Proof For every edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The region hypothesis gives $\kappa(M_e)$, and the comparison gives $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. Vertex injectivity gives $\mathrm{inj}(T_{A,u})$ and $\mathrm{inj}(T_{A,v})$. Thus

$$\mathrm{inj}(T_{A,u}) \wedge \mathrm{inj}(T_{A,M_e}) \wedge \mathrm{inj}(T_{A,v}).$$

□

Theorem 15.91 (Edge-middle tensor from singleton regions, nonempty middle case). Lean ✓ Stmt *Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. Assume $M_e \neq \emptyset$. Assume also singleton comparison, union closure for κ, and the comparison $\kappa(M_e) \implies \mathrm{inj}(T_{A,M_e})$. If A is vertex-injective, then the edge-middle tensor family T_{A,M_e} is injective.*

Proof. ✓ Proof The finite-region theorem applied to $M_e = \bigcup_{w \in M_e}\{w\}$ gives $\kappa(M_e)$. The edge-middle comparison sends this to $\mathrm{inj}(T_{A,M_e})$. $\square$

Theorem 15.92 (One edge-blocked chain from singleton regions, nonempty middle case)**.** Lean ✓ Stmt *Let $e = (u,v)$ and put $M_e = V \setminus \{u,v\}$. Under the hypotheses of the preceding theorem, vertex injectivity of A gives injectivity of the edge-blocked three-site chain at e.*

Proof. ✓ Proof The preceding theorem gives $\mathrm{inj}(T_{A,M_e})$. The endpoint reduction combines this middle-tensor injectivity with vertex injectivity at u and v. $\square$

Theorem 15.93 (All edge-blocked chains from singleton regions, nonempty middle case)**.** Lean ✓ Stmt *If $M_e = V \setminus \{u,v\}$ is nonempty for every edge $e = (u,v)$, then the preceding one-edge statement applies at every edge. Thus vertex injectivity of A gives injectivity of every edge-blocked three-site chain.*

Proof. ✓ Proof Apply the one-edge statement separately to each edge e. $\square$

Theorem 15.94 (All edge-blocked chains from singleton regions, cardinal form)**.** Lean Lean ✓ Stmt *Assume singleton comparison, union closure for κ, the comparison $\kappa(M_e) \Longrightarrow \mathrm{inj}(T_{A,M_e})$, and vertex injectivity of A. If $2 < |V|$, then $V \setminus \{u,v\} \neq \emptyset$ for every edge $e = (u,v)$, and these hypotheses give $\mathrm{inj}(T_{A,V\setminus\{u,v\}})$ and edge-blocked three-site injectivity at every edge.*

Proof. ✓ Proof For each $e = (u,v)$, Theorem 15.40 gives $V \setminus \{u,v\} \neq \emptyset$. Apply the preceding all-edge singleton theorem. $\square$

Definition 15.95 (Finite kernel descent datum)**.** Lean ✓ Stmt Let V be a vertex type with decidable equality. A finite kernel descent datum is a predicate K on finite subsets $S \subseteq V$ together with implications

$$j \in S \quad \Longrightarrow \quad K(S) \Longrightarrow K(S \setminus \{j\}).$$

In the edge-blocking proof, $K(S)$ is the assertion that the boundary coefficients c_ρ, with the virtual indices exposed at that stage, give zero after the tensors in S have been contracted.

Lemma 15.96 (Finite descent of kernel conditions)**.** Lean Lean ✓ Stmt *For a finite kernel descent datum and any finite $R \subseteq V$,*

$$K(R) \Longrightarrow K(\emptyset).$$

More generally, if $K(\emptyset) \Longrightarrow Q$, then $K(R) \Longrightarrow Q$.

Proof. ✓ Proof Induct on the finite set R. If $R = \emptyset$, there is nothing to prove. Otherwise write $R = S \cup \{j\}$ with $j \notin S$. The deletion implication gives $K(R) \Longrightarrow K(S)$, and the induction hypothesis gives $K(S) \Longrightarrow K(\emptyset)$. $\square$

Definition 15.97 (Edge-middle kernel descent data)**.** Lean Lean ✓ Stmt Let $e = (u,v)$ and $M_e = V \setminus \{u,v\}$. An edge-middle kernel descent datum is indexed by boundary labels ρ consisting of the two residual endpoint configurations. It assigns to every finitely supported coefficient family $c = (c_\rho)_\rho$ a finite kernel descent datum $K_c(S)$ such that

$$\left(\forall \tau, \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0\right) \Longrightarrow K_c(M_e),$$

and

$$K_c(\emptyset) \Longrightarrow \forall \rho, c_\rho = 0.$$

Theorem 15.98 (Kernel descent gives edge-middle injectivity). Lean Lean ✓ Stmt *Let $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. If an edge-middle kernel descent datum is given, then T_{A,M_e} is injective. Consequently, if A is vertex-injective, the edge-blocked three-site chain at e is injective.*

Proof. ✓ Proof Let $c = (c_\rho)_\rho$ be a finite linear relation among the middle tensor vectors:

$$\forall \tau, \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0.$$

The initial part of the descent datum gives $K_c(M_e)$. The finite descent lemma gives $K_c(M_e) \implies K_c(\emptyset)$, and the terminal part gives $\forall \rho, c_\rho = 0$. Thus every finite relation is trivial, which is $\operatorname{inj}(T_{A,M_e})$. Vertex injectivity at u and v gives $\operatorname{inj}(T_{A,u})$ and $\operatorname{inj}(T_{A,v})$; the edge-blocked three-site injectivity at e follows. □

Theorem 15.99 (Edge blocking preserves injectivity). Lean ✓ Stmt *For a tensor map or blocked tensor family T, write $\operatorname{inj}(T)$ for the assertion that T is injective. If A is vertex-injective, then for every edge $e = (u, v)$,*

$$\operatorname{inj}(T_{A,u}), \qquad \operatorname{inj}(T_{A,V\setminus\{u,v\}}), \qquad \operatorname{inj}(T_{A,v}).$$

Here $T_{A,u}$ and $T_{A,v}$ are the two endpoint tensor maps, and $T_{A,V\setminus\{u,v\}}$ is the blocked middle tensor family. This is the precise assertion after the edge-blocking diagram in [MGRPG+18, Section 3].

Proof. Fix $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. From vertex injectivity, construct the edge-middle descent datum required by Theorem 15.98: for each finite relation

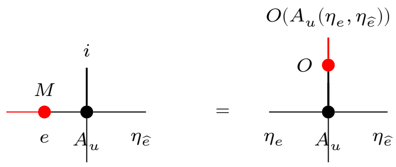
$$\forall \tau, \sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0,$$

the associated conditions $K_c(S)$ satisfy $K_c(M_e)$, $j \in S \implies K_c(S) \implies K_c(S \setminus \{j\})$, and $K_c(\emptyset) \implies \forall \rho, c_\rho = 0$. The descent implication is the local contraction identity: for $j \in S$, vertex injectivity at j gives a left inverse L_j, and applying L_j to the relation contracts away that tensor,

$$j \in S, K_c(S) \quad \overset{L_j}{\implies} \quad K_c(S \setminus \{j\}).$$

The kernel descent theorem gives $\operatorname{inj}(T_{A,M_e})$. The same theorem, together with vertex injectivity at u and v, gives $\operatorname{inj}(T_{A,u})$, $\operatorname{inj}(T_{A,M_e})$, and $\operatorname{inj}(T_{A,v})$ as the edge-blocked three-site injectivity assertion. □

Theorem 15.100 (Local virtual insertion realized physically on either endpoint). Lean ✓ Stmt *Let A be vertex-injective and let $e = (u, v)$ be an edge. For every matrix M on the bond space of e, the one-leg virtual action induced by M on the local tensor at u, and likewise at v, is realized by a physical operator on that endpoint's physical space.*



Proof. ✓ Proof Apply the one-edge realization theorem to the two endpoint incidences of the distinguished edge. □

Theorem 15.101 (Left-endpoint projected recovery). Lean ✓ Stmt *Let $e = (u, v)$. A physical operator at the left endpoint pulls back to the one-edge virtual action of M^{T} if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. ✓ Proof This is the projected local recovery equivalence applied to the left incident edge of e, with the transpose convention used by the left-endpoint coefficient identity. □

Theorem 15.102 (Right-endpoint projected recovery). Lean ✓ Stmt *Let $e = (u, v)$. A physical operator at the right endpoint pulls back to the one-edge virtual action of M if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. ✓ Proof This is the projected local recovery equivalence applied to the right incident edge of e. □

Theorem 15.103 (Endpoint recovery of a common virtual insertion). Lean ✓ Stmt *Let $e = (u, v)$. If $P_u O_1 P_u = \Phi_u T_{u,e}(M^{\mathsf{T}})\Psi_u$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M)\Psi_v$, then $\Psi_u O_1 \Phi_u = T_{u,e}(M^{\mathsf{T}})$ and $\Psi_v O_2 \Phi_v = T_{v,e}(M)$. Here $T_{w,e}(N)$ denotes the virtual operator on the legs incident to w which acts by N on the leg belonging to e and by the identity on the other incident legs.*

Proof. ✓ Proof The endpoint equivalences give $P_u O_1 P_u = \Phi_u T_{u,e}(M^{\mathsf{T}})\Psi_u \iff \Psi_u O_1 \Phi_u = T_{u,e}(M^{\mathsf{T}})$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M)\Psi_v \iff \Psi_v O_2 \Phi_v = T_{v,e}(M)$. The two hypotheses are the left sides of these equivalences. □

Theorem 15.104 (Endpoint physical-to-virtual recovery under image preservation). Lean ✓ Stmt *Let $e = (u, v)$. Suppose that $P_u O_1 P_u = \Phi_u T_{u,e}(M^{\mathsf{T}})\Psi_u$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M)\Psi_v$. Suppose also that O_1 maps the image of Φ_u into itself and O_2 maps the image of Φ_v into itself. Then*

$$O_1 \Phi_u = \Phi_u T_{u,e}(M^{\mathsf{T}}), \qquad O_2 \Phi_v = \Phi_v T_{v,e}(M).$$

Proof. ✓ Proof The preceding endpoint theorem identifies the two virtual pullbacks with the one-edge actions of M^{T} and M. The pullback specification gives the projected physical actions, and the image-preservation hypotheses remove the endpoint projectors. □

Theorem 15.105 (Right-endpoint physical realization of the inserted coefficient). Lean ✓ Stmt *Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on v realizes the one-edge virtual matrix action on the distinguished incident edge, then the inserted-edge coefficient at σ is obtained by leaving the left endpoint and open middle contraction explicit and applying that physical operator to the right endpoint tensor.*

Proof. ✓ Proof Expand the physical realization on a basis local configuration at the right endpoint. The resulting sum over the right bond index is then reindexed together with the ordinary edge-boundary data. □

Theorem 15.106 (Left-endpoint physical realization of the inserted coefficient). Lean ✓ Stmt *Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on u realizes the one-edge virtual action of M^{T} on the distinguished incident edge, then the inserted-edge coefficient for M at σ is obtained by applying that physical operator to the left endpoint tensor and leaving the open middle contraction and the right endpoint explicit.*

Proof. ✓ Proof Expand the physical realization on a basis local configuration at the left endpoint. The transpose converts the ordinary right bond index and the new left bond index into the coefficient M_{ij}; the resulting finite sum is then reindexed with the ordinary edge-boundary data. □

Theorem 15.107 (Endpoint physical realizations of the inserted coefficient). Lean ✓ Stmt *If the PEPS tensor is vertex-injective, then for every physical configuration σ, each inserted edge matrix is realized by physical operators at the endpoint tensors: the left endpoint realizes M^{T}, the right endpoint realizes M, and both realizations reproduce the same inserted-edge coefficient at σ in the edge-centered three-site decomposition.*

Proof. ✓ Proof Apply the local physical-realization theorem at the two endpoints and then use the two coefficient identities above. □

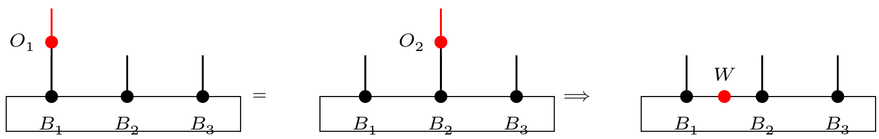
Theorem 15.108 (Equal neighboring physical actions recover a virtual insertion). Lean ✓ Stmt *Let $e = (u, v)$. Assume that the edge-blocked three-site chain at e is injective. Let O_1, O_2 be physical operators at the endpoint tensors, and write Φ_u, Φ_v for the two local tensor maps. Suppose that, for every physical configuration σ,*

$$\sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u} C_{\mathrm{mid}}(\sigma; \beta_{\mathrm{L}}, \beta_{\mathrm{R}}) A_v(\beta_v)_{\sigma_v} = \sum_{\beta} A_u(\beta_u)_{\sigma_u} C_{\mathrm{mid}}(\sigma; \beta_{\mathrm{L}}, \beta_{\mathrm{R}}) O_2(A_v(\beta_v))_{\sigma_v}.$$

Then

$$\exists M \in M_{D(e)}(\mathbb{C}), \qquad O_1\Phi_u = \Phi_u T_{u,e}(M^{\mathsf{T}}), \qquad O_2\Phi_v = \Phi_v T_{v,e}(M).$$

The same implication is represented diagrammatically by the $O_1, O_2 \mapsto W$ step:



This is the converse part of the insertion-algebra lemma in [MGRPG+18, Section 3], where the recovered matrix is denoted W.

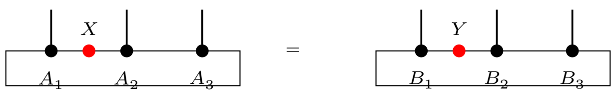
Definition 15.109 (Edge-blocked insertion algebra correspondence). Lean ✓ Stmt For two PEPS tensors A and B and an edge e, this property is

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\mathrm{alg}} M_{D_B(e)}(\mathbb{C}), \qquad \forall \sigma,\ \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

Theorem 15.110 (Edge-blocked insertion algebra isomorphism). Lean ✓ Stmt *Suppose the two edge-blocked three-site chains associated to A and B at the edge e are injective, and suppose the two PEPS tensors define the same state. Then*

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\mathrm{alg}} M_{D_B(e)}(\mathbb{C}), \qquad \forall \sigma,\ \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

This is the edge-blocked form of the algebra isomorphism $X \mapsto Y$ in [MGRPG+18, Section 3]:



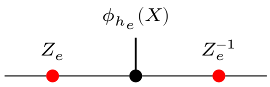
Theorem 15.111 (Matrix-algebra isomorphisms preserve size). Lean ✓ Stmt *If the full matrix algebras $M_D(\mathbb{C})$ and $M_E(\mathbb{C})$ are isomorphic as $\mathbb{C}$-algebras, then $D = E$.*

Proof. ✓ Proof The isomorphism preserves complex vector-space dimension, and $\dim_{\mathbb{C}} M_D(\mathbb{C}) = D^2$, $\dim_{\mathbb{C}} M_E(\mathbb{C}) = E^2$, so $D = E$. □

Theorem 15.112 (Matrix-algebra isomorphisms are inner after identifying size). *Lean* *✓ Stmt* *Let $\varphi : M_D(\mathbb{C}) \to M_E(\mathbb{C})$ be a $\mathbb{C}$-algebra isomorphism. Then $D = E$; let h denote this equality and $\iota_h : M_D(\mathbb{C}) \to M_E(\mathbb{C})$ the canonical index-relabelling isomorphism. There is $X \in \mathrm{GL}_E(\mathbb{C})$ with $\varphi(M) = X\,\iota_h(M)\,X^{-1}$ for all $M \in M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 15.111, $D = E$, so ι_h is defined. Then $\varphi \circ \iota_h^{-1}$ is a $\mathbb{C}$-algebra automorphism of $M_E(\mathbb{C})$, and Skolem–Noether (Theorem 3.11) gives $X \in \mathrm{GL}_E(\mathbb{C})$ with $(\varphi \circ \iota_h^{-1})(N) = XNX^{-1}$; evaluating at $N = \iota_h(M)$ gives $\varphi(M) = X\,\iota_h(M)\,X^{-1}$. □

Theorem 15.113 (Edge gauge from the insertion algebra isomorphism). *Lean* *✓ Stmt* *Suppose the edge insertion correspondence is a $\mathbb{C}$-algebra isomorphism between the two full matrix algebras on the chosen bond. Then the two bond dimensions on that edge are equal. If $h_e : D_A(e) = D_B(e)$ is this equality, and if $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_B(e)}(\mathbb{C})$ is the induced algebra isomorphism, then the insertion correspondence is $Y = Z_e\,\phi_{h_e}(X)\,Z_e^{-1}$ for an invertible edge matrix Z_e.*



This is the finite-dimensional matrix-algebra step in [MGRPG+18, Section 3].

Proof. ✓ Proof Apply Theorem 15.112 to the full matrix algebras determined by the two edge bond spaces. □

Definition 15.114 (Factorized local gauge datum). Lean ✓ Stmt Fix A, B, a bond-dimension equality, and a vertex v. Let $\mathcal{T}_v^A$ be the local tensor map of A, let L_v^A be the chosen left inverse of $\mathcal{T}_v^A$, and let $\Pi_v^A = \mathcal{T}_v^A L_v^A$. The factorized local gauge datum consists of the following two identities for local virtual configurations η, η':

$$\Pi_v^A B_v(\eta, -) = B_v(\eta, -), \qquad L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e),$$

with $X_e \in \mathrm{GL}(D_A(e))$ on each incident edge. This is the local output required from the blocked-middle reduction in [MGRPG+18, Section 3].

Definition 15.115 (Blocked-middle local gauge formula). Lean ✓ Stmt The blocked-middle local gauge formula at v is the assertion that there are matrices $X_e \in \mathrm{GL}(D_A(e))$, one on each incident edge, such that for all local virtual configurations η and physical indices σ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Theorem 15.116 (Blocked-middle formula implies factorized local gauge). *Lean* *✓ Stmt* *If the blocked-middle local gauge formula holds at v with matrices X_e, then the factorized local gauge datum holds at v with the same matrices.*

Proof. ✓ Proof For fixed η, set $c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. The displayed blocked-middle formula is precisely $B_v(\eta, -) = \mathcal{T}_v^A c_\eta$. Hence

$$\Pi_v^A B_v(\eta, -) = \Pi_v^A \mathcal{T}_v^A c_\eta = \mathcal{T}_v^A c_\eta = B_v(\eta, -),$$

and

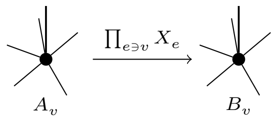
$$L_v^A B_v(\eta, -)(\eta') = L_v^A \mathcal{T}_v^A c_\eta(\eta') = c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e).$$

□

Theorem 15.117 (Blocked-middle formula gives a local gauge formula). Lean ✓ Stmt *Suppose that the edge-centred three-site reduction supplies invertible matrices X_e on the incident edges of v such that, for all local virtual configurations η and physical indices σ,*

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Then the same family X_e is a local gauge at v:

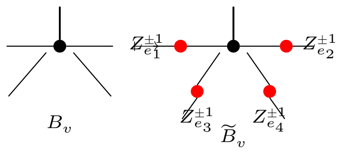


Proof. ✓ Proof By the preceding theorem, $L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. Since $\Pi_v^A B_v(\eta, -) = B_v(\eta, -)$ and $\Pi_v^A = \mathcal{T}_v^A L_v^A$,

$$B_v(\eta, \sigma) = \sum_{\eta'} L_v^A B_v(\eta, -)(\eta') A_v(\eta', \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

□

Theorem 15.118 (Absorbing edge gauges into the second PEPS). Lean ✓ Stmt *Given edge gauges X_e, set $\widetilde{B} = B^X$. Then $D_{\widetilde{B}} = D_B$ and, for every vertex v, $\widetilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$, where $M_{ie} = X_e$ or $M_{ie} = (X_e^{-1})^t$ according to the orientation of the endpoint ie.*



After this absorption, arbitrary insertions of the same matrix on the same edge are compared in the first PEPS and the modified second PEPS.

Proof. ✓ Proof With $\widetilde{B} = B^X$, Theorem 15.11 gives $D_{\widetilde{B}} = D_B$, and Theorem 15.12 gives $\widetilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$. □

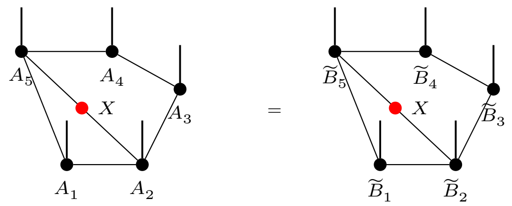
Definition 15.119 (Post-absorption inserted-edge comparison). Lean ✓ Stmt For tensor families $A, \widetilde{B}$, the post-absorption comparison holds when $h : D_A = D_{\widetilde{B}}$ and the following equality holds. For each edge e, write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_{\widetilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by $h_e : D_A(e) = D_{\widetilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ,

$$C_A(e, X; \sigma) = C_{\widetilde{B}}(e, \phi_{h_e}(X); \sigma).$$

Theorem 15.120 (Post-absorption edge insertion equality). Lean ✓ Stmt *Let A and B be vertex-injective PEPS with the same state, and assume the bond-dimension equality $h : D_A = D_B$. After the edge gauges have been absorbed into the second tensor family, there are gauges Z_e and $\widetilde{B} = B^Z$ such that $D_{\widetilde{B}} = D_A$ and, for every edge e, write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \to M_{D_{\widetilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by the corresponding equality $h_e : D_A(e) = D_{\widetilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ,*

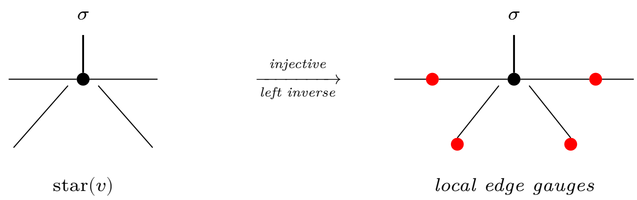
$$C_A(e, X; \sigma) = C_{\widetilde{B}}(e, \phi_{h_e}(X); \sigma),$$

Diagrammatically, this is the equality



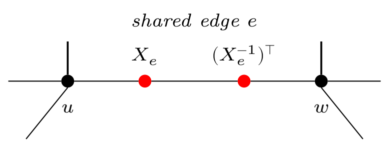
from [MGRPG+18, Section 3, lines 1037–1065].

Theorem 15.121 (Local gauge extraction)**.** Lean Lean ✓ Stmt *Under the factorized local gauge hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. The physical-realization and physical-to-virtual steps in that argument are recorded separately as Theorems 15.100 and 15.108. After Theorem 15.120 supplies the factorized local gauge hypothesis, injectivity on the star neighborhood provides a left inverse that isolates the local edge gauges:*



Proof. ✓ Proof The factorized local gauge datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity. □

Theorem 15.122 (Gauge consistency)**.** Lean Not ready *Choose an edge $e = (u, v)$ and block all other vertices into one middle tensor. The three resulting tensors form an injective chain, so the one-dimensional injective comparison assigns an invertible matrix to the edge e. Repeating this for every edge and absorbing these matrices into the second tensor family gives the post-absorption insertion equality of [MGRPG+18, Section 3]: for every edge and every matrix, inserting that matrix on the edge in the first PEPS gives the same state as inserting it on the same edge in the modified second PEPS. Applying the local extraction at each vertex then gives one edge gauge family whose endpoint actions transform the first tensor family into the second:*



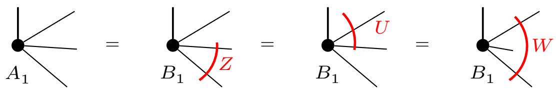
Remark 15.123. The PEPS diagrams in this section record the gauge steps used in [MGRPG+18, Section 3]. The edge-orientation and vertex-action diagrams fix the endpoint convention for the edge gauges obtained by blocking the PEPS to a three-site injective MPS. The cancellation diagram is the converse contracted identity: inserting opposite endpoint actions on an internal edge leaves the PEPS coefficient unchanged. The edge-gauge absorption diagram records the formation of $\widetilde{B}$ before the post-absorption insertion equality. The local-extraction diagram summarizes the comparison of one-site-enlarged injective regions with injective complements, after the edge gauges have been absorbed into $\widetilde{B}$. The global-consistency diagram summarizes repeating the edge blocking over all edges, so the two endpoint actions on a shared edge are one edge gauge. The two-injective-tensor diagram records the two-block comparison from [MGRPG+18, Section 3], and the final one-vertex-complement diagram records its application to one vertex and its complement. These pictures use the blueprint tensor-dot convention, but each one is attached to the same contraction, insertion, or blocking step as the corresponding argument in the paper.

Definition 15.124 (Left-identity three-leg residual form). Lean ✓ Stmt For three virtual leg spaces α, β, γ and an endomorphism Z of $\beta \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as the identity on the α leg and as Z on the remaining two legs. This is the $I_\alpha \otimes Z$ form appearing after the first tensor is inverted in [MGRPG+18, Section 3].

Definition 15.125 (Right-identity three-leg residual form). Lean ✓ Stmt For an endomorphism U of $\alpha \otimes \beta$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as U on the first two legs and as the identity on the γ leg. This is the $U \otimes I_\gamma$ form in [MGRPG+18, Section 3].

Definition 15.126 (Middle-identity three-leg residual form). Lean ✓ Stmt For an endomorphism W of $\alpha \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as W on the outer two legs and as the identity on the middle β leg. This is the third residual form in [MGRPG+18, Section 3].

Theorem 15.127 (Scalar reduction of the residual two-tensor gauges). *Lean* ✓ *Stmt* *In the proof of the generalized two-injective-tensor comparison, applying the inverse of the second injective tensor leaves three possible virtual operators on the exposed legs of the first tensor:*



Inverting the first injective tensor then shows that the corresponding residual operator on the three exposed virtual legs has the three decompositions

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}$$

Equivalently, for nonzero virtual leg spaces α, β, γ, if one endomorphism of $\alpha \otimes \beta \otimes \gamma$ lies simultaneously in the three subspaces $I_\alpha \otimes \mathrm{End}(\beta \otimes \gamma)$, $\mathrm{End}(\alpha \otimes \beta) \otimes I_\gamma$, and the analogous subspace with I_β in the middle leg, then it is a scalar multiple of the identity. Thus the residual virtual operators are scalar multiples of the identity. This is the scalar step in the proof of [MGRPG+18, Section 3].

Proof. ✓ Proof Write the common operator as T. The equality $T = I_\alpha \otimes Z = U \otimes I_\gamma$ gives, for all indices,

$$\delta_{aa'} Z_{bc,b'c'} = U_{ab,a'b'} \delta_{cc'}.$$

If $c \neq c'$, the right hand side is zero; choosing $a = a'$ gives $Z_{bc,b'c'} = 0$. Hence Z is diagonal in the γ-index. Similarly, the equality $T = I_\alpha \otimes Z = W_{\alpha\gamma} \otimes I_\beta$ gives

$$\delta_{aa'} Z_{bc,b'c'} = W_{ac,a'c'} \delta_{bb'},$$

so $Z_{bc,b'c'} = 0$ whenever $b \neq b'$. Thus $Z_{bc,b'c'} = 0$ unless $b = b'$ and $c = c'$.

It remains to compare the surviving diagonal entries. From $\delta_{aa'} Z_{bc,bc} = U_{ab,a'b}$, the value $Z_{bc,bc}$ is independent of c. From the middle-identity equality it is also independent of b. Therefore $Z = \lambda I_{\beta\otimes\gamma}$. Substitution into

$$I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta$$

gives $U = \lambda I_{\alpha\otimes\beta}$ and $W_{\alpha\gamma} = \lambda I_{\alpha\otimes\gamma}$. ∎

Theorem 15.128 (Pointwise product cancellation for two injective blocks). Lean ✓ Stmt *Let A_1, A_2, B_1, B_2 be tensors with matching external, shared-boundary, and physical index sets. Assume that the two external boundary spaces and the shared-boundary configuration space are nonempty, and that A_1 and A_2 are injective. Suppose that, for every pair of external indices, every pair of shared-boundary configurations, and every pair of physical indices,*

$$A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1, \mu, \sigma_1)B_2(\eta_2, \nu, \sigma_2).$$

Then there exists $\lambda \in \mathbb{C}^\times$ such that

$$A_1 = \lambda B_1, \qquad A_2 = \lambda^{-1} B_2.$$

Proof. ✓ Proof Choose external and shared-boundary indices, and use injectivity of A_1 and A_2 to choose physical indices for which the two chosen coefficients are nonzero. The product equality then shows that the corresponding coefficients of B_1 and B_2 are also nonzero. Define

$$\lambda = \frac{B_2(\eta_2^0, \nu^0, \sigma_2^0)}{A_2(\eta_2^0, \nu^0, \sigma_2^0)}.$$

For every (η_1, μ, σ_1), the product identity with $(\eta_2^0, \nu^0, \sigma_2^0)$ fixed gives

$$A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2^0, \nu^0, \sigma_2^0) = B_1(\eta_1, \mu, \sigma_1)B_2(\eta_2^0, \nu^0, \sigma_2^0).$$

Dividing by $A_2(\eta_2^0, \nu^0, \sigma_2^0) \neq 0$ gives $A_1 = \lambda B_1$. Fixing $(\eta_1^0, \mu^0, \sigma_1^0)$ and substituting $A_1 = \lambda B_1$ into the product identity gives

$$\lambda\, B_1(\eta_1^0, \mu^0, \sigma_1^0)A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1^0, \mu^0, \sigma_1^0)B_2(\eta_2, \nu, \sigma_2).$$

Dividing by $B_1(\eta_1^0, \mu^0, \sigma_1^0) \neq 0$ gives $A_2 = \lambda^{-1} B_2$. □

Definition 15.129 (Matrix unit for coefficient extraction). Lean ✓ Stmt For indices i and j, let E_{ij} denote the matrix whose only nonzero entry is the (i, j) entry, where it is equal to 1.

Theorem 15.130 (Coefficient extraction for one shared bond). Lean ✓ Stmt *Suppose that the two blocks share a single virtual bond. Then inserting the matrix unit $E_{\mu,\nu}$ in that bond extracts the corresponding product of tensor coefficients:*

$$C_{A_1,A_2}(b, E_{\mu,\nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2, \nu, \sigma_2).$$

Proof. ✓ Proof Since there is only one shared bond, the condition that the two shared configurations agree away from the distinguished bond is vacuous, and

$$C_{A_1,A_2}(b, E_{\mu,\nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\mu',\nu'} (E_{\mu,\nu})_{\mu'(b),\nu'(b)} A_1(\eta_1, \mu', \sigma_1)A_2(\eta_2, \nu', \sigma_2).$$

The entries of the matrix unit satisfy

$$(E_{\mu,\nu})_{i,j} = \delta_{i,\mu(b)}\delta_{j,\nu(b)}.$$

Thus the only nonzero summand is the one with $(\mu', \nu') = (\mu, \nu)$. □

Theorem 15.131 (One-bond two-injective comparison). Lean ✓ Stmt *Suppose that the two injective blocks have exactly one shared virtual bond and nonempty external boundary spaces. If all matrix insertions on that bond give equal two-block coefficients for the pairs (A_1, A_2) and (B_1, B_2), then there exists $\lambda \in \mathbb{C}^\times$ such that*

$$A_1 = \lambda B_1, \qquad A_2 = \lambda^{-1} B_2.$$

Proof. ✓ Proof Insert the matrix unit $E_{\mu,\nu}$ and use the coefficient-extraction theorem to obtain the pointwise product equality

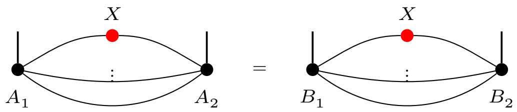
$$A_1(\eta_1,\mu,\sigma_1)A_2(\eta_2,\nu,\sigma_2) = B_1(\eta_1,\mu,\sigma_1)B_2(\eta_2,\nu,\sigma_2)$$

for all indices. The pointwise product cancellation theorem gives the reciprocal scalar proportionality. $\square$

Theorem 15.132 (Generalized two-injective-tensor comparison)**.** Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let A_1, A_2, B_1, B_2 be injective tensors with the same shared virtual bonds and nonempty external boundary spaces, and assume that there is at least one shared virtual bond. For the set $\mathcal{B}$ of shared bonds, a bond $b \in \mathcal{B}$, and $X \in M_{D_b}(\mathbb{C})$, set*

$$C_{A_1,A_2}(b,X;\eta_1,\eta_2,\sigma_1,\sigma_2) = \sum_{\substack{\mu,\nu \\ \mu|_{\mathcal{B}\setminus\{b\}}=\nu|_{\mathcal{B}\setminus\{b\}}}} X_{\mu_b,\nu_b} A_1(\eta_1,\mu,\sigma_1)A_2(\eta_2,\nu,\sigma_2).$$

Assume $C_{A_1,A_2}(b,X;\eta_1,\eta_2,\sigma_1,\sigma_2) = C_{B_1,B_2}(b,X;\eta_1,\eta_2,\sigma_1,\sigma_2)$ for every $b, X, \eta_1, \eta_2, \sigma_1, \sigma_2$:



Then

$$\exists \lambda \in \mathbb{C}^\times, \qquad A_1 = \lambda B_1, \qquad A_2 = \lambda^{-1} B_2.$$

This is the blueprint form of the two-injective tensor comparison in [MGRPG+18, Section 3].

Proof. The proof is the comparison lemma of [MGRPG+18, Section 3]. After grouping the relevant virtual spaces, reduce to three nonzero virtual legs α, β, γ. Applying the inverse of A_2 to the insertion equalities obtained by freeing, respectively, the (β,γ), (α,β), and (α,γ) legs gives endomorphisms Z, U, and W such that, for every external index η, shared-boundary index μ, and physical index σ of the first block,

$$A_1(\eta,\mu,\sigma) = \sum_\nu B_1(\eta,\nu,\sigma)(I_\alpha \otimes Z)_{\nu,\mu} = \sum_\nu B_1(\eta,\nu,\sigma)(U \otimes I_\gamma)_{\nu,\mu} = \sum_\nu B_1(\eta,\nu,\sigma)(W_{\alpha\gamma} \otimes I_\beta)_{\nu,\mu}.$$

Injectivity of B_1 gives the residual equality

$$I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta.$$

Write $Z = \sum_i Z_i^{(1)} \otimes Z_i^{(2)}$, $U = \sum_i U_i^{(1)} \otimes U_i^{(2)}$, and $W = \sum_i W_i^{(1)} \otimes W_i^{(2)}$. The previous display is exactly

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}.$$

Theorem 15.127 gives $Z = \lambda I_{\beta\otimes\gamma}$, $U = \lambda I_{\alpha\otimes\beta}$, and $W_{\alpha\gamma} = \lambda I_{\alpha\otimes\gamma}$. Hence $A_1(\eta,\mu,\sigma) = \lambda B_1(\eta,\mu,\sigma)$. If $\lambda = 0$, then $A_1 = 0$, contradicting injectivity of A_1; hence $\lambda \neq 0$.

Put $X = I_{D_b}$ in the insertion identity. Since $A_1 = \lambda B_1$, for every boundary and physical index one has

$$\sum_\mu B_1(\eta_1,\mu,\sigma_1)(A_2(\eta_2,\mu,\sigma_2) - \lambda^{-1}B_2(\eta_2,\mu,\sigma_2)) = 0.$$

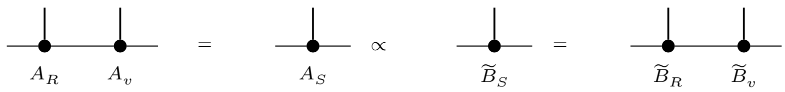
Injectivity of B_1 gives $A_2(\eta_2,\mu,\sigma_2) = \lambda^{-1}B_2(\eta_2,\mu,\sigma_2)$. $\square$

Theorem 15.133 (One-vertex versus complement comparison). Lean ✓ Stmt *Let the shared bond type and the two external boundary spaces be nonempty. After the edge gauges have been absorbed into the second tensor family $\widetilde{B}$, block one vertex v against its complement. Assume that $A_v, A_{v^c}, \widetilde{B}_v, \widetilde{B}_{v^c}$ are injective as two-block tensors. The post-absorption insertion equality gives, for every shared bond b, every matrix $X \in M_{D_b}(\mathbb{C})$, and all external and physical indices,*

$$C_{A_v,A_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{\widetilde{B}_v,\widetilde{B}_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Then

$$\exists \lambda_v \in \mathbb{C}^\times, \qquad A_v = \lambda_v \widetilde{B}_v.$$



This is the final local comparison following the post-absorption insertion equality in [MGRPG+18, Section 3].

Proof. ✓ Proof With $A_1 = A_v$, $A_2 = A_{v^c}$, $B_1 = \widetilde{B}_v$, and $B_2 = \widetilde{B}_{v^c}$, the post-absorption equality is precisely

$$\forall b, X, \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}, \quad C_{A_1,A_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{B_1,B_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Theorem 15.132 gives

$$\exists \lambda_v \in \mathbb{C}^\times, \qquad A_v = \lambda_v \widetilde{B}_v, \qquad A_{v^c} = \lambda_v^{-1} \widetilde{B}_{v^c}.$$

The desired local proportionality is the first equality. □

Remark 15.134 (Normal-region form). The normal-region application in [MGRPG+18, Section 3] uses the same scalar comparison after comparing two injective regions $R \subset S = R \cup \{v\}$. If the region comparisons give $A_R = \lambda_R \widetilde{B}_R$ and $A_S = \lambda_S \widetilde{B}_S$, then the source-paper diagram reads

$$A_R A_v = A_S = \lambda_S \widetilde{B}_S = \lambda_S \widetilde{B}_R \widetilde{B}_v.$$

Applying the left inverse supplied by injectivity of the block $A_R = \lambda_R \widetilde{B}_R$ gives $A_v = (\lambda_S/\lambda_R)\widetilde{B}_v$.

Theorem 15.135 (Fundamental Theorem for injective PEPS, conditional on bond-dimension equality). Lean Not ready *Suppose that the corresponding virtual edge spaces of A and B are already identified. If A and B are vertex-injective and give the same PEPS state, then there are invertible matrices X_e, one for each edge, such that at every vertex the tensor B_v is obtained from A_v by the oriented endpoint action of the incident matrices.*

Theorem 15.136 (Fundamental Theorem for injective PEPS). Lean Not ready *Let A and B be vertex-injective PEPS tensors on a finite simple graph. If they give the same PEPS state, then their corresponding virtual edge spaces have the same dimensions and, after identifying these spaces, there are invertible matrices X_e, one for each edge, such that B_v is obtained from A_v by the oriented endpoint action of the matrices incident to v. This is Theorem 2 of [MGRPG+18].*

15.1 Normal PEPS

Definition 15.137 (Four-region decomposition for a union of regions). Lean Lean Lean Lean Lean ✓ Stmt Let V be a finite vertex set. For two regions $A, B \subseteq V$, define the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $V \setminus (A \cup B)$. These are the four regions used in the proof of Lemma 15.143. They also form an indexed family over $\{0, 1, 2, 3\}$ in the order $A \setminus B$, $A \cap B$, $B \setminus A$, $V \setminus (A \cup B)$.

Lemma 15.138 (Identities for the four-region decomposition). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let V be a finite vertex set and let $A, B \subseteq V$. The left-only and overlap regions reconstruct A, the overlap and right-only regions reconstruct B, the three inside regions reconstruct $A \cup B$, and all four regions reconstruct V. The same identities also hold in the indexed four-region family: pieces 0 and 1 reconstruct A, pieces 1 and 2 reconstruct B, and pieces $0, 1, 2$ reconstruct $A \cup B$. These same reconstructions are available as finite unions over the corresponding subfamilies of the indexed four-region family. Piece 3 is the complement of $A \cup B$, equivalently the complement of the three inside indexed pieces, and also $A^c \cap B^c$. The singleton subfamily $\{3\}$ gives the outside block. Pieces $2, 3$ reconstruct A^c, equivalently the complement of pieces $0, 1$, and pieces $0, 3$ reconstruct B^c, equivalently the complement of pieces $1, 2$, again both as explicit unions and as finite unions over the indexed subfamilies. Conversely, pieces $0, 1$ are the complement of pieces $2, 3$, pieces $1, 2$ are the complement of pieces $0, 3$, and adding all four pieces reconstructs V. Equivalently, each vertex lies in one of the four regions, and the indexed four-region family has union V. In fact, every vertex belongs to exactly one indexed piece.*

Proof. ✓ Proof Check membership of an arbitrary vertex in A and in B. □

Lemma 15.139 (Disjointness in the four-region decomposition). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the four-region decomposition, the left-only and right-only parts are disjoint from the overlap and from each other. Each inside part, and hence the union of the three inside regions, is disjoint from the outside region. In indexed form, the pieces forming A are disjoint from the pieces forming A^c, and the pieces forming B are disjoint from the pieces forming B^c, both as explicit binary unions and as finite unions over the corresponding selected indexed subfamilies. The indexed four-region family is pairwise disjoint.*

Proof. ✓ Proof Use the standard disjointness identities for set difference and intersection, together with the reconstruction of $A \cup B$. □

Definition 15.140 (Contraction maps for the union injectivity proof). Lean ✓ Stmt Let K, E_{012}, E_2, and E_{12} be complex vector spaces. A contraction-map family for the proof of [MGRPG+18, Section 3] consists of three linear maps $\mathcal{C}_{\{0,1,2\}} : K \to E_{012}$, $\mathcal{C}_{\{2\}} : K \to E_2$, and $\mathcal{C}_{\{1,2\}} : K \to E_{12}$.

Definition 15.141 (Contraction chain for the union injectivity proof). Lean ✓ Stmt A contraction chain consists of the three contraction maps of Definition 15.140, together with the three implications used in Lemma 15.142.

Lemma 15.142 (Kernel criterion for the union contraction chain). Lean Lean ✓ Stmt *Let $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Let K be the boundary vector space attached to P_3, and let $\mathcal{C}_{\{0,1,2\}} : K \to E_{012}$, $\mathcal{C}_{\{2\}} : K \to E_2$, and $\mathcal{C}_{\{1,2\}} : K \to E_{12}$ be the three contraction maps in the proof of [MGRPG+18, Section 3]. If*

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{2\}}(X) = 0, \qquad \mathcal{C}_{\{2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{1,2\}}(X) = 0,$$

and $\mathcal{C}_{\{1,2\}}(X) = 0 \Longrightarrow X = 0$, then $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$.

Proof. ✓ Proof For $X \in K$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, apply the three implications in sequence:

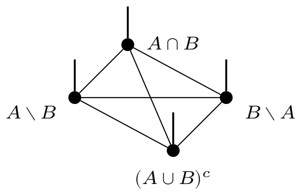
$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{1,2\}}(X) = 0 \Longrightarrow X = 0.$$

$\square$

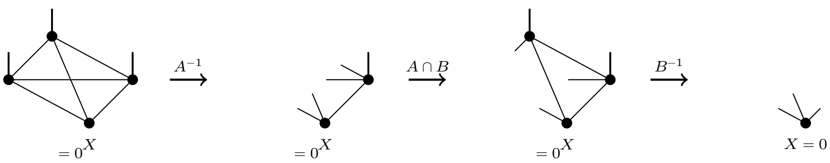
Lemma 15.143 (Union of injective regions)**.** Not ready *Let V be a finite vertex set. Write* $\operatorname{inj}(R)$ *for the assertion that the PEPS tensor blocked to the region $R \subseteq V$ is injective. For two possibly overlapping regions $A, B \subseteq V$,*

$$\operatorname{inj}(A) \wedge \operatorname{inj}(B) \Longrightarrow \operatorname{inj}(A \cup B).$$

The proof in [MGRPG+18, Section 3] blocks the network into four parts $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$, and applies the two available left inverses in sequence:



Taking a boundary tensor X on $(A \cup B)^c$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, injectivity of A gives $\mathcal{C}_{\{2\}}(X) = 0$; reinserting $A \cap B$ and applying injectivity of B then gives $X = 0$:



Proof. Put $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Then $P_0 \cup P_1 = A$, $P_1 \cup P_2 = B$, and $P_0 \cup P_1 \cup P_2 = A \cup B$. Let $\mathcal{C}_I$ denote contraction of the blocked tensor over the region $\bigcup_{i \in I} P_i$. To prove injectivity of $A \cup B$, take a boundary tensor X on P_3 such that $\mathcal{C}_{\{0,1,2\}}(X) = 0$. Since $P_0 \cup P_1 = A$ is injective, its left inverse gives

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \quad \Longrightarrow \quad \mathcal{C}_{\{2\}}(X) = 0.$$

The remaining two steps are

$$\begin{aligned} \mathcal{C}_{\{2\}}(X) = 0 \quad &\Longrightarrow \quad \mathcal{C}_{\{1,2\}}(X) = 0, \\ \mathcal{C}_{\{1,2\}}(X) = 0 \quad &\Longrightarrow \quad X = 0. \end{aligned}$$

The first implication is obtained by contracting with the tensor on P_1, and the second is the left inverse for the injective region $P_1 \cup P_2 = B$. Thus $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$, and therefore $P_0 \cup P_1 \cup P_2 = A \cup B$ is injective. $\square$

Definition 15.144 (Injectivity predicate for blocked regions)**.** Lean Lean ✓ Stmt A region-injectivity hypothesis assigns to each finite vertex region the assertion that the tensor obtained by blocking that region is injective. This predicate is used for regions such as R, S, T, and their complements in the normal PEPS proof. The union-closure hypothesis records the conclusion supplied by the source union-of-injective-regions lemma: the union of two injective regions is injective.

Lemma 15.145 (Finite unions of injective regions). Lean ✓ Stmt *Under the union-closure hypothesis for region injectivity, any nonempty finite union of injective finite regions is injective.*

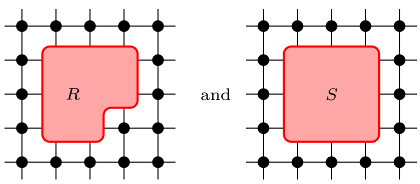
Proof. ✓ Proof Induct on the nonempty finite indexing set. The singleton case is the given injectivity hypothesis, and the induction step is one application of binary union closure. □

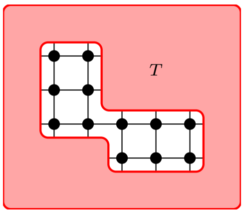
Lemma 15.146 (Indexed inside union under closure). Lean Lean ✓ Stmt *Under the union-closure hypothesis for region injectivity, if the indexed subfamilies* $\{0, 1\}$ *and* $\{1, 2\}$ *in the four-region decomposition are injective, then the inside subfamily* $\{0, 1, 2\}$ *is injective. Equivalently, injectivity of the two source-paper regions* $(A \setminus B) \cup (A \cap B)$ *and* $(A \cap B) \cup (B \setminus A)$ *gives injectivity of* $(A \setminus B) \cup (A \cap B) \cup (B \setminus A) = A \cup B$*.*

Proof. ✓ Proof Rewrite the three selected indexed unions as A, B, and $A \cup B$, then apply binary union closure. □

Definition 15.147 (Complement of a finite region). Lean Lean ✓ Stmt For a finite vertex region $R \subseteq V$, its complement is $V \setminus R$. Equivalently, a vertex lies in the complement exactly when it does not lie in R.

Definition 15.148 (Square-lattice coordinate regions). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The finite $n \times m$ square lattice is represented by pairs of coordinates. The corresponding square-lattice graph has horizontal and vertical nearest-neighbor edges; the two elementary adjacency lemmas record the right-neighbor and upper-neighbor cases used for translated edge blockings, and the corresponding coordinate edges name those two nearest-neighbor edges as elements of the edge set. These named right and upward edges have the expected horizontal and vertical orientations. Every edge of this graph is horizontal or vertical, but not both. Since edges are represented by ordered endpoint pairs, a horizontal edge is oriented from left to right and is exactly the coordinate right edge from its left endpoint; similarly, a vertical edge is oriented from bottom to top and is exactly the coordinate upward edge from its lower endpoint. A coordinate rectangle is a product of a horizontal coordinate set and a vertical coordinate set. The 2×3 and 3×2 coordinate-product predicates assert only that these two coordinate sets have cardinalities $2, 3$ and $3, 2$, respectively. Contiguous coordinate rectangles are products of contiguous coordinate intervals, and the contiguous 2×3 and 3×2 predicates name the rectangular blocks used in the normal square-lattice theorem. The displayed region R is the union of a 2×3 contiguous rectangle with the 3×2 contiguous rectangle occupying its upper two rows and extending one column to the right. The displayed region S is the union of two 2×3 contiguous rectangles shifted by one horizontal coordinate, hence the full 3×3 square spanned by them. The displayed local-window model for T is the complement, inside a local 5×6 coordinate window, of the vertical 2×3 edge block and the horizontal 3×2 edge block shown in the source picture. The finite-lattice complementary block is the whole finite square lattice with those two edge blocks removed; this is the block denoted A_3 in the proof of [MGRPG+18, Theorem 3]. The source shapes are:





Lemma 15.149 (Basic identities for square-lattice coordinate regions). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Membership in a coordinate rectangle is coordinatewise membership, the contiguous coordinate interval has the expected coordinate inequalities, membership in a contiguous coordinate rectangle is the conjunction of the four bounding inequalities, and membership in R is the disjunction of membership in the 2×3 lower-left piece or in the shifted 3×2 upper piece. Membership in S is the disjunction of membership in the left 2×3 piece or in the horizontally shifted 2×3 piece. Membership in the local-window T region is membership in the local 5×6 window together with nonmembership in the union of the two displayed edge blocks. Bounded contiguous coordinate rectangles of sizes 2×3 and 3×2 satisfy, respectively, the predicates for contiguous 2×3 and contiguous 3×2 square-lattice rectangles. The cardinality of a coordinate rectangle is the product of the two coordinate cardinalities, and the full coordinate rectangle is the whole finite vertex set.*

Proof. ✓ Proof The coordinate-rectangle statements follow from membership and cardinality identities for Cartesian products of finite coordinate sets. The contiguous-rectangle statement then unfolds the two coordinate intervals. The membership statements for R, S, and the two edge blocks removed from T additionally unfold membership in a union of two such rectangles, while membership in T also unfolds the set difference from the local 5×6 window. The shape predicates follow by using the displayed lower-left corners as witnesses. ☐

Lemma 15.150 (The normal regions R and S are unions of contiguous rectangles). Lean Lean ✓ Stmt *If the displayed 3×3 window containing R and S lies in the finite square lattice, then R is the union of a contiguous 2×3 rectangle and a contiguous 3×2 rectangle, while S is the union of two contiguous 2×3 rectangles.*

Proof. ✓ Proof This is the defining description of R and S, together with the coordinate bounds which make the displayed rectangles contiguous 2×3 and 3×2 rectangles. ☐

Lemma 15.151 (The edge blocks removed from T are contiguous rectangles). Lean Lean Lean ✓ Stmt *If the two edge blocks cut out of the displayed T-region lie in the finite square lattice, then they are respectively a contiguous 2×3 rectangle and a contiguous 3×2 rectangle.*

Proof. ✓ Proof This is the defining description of the two removed edge blocks, together with the coordinate bounds which make them valid contiguous rectangles. ☐

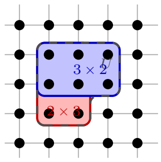
Lemma 15.152 (The local T-region and its removed edge blocks fill the window). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The two edge blocks removed from the displayed local T-region lie inside the local 5×6 coordinate window. They are disjoint from each other and from this local T region, and their union with it is exactly that local window.*

Proof. ✓ Proof This follows by unfolding the definition of T as the set-theoretic complement of the two displayed edge blocks inside the local window. ☐

Lemma 15.153 (The edge-complementary block fills the finite lattice). Lean Lean Lean Lean ✓ Stmt *The finite-lattice complementary block around the edge is disjoint from the two displayed edge blocks, and its union with them is the full finite square lattice. The local-window T region is contained in this finite-lattice complementary block.*

Proof. ✓ Proof This is the set-theoretic complement of the two displayed edge blocks in the full finite square lattice. The containment of the local-window T region follows because its points also avoid the two removed edge blocks. □

Definition 15.154 (Rectangular covers of square-lattice regions). Lean Lean Lean Lean Lean ✓ Stmt A rectangular cover of a square-lattice region is a nonempty finite family of regions whose union is exactly the given region, and each member is a contiguous 2×3 or 3×2 rectangle. The two cases used in the proof are the cover of the local displayed T-region and the cover of the finite-lattice complementary block A_3.



$U = \bigcup_j C_j, \quad C_j \cong 2 \times 3$ or 3×2

Lemma 15.155 (The present local T-window and A_3 cover obstructions). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The present origin-based local-window model for the displayed T-region admits no nonempty finite cover by contained contiguous 2×3 and 3×2 rectangles whenever the ambient lattice contains the 5×6 window. The origin-based rotated local-window model used for vertical edges has the analogous obstruction whenever the ambient lattice contains the 6×5 window. The normalized 5×6 and 6×5 cases are recorded as corollaries. The present origin-based horizontal edge-complement model has the analogous obstruction whenever the ambient lattice contains the 5×7 frame; the normalized 5×7 case is again recorded as a corollary. The present origin-based vertical edge-complement model has the corresponding obstruction whenever the ambient lattice contains the 7×5 frame; the normalized 7×5 case is recorded as a corollary.*

Proof. ✓ Proof The lower-left point of the local window belongs to the present T model. Any contiguous 2×3 rectangle containing it also contains a point of the removed vertical edge block, while any contiguous 3×2 rectangle containing it also contains a point of the removed horizontal edge block. The rotated statement is the same coordinate argument with the two shapes interchanged. The horizontal edge-complement statement uses the same lower-left point, and so does the normalized vertical edge-complement statement. These cases are all instances of the same point-forcing obstruction to an exact rectangular cover. □

Lemma 15.156 (Injectivity from a rectangular cover). Lean ✓ Stmt *If a square-lattice region has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that the region is injective.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. □

Lemma 15.157 (The normal regions R and S differ in one site). Lean Lean ✓ Stmt *For the displayed regions R and S with the same lower-left corner, R is contained in S. The difference $S \setminus R$ is exactly the lower-right site of the 3×3 square: in coordinates, the site one obtains by shifting two steps horizontally and zero steps vertically from the lower-left corner.*

Proof. ✓ Proof Expand the two displayed unions. The only point of the second 2×3 piece of S which is not already in the union defining R is its lower-right point. □

Definition 15.158 (Normal rectangular injectivity hypotheses). Lean Lean ✓ Stmt In the square-lattice normal theorem, every 2×3 and 3×2 rectangular region is assumed injective. The corresponding abstract hypotheses specify which finite regions have these two shapes and assert injectivity for all of them. In the coordinate square-lattice specialization, these two families are the contiguous coordinate 2×3 and 3×2 rectangles.

Lemma 15.159 (Coordinate rectangular injectivity gives abstract rectangular injectivity). Lean ✓ Stmt *The coordinate square-lattice rectangular injectivity hypotheses determine abstract rectangular injectivity hypotheses, with the two abstract rectangular families taken to be the contiguous coordinate* 2×3 *and* 3×2 *rectangles.*

Proof. ✓ Proof This is the direct specialization of the abstract rectangular families to the two contiguous coordinate-rectangle predicates. □

Lemma 15.160 (Coordinate rectangular injectivity gives injective rectangles). Lean Lean ✓ Stmt *Under the coordinate square-lattice rectangular injectivity hypotheses, every bounded contiguous* 2×3 *rectangle and every bounded contiguous* 3×2 *rectangle is injective.*

Proof. ✓ Proof This is the defining rectangular injectivity hypothesis applied to the coordinate witnesses for the two rectangle shapes. □

Lemma 15.161 (Rectangular injectivity gives injective T-edge blocks). Lean Lean Lean ✓ Stmt *For the displayed local* T*-region inside its* 5×6 *coordinate window, if the smaller edge-block bounds* $x_0 + 5 \leq n$ *and* $y_0 + 5 \leq m$ *hold in the finite* $n \times m$ *square lattice, then the coordinate square-lattice rectangular injectivity hypotheses imply that the vertical* 2×3 *block and horizontal* 3×2 *block removed from the window are injective. If, in addition, region injectivity is closed under unions, then the union of these two removed edge blocks is injective.*

Proof. ✓ Proof Apply the rectangular injectivity hypotheses to the already identified contiguous rectangle shapes of the two edge blocks. The final assertion is one application of union closure to these two injective blocks. □

Lemma 15.162 (Injectivity from a rectangular cover of the local T-region). Lean Lean Lean ✓ Stmt *If the displayed local* T*-region has a nonempty finite cover by contiguous* 2×3 *and* 3×2 *rectangles, then rectangular injectivity and finite union closure imply that* T *is injective. The rotated vertical local* T*-region has the same conditional injectivity criterion.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. □

Lemma 15.163 (Injectivity from a rectangular cover of the edge complement). Lean Lean Lean ✓ Stmt *If the finite-lattice complementary block* A_3 *has a nonempty finite cover by contiguous* 2×3 *and* 3×2 *rectangles, then rectangular injectivity and finite union closure imply that* A_3 *is injective. The same criterion applies to every vertical edge-complementary block: a nonempty finite cover of that block by contiguous* 2×3 *and* 3×2 *rectangles implies injectivity of the vertical edge-complementary block. In the normalized horizontal-edge* 5×7 *frame, it is enough to give such a cover of the local* T*-region, since adjoining the two top-collar* 3×2 *rectangles gives a cover of* A_3*.*

Proof. ✓ Proof Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. For the normalized 5×7 edge complement, first extend the local T-cover by the two top-collar rectangles. ☐

Lemma 15.164 (A cover of T extends to a cover of the edge complement)**.** Lean Lean ✓ Stmt *In the normalized horizontal-edge 5×7 frame with lower-left corner $(0,0)$, any rectangular cover of the local T-region extends to a rectangular cover of the edge-complementary block A_3 by adjoining the two top-collar contiguous 3×2 rectangles. In the normalized vertical-edge 7×5 frame with lower-left corner $(0,0)$, any rectangular cover of the rotated local T-region extends to a rectangular cover of the vertical edge-complementary block by adjoining the two right-collar contiguous 2×3 rectangles.*

Proof. ✓ Proof Index the new cover by the old cover indices together with two extra indices for the collar rectangles. For the horizontal block, the decomposition is

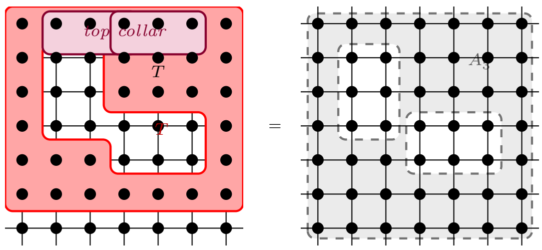
$$A_3 = T \cup (\{0,1,2\} \times \{5,6\}) \cup (\{2,3,4\} \times \{5,6\}).$$

For the vertical block, the decomposition is

$$A_3^{\mathrm{v}} = T^{\mathrm{v}} \cup (\{5,6\} \times \{0,1,2\}) \cup (\{5,6\} \times \{2,3,4\}).$$

☐

Lemma 15.165 (The normalized edge complements have collars)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the normalized horizontal-edge 5×7 frame, the finite-lattice complementary block A_3 is the local-window T-region together with two top-collar contiguous 3×2 rectangles. Consequently, if the local T-region is injective, then rectangular injectivity and union closure imply that A_3 is injective. There is also a transposed 7×5 horizontal-frame collar identity with two contiguous 2×3 right-collar rectangles. For the rotated vertical edge-blocking regions, the corresponding complement is the rotated local T-region together with the same two right-collar rectangles; hence rotated T-injectivity, and hence a rectangular cover of rotated T, implies injectivity of the vertical complementary block.*



Proof. ✓ Proof The two set identities follow by expanding the coordinate inequalities. The injectivity statements apply union closure first to the local T-region and the first collar rectangle, and then to the second collar rectangle. ☐

Lemma 15.166 (Normalized edge-blocking data)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *In the normalized 5×7 horizontal-edge frame, the red block is the vertical 2×3 block, the blue block is the horizontal 3×2 block, and the complementary block is the finite-lattice edge complement A_3. The distinguished edge is a horizontal edge of the square-lattice graph. The left endpoint of the distinguished edge lies in the red block, the right endpoint lies in the blue block, the three regions are pairwise disjoint, and their union is the whole finite square lattice.*

If the local T-region is injective, rectangular injectivity and union closure give injectivity of all three blocks; in particular a rectangular cover of T suffices. The normalized 7×5 vertical-edge frame has the analogous red and blue rectangular blocks, a distinguished vertical edge, and the same set-theoretic blocking data; at this stage its complementary-block injectivity is either kept as an explicit hypothesis or obtained from rotated T-injectivity, and therefore from a rectangular cover of rotated T, by the vertical collar lemma. In both cases the named blocking datum supplies the corresponding three injective regions for the edge-blocked chain.

Proof. ✓ Proof The endpoint memberships are coordinate checks. Rectangular injectivity gives the rectangular red and blue blocks. For the horizontal edge, the collar lemma gives the complement, and the rectangular cover criterion removes the direct T-injectivity input when such a cover is supplied. For the vertical edge, complement injectivity is an explicit input or is obtained from the rotated T-cover. The disjointness and cover statements are finite-set complement identities. □

Lemma 15.167 (Translated horizontal edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose horizontal blocking window lies in the finite square lattice, there is a distinguished horizontal edge with the same relative position as in the normalized picture. The translated red region is the 2×3 block, the translated blue region is the 3×2 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity. In both forms the datum supplies the three injective regions used in the edge-blocked chain. The distinguished edge is the corresponding coordinate right edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized horizontal picture.*

Proof. ✓ Proof The endpoint memberships and horizontal orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. □

Lemma 15.168 (Translated vertical edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose vertical blocking window lies in the finite square lattice, there is a distinguished vertical edge with the same relative position as in the normalized vertical picture. The translated red region is the 3×2 block, the translated blue region is the 2×3 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity. In both forms the datum supplies the three injective regions used in the edge-blocked chain. The distinguished edge is the corresponding coordinate upward edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized vertical picture.*

Proof. ✓ Proof The endpoint memberships and vertical orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. □

Definition 15.169 (Translated edge windows). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean

Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt A translated edge window records that a chosen edge is one of the translated horizontal or vertical edge pictures, together with a rectangular cover of the complementary block. Such a window gives the one-edge blocking datum for that edge, hence the three injective regions used in the edge-blocked chain. The coordinate right and upward edges obtain such windows directly from the corresponding translated picture. Equivalently, an actual coordinate right or upward edge obtains such a window when it has the necessary room around it for the translated 5×7 or 7×5 frame and the complementary cover is supplied. The same criterion applies to an arbitrary horizontal or vertical square-lattice edge after identifying it with its coordinate right or upward representative. The oriented margin-and-cover datum is the same criterion packaged as per-edge data in the rectangular coordinate model; it directly supplies the one-edge blocking datum, its three injective regions, and the identities $u_e \in R_e$, $v_e \in B_e$, $R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset$, and $R_e \cup B_e \cup C_e = V$. The boundary obstruction lemmas show that these translated-window and margin-cover criteria are interior criteria, not the final every-edge construction in the open square-lattice model.

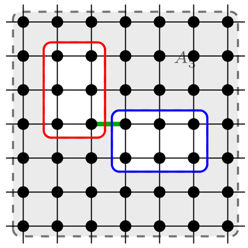
Theorem 15.170 (Construction from translated edge windows). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Given rectangular injectivity hypotheses, finite union closure, and a translated edge window at every edge of the finite square lattice, the corresponding one-edge blockings determine the normal edge-blocking hypotheses. This is the conditional construction; constructing such windows for all edges of the 7×7 lattice remains separate; the current open rectangular translated-window criterion does not by itself supply such a family. The one-edge datum recovered from the resulting hypotheses is the datum supplied by the corresponding translated window, and the resulting hypotheses give the three injective regions at every edge, together with $u_e \in R_e$, $v_e \in B_e$, $R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset$, and $R_e \cup B_e \cup C_e = V$. Equivalently, it is enough to supply the oriented margin-and-cover data at every edge; the resulting hypotheses recover the corresponding datum, give the same three injectivity conclusions, and give the same endpoint, disjointness, and covering conclusions. The obstruction theorem records that this interior margin-cover criterion does not by itself supply data at every edge of the open 7×7 lattice.*

Proof. ✓ Proof Apply the general construction for one-edge blocking data to the datum supplied by the chosen translated window at each edge. □

Lemma 15.171 (Conditional injectivity of the normal R and S regions). Lean Lean ✓ Stmt *If region injectivity is closed under unions, then the coordinate square-lattice rectangular injectivity hypotheses imply that the displayed regions R and S are injective.*

Proof. ✓ Proof Region R is the union of one injective 2×3 rectangle and one injective 3×2 rectangle. Region S is the union of two injective 2×3 rectangles. Apply union closure in each case. □

Definition 15.172 (Normal edge-blocking hypotheses). Lean Lean ✓ Stmt Around one edge, the normal proof uses a blocking into three injective regions: a red region containing one endpoint, a blue region containing the other endpoint, and an injective complementary region. The three regions are pairwise disjoint and cover the whole vertex set. The edge-blocking hypotheses are precisely this one-edge datum at every edge; the red, blue, and complementary regions are read off from it. The local blocking picture used in the proof of [MGRPG+18, Theorem 3]:



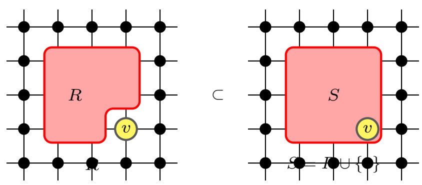
Theorem 15.173 (Injectivity supplied by normal edge blocking). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *A family of one-edge blocking data is precisely the normal edge-blocking hypotheses. If H is assembled from the family d_e, then the one-edge datum that H associates with e is d_e. Write the endpoints of e as $u_e < v_e$ in the canonical vertex ordering. If the three regions are R_e, B_e, and C_e, then*

$$u_e \in R_e, \qquad v_e \in B_e, \qquad R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset, \qquad R_e \cup B_e \cup C_e = V,$$

and the underlying PEPS tensor is injective on R_e, B_e, and C_e.

Proof. ✓ Proof By construction, the one-edge datum at each edge e is d_e. The displayed endpoint, disjointness, coverage, and injectivity statements are exactly the corresponding assumptions of that one-edge datum. □

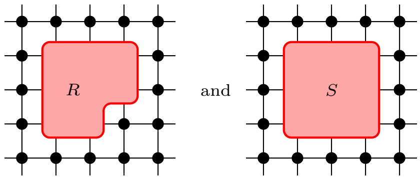
Definition 15.174 (Normal one-site separation hypotheses). Lean ✓ Stmt For each site, the normal theorem assumes two injective regions with injective complements that differ exactly at that site. This is the one-site comparison used after the edge gauges have been obtained. Diagrammatically, in the square-lattice proof this is the comparison between the source regions R and $S = R \cup \{v\}$:

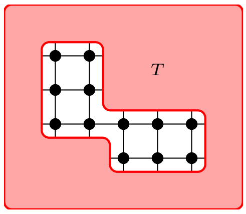


Theorem 15.175 (Membership in a one-site separated region). Lean ✓ Stmt *For the two regions associated to a site v, a vertex w belongs to the region containing v if and only if either $w = v$, or w belongs to the comparison region omitting v.*

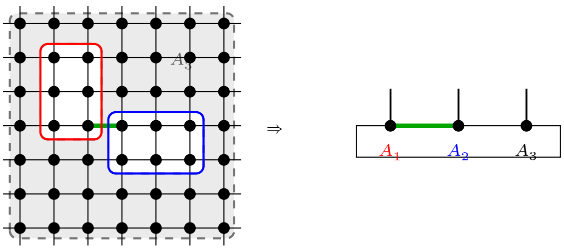
Proof. ✓ Proof If $w = v$, use the distinguished-site membership and nonmembership assumptions. If $w \neq v$, use the equality of the two regions away from the distinguished site. □

Definition 15.176 (Normal square-lattice blocking regions). Lean Lean ✓ Stmt In the translationally invariant square-lattice proof, the source paper assumes that every 2×3 and 3×2 region is injective. The formal statement here is the corresponding abstract, scope-restricted hypothesis bundle: it records the rectangular injectivity assumptions, the size bound $n, m \geq 7$ with $|V| = nm$, and three finite regions R, S, and T whose injectivity is assumed directly. It does not by itself assert that these regions are the displayed square-lattice unions, nor that their translated variants give the edge-centred blocking around every edge. Conversely, for the coordinate square lattice with $n, m \geq 7$, the rectangular hypotheses, union closure, and a rectangular cover of the displayed T-region yield this abstract R, S, T structure, with R and S obtained from their rectangular decompositions and T obtained from the supplied cover. The source regions are:

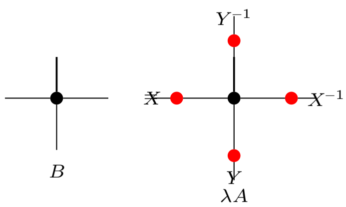




Theorem 15.177 (Normal edge blocking to a three-site injective chain). Not ready *Under the square-lattice normality hypotheses of [MGRPG⁺18, Theorem 3], the regions R, S, and T, together with their translated variants, give an edge-centred blocking into red, blue, and complementary injective regions. These are the regions on which the three-site injective-chain insertion argument is applied:*

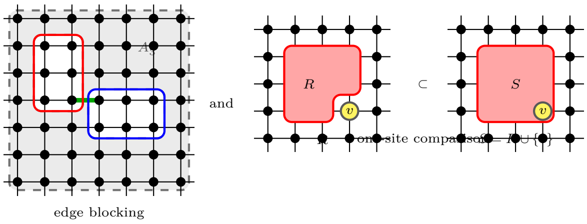


Theorem 15.178 (Translationally invariant normal gauge absorption). Not ready *In the translationally invariant normal square-lattice case, the edge gauges obtained after blocking are represented by one horizontal matrix X and one vertical matrix Y, unique up to scalar. Absorbing these matrices into B gives a tensor $\widetilde{B}$, equivalently the final local gauge formula of [MGRPG⁺18, Theorem 3]:*



Theorem 15.179 (Fundamental Theorem for normal square-lattice PEPS). Not ready *Let A and B be translationally invariant normal square-lattice PEPS tensors such that every 2×3 and 3×2 region is injective. If they generate the same state on an $n \times m$ region with $n, m \geq 7$, then they are related by horizontal and vertical gauges X, Y, up to a scalar λ satisfying $\lambda^{nm} = 1$. This is [MGRPG⁺18, Theorem 3].*

Definition 15.180 (Normal PEPS blocking hypotheses). Lean Lean Lean Lean Lean ✓ Stmt The general normal theorem assumes both the edge-centred blockings into three injective regions and the one-site comparison regions with injective complements. These two assumptions are collected as the normal PEPS blocking hypotheses; the same hypotheses give the corresponding edge and one-site consequences without restating the separate edge and one-site assumptions.



Theorem 15.181 (Square-lattice normal blocking hypotheses from margin covers). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Given rectangular injectivity hypotheses, finite union closure, oriented margin-and-cover hypotheses at every edge, and one-site separation, the finite square lattice satisfies the general normal blocking hypotheses. The edge-blocking hypotheses are exactly the margin-cover construction, the one-edge datum at an edge is the datum supplied there, and the one-site separation is exactly the supplied one-site separation. Hence these hypotheses give the three injective regions, the endpoint-disjoint-cover facts at each edge, and the one-site membership formula as consequences of the same normal blocking hypotheses.*

Proof. ✓ Proof Combine the edge-blocking hypotheses constructed from the margin-cover hypotheses with the given one-site separation hypotheses. □

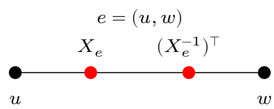
Theorem 15.182 (Fundamental Theorem for normal PEPS). Not ready *Suppose two normal PEPS generate the same state, can be blocked into three-site injective chains around every edge, and for every site admit two injective regions with injective complements that differ exactly at that site. Then their defining tensors are related by a local gauge, and the gauges are unique up to the scalar freedom allowed by the graph. This is the general normal-PEPS theorem stated after [MGRPG+18, Theorem 3].*

Definition 15.183 (Oriented endpoint scalar). Lean ✓ Stmt An edge-scalar family $c = (c_e)_{e\in E}$ determines the endpoint scalar $c_{v,e}$ by $c_{v,e} = c_e$ at the lower endpoint of e and $c_{v,e} = c_e^{-1}$ at the upper endpoint.

Definition 15.184 (Vertex-balanced edge scalars). Lean ✓ Stmt An edge-scalar family is vertex-balanced when $\prod_{e\ni v} c_{v,e} = 1$ for every vertex v.

Definition 15.185 (Gauge equivalence modulo balanced edge scalars). Lean ✓ Stmt Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar c_e on each edge such that the corresponding endpoint actions satisfy $X_{v,e} = c_{v,e}Y_{v,e}$, and $\prod_{e\ni v} c_{v,e} = 1$ for every vertex v.

Theorem 15.186 (Multiplication of oriented edge scalars). Lean ✓ Stmt *If two edge-scalar families are multiplied, then their endpoint scalars satisfy $(cd)_{v,e} = c_{v,e}d_{v,e}$ on every oriented half-edge.*



Proof. ✓ Proof At the lower endpoint this is ordinary multiplication of the edge scalars. At the upper endpoint, $(c_e d_e)^{-1} = c_e^{-1} d_e^{-1}$ because the edge scalars lie in $\mathbb{C}^\times$. $\square$

Theorem 15.187 (Inversion of oriented edge scalars). Lean ✓ Stmt *Inverting an edge-scalar family gives $(c^{-1})_{v,e} = c_{v,e}^{-1}$ on every oriented endpoint.*

Proof. ✓ Proof At the lower endpoint, $(c^{-1})_{v,e} = c_e^{-1} = c_{v,e}^{-1}$. At the upper endpoint, $(c^{-1})_{v,e} = (c_e^{-1})^{-1} = c_{v,e}^{-1}$. $\square$

Theorem 15.188 (Nonzero oriented edge scalars). Lean ✓ Stmt *Every oriented endpoint scalar satisfies $c_{v,e} \neq 0$.*

Proof. ✓ Proof Edge scalars are units, and the inverse of a unit is again a unit. $\square$

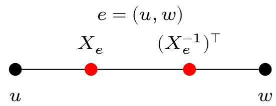
Theorem 15.189 (Product of balanced edge scalars). Lean ✓ Stmt *The pointwise product of two vertex-balanced edge-scalar families is again vertex-balanced.*

Proof. ✓ Proof For every vertex v, $\prod_{e\ni v}(cd)_{v,e} = (\prod_{e\ni v} c_{v,e})(\prod_{e\ni v} d_{v,e}) = 1 \cdot 1 = 1$. $\square$

Theorem 15.190 (Inverse of balanced edge scalars). Lean ✓ Stmt *The pointwise inverse of a vertex-balanced edge-scalar family is again vertex-balanced.*

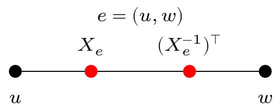
Proof. ✓ Proof For every vertex v, $\prod_{e\ni v}(c^{-1})_{v,e} = (\prod_{e\ni v} c_{v,e})^{-1} = 1^{-1} = 1$. $\square$

Theorem 15.191 (Reflexivity of balanced edge-scalar equivalence). Lean ✓ Stmt *Every edge-gauge family is equivalent to itself modulo balanced edge scalars, witnessed by $c_e = 1$ for every edge e.*



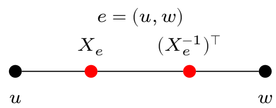
Proof. ✓ Proof If $c_e = 1$, then $c_{v,e} = 1$, $X_{v,e} = c_{v,e} X_{v,e}$, and $\prod_{e\ni v} c_{v,e} = 1$. $\square$

Theorem 15.192 (Symmetry of balanced edge-scalar equivalence). Lean ✓ Stmt *If one gauge family differs from another by balanced edge scalars, then the second differs from the first by the inverse balanced edge-scalar family.*



Proof. ✓ Proof If $X_{v,e} = c_{v,e} Y_{v,e}$ with $c_{v,e} \neq 0$, then $Y_{v,e} = c_{v,e}^{-1} X_{v,e}$. The inverse scalar family is balanced by Theorem 15.190. $\square$

Theorem 15.193 (Transitivity of balanced edge-scalar equivalence). Lean ✓ Stmt *If one gauge family differs from a second by balanced edge scalars, and the second differs from a third by balanced edge scalars, then the first differs from the third by the pointwise product of the two scalar families.*



Proof. ✓ Proof From $X_{v,e} = c_{v,e}Y_{v,e}$ and $Y_{v,e} = d_{v,e}Z_{v,e}$ one obtains $X_{v,e} = (c_{v,e}d_{v,e})Z_{v,e}$. The product scalar family is balanced by Theorem 15.189. ☐

Theorem 15.194 (The balanced edge-scalar quotient is an equivalence relation)**.** Lean ✓ Stmt *Gauge equivalence modulo balanced edge scalars is reflexive, symmetric, and transitive.*

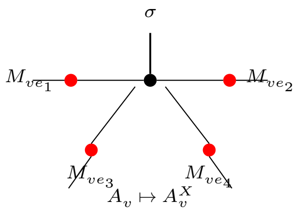
Proof. ✓ Proof Reflexivity is witnessed by 1, symmetry by c^{-1}, and transitivity by cd. ☐

Definition 15.195 (The balanced edge-scalar quotient relation)**.** Lean ✓ Stmt The preceding equivalence relation is the quotient relation on edge-gauge families modulo vertex-balanced edge scalars.

Theorem 15.196 (Balanced edge scalars preserve the local gauge action)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then they produce the same gauged tensor at every vertex.*

Proof. ✓ Proof In each summand of the gauged tensor at v, the edge scalars contribute the factor $\prod_{e \ni v} c_{v,e} = 1$. Hence the summand, and therefore the whole local tensor, is unchanged. ☐

Theorem 15.197 (Balanced edge scalars preserve the gauged tensor)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then applying them to the same PEPS tensor produces the same gauged tensor.*



Proof. ✓ Proof For every vertex v, virtual index η, and physical index σ, $(A^X)_v(\eta, \sigma) = (A^Y)_v(\eta, \sigma)$ for the two gauged tensors by Theorem 15.196. Hence the two gauged tensors are equal. ☐

Theorem 15.198 (Balanced edge scalars preserve the gauged state)**.** Lean ✓ Stmt *If two gauge families differ by a balanced edge-scalar reweighting, then the PEPS tensors obtained by applying those gauge families represent the same state.*

Proof. ✓ Proof The two gauged tensors are equal, hence all of their state coefficients are equal. ☐

Theorem 15.199 (Gauge uniqueness modulo balanced edge scalars)**.** Lean Not ready *If two gauge families relate the same vertex-injective PEPS tensor to the same target tensor, then their edge gauges differ by nonzero scalars c_e whose oriented product at every vertex is* 1*. Thus the correct graph quotient is by balanced edge scalars.*

Remark 15.200. The local left inverse Ψ_v, the realization map $T \mapsto O_T$, the split at one incident edge, and the partition $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$ isolate the local indices and maps needed for the edge-centered reduction of [MGRPG+18, Section 3]. The middle-region blocked coefficient is constructed by the preceding definitions; the remaining local-gauge step is the comparison with the corresponding three-site injective chain. The corrected uniqueness statement is a quotient by balanced edge scalars; its proof still requires the same blocking construction together with the corresponding tensor-factor uniqueness statement at a single vertex.

Chapter 16

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N-site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \to \mathbb{C},$$

rather than as an explicit tensor product.

Definition 16.1 (N-site coefficient space)**.** Lean ✓ Stmt For local dimension d and chain length N, the ambient coefficient space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \to \mathbb{C}.$$

This is the concrete model used throughout the chapter for periodic-chain states.

16.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \cdots A^{i_L}$. The local ground-space map is

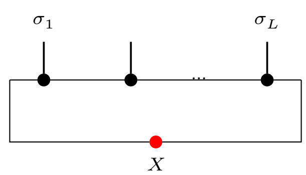
$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}), \qquad \Gamma_L(X)(\sigma) := \mathrm{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \cdots A^{\sigma(L-1)}$.

Definition 16.2 (Ground-space map)**.** Lean ✓ Stmt The map Γ_L above is a $\mathbb{C}$-linear map

$$\Gamma_L : M_D(\mathbb{C}) \to (\{0, \dots, d-1\}^L \to \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 16.3 (Evaluation formula). Lean ✓ Stmt *For every boundary matrix $X \in M_D(\mathbb{C})$ and every length-L word σ,*

$$\Gamma_L(X)(\sigma) = \mathrm{tr}(A^\sigma X).$$

Definition 16.4 (Local ground space). Lean ✓ Stmt The *local ground space* is the image

$$G_L(A) := \mathrm{im}(\Gamma_L) \subseteq (\{0, \dots, d{-}1\}^L \to \mathbb{C}).$$

Lemma 16.5 (Dimension bound). Lean ✓ Stmt *For every L,*

$$\dim G_L(A) \le D^2.$$

Proof. ✓ Proof Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \le \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 16.6 (Ambient-space dimension). Lean ✓ Stmt *The coefficient-function model satisfies*

$$\dim(\{0, \dots, d{-}1\}^L \to \mathbb{C}) = d^L.$$

Lemma 16.7 (Nontriviality criterion). Lean ✓ Stmt *If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:*

$$G_L(A) \ne (\{0, \dots, d{-}1\}^L \to \mathbb{C}).$$

Proof. ✓ Proof If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 16.6; Lemma 16.5 gives the contradiction $d^L \le D^2$. □

16.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L-site Euclidean space.

Definition 16.8 (Euclidean transport of the local ground space). Lean ✓ Stmt Via the canonical identification between coefficient functions and the Euclidean space $\ell^2(\{0, \dots, d{-}1\}^L)$, one transports the local ground space to a Hilbert-space subspace

$$G_L^{\mathrm{ES}}(A) \subseteq \ell^2(\{0, \dots, d{-}1\}^L).$$

Lemma 16.9 (Membership in the transported local ground space). Lean ✓ Stmt *A Euclidean vector lies in $G_L^{\mathrm{ES}}(A)$ exactly when transporting it back through the canonical coefficient-space equivalence gives an element of the original local ground space $G_L(A)$.*

Proof. ✓ Proof This is just the definition of $G_L^{\mathrm{ES}}(A)$ as the image of $G_L(A)$ under the inverse of the canonical equivalence. □

Definition 16.10 (Parent interaction). Lean ✓ Stmt The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L-site space.

Lemma 16.11 (Parent interaction kills ground-space elements). Lean ✓ Stmt *For any $v \in G_L(A)$, we have $h_L(A)\, v = 0$.*

Proof. ✓ Proof Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. □

Definition 16.12 (Periodic window extraction). Lean ✓ Stmt For a configuration $\sigma \in \{0, \dots, d{-}1\}^N$ on the periodic chain and a starting site i, the extracted word

$$\mathrm{extractWindow}_{L,i}(\sigma) \in \{0, \dots, d{-}1\}^L$$

reads the entries of σ on the cyclic interval $i, i+1, \dots, i+L-1$ modulo N.

Definition 16.13 (Periodic window replacement). Lean ✓ Stmt If $L \le N$, the configuration

$$\mathrm{replaceWindow}_{L,i}(\sigma, \tau)$$

is obtained from σ by replacing the cyclic window starting at i by the length-L word τ.

Lemma 16.14 (Extracting a replaced window). Lean ✓ Stmt *If $L \le N$, then extracting the same cyclic window after replacement recovers the inserted word:*

$$\mathrm{extractWindow}_{L,i}(\mathrm{replaceWindow}_{L,i}(\sigma, \tau)) = \tau.$$

Lemma 16.15 (Replacing an extracted window). Lean ✓ Stmt *If $L \le N$, then replacing the cyclic window at i in σ by the values extracted from that same σ leaves σ unchanged:*

$$\mathrm{replaceWindow}_{L,i}(\sigma, \mathrm{extractWindow}_{L,i}(\sigma)) = \sigma.$$

Proof. ✓ Proof Case-split on whether the offset $(k{-}i{+}N) \bmod N$ lies in $[0, L)$: inside the window the value comes from extractWindow, which returns σ at the same position; outside the window both sides already agree with σ. □

For a periodic chain of length N, the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 16.16 (Translated local term). Lean ✓ Stmt For each site index $i \in \{0, \dots, N{-}1\}$, the translated local term h_i embeds the parent interaction into the full N-site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 16.17 (Parent Hamiltonian). Lean ✓ Stmt The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

Definition 16.18 (Frustration-free condition). Lean ✓ Stmt A state ψ is frustration-free if each local term annihilates it:

$$\forall\, i \in \{0, \dots, N{-}1\}, \quad h_i \psi = 0.$$

Lemma 16.19 (MPV window membership). Lean ✓ Stmt *For any window of L consecutive sites starting at position i in an N-site periodic chain ($L \le N$), the restriction of the MPV state to that window lies in $G_L(A)$.*

Proof. ✓ Proof The witness is the product of A-matrices on the complement sites. Trace cyclicity rotates the full N-site product so that window indices come first, matching the ground-space map definition. □

Lemma 16.20 (Local term annihilates the MPV). Lean ✓ Stmt *For each site i, $h_i V^{(N)}(A) = 0$.*

Proof. ✓ Proof Combines MPV window membership with the fact that the parent interaction kills ground-space elements. □

Lemma 16.21 (Parent Hamiltonian annihilates the MPV). Lean ✓ Stmt *For every N, the MPV state satisfies*

$$H_N(A, L)\, V^{(N)}(A) = 0.$$

Proof. ✓ Proof Sum of zeros, since each local term annihilates the MPV. □

Lemma 16.22 (Frustration-freeness of the MPV). Lean ✓ Stmt *The MPV state is frustration-free for the parent model.*

Proof. ✓ Proof Immediate from Lemma 16.20. □

16.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

16.3.1 Restriction maps

Given a state ψ on $L+1$ sites, we define two restriction maps.

Remark 16.23 (Word and restriction auxiliary formulas). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction operators are the associated linear maps on coefficient spaces, together with their pointwise formulas.

Definition 16.24 (Left restriction). Lean ✓ Stmt Fix the last physical index j:

$$\sigma \longmapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

Definition 16.25 (Right restriction). Lean ✓ Stmt Fix the first physical index i:

$$\sigma \longmapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

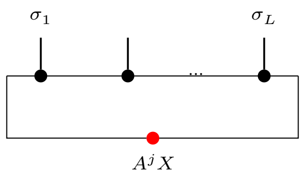
Definition 16.26 (Left ground membership). Lean ✓ Stmt A state ψ on $L+1$ sites satisfies the left ground condition if for every j, the state obtained by fixing the last index to j lies in $G_L(A)$.

Definition 16.27 (Right ground membership). Lean ✓ Stmt A state ψ on $L+1$ sites satisfies the right ground condition if for every i, the state obtained by fixing the first index to i lies in $G_L(A)$.

16.3.2 Forward direction: ground space restricts

Lemma 16.28 (Left restriction). Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$ for some X, then for each j, the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.*

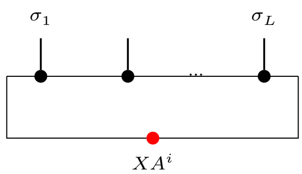
Proof. ✓ Proof The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \mathrm{tr}(A^{\sigma_1} \cdots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. ☐

Lemma 16.29 (Right restriction). Lean ✓ Stmt *If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \mathrm{tr}(A^\sigma X)$, then for each i, the state obtained by fixing the first index to i lies in $G_L(A)$ with witness XA^i.*

Proof. ✓ Proof On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \mathrm{tr}(A^i A^\sigma X) = \mathrm{tr}(A^\sigma \cdot XA^i) = \Gamma_L(XA^i)(\sigma)$. ☐

16.3.3 Injectivity of the ground-space map

Theorem 16.30 (Ground-space map injectivity). Lean ✓ Stmt *If A is injective and $L \geq 1$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$ then $\mathrm{tr}(A^\sigma X) = 0$ for every length-L word σ. Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. ☐

Theorem 16.31 (Ground-space map injectivity from word span). Lean ✓ Stmt *If the length-L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.*

Proof. ✓ Proof If $\Gamma_L(X) = 0$, then $\mathrm{tr}(A^\sigma X) = 0$ for every word σ of length L. The spanning hypothesis turns this into $\mathrm{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. ☐

Lemma 16.32 (Ground-space map injectivity for block-injective tensors). Lean ✓ Stmt *If A is L_0-block-injective, then Γ_{L_0} is injective.*

Proof. ✓ Proof By definition, L_0-block injectivity says precisely that the length-L_0 products span $M_D(\mathbb{C})$. Apply Theorem 16.31. ☐

Theorem 16.33 (Ground-space dimension). Lean ✓ Stmt *If A is injective and $L \geq 1$, then* $\dim G_L(A) = D^2$.

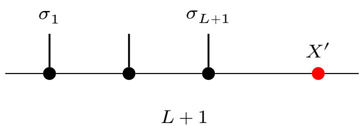
Proof. ✓ Proof The upper bound $\dim G_L(A) \leq D^2$ is Lemma 16.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. ☐

16.3.4 The intersection property

Theorem 16.34 (Intersection property). Lean ✓ Stmt *If A is injective and $L \geq 2$, then a state ψ on $L+1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\mathrm{left}} \cap G_L^{\mathrm{right}} \Longrightarrow \psi \in G_{L+1}(A).$$

Proof. ✓ Proof The "inverting and growing back" argument compares the left and right restrictions on their common $(L-1)$-site overlap and then reconstructs the full $(L+1)$-site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i,\sigma) = \mathrm{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j)$.
3. Matching on the $(L-1)$-site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j.
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.28), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \mathrm{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 16.35 (Intersection characterization). Lean ✓ Stmt *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\mathrm{left}} \cap G_L^{\mathrm{right}}$.*

Proof. ✓ Proof Immediate from the two forward-direction lemmas and Theorem 16.34. □

16.3.5 Unique ground state

Contiguous and cyclic window operators

Remark 16.36 (Contiguous window operators). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For a non-wrapping interval $[s, s+M-1] \subseteq \{0, \dots, N-1\}$, one can insert a length-M word into that interval and keep the complementary sites fixed. The associated restriction map evaluates an N-site state on those configurations. The auxiliary lemmas state the full-window case $s = 0$, $M = N$, the compatibility of this restriction with deleting the last or first site, and the fact that fixing the first K entries of a contiguous $(K+L)$-window leaves the contiguous L-window beginning at $s + K$.

Remark 16.37 (Cyclic window operators). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt
On the periodic chain, a cyclic window starting at i reads the sites $i, i+1, \dots, i+L-1$ modulo N. Deleting the final site of a cyclic $(L+1)$-window reads the value at cyclic offset L in the outside configuration and gives the L-window starting at the same site. Deleting the first site, in the non-repeating case $L+1 \le N$, reads the value at cyclic offset 0 in the outside configuration and gives the L-window starting one site later. When this window does not wrap around the ring, the cyclic construction agrees with the contiguous one, and so do the two restriction maps.

Lemma 16.38 (Cyclic-window boundary-matrix transport)**.** Lean Lean Lean ✓ Stmt *Suppose a cyclic $(L+1)$-window is represented by a boundary matrix Y. Fixing its last site to the letter b gives the initial length-L window represented by $A^b Y$. In the non-repeating regime $L+1 \le N$, fixing the first site to the letter a gives the shifted length-L window represented by YA^a. Consequently, for two adjacent cyclic $(L+1)$-windows in this regime, if the two completed outside assignments obtained after these one-site restrictions are identical as assignments on the N-site chain, then L-block injectivity gives*

$$Y_1 A^a = A^b Y_2.$$

Proof. ✓ Proof The first two identities are the cyclic first- and last-site deletion identities, followed by the corresponding ground-space restriction formulas. For adjacent windows, equality of the completed outside assignments makes the two restricted states on the common overlap equal. Injectivity of the length-L ground-space map then identifies their boundary matrices. □

Lemma 16.39 (Iterated contiguous-window intersection)**.** Lean ✓ Stmt *If an N-site state satisfies the L-site ground condition on every non-wrapping contiguous window (with $L \ge 2$), then it lies in $G_N(A)$.*

Proof. ✓ Proof The induction repeatedly merges two neighbouring admissible windows $[i, i+L-1]$ and $[i+1, i+L]$ into the larger window $[i, i+L]$ by applying Theorem 16.34 to their common overlap $[i+1, i+L-1]$. Strong induction on window size, combining adjacent windows via Theorem 16.34 at each step. □

Definition 16.40 (Suffix restriction)**.** Lean ✓ Stmt Fix a prefix $u : \{0, \dots, K-1\} \to \{0, \dots, d-1\}$. The suffix restriction of an $(K+L)$-site state ψ is the L-site state given by

$$(\mathrm{tailRestrict}_u\, \psi)(\sigma) = \psi(u, \sigma).$$

Definition 16.41 (Suffix ground membership)**.** Lean ✓ Stmt A state ψ on $K+L$ sites lies in the suffix ground condition if every fixed prefix u gives a suffix state in $G_L(A)$.

Lemma 16.42 (Ground-space suffix restriction)**.** Lean Lean Lean Lean ✓ Stmt *If $\psi \in G_{K+L}(A)$, then every fixed-prefix suffix restriction of ψ lies in $G_L(A)$. For a vector in $G_{L+1}(A)$, fixing either endpoint gives the expected left or right multiplication of its boundary matrix.*

Proof. ✓ Proof Write $\psi(\rho) = \mathrm{tr}(A^\rho X)$ for a boundary matrix X. Fixing the final site gives the boundary matrix $A^j X$, while fixing the first site gives XA^i by cyclicity of the trace. Fixing a prefix u gives

$$\psi(u, \sigma) = \mathrm{tr}(A^u A^\sigma X) = \mathrm{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector. □

Lemma 16.43 (Left-tail compatibility)**.** Lean ✓ Stmt *Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K. Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ such that*

$$\psi(\sigma, j) = \mathrm{tr}(A^\sigma Z_j), \qquad \psi(u, \tau) = \mathrm{tr}(A^\tau Y_u),$$

and for all j and u,

$$Z_j A^u = A_j Y_u.$$

Proof. ✓ Proof Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. Restricting the suffix state at its last site gives the boundary matrix $A_j Y_u$, while first fixing the last site and then taking the suffix gives $Z_j A^u$. The two restrictions are the same because appending the fixed prefix and then the last letter gives the same configuration as appending the last letter to the suffix and then placing the prefix in front. Injectivity of Γ_{L_0}, obtained from L_0-block injectivity, yields $Z_j A^u = A_j Y_u$. □

Theorem 16.44 (Common right factor from word compatibility)**.** Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy*

$$Z_j A^\sigma = A_j Y_\sigma$$

for every word σ of length K, then there exists a matrix X such that $Z_j = A_j X$ for all j.

Proof. ✓ Proof For $K = 1$, the hypothesis gives $Z_j A_i = A_j Y_i$ for every letter i. Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A_j X$ for a common matrix X. For larger K, strip the first letter: for each i, the matrices $Z_j A_i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. □

Theorem 16.45 (Open-chain right extension at the injectivity length)**.** Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K, then $\psi \in G_{K+L_0+1}(A)$.*

Proof. ✓ Proof Lemma 16.43 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A_j Y_u$ for every prefix word u of length K. Theorem 16.44 gives a common right factor X such that $Z_j = A_j X$ for all j. Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground- space vector determined by X. Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$-tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. □

Theorem 16.46 (Open-chain normal range reduction)**.** Lean ✓ Stmt *Let A be L_0-block injective with $D \geq 1$ and $L_0 > 0$. If an N-site state satisfies the ground-space constraint on every non-wrapping contiguous window of length $L_0 + 1$, for every choice τ of the complementary outside configuration, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.*

Proof. ✓ Proof Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 16.40 and the contiguous-window transport above identify every fixed-prefix suffix with one of the assumed length-$L_0 + 1$ windows. Theorem 16.45 then grows back the rightmost site. □

Lemma 16.47 (Cyclic-window range antitonicity)**.** Lean Lean ✓ Stmt *On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.*

Proof. ✓ Proof Fix a cyclic window of length $L+1$ and restrict its last site. The resulting length-L vector is again in the ground space, so one-site peeling gives the inclusion for consecutive ranges. Iteration gives the general antitonicity. □

Theorem 16.48 (Cyclic-window normal open-chain reduction). Lean ✓ Stmt *Let A be L_0-block injective with $L_0 > 0$ and $D \geq 1$. If an N-site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.*

Proof. ✓ Proof Lemma 16.47 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 16.46 applies. ☐

Lemma 16.49 (Last cyclic-window one-sided compatibility at the boundary). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$ and $M \geq L_0$. If the reduced cyclic window used in closing the boundary at the last site has boundary matrices Y_τ, then for every physical letter j one has*

$$C_\tau^+ A^j X = Y_\tau A^j,$$

where C_τ^+ is the complementary word seen by that cyclic window.

Proof. ✓ Proof The trace identity for this cyclic window is tested against all words of length L_0. Injectivity of Γ_{L_0}, which follows from L_0-block injectivity, strips the test word and gives the displayed matrix identity. ☐

Lemma 16.50 (Opposite cyclic-window one-sided compatibility at the boundary). Lean ✓ Stmt *Assume A is L_0-block injective with $L_0 > 0$ and $M \geq L_0$. If the opposite reduced cyclic window used in closing the boundary has boundary matrices Y_τ, then*

$$X A^j C_\tau^- = A^j Y_\tau$$

for every physical letter j, where C_τ^- is the complementary word seen from the opposite cyclic position.

Proof. ✓ Proof Factor the cyclic word at the opposite cyclic position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length-L_0 test word. ☐

Theorem 16.51 (Reduced cyclic-window compatibilities at the boundary). Lean ✓ Stmt *Let A be L_0-block injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X. Then there are two families of boundary matrices Y_τ^+ and Y_τ^-, indexed by outside configurations τ, such that for every physical letter j,*

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \qquad X A^j C_\tau^- = A^j Y_\tau^-,$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two opposite cyclic windows used in closing the boundary, both of reduced length $L_0 + 1$.

Proof. ✓ Proof First use Lemma 16.47 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the two cyclic-window compatibility lemmas at the two boundary positions to obtain the displayed one-sided identities. ☐

Lemma 16.52 (Middle-word backgrounds for the two cyclic-window complements). Lean Lean Lean Lean ✓ Stmt *Fix a dummy physical letter η and a middle word μ of length $N - (L_0 + 1)$. Write $\tau_\eta^+(\mu)$ for the last-site cyclic-window background and $\tau_\eta^-(\mu)$ for the opposite cyclic-window background. Their exposed complements are both exactly μ.*

Proof. ✓ Proof With the same dummy letter η on the remaining sites, fill the physical sites $L_0, \ldots, N-2$ with μ for the last-site cyclic window, and fill the sites $1, \ldots, N - L_0 - 1$ with μ for the opposite cyclic window. The two complement-extraction identities are then immediate from the index arithmetic. ☐

Lemma 16.53 (One boundary-matrix family from compared boundary matrices). Lean ✓ Stmt *Fix a dummy physical letter η. Suppose the two one-sided boundary matrix families Y^+ and Y^- have been extracted from the cyclic windows. If, for every middle word μ, these boundary matrices agree on the backgrounds built from η,*

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)},$$

then there is a single family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu$$

for all letters a, b.

Proof. ✓ Proof Define Y_μ to be the first boundary matrix evaluated on the background $\tau^+_\eta(\mu)$. Lemma 16.52 rewrites the first cyclic-window complement as μ, while the assumed comparison at $\tau^-_\eta(\mu)$ rewrites the opposite boundary matrix to the same Y_μ. □

Lemma 16.54 (Common-middle commutation with one boundary family). Lean ✓ Stmt *Suppose a single family of boundary matrices Y_μ, indexed by middle words μ, satisfies the two one-sided identities*

$$A^\mu A^b X = Y_\mu A^b, \qquad X A^a A^\mu = A^a Y_\mu$$

for every pair of physical letters a, b. Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the common middle.

Proof. ✓ Proof Multiply the second identity on the right by A^b and the first identity on the left by A^a:

$$X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

□

Theorem 16.55 (Boundary matrix commutation from the wrapping window). Lean ✓ Stmt *Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then, for every $j = 0, \ldots, d-1$,*

$$X A^j = A^j X$$

Proof. ✓ Proof The wrapping window across the boundary produces matrix identities of the form $XMW = MY$ for arbitrary test matrices M and complementary words W. Spanning first over the length-$(L-1)$ tails and then over the complementary words, injectivity forces $(XM - MX)W = 0$ for every W, hence $XM = MX$ for every matrix M. In particular, X commutes with each letter A^j. □

Definition 16.56 (MPV submodule). Lean ✓ Stmt The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}})$.

Lemma 16.57 (The MPV is the ground-space map at the identity). Lean ✓ Stmt *For every chain length N,*

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 16.58 (MPV lies in the ground space). Lean ✓ Stmt $V^{(N)}(A) \in G_N(A)$ *for every N, with boundary matrix $X = I$.*

Proof. ✓ Proof $V^{(N)}(A)(\sigma) = \mathrm{tr}(A^\sigma) = \mathrm{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

Definition 16.59 (Unique ground state). Lean ✓ Stmt A submodule S has a unique ground state if $\dim S = 1$.

Theorem 16.60 (One-dimensionality criterion). Lean ✓ Stmt *For finite-dimensional S, $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c\,\psi_0$.*

Proof. ✓ Proof Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. □

Definition 16.61 (Periodic chain ground space). Lean ✓ Stmt For $N > 0$ and $L \le N$, define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0,\dots,d-1\}^N \to \mathbb{C}) : \forall i \in \{0,\dots,N-1\},\ \psi|_{[i,i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i,i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 16.62 (Cyclic-window witnesses inside the chain ground space). Lean ✓ Stmt *If $0 < N$, $L \le N$, and $\psi \in \mathcal{G}_{N,L}(A)$, then for every cyclic window and every outside assignment there is a boundary matrix Y such that the corresponding restricted L-site state is $\Gamma_L(Y)$.*

Proof. ✓ Proof This is the defining cyclic-window condition for membership in $\mathcal{G}_{N,L}(A)$, together with the definition of the local ground space as the range of Γ_L. □

Lemma 16.63 (The MPV satisfies every cyclic window constraint). Lean ✓ Stmt *If $0 < N$ and $L \le N$, then*

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 16.64 (MPV nonvanishing for block-injective tensors). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $D \ge 1$, and $N \ge L_0 + 1$, then $V^{(N)}(A) \ne 0$ in the N-site space.*

Proof. ✓ Proof Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\mathrm{tr}(A^\sigma) = 0$ for every word σ of length N. For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\mathrm{tr}(A^w A^u) = 0$. Since A is L_0-block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\mathrm{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. □

16.3.6 Block-word stripping

Theorem 16.65 (Common right factor from block-word compatibility). Lean ✓ Stmt *Assume A is L_0-block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0. If for some $K \ge 0$ and every suffix word w of length K there exists $Y_w \in M_D(\mathbb{C})$ such that, for every word u of length L_0,*

$$Z_u A^w = A^u Y_w$$

then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0.

Proof. ✓ Proof Induct on K. For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length-L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X. □

Theorem 16.66 (Block-word commutation from long-word commutation). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length-m product A^ω, then X already commutes with every length-L_0 product A^u.*

Proof. ✓ Proof Write each length-m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$XA^uA^w = A^uA^wX = A^u(A^wX).$$

Thus the family $Z_u := XA^u$ satisfies the hypothesis of Theorem 16.65. One obtains $XA^u = A^uR$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length-L_0 words shows $R = X$, hence $XA^u = A^uX$. □

Theorem 16.67 (Centrality from long-word commutation). Lean ✓ Stmt *If A is L_0-block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length-m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 16.66, X commutes with every length-L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. □

Lemma 16.68 (Amplifying fixed-length word commutation). Lean ✓ Stmt *If a boundary matrix X commutes with every word product of length m, then it commutes with every word product whose length is a multiple $q\,m$.*

Proof. ✓ Proof Proceed by induction on q. For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q+1)m$ be given and write it as a length-m prefix followed by a length-$q\,m$ suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. □

Lemma 16.69 (Padded complement annihilation). Lean ✓ Stmt *If A is L_0-block-injective and $q \geq 1$, then an operator Z that annihilates every length-k word product on the left is zero whenever $k \leq qL_0$:*

$$ZA^w = 0 \quad (|w| = k) \qquad \Longrightarrow \qquad Z = 0.$$

Proof. ✓ Proof The hypothesis $\operatorname{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0. Each length-qL_0 word factors as a length-k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. □

Lemma 16.70 (One-sided boundary witnesses are unique). Lean Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If two boundary matrices have the same product with every one-site tensor on one fixed side, then the two boundary matrices are equal.*

Proof. ✓ Proof If $Y_1A^j = Y_2A^j$ for every physical index j, then for every nonempty word w, induction on the word gives

$$Y_1A^w = Y_2A^w$$

Since A is L_0-block-injective,

$$\operatorname{span}\{A^w : |w| = L_0\} = M_D(\mathbb{C}).$$

Thus the left multiplication maps $M \mapsto Y_1M$ and $M \mapsto Y_2M$ are equal on $M_D(\mathbb{C})$, and evaluating on the identity gives $Y_1 = Y_2$. The other one-sided statement is the same argument, with right multiplication in place of left multiplication. □

Lemma 16.71 (Generator commutation from long-word commutation). Lean ✓ Stmt *Under the hypotheses of Theorem 16.67, one has $XA^i = A^iX$ for every physical index i.*

Proof. ✓ Proof Apply Theorem 16.67 with $M = A^i$. □

Theorem 16.72 (Boundary contraction in the MPV span from long-word commutation). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and let $m \geq L_0$. If a boundary matrix X commutes with every length-m word product, then $\Gamma_N(X)$ lies in the MPV line* $\operatorname{span}\{V^{(N)}(A)\}$.

Proof. ✓ Proof Theorem 16.67 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbb{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. □

Theorem 16.73 (Boundary contraction in the MPV span from positive-length commutation). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If X commutes with every word product of some positive length m, then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof Lemma 16.68 amplifies the length-m commutation hypothesis to length $L_0\, m \geq L_0$. Then Theorem 16.72 applies. □

Theorem 16.74 (Boundary contraction in the MPV span from one middle boundary family). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. If the common-middle one-boundary-family identities of Lemma 16.54 hold for X, then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof The common-middle identities give commutation with all words of length $|\mu| + 2$, a positive length. The positive-length containment theorem then applies. □

Theorem 16.75 (Boundary contraction in the MPV span from wrapped-witness comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$, and fix a dummy physical letter η. If the two one-sided boundary matrix families satisfy*

$$Y^+_{\tau^+_\eta(\mu)} = Y^-_{\tau^-_\eta(\mu)}$$

for every middle word μ, then $\Gamma_N(X)$ lies in the MPV line.

Proof. ✓ Proof Lemma 16.53 uses the comparison at the two η-backgrounds to turn the two boundary matrices into a single common-middle boundary family. Then Theorem 16.74 applies. □

Theorem 16.76 (Conditional normal-range reduction from witness comparison). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every dummy letter η and every boundary matrix obtained from an N-site chain ground state, the two boundary-matrix families supplied by Theorem 16.51 agree at $\tau^+_\eta(\mu)$ and $\tau^-_\eta(\mu)$ for every middle word μ. Then the range-L chain ground space is contained in the MPV line.*

Proof. ✓ Proof The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The reduced cyclic-window compatibility theorem supplies the two one-sided boundary-matrix families for this X. The comparison hypothesis, applied to a dummy letter η, says that these boundary matrices agree at $\tau^+_\eta(\mu)$ and $\tau^-_\eta(\mu)$ for every middle word μ. Theorem 16.75 finishes. □

Lemma 16.77 (Right products determine the compared boundary matrix). Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$. Fix two boundary-matrix families Y^+ and Y^- and the two reindexed backgrounds $\tau^+_\eta(\mu)$ and $\tau^-_\eta(\mu)$. If, for every physical letter j,*

$$Y^+_{\tau^+_\eta(\mu)} A^j = Y^-_{\tau^-_\eta(\mu)} A^j,$$

then

$$Y^{+}_{\tau^{+}_{\eta}(\mu)} = Y^{-}_{\tau^{-}_{\eta}(\mu)}.$$

Proof. ✓ Proof Apply Lemma 16.70 to the two boundary matrices $Y^{+}_{\tau^{+}_{\eta}(\mu)}$ and $Y^{-}_{\tau^{-}_{\eta}(\mu)}$. □

Theorem 16.78 (Boundary matrices agree after closing the boundary)**.** Lean ✓ Stmt *Let A be L_0-block-injective with $L_0 > 0$ and $D \geq 1$, let $N \geq 2$ and $L_0 < L \leq N$, and let $\psi \in \mathcal{G}_{N,L}(A)$ have an open-chain representation $\psi = \Gamma_N(X)$. Suppose Y^+ and Y^- are the two boundary-matrix families extracted from the two cyclic windows used in closing the periodic boundary, and that they satisfy, for every physical letter j and every background configuration τ,*

$$A^{\tau_{L_0}\cdots\tau_{N-2}}A^jX = Y^+_\tau A^j, \qquad XA^jA^{\tau_1\cdots\tau_{N-L_0-1}} = A^jY^-_\tau.$$

For every physical letter $\eta \in \{0,\dots,d-1\}$ and every middle word μ of length $N-(L_0+1)$, define two background configurations by

$$\tau^+_\eta(\mu)_i = \begin{cases}\mu_{i-L_0}, & L_0 \leq i < N-1,\\ \eta, & \textit{otherwise},\end{cases} \qquad \tau^-_\eta(\mu)_i = \begin{cases}\mu_{i-1}, & 1 \leq i < N-L_0,\\ \eta, & \textit{otherwise}.\end{cases}$$

Then the two boundary matrices agree on these two backgrounds:

$$Y^{+}_{\tau^{+}_{\eta}(\mu)} = Y^{-}_{\tau^{-}_{\eta}(\mu)}.$$

This is the boundary-closing comparison corresponding to the closure property in [CPGSV21, lines 2078–2090].

Proof. Not ready The reduced cyclic-window compatibilities at the boundary give, for every physical letter j,

$$A^\mu A^jX = Y^+_{\tau^+_\eta(\mu)}A^j, \qquad XA^jA^\mu = A^jY^-_{\tau^-_\eta(\mu)}.$$

By Lemma 16.77, it is enough to prove the right-product identities

$$Y^+_{\tau^+_\eta(\mu)}A^j = Y^-_{\tau^-_\eta(\mu)}A^j$$

for every j. Write $N = M+1$, so the wrapped boundary window starts at M and the mirror boundary window starts at $M+1-L_0$. Let $Y_i(\tau)$ denote the cyclic-window witness supplied by Lemma 16.62 at the window beginning at i. The first step identifies the two abstract boundary-condition families with the two boundary-crossing witnesses:

$$Y^+_{\tau^+_\eta(\mu)} = Y_M(\tau^+_\eta(\mu)), \qquad Y^-_{\tau^-_\eta(\mu)} = Y_{M+1-L_0}(\tau^-_\eta(\mu)).$$

The remaining boundary-closing statement is therefore, for every physical letter j,

$$Y_M(\tau^+_\eta(\mu))A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

For consecutive cyclic windows, Lemma 16.38 gives the adjacent overlap identity

$$Y_i(\tau)A^a = A^bY_{i+1}(\tau')$$

whenever the two completed assignments

$$\hat{\tau}^a_{i,0}(k) = \begin{cases}a, & k=i,\\ \tau(k), & k\neq i,\end{cases} \qquad \widehat{\tau'}^{\,b}_{i+1,L_0}(k) = \begin{cases}b, & k \equiv i+L_0+1 \pmod N,\\ \tau'(k), & k \not\equiv i+L_0+1 \pmod N,\end{cases}$$

agree, that is, as functions on the N-site chain:

$$\hat{\tau}^{a}_{i,0} = \widehat{\tau'}^{b}_{i+1,L_0}.$$

The available adjacent identities therefore have the form

$$Y_{i_r}(\tau_r)A^{a_r} = A^{b_r}Y_{i_{r+1}}(\tau_{r+1}) \qquad (0 \le r < q),$$

only at pairs whose completed assignments are equal. They do not by themselves give a telescope from $\tau^-_\eta(\mu)$ to $\tau^+_\eta(\mu)$ with the same letter j attached on the right throughout; the two endpoint backgrounds need not agree away from a single adjacent overlap. What remains is a separate cyclic boundary-transport argument. It must use the local identities for genuinely equal completions, together with the fact that all witnesses come from the same periodic-chain state, to obtain, for each fixed final letter j,

$$Y_M(\tau^+_\eta(\mu))A^j = Y_{M+1-L_0}(\tau^-_\eta(\mu))A^j.$$

□

Theorem 16.79 (Chain ground space equals the MPV line in the injective case)**.** Lean ✓ Stmt *If A is injective, $D \ge 1$, $N \ge 2$, and $1 < L \le N$, then*

$$\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 16.39 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X. The wrapping window condition now applies Theorem 16.55 and forces X to commute with every letter A^j. Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 16.63. □

Theorem 16.80 (Normal-range reduction: containment in the MPV line)**.** Lean ✓ Stmt *If A is normal, $D \ge 1$, and L_0-block-injective for some $L_0 > 0$, with $N \ge 2$ and $L_0 < L \le N$, then*

$$\mathcal{G}_{N,L}(A) \subseteq \operatorname{span}\{V^{(N)}(A)\}.$$

Proof. Not ready The closure property still needs the boundary-matrix comparison corresponding to [CPGSV21, lines 2078–2090]. Theorem 16.51 supplies the two one-sided cyclic-window identities for the open-chain boundary matrix X. The padded-annihilation lemma states that exact complement-span length mismatches are no longer the obstruction. The new conditional reduction theorem shows that the proof would finish as soon as the two actual boundary matrices are compared after reindexing their complements to a common middle word. What is still missing is precisely that comparison after closing the boundary. □

Theorem 16.81 (Chain ground space equals the MPV line in the normal case)**.** Lean ✓ Stmt *If A is normal, $D \ge 1$, and L_0-block-injective for some $L_0 > 0$, with $N \ge 2$ and $L_0 < L \le N$, then*

$$\mathcal{G}_{N,L}(A) = \operatorname{span}\{V^{(N)}(A)\}.$$

Theorem 16.82 (Unique ground state on the periodic chain)**.** Lean ✓ Stmt *For an injective tensor A with $D \ge 1$, if $N \ge 2$ and $1 < L \le N$, then the periodic chain ground space is one-dimensional.*

Proof. ✓ Proof By Theorem 16.79, the ground space is the line $\operatorname{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 16.57 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 16.30 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. □

Theorem 16.83 (Unique ground state for block-injective tensors). Lean Not ready *If A is L_0-block-injective with $L_0 > 0$ and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on the periodic chain.*

Theorem 16.84 (Optimal unique ground state for normal tensors). Lean Not ready *If A is normal (hence L_0-block-injective for some $L_0 > 0$), $D \geq 1$, and in normal form, the interaction range can be reduced to $L_0 + 1$.*

16.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

Definition 16.85 (Commuting parent Hamiltonian). Lean ✓ Stmt The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. See Ref. [CPGSV16, Definition 3.9].

Definition 16.86 (Nearest-neighbor commuting parent Hamiltonian). Lean ✓ Stmt A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length $L = 2$. See Ref. [CPGSV16, Definition 3.9].

Remark 16.87 (Elementary consequences of the commuting definition). Lean Lean Lean ✓ Stmt The nearest-neighbor condition is simply the length-2 specialization of the general commuting condition; the latter is symmetric in the two sites, and it implies that the full parent Hamiltonian commutes with each local interaction term.

Theorem 16.88 (RFP implies NNCPH). Lean ✓ Stmt *If A is a normal renormalization fixed point, then for every chain length $N \geq 2$ the parent Hamiltonian with $L = 2$ has commuting local terms. This implication is cited here from the product-of-entangled-pairs description in Appendix B of Theorem 3.10 in [CPGSV16].*

Remark 16.89 (Product-pair data for NNCPH). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt On an even chain, one can extract the p-th adjacent two-site word of a configuration and compare the MPS coefficients with a repeated two-site pair state. The one-pair case recovers the original two-site amplitude. For an Appendix B structural witness, the gauge-removed tensor ΛU_i has the same MPV coefficients as A, and the structural two-site amplitude is exactly the two-site MPV coefficient. Hence the coefficient part of the extraction reduces to proving the repeated-pair identity for this gauge-removed tensor. The corresponding data also include a family of idempotent chain-level projectors whose values are the nearest-neighbor local terms. The final theorem in these data shows that these local terms commute. Consequently the same data yields the nearest-neighbor commuting condition on every finite chain. The Appendix B construction extracts the structural decomposition from normal left-canonical RFP data. The missing mathematical input is the product-pair extraction through the two-site amplitude determined by that structural witness.

Theorem 16.90 (NNCPH implies RFP). Lean ✓ Stmt *If A is normal, in left-canonical form, and its parent Hamiltonian with $L = 2$ has commuting local terms for every chain length $N \geq 2$, then A is a renormalization fixed point. This implication is cited here from Beigi's ground-space characterization of one-dimensional commuting nearest-neighbor Hamiltonians; the left-canonical hypothesis fixes the normalization of the transfer map.*

16.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following [CPGSV16, Appendix D, Section D.2].

Definition 16.91 (Decorrelation). Lean ✓ Stmt Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A$, $\mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A$, $O_B \in \mathcal{O}_B$.

Definition 16.92 (Commuting parent Hamiltonian structure). Lean ✓ Stmt A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 16.93 (Idempotence gives a trivial commuting parent Hamiltonian). Lean ✓ Stmt *If*

$$P_K \circ P_K = P_K,$$

then P_K has a commuting parent Hamiltonian structure.

Proof. ✓ Proof Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K. □

Lemma 16.94 (Cancellation for commuting idempotents). Lean ✓ Stmt *If P and Q are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. ✓ Proof Expand the middle factor, use commutativity to rewrite $QPQP$ as $QPPQ$, and then apply idempotence of P and Q. □

Theorem 16.95 (Idempotence of the product projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_K = P_K.$$

Proof. ✓ Proof Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 16.96 (Absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{AX} \circ P_K = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX},

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 16.97 (Absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{XB} = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB},

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 16.98 (Cross-absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_K = P_K.$$

Proof. ✓ Proof By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. □

Theorem 16.99 (Cross-absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{AX} = P_K.$$

Proof. ✓ Proof By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. □

Theorem 16.100 (Reverse product identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. ✓ Proof This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. □

Theorem 16.101 (Commuting complementary projectors). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition and*

$$Q_{AX} = 1 - P_{AX}, \qquad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. ✓ Proof Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. □

Theorem 16.102 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt *If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A-observables commute with P_{XB} while B-observables commute with P_{AX}, then regions A and B are decorrelated w.r.t. K. This is the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].*

Proof. ✓ Proof The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX}, the five-fold composition factors through this zero term. □

Theorem 16.103 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt *Corollary of Theorem 16.102: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K.*

Lemma 16.104 (Frustration-free identity). Lean ✓ Stmt *For any endomorphisms P and Q,*

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. ✓ Proof Expanding the left-hand side and cancelling gives the result. □

Theorem 16.105 (Commuting parent Hamiltonian identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV16, Appendix D, Section D.2].

Proof. ✓ Proof Applying Lemma 16.104 to P_{AX} and P_{XB}, and substituting $P_{AX} \circ P_{XB} = P_K$. □

Theorem 16.106 (Ground-space membership). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then*

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV16]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. ✓ Proof ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$ by left absorption, and similarly for P_{XB}.

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$. □

16.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N. For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

Definition 16.107 (Transported parent ground space). Lean ✓ Stmt Denote by $\mathcal{G}^{\mathrm{ES}}_{N,L}(A)$ the parent ground space after identifying the coefficient space with its Euclidean ℓ^2-model.

Definition 16.108 (Transported parent Hamiltonian). Lean ✓ Stmt Denote by $H^{\mathrm{ES}}_{N,L}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and its Euclidean avatar.

Theorem 16.109 (Euclidean transport of the parent ground space). Lean ✓ Stmt *Under the canonical ℓ^2 transport of the N-site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.*

Proof. ✓ Proof This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. □

Lemma 16.110 (Membership in the transported parent ground space). Lean ✓ Stmt *A vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.*

Remark 16.111 (Euclidean local projector ingredients). Lean Lean Lean Lean ✓ Stmt Let $P_L^{\mathrm{ES}}(A)$ denote the orthogonal projector onto $(G_L^{\mathrm{ES}}(A))^\perp$ on the Euclidean L-site space. For a cyclic window starting at i and an outside configuration τ, the transported restriction map $R_{i,\tau}$ evaluates an N-site Euclidean vector on that window. The transported local term $h_{i,\mathrm{ES}}$ is the corresponding N-site local projector, and $R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 16.112 (Positivity of the Euclidean local summands). Lean Lean Lean ✓ Stmt *The Euclidean local projector $P_L^{\mathrm{ES}}(A)$ is positive. Consequently, every summand $R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}$ is positive, and so is the finite sum over τ.*

Proof. ✓ Proof Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. □

Lemma 16.113 (Cyclic averaging for the transported ES local terms). Lean Lean ✓ Stmt *The transported parent Hamiltonian is the sum of the transported local terms $H_{N,\mathrm{ES}} = \sum_i h_{i,\mathrm{ES}}$, and each transported local term admits the exact cyclic-averaging formula*

$$h_{i,\mathrm{ES}} = d^{-L} \sum_\tau R_{i,\tau}^\dagger P_L^{\mathrm{ES}}(A) R_{i,\tau}.$$

Proof. ✓ Proof Unfold the transported local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ. □

Lemma 16.114 (Positivity of the transported ES Hamiltonian). Lean Lean ✓ Stmt *Each transported local term $h_{i,\mathrm{ES}}$ is positive, hence the full transported parent Hamiltonian $H_{N,\mathrm{ES}}$ is positive.*

Proof. ✓ Proof Apply the averaging formula above: $h_{i,\mathrm{ES}}$ is a non-negative scalar multiple of a finite sum of positive conjugates of $P_L^{\mathrm{ES}}(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\mathrm{ES}}$. □

Lemma 16.115 (Symmetric-projection structure of transported ES local terms). Lean Lean ✓ Stmt *The Euclidean local projector $P_L^{\mathrm{ES}}(A)$ is a symmetric projection, and every transported local term $h_{i,\mathrm{ES}}$ on the N-site chain is a symmetric projection.*

Proof. ✓ Proof The local operator $P_L^{\mathrm{ES}}(A)$ is an orthogonal projection onto $(G_L^{\mathrm{ES}}(A))^\perp$. For $L \le N$, restricting $h_{i,\mathrm{ES}}v$ to a cyclic window gives $P_L^{\mathrm{ES}}(A)$ applied to the corresponding restriction of v; applying $h_{i,\mathrm{ES}}$ a second time therefore applies $(P_L^{\mathrm{ES}}(A))^2 = P_L^{\mathrm{ES}}(A)$ on that window. The positivity result above supplies symmetry, and the case $L > N$ is the zero projection by definition. □

Lemma 16.116 (Kernel of the Euclidean parent interaction). Lean ✓ Stmt *The Euclidean local interaction $P_L^{\mathrm{ES}}(A)$ annihilates exactly the Euclidean local ground space:*

$$P_L^{\mathrm{ES}}(A)v = 0 \quad \Longleftrightarrow \quad v \in G_L^{\mathrm{ES}}(A).$$

Proof. ✓ Proof Since $P_L^{\mathrm{ES}}(A)$ is the orthogonal projector onto $(G_L^{\mathrm{ES}}(A))^\perp$, its kernel is the original subspace $G_L^{\mathrm{ES}}(A)$. □

Lemma 16.117 (Local-term kernel via cyclic restrictions). Lean Lean ✓ Stmt *For $L \le N$, a transported local term $h_{i,\mathrm{ES}}$ annihilates a vector v iff every boundary-filled restriction of v to the cyclic L-window starting at i lies in $G_L^{\mathrm{ES}}(A)$.*

Proof. ✓ Proof Restricting $h_{i,\mathrm{ES}}v$ to the same cyclic window gives $P_L^{\mathrm{ES}}(A)$ applied to the corresponding restriction of v. Lemma 16.116 identifies the vanishing of this local projector with membership in $G_L^{\mathrm{ES}}(A)$. □

Theorem 16.118 (Adjacent local kernels realize the intersection property). Lean Lean Lean ✓ Stmt *If A is injective and $L \ge 2$, then on an $(L+1)$-site chain the kernels of the two adjacent transported L-site local terms are exactly the transported $(L+1)$-site MPS ground space:*

$$h_{0,\mathrm{ES}}v = 0 \quad \textit{and} \quad h_{1,\mathrm{ES}}v = 0 \quad \iff \quad v \in G_{L+1}^{\mathrm{ES}}(A).$$

Proof. ✓ Proof Lemma 16.117 translates the two local kernel assumptions into the left and right L-site ground conditions. The open-chain intersection property, Theorem 16.34, then grows back the shared $(L-1)$-site overlap to the $(L+1)$-site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization. □

Lemma 16.119 (Non-overlap positivity for transported ES local terms). Lean Lean Lean Lean Lean ✓ Stmt *Let $L \le N$ and let $h_{i,\mathrm{ES}}, h_{j,\mathrm{ES}}$ be transported local terms whose cyclic L-windows are site-disjoint: no site of the periodic chain has cyclic offset $< L$ from both i and j. Then the terms commute pointwise: for every vector v,*

$$h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v) = h_{j,\mathrm{ES}}(h_{i,\mathrm{ES}}v),$$

and their ordered cross term is non-negative:

$$0 \le \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof In coordinates, $h_{i,\mathrm{ES}}$ applies $P_L^{\mathrm{ES}}(A)$ only to the i-window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i-window does not change the extracted j-window, and the i- and j-window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the non-negative ordered cross term. □

Lemma 16.120 (Quadratic form to norm bound). Lean ✓ Stmt *Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v,*

$$\gamma\,\langle v, Hv\rangle \le \langle Hv, Hv\rangle$$

i.e. $H^2 \ge \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma\,\|v\| \le \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ. The positive-semidefiniteness hypothesis is essential: without it, $H = -\mathrm{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. ✓ Proof By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k. For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma\,\lambda_k \le \lambda_k^2$, whence $\lambda_k \le 0$ or $\lambda_k \ge \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \ge \gamma$. Writing $v = \sum_k c_k\psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \ge \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2\lambda_k^2 \ge \gamma^2\sum_k |c_k|^2 = \gamma^2\|v\|^2$. □

Lemma 16.121 (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt *Let $H^{\mathrm{ES}}_{N,L}(A)$ be the transported parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate*

$$\gamma \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, v\rangle \le \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}(A)v, H^{\mathrm{ES}}_{N,L}(A)v\rangle$$

for every $N \ge 2L$ and every vector v, then

$$\gamma \|v\| \le \|H^{\mathrm{ES}}_{N,L}(A)v\|$$

for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate is supplied.

Proof. ✓ Proof The transported Hamiltonian is positive by Lemma 16.114, and its transported ground space is its kernel by Theorem 16.109. Therefore the abstract spectral-theorem conversion of Lemma 16.120 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. □

Remark 16.122 (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\operatorname{Re}\langle Pv, Pv\rangle = \operatorname{Re}\langle Pv, v\rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\operatorname{Re}\langle Pv, Qv\rangle$ for commuting symmetric projections P, Q; the Cauchy–Schwarz conversion from $\|PQv\| \le c\|Pv\|$ to $\operatorname{Re}\langle Pv, Qv\rangle \ge -c \operatorname{Re}\langle Pv, v\rangle$; finite-sum expansions of $\operatorname{Re}\langle(\sum_i P_i)v, v\rangle$ and $\operatorname{Re}\langle(\sum_i P_i)v, (\sum_i P_i)v\rangle$; the diagonal/off-diagonal split of the ordered double sum; and the aggregate off-diagonal estimate obtained from row sums $\sum_{j\ne i} c_{ij} \le 1$.

Lemma 16.123 (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey*

$$\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)c_{ij} \operatorname{Re}\langle P_i v, v\rangle \qquad (i \ne j),$$

with row sums $\sum_{j\ne i} c_{ij} \le 1$ and $\gamma \le 1$, then $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Expand $\operatorname{Re}\langle Hv, Hv\rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\operatorname{Re}\langle P_i v, v\rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1-\gamma)\sum_i \operatorname{Re}\langle P_i v, v\rangle$. □

Lemma 16.124 (Finite-overlap projection row-sum reduction). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m. If noninteracting ordered cross terms are non-negative and, on each interacting pair,*

$$\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)\frac{1}{m} \operatorname{Re}\langle P_i v, v\rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form.

Proof. ✓ Proof Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j\ne i} c_{ij} \le 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 16.123. □

Lemma 16.125 (Finite-overlap projection norm-compression reduction). Lean Lean ✓ Stmt *Let P_i be a finite family of symmetric projections and let $\gamma \le 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are non-negative, and interacting pairs satisfy the norm-compression estimate*

$$\|P_i P_j v\| \le (1-\gamma)\frac{1}{m}\|P_i v\|.$$

Then $H = \sum_i P_i$ satisfies $H^2 \ge \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \le (1-\gamma)/m$.

Proof. ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\operatorname{Re}\langle P_i v, P_j v\rangle \ge -(1-\gamma)m^{-1}\operatorname{Re}\langle P_i v, v\rangle$. If a separate coefficient η is used, the assumption $\eta \le (1-\gamma)/m$ first weakens that estimate to the displayed one. The finite-overlap row-sum reduction then applies. □

Lemma 16.126 (Parent-Hamiltonian ordered row-sum quadratic-form reduction). Lean ✓ Stmt *For a fixed chain length N, suppose the transported local terms $h_{i,\mathrm{ES}}$ satisfy the ordered row-sum estimate*

$$\operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1-\gamma)c_{ij}\operatorname{Re}\langle h_{i,\mathrm{ES}}v, v\rangle \qquad (i \ne j),$$

with $\sum_{j\ne i} c_{ij} \le 1$ and $\gamma \le 1$. Then the transported parent Hamiltonian satisfies

$$\gamma \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}v, v\rangle \le \operatorname{Re}\langle H^{\mathrm{ES}}_{N,L}v, H^{\mathrm{ES}}_{N,L}v\rangle$$

for every vector v.

Proof. ✓ Proof Lemma 16.115 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\mathrm{ES}}$. Apply Lemma 16.123 to this family and use $H^{\mathrm{ES}}_{N,L} = \sum_i h_{i,\mathrm{ES}}$. □

Lemma 16.127 (Parent-Hamiltonian gap from ordered local row bounds). Lean ✓ Stmt *Assume uniformly for every $N \ge 2L$ that the transported local terms satisfy the ordered row-sum estimate above with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,*

$$\frac{1}{4L}\|v\| \le \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N. The positivity and kernel identification of the transported Hamiltonian then allow Lemma 16.121 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

Lemma 16.128 (Parent-Hamiltonian finite-overlap Friedrichs quadratic reduction). Lean ✓ Stmt *For a fixed chain length N, let m be a positive integer and let $\gamma \le 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have non-negative ordered cross terms, overlapping local terms satisfy the finite-overlap Friedrichs estimate*

$$\operatorname{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \ge -(1-\gamma)\frac{1}{m}\operatorname{Re}\langle h_{i,\mathrm{ES}}v, v\rangle,$$

and the transported local terms are symmetric projections, then the transported Hamiltonian satisfies ${H^{\mathrm{ES}}_{N,L}}^2 \ge \gamma H^{\mathrm{ES}}_{N,L}$ as a quadratic form.

Proof. ✓ Proof Apply the finite-overlap projection reduction with $P_i = h_{i,\mathrm{ES}}$. □

Lemma 16.129 (Finite-overlap norm-compression reduction). Lean Lean ✓ Stmt *For a fixed chain length N, let $m > 0$ and $\gamma \le 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row, non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \le (1-\gamma)\frac{1}{m}\|h_{i,\mathrm{ES}}v\|.$$

Then ${H^{\mathrm{ES}}_{N,L}}^2 \ge \gamma H^{\mathrm{ES}}_{N,L}$ as a quadratic form. The same conclusion holds from a separate coefficient η satisfying $\eta \le (1-\gamma)/m$.

Proof. ✓ Proof Apply the finite-overlap norm-compression projection reduction to the family $P_i = h_{i,\mathrm{ES}}$. If a separate coefficient η is used, first weaken the compression estimate using $\eta \le (1-\gamma)/m$. □

Definition 16.130 (Cyclic-window support). Lean ✓ Stmt For a length-L cyclic window on $\{0,\dots,N-1\}$ starting at i, its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod N,$$

with repeated visits counted once.

Lemma 16.131 (Cyclic-window support offsets). Lean ✓ Stmt *If $L \le N$, then a site k belongs to the cyclic support S_i exactly when its cyclic offset from i is below L:*

$$k \in S_i \iff (k-i) \bmod N < L.$$

Proof. ✓ Proof Write $k = i + r \pmod N$ for a representative offset r. Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. □

Definition 16.132 (Cyclic-window overlap). Lean ✓ Stmt Two length-L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \ne \emptyset$.

Lemma 16.133 (Concrete non-overlap positivity for cyclic supports). Lean Lean ✓ Stmt *If $L \le N$ and the cyclic supports S_i and S_j do not meet, then the transported local ES terms have non-negative ordered cross term:*

$$0 \le \mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle.$$

Proof. ✓ Proof The support-offset characterization converts failure of $S_i \cap S_j \ne \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. □

Lemma 16.134 (Cyclic-window overlap row cardinality). Lean ✓ Stmt *If $2L \le N$ and $L > 1$, then for every $i \in \{0,\dots,N-1\}$,*

$$\#\{j \ne i : i \sim_L j\} \le 2(L-1).$$

Proof. ✓ Proof If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod N$. Since $2L \le N$, the offset from i to j is either the clockwise distance $a - b \in \{1,\dots,L-1\}$ or the counterclockwise distance $b-a \in \{1,\dots,L-1\}$; the zero distance is excluded by $j \ne i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. □

Lemma 16.135 (Parent-Hamiltonian gap from finite-overlap Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ that the hypotheses of the fixed-chain finite-overlap Friedrichs reduction hold with $\gamma = 1/(4L)$ and this value of m. Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.*

Proof. ✓ Proof The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N. Lemma 16.121 converts that estimate to the norm lower bound. □

Lemma 16.136 (Parent-Hamiltonian gap from cyclic-window Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \geq -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle.$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof Lemma 16.134 gives the row-cardinality bound $m = 2(L-1)$ for the concrete cyclic-window overlap relation, and Lemma 16.115 supplies the symmetric projection structure of each transported local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 16.133 gives the required non-negative ordered cross term. Apply Lemma 16.135 with these facts and the displayed overlapping-window estimate. □

Lemma 16.137 (Parent-Hamiltonian gap from overlapping cyclic Friedrichs data). Lean ✓ Stmt *Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length-L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy*

$$\mathrm{Re}\langle h_{i,\mathrm{ES}}v, h_{j,\mathrm{ES}}v\rangle \geq -(1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\,\mathrm{Re}\langle h_{i,\mathrm{ES}}v, v\rangle.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof Apply Lemma 16.136. Its non-overlap positivity and finite-row hypotheses are already internal to that cyclic-window reduction; the displayed overlapping estimate supplies its remaining ordered cross-term condition. □

Lemma 16.138 (Parent-Hamiltonian gap from overlapping cyclic norm-compression data). Lean ✓ Stmt *Fix $L > 1$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq (1 - \tfrac{1}{4L})\frac{1}{2(L-1)}\|h_{i,\mathrm{ES}}v\|.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$.

Proof. ✓ Proof The norm-compression condition is converted to the ordered Friedrichs lower bound by the projection-geometry Cauchy–Schwarz lemma. The preceding overlapping-cyclic Friedrichs reduction then gives the norm lower bound. □

Lemma 16.139 (Parent-Hamiltonian gap from a cyclic overlap compression constant). Lean Lean ✓ Stmt *Fix $L > 1$, and let $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that*

$$\|h_{i,\mathrm{ES}}(h_{j,\mathrm{ES}}v)\| \leq \eta\,\|h_{i,\mathrm{ES}}v\|.$$

If $\gamma > 0$, $\gamma \leq 1$, and $\eta \leq (1-\gamma)/m$, then $0 < \gamma$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

The assertion $0 < \gamma$ repeats the hypothesis, in the same shape as the strict statement below. In particular, if $\eta \geq 0$ and $\eta m < 1$, then $0 < 1 - \eta m$ and, for every admissible N and every $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$,

$$(1-\eta m)\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

Proof. ✓ Proof The cyclic-window cardinality lemma gives at most $m = 2(L-1)$ overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen γ, and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with $\gamma = 1 - \eta m$. □

Theorem 16.140 (Martingale criterion for spectral gap)**.** Not ready *Consider a frustration-free Hamiltonian*

$$H = \sum_{i=1}^{n} h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i,j) and that $\sum_{j:\, j\neq i} c_{ij} \leq 1$ for every i. Then the smallest positive eigenvalue of H satisfies $\lambda^{+}_{\min}(H) \geq \gamma$.

Proof. Not ready The relevant geometry is a family of translated length-L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i,j)$ is smaller than L. From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i<j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i<j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0. For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j)$. Summing over unordered pairs $\{i,j\}$:

$$\sum_{\{i,j\},\text{overlap}} (h_i h_j + h_j h_i) \geq -(1-\gamma) \sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i,j\},\text{overlap}} c_{ij}(h_i + h_j) = \sum_i \Bigg(\sum_{\substack{j\neq i \\ \text{overlap}}} c_{ij} \Bigg) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j\neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1-\gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. □

Lemma 16.141 (Martingale condition for MPS). Not ready *Let A be injective and fix an interaction range $L \geq 2$ as in Definition 16.10. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L-site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 16.16. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i,j) < L$, where $d_N(i,j) := \min(|i-j|, N-|i-j|)$ is the cyclic distance on the ring (i.e. whose L-site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 16.140 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N). Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C] obtain this uniformity by the martingale method, invoking finite-chain-state estimates to control the constants. For the explicit cyclic-window constants displayed here, the remaining quantitative step is the overlapping-window Friedrichs-angle bound in Theorem 16.142; the preceding lemmas reduce the martingale inequality and the spectral gap to that estimate. (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)*

Proof. Not ready Two terms h_i, h_j with $0 < d_N(i,j) < L$ have supports overlapping in $L - d_N(i,j)$ sites (where d_N is the cyclic distance on the N-site ring). Since A is injective, the intersection property (Theorem 16.34) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i,j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i,j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i,j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i,j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L-1$. Let $\alpha := \max_{1 \leq s < L} \cos\theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i,j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i,j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 16.140.

Positivity of γ. The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha\,(h_i + h_j)$. The martingale inequality holds with $\gamma = 1 - 2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. Thus the remaining positivity of γ is the overlapping-window estimate stated in Theorem 16.142. This is the quantitative content of the martingale condition of Theorem 16.140: it gives a lower bound on the nonzero angles between the overlapping local ground spaces, with constants depending on A and L but independent of N. At this point it remains to compare this explicit cyclic-window bound with the principal-angle estimates cited in [FNW92, Nac96, KL18]. Either the cited estimates supply exactly this bound under the cyclic-window convention, or the displayed theorem is a stronger replacement estimate. □

Theorem 16.142 (Explicit Friedrichs-angle gap bound). Lean ✓ Stmt *For an injective tensor A and a range $L > 1$, the theorem asserts the analytic estimate with the explicit constant*

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, it asserts that if $N \geq 2L$ and $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^\perp$, then

$$\gamma_L \|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A) v\|.$$

Proof. Not ready The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 16.113, 16.114, and 16.115. Lemma 16.121 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 16.127 establishes the finite-sum algebra from ordered cross-term row bounds, and Lemmas 16.128 and 16.135 state the common finite-overlap specialization with $m = 2(L-1)$. Lemma 16.119 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 16.133 translates this to the concrete support-disjoint cyclic windows. Lemma 16.134 gives the finite-overlap row-cardinality bound. Lemma 16.136 combines these ingredients with the finite-overlap reduction, so the remaining analytic task is the Friedrichs-angle estimate for overlapping cyclic windows. Lemma 16.137 states the same overlap-only formulation, and Lemma 16.138 states a sufficient norm-compression formulation. ☐

Theorem 16.143 (Spectral gap for MPS parent Hamiltonians)**.** Lean ✓ Stmt *For an injective tensor A and a range $L > 1$, there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}^{\mathrm{ES}}_{N,L}(A))^{\perp}$ one has*

$$\gamma\|v\| \leq \|H^{\mathrm{ES}}_{N,L}(A)v\|.$$

By Lemma 16.120, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. Not ready This follows immediately from Theorem 16.142 by taking as γ the explicit constant produced there, which yields the required existential spectral-gap constant. ☐

16.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 10 [FNW92, PGVWC07].

Definition 16.144 (Assembled parent ground space)**.** Not ready For a family of blocks $A = (A_j)_{j=1}^{r}$ and weights μ, write $\mathrm{ParentGround}_{N,L}(A)$ for the periodic chain ground space of the block-diagonal tensor assembled from the blocks.

Lemma 16.145 (Definition by assembled chain ground space)**.** *Not ready By definition, this subspace is exactly the chain ground space of the assembled tensor.*

Definition 16.146 (Span of BNT block states)**.** Not ready The BNT span is the linear span of the periodic MPV states of the individual blocks:

$$\mathrm{BNTSpan}_N(A) := \mathrm{span}\{V^{(N)}(A_j) : j = 1, \dots, r\}.$$

Definition 16.147 (Route B use of the product-word span condition)**.** Lean ✓ Stmt For a fixed length m, the Route B product-word span condition is the finite-length spanning condition already stated in Remark 23.42: the simultaneous block-word evaluations

$$\omega \longmapsto (A_j^{\omega})_j$$

span the whole product algebra $\prod_j M_{D_j}(\mathbb{C})$. The remaining mathematical step is to derive this condition for some positive length from the separated canonical-form/BNT hypotheses. This is the block-injective span requirement in [CPGSV21, lines 2120–2128]: after blocking, the parent ground space is generated by the basis of normal tensors, with the improved bound coming from block-diagonal boundary conditions.

Lemma 16.148 (Each BNT MPV lies in the assembled parent ground space). Not ready *If $1 < L$ and $N \geq L+1$, then for each block A_j of a canonical-form/BNT decomposition, the corresponding MPV state lies in the assembled parent ground space.*

Proof. Not ready One first applies Lemma 16.63 to the individual block. The proof then inserts the resulting boundary matrix into the j-th diagonal block of the assembled tensor, with the compensating scalar factor coming from the nonzero weight μ_j. □

Lemma 16.149 (Periodic block lift). Not ready *For each block index j with $\mu_j \neq 0$, the chain ground space of the block A_j embeds into the chain ground space of the assembled tensor:*

$$\mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ChainGround}_{N,L}(\mathrm{Assemble}(\mu, A)).$$

Proof. Not ready The cyclic-window definition of the chain ground space reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window of length L lifts to the assembled tensor by inserting the block-diagonal boundary matrix with the compensating scalar factor μ_j^{-L}. □

Lemma 16.150 (Periodic block decomposition: forward inclusion). Not ready *Assume $\mu_j \neq 0$ for every block index j. The supremum of the blockwise periodic-chain ground spaces is contained in the assembled tensor's periodic-chain ground space:*

$$\bigsqcup_j \mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ChainGround}_{N,L}(\mathrm{Assemble}(\mu, A)),$$

where j ranges over all block indices.

Proof. Not ready Aggregate the block lift over all block indices. □

Lemma 16.151 (Forward inclusion for assembled parent ground spaces). Not ready *For a canonical-form/BNT family, the supremum of the blockwise periodic-chain ground spaces is contained in the assembled parent ground space:*

$$\bigsqcup_j \mathrm{ChainGround}_{N,L}(A_j) \subseteq \mathrm{ParentGround}_{N,L}(A).$$

Proof. Not ready The canonical-form/BNT hypothesis supplies $\mu_j \neq 0$ for every j, and the assembled parent ground space is the chain ground space of the assembled tensor by definition. □

Lemma 16.152 (Block-diagonal commutant reduction from span extension). Not ready *Suppose the virtual sector projections P_j of a dependent direct-sum bond space lie in the span of a family of matrices. Any boundary matrix commuting with that family commutes with each P_j, hence is block diagonal. Applied after pulling an assembled tensor back along the direct-sum reindexing, length-m word commutation plus the corresponding projection-span hypothesis forces all off-block entries of the pulled-back boundary matrix to vanish.*

Proof. Not ready Commutation is linear in the matrix from the spanning family, so it extends from the generators to their span. Since each P_j lies in that span, the boundary matrix commutes with every sector projection. Lemma 2.38 then gives block diagonality, and Lemma 2.37 gives the entrywise off-block-zero formulation. □

Lemma 16.153 (Block projections from product-algebra span). Not ready *Let $T_a = (T_a(i))_i$ be a family of tuples spanning the full product algebra $\prod_i M_{n_i}(\mathbb{C})$, and let $c_i \neq 0$ be nonzero scalars. Then, for every sector k, the sector projection P_k lies in the span of the scaled dependent block-diagonal matrices*

$$\bigoplus_i c_i T_a(i).$$

Proof. Not ready Apply the linear map $(M_i)_i \mapsto \bigoplus_i c_i M_i$ to the full-span product-algebra element with $(c_k)^{-1} I$ in the k-th component and zero in every other component. Its image is exactly P_k. ☐

Lemma 16.154 (Projection span from product-word span). Not ready *If the simultaneous length-m block-word tuples*

$$\omega \longmapsto (A_j^{\omega})_j$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ and every μ_j is nonzero, then the pulled-back length-m word products of Assemble(μ, A) *span every virtual sector projection P_j. Consequently, any boundary matrix commuting with all those assembled words has block-diagonal pulled-back form, and its off-block entries vanish.*

Proof. Not ready First embed the product algebra linearly as dependent block-diagonal matrices, scaling the j-th block by μ_j^m. Since $\mu_j^m \neq 0$, the product-algebra span contains the tuple with $(\mu_j^m)^{-1} I$ in one component and zero elsewhere; its scaled diagonal image is exactly P_j. The word-evaluation formula for the assembled tensor identifies these scaled diagonal matrices with the pulled-back assembled word products, and the commutant conclusion follows from Lemma 16.152. ☐

Definition 16.155 (Pairwise block-separating word polynomials). Lean Lean ✓ Stmt A length-S partial selector for block k on a finite target set T is a tuple in the span of simultaneous length-S block-word evaluations which is I on block k and 0 on every block in T. The pairwise version asks for such a partial selector for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 16.156 (Pair trace-separation criterion). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For two blocks A and B, the length-S pair word tuple is $w \mapsto (A^w, B^w)$. The pair product-span condition says that these tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length-S tuple vanishes is $(0, 0)$. The cumulative variant replaces one fixed length by all word lengths at most S, and the all-length variant allows every finite word.

Lemma 16.157 (Three-block trace reduction from a nonzero middle matrix). Lean Lean ✓ Stmt *Let A be L-block injective, and let $\Delta_A \neq 0$. Suppose that, for all words u, v, t of length L,*

$$\operatorname{tr}(\Delta_A A_v A_t A_u) + \operatorname{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every $Z \in M_{D_A}(\mathbb{C})$ there is a matrix $W \in M_{D_B}(\mathbb{C})$ such that, for all words t of length L,

$$\operatorname{tr}(Z A_t) + \operatorname{tr}(W B_t) = 0.$$

This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [PGVWC07].

Proof. ✓ Proof Since $\Delta_A \neq 0$, Lemma 8.11 writes Z as a linear combination of matrices $A_u \Delta_A A_v$. For one such term, cyclicity of the trace rewrites the three-block relation as

$$\operatorname{tr}(A_u \Delta_A A_v A_t) + \operatorname{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices $B_u \Delta_B B_v$ gives the required W, and linearity gives the general case. □

Lemma 16.158 (Finite-chain image inclusion from a three-block relation)**.** Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Under the hypotheses of Lemma 16.157, the length-L image spaces satisfy*

$$\mathcal{G}_L^A \subseteq \mathcal{G}_L^B.$$

If the corresponding nonzero middle matrix exists on both sides, then $\mathcal{G}_L^A = \mathcal{G}_L^B$.

Proof. ✓ Proof Write a vector in $\mathcal{G}_L^A$ as $t \mapsto \operatorname{tr}(A_t Z)$. The preceding lemma supplies W with $\operatorname{tr}(ZA_t) + \operatorname{tr}(WB_t) = 0$ for every word t. Cyclicity of the trace gives

$$\operatorname{tr}(A_t Z) = \operatorname{tr}(ZA_t) = -\operatorname{tr}(WB_t) = \operatorname{tr}(B_t(-W)),$$

so the vector lies in $\mathcal{G}_L^B$. The equality statement follows by applying the same argument with the two blocks exchanged. □

Lemma 16.159 (Dimension step for two-block direct-sum image spaces)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If A and B are length-L block-injective, the map $Z \mapsto (t \mapsto \operatorname{tr}(A_t Z))$ has image dimension D_A^2, and similarly for B. Therefore an inclusion $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$, together with $D_B \leq D_A$, forces $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$. In particular, the three-block trace relation from Lemma 16.158 gives this equality whenever the source-paper size ordering is available. If the inclusion comes from the strictly larger block, $D_B < D_A$, then the same dimension step already contradicts the existence of the three-block trace relation. More generally, if an external source input rules out the simultaneous conclusion $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$, then the three-block image spaces $\mathcal{G}_{3L}^A$ and $\mathcal{G}_{3L}^B$ have zero intersection. With injectivity at the three-block length, the strict-size branch itself already gives homogeneous pair trace separation at length $3L$. One such source input is non-proportionality of the periodic MPV states at a sufficiently long chain length, equivalently distinctness of the corresponding periodic MPV lines: equality of the local image spaces gives equality of the periodic-chain constraint spaces, and injective uniqueness identifies those spaces with the two MPV lines. For same-dimensional separated BNT blocks, the non-equivalence condition supplies such non-proportional MPV states at arbitrarily large chain lengths; the direct-sum injectivity hypotheses remain explicit. Zero intersection of the two three-block image spaces also gives homogeneous pair trace separation at that length once the boundary-to-chain maps at the three-block length are injective. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common three-block trace-separation length, and hence the positive-length product-word span required by the selector construction. Since canonical-form/BNT blocks are one-site injective, one may take $L = 2$; injectivity at lengths 2 and 6 then follows by passing from a fixed injective length to positive multiples. This length-6 trace separation is equivalently a full homogeneous pair span; concatenating one-letter words propagates fullness to all later homogeneous lengths, so the direct-sum witness also gives a uniform finite pair-span period window. Combining this window with the conditional period-window homogenization theorem gives the selector-format homogeneous word span without separately assuming homogeneous identity pairs. The same conclusion applies to normal-CF-BNT data once one-site injectivity is supplied explicitly, since the normal-CF-BNT predicate carries the irreducibility and normalization hypotheses used in the direct-sum comparison.*

Proof. ✓ Proof Block injectivity makes the boundary-to-chain map injective, so $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$. Inclusion gives $D_A^2 \leq D_B^2$, while $D_B \leq D_A$ gives the reverse inequality. Hence

$D_A = D_B$, and equality of dimensions inside the inclusion gives equality of the image spaces. In the strict-size branch this contradicts $D_B < D_A$. For the conditional directness statement, a vector in the intersection of the two three-block image spaces has two boundary representations. If the first boundary matrix is zero, the vector is zero. Otherwise the difference of the two coefficient formulas is exactly the homogeneous three-block trace relation, so the dimension step gives the forbidden collapse. In the equal-size branch, equal local image spaces impose the same cyclic local constraints on a periodic chain. The injective uniqueness theorem then identifies both cyclic ground spaces with the spans of their MPV states, contradicting non-proportionality of those states. □

Lemma 16.160 (All-length pair trace separation has finite cumulative length). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If no nonzero trace functional vanishes on all finite pair words, then there is a finite S such that vanishing on every word of length at most S already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.*

Proof. ✓ Proof The trace condition is dual to the span of all finite pair words. If that span is the whole product algebra, every vector lies in some cumulative span. Since the product algebra is finite-dimensional, the increasing sequence of cumulative spans stabilizes; at the stabilization index it is already the whole product algebra. Applying the finite trace-duality criterion gives the stated finite bound. This proves the cumulative finite-dimensional step; the remaining BNT step is to convert it to a single homogeneous length. Cumulative span-top alone is not identity padding, because the length-zero word already contributes the identity. □

Lemma 16.161 (Pair algebra projections are full). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If one of the two blocks is injective, then the corresponding coordinate projection of the pair algebra generated by the simultaneous one-letter pairs is the full matrix algebra. Moreover, every finite pair word lies in this pair algebra, and the all-word linear span is the submodule generated by the pair algebra. If the pair algebra is the graph of an algebra automorphism of the matrix algebra, Skolem–Noether turns this graph case into gauge-phase equivalence, so under non-gauge-equivalence the graph branch is impossible. For two blocks of unequal bond dimensions, the graph branch is also impossible because it would identify matrix algebras with different finite dimensions over $\mathbb{C}$.*

Proof. ✓ Proof Projecting the pair generators to one coordinate gives exactly the one-block generator family. Injectivity says that the linear span of this family is the full matrix algebra, and the linear span is contained in the unital algebra it generates. The pair-word membership follows by induction on the word, using closure of the generated algebra under multiplication. Conversely, the all-word linear span is closed under componentwise multiplication, so it contains the algebra generated by the pair generators. In the graph case, each generator satisfies $B_i = \varphi(A_i)$; applying Skolem–Noether to φ gives an ordinary gauge, hence a gauge-phase equivalence with scalar 1. □

Lemma 16.162 (Conditional padding from exact homogeneous identity witnesses). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Homogeneous identity-padding witnesses multiply by concatenation of words. Therefore, if one has a positive period and a full residue window of homogeneous pair spans, then the simultaneous identity pair lies in all sufficiently large homogeneous pair spans. Combined with all-length pair trace separation, this period-window input gives one homogeneous pair trace-separation length. This does not prove the direct-sum input* *of [PGVWC07] by itself: fullness of spans up to a bounded length is weaker than the exact homogeneous input used here.*

Proof. ✓ Proof The length-zero word gives the identity only at length zero, so it is not a positive-length padding theorem. The useful padding facts are homogeneous: if the identity pair is in the span at lengths m and n, componentwise multiplication puts it in the span at length $m+n$. A period witness and a complete residue window therefore give homogeneous padding at every sufficiently large length, which is the input needed by the all-length-to-fixed-length implication. □

Lemma 16.163 (Pair trace separation gives pairwise selectors)**.** Lean Lean Lean Lean Lean Lean ✓ Stmt *At a fixed length S, the pair trace-separation criterion is dual to full pair product-span. Therefore a pair trace-separation witness supplies a word polynomial that is identity on the first block and zero on the second. A common witness for all ordered distinct pairs gives pairwise block-separating words at the same length.*

Proof. ✓ Proof If the pair span were proper, a nonzero linear functional would vanish on it. Nondegeneracy of the matrix trace pairing represents that functional as (Δ_A, Δ_B), contradicting pair trace separation. Once the pair span is the whole product algebra, the tuple $(I, 0)$ is in the span; using the same coefficients on the ambient block family gives the required partial selector. □

Lemma 16.164 (Block selectors from pairwise block separation)**.** Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If every ordered pair of distinct blocks admits a common-length pairwise separator, then full block-selector words exist. More precisely, multiplying the pairwise separators for a fixed block k against all other blocks gives a selector of length $(r-1)S$ for k. Hence common block injectivity plus pairwise separators implies the finite product-word span in the block-injectivity proposition in [CPGSV16, lines 340–345] and the block-injective canonical-form trace-separation condition.*

Proof. ✓ Proof The empty word gives the identity tuple, selecting on the empty target set. The span of word tuples is closed under pointwise multiplication because the product of a length-L word and a length-S word is the concatenated length-$(L+S)$ word. Inducting over the finite set of blocks different from k and multiplying the corresponding pairwise separators keeps the value I on k and introduces one additional zero block at each step. Extracting coefficients from the final tuple-span membership gives the coefficient form of block-selector words. □

Lemma 16.165 (Product-word span from block-injective selector words)**.** Not ready *Let (A_j) be a finite family of block-injective tensors. Suppose there are length-S block-selector words: for each sector k, a linear combination of length-S words evaluates to I on A_k and to 0 on every other block. Then the simultaneous length-$(1+S)$ block-word evaluations span $\prod_j M_{D_j}(\mathbb{C})$, with positive length. In particular, a canonical form with BNT separation supplies the required block injectivity, but the selector-word conclusion does not require the full canonical-form/BNT data. It is enough, by Lemma 16.164, to construct pairwise separators for every ordered pair of distinct blocks. Lemma 16.163, a common pair trace-separation length is one sufficient finite-dimensional witness. This witness remains homogeneous: either it is supplied directly as fixed-length pair separation, or it is obtained from an explicit period-window hypothesis as in Lemma 16.162. If the assembly weights are all nonzero, then the pulled-back words at the resulting length for* $\mathrm{Assemble}(\mu, A)$ *span every virtual sector projection, and the corresponding commutant reduction is available.*

Proof. Not ready Block injectivity writes each target sector matrix as a fixed-length linear combination of word evaluations. For a canonical-form/BNT family, Definition 10.1 gives one-site injectivity of each block, hence 1-block injectivity by Lemma 2.30. For a target tuple $(M_j)_j$, write M_k as a one-letter linear combination in block k, multiply by the selector for k, and sum over k. The selector kills all off-target blocks, while it is the identity on block k, so the length-$(1+S)$ word tuples span the full product algebra. If one starts with pairwise separators instead, first use Lemma 16.164 to multiply them into full selectors; if one starts from the pair trace-separation criterion, first use Lemma 16.163. The projection-span and commutant consequences then follow from Lemma 16.154. □

Lemma 16.166 (Conditional boundary-matrix block split from projection span)**.** Not ready *Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L+1$, and $0 < m$. Assume every virtual sector projection lies in the pulled-back span of the length-m word products of the assembled tensor. If a boundary matrix for the assembled tensor commutes with all those length-m words, then the vector obtained by applying the assembled ground-space map to that boundary matrix lies in $\bigsqcup_j \mathcal{G}_{N,L}(A_j)$. Consequently, if every assembled periodic-chain ground vector admits such a commuting boundary matrix representation (the result of the wrapping-window comparison), then*

$$\mathcal{G}_{N,L}(\mathrm{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j),$$

and together with the already-proved forward inclusion this gives the corresponding conditional equality. Composing the reverse inclusion with blockwise injective uniqueness gives the conditional containment of the assembled parent ground space in the BNT span; this lemma keeps the projection-span and boundary-representation hypotheses explicit.

Proof. Not ready The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Expanding the assembled ground-space map on a block-diagonal boundary matrix gives a finite sum of scalar multiples of the block MPV states, and each block MPV lies in its own periodic chain ground space. The reverse-inclusion, equality, and BNT-span forms are then reformulations of this boundary-matrix calculation, combined with the forward inclusion and the blockwise uniqueness theorem. □

Lemma 16.167 (Conditional Route B block split from product-word span)**.** Not ready *Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L+1$, and $0 < m$. If the length-m product-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ and every assembled periodic-chain ground vector comes with a boundary matrix representation commuting with all length-m assembled words, then*

$$\mathcal{G}_{N,L}(\mathrm{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\mathrm{BNTSpan}_N(A)$*. The product-word span and boundary representation remain hypotheses here; they are not derived by this composition lemma.*

Proof. Not ready The product-word span witness gives the sector-projection span by Lemma 16.154; the nonzero weights required there are supplied by the canonical-form/BNT hypothesis. The conditional boundary-matrix block split from Lemma 16.166 then gives the reverse inclusion. The equality and BNT-span forms are the corresponding projection-span results after this product-word-to-projection-span reduction, followed by the blockwise uniqueness argument already built into the projection-span theorem. □

Lemma 16.168 (Conditional Route B block split from BNT selector words)**.** Not ready *Let A be in canonical-form/BNT, assume $1 < L$ and $N \geq L+1$, and suppose length-S block-selector words are given. If every assembled periodic-chain ground vector has a boundary matrix representation commuting with all length-$(1+S)$ assembled words, then*

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional block-splitting equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\text{BNTSpan}_N(A)$*. The selector-word hypothesis and the boundary representation remain explicit: this lemma does not derive selector existence from BNT separation, nor the wrapping/open-chain boundary representation. It is the formal composition matching the block-injective ground-space route of [CPGSV21, lines 2120–2128], where the growing-back argument is restricted to block-diagonal boundary conditions.*

Proof. Not ready Lemma 16.165 turns the canonical-form/BNT hypotheses plus length-S selector words into the product-word span condition at the positive length $1+S$. Lemma 16.167 then applies Route B with the boundary matrix commuting with the same length-$(1+S)$ assembled words. The equality and BNT-span containment are exactly the corresponding product-word-span consequences after this selector-to-product-span replacement. □

Theorem 16.169 (Containment in the BNT span by block decomposition)**.** Not ready *If A is in canonical-form/BNT and $1 < L$ and $N \geq L+1$, every vector in the assembled parent ground space should decompose into blockwise chain ground states, and therefore lie in the BNT span.*

Proof. Not ready The remaining structural step is the periodic block decomposition of the assembled chain ground space. Lemma 16.164 reduces finite selector words to pairwise block-separating word polynomials. Lemma 16.168 then gives the conditional Route B composition once those selector words and the wrapping-window commuting boundary matrix are available. What remains open is to derive the needed separation data and to supply the compatible boundary representation. With the block decomposition available, one applies the blockwise injective uniqueness theorem to each block separately and recovers the BNT span. □

Theorem 16.170 (Ground space equals span of BNT)**.** Not ready *If $1 < L$ and $N \geq L+1$, then for a canonical-form/BNT family the assembled parent ground space equals the BNT span:*

$$\text{ParentGround}_{N,L}(A) = \text{BNTSpan}_N(A).$$

Proof. Not ready We prove the two inclusions separately. The inclusion

$$\text{ParentGround}_{N,L}(A) \subseteq \text{BNTSpan}_N(A)$$

is exactly Theorem 16.169. For the reverse inclusion, Lemma 16.148 shows that each BNT MPV lies in $\text{ParentGround}_{N,L}(A)$. Since the parent ground space is a linear subspace, it therefore contains every linear combination of such vectors, and hence contains $\text{BNTSpan}_N(A)$. Thus

$$\text{BNTSpan}_N(A) \subseteq \text{ParentGround}_{N,L}(A),$$

and the two spaces are equal. □

Remark. A separate kernel-identification statement for these assembled block families would convert the chain-ground-space statement into the corresponding statement for $\ker(H_N)$.

Chapter 17

Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 16), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict transfer-operator gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

17.1 Connected correlators from the transfer map

Definition 17.1 (One-point expectation). Lean ✓ Stmt Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R. Under the hypotheses of Theorem 6.11, that fixed point exists and is unique up to scaling; we normalize it so that $\mathrm{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \mathrm{tr}(X\rho^R).$$

Definition 17.2 (Two-point expectation at distance n). Lean ✓ Stmt For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \mathrm{tr}\big(Y\, \mathcal{E}_A^n(X\rho^R)\big)\,.$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 17.3 (Connected correlator). Lean ✓ Stmt The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle\langle Y_0 \rangle.$$

17.2 Spectral expansion and exponential decay

Theorem 17.4 (Sum of exponentials (spectral expansion interface)). Lean ✓ Stmt *If coefficients $c_j \in \mathbb{C}$ and eigenvalues $\lambda_j \in \mathbb{C}$ ($j = 1, \dots, D^2 - 1$) satisfy, for all $n \geq 0$,*

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \, \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [CPGSV21, Section II.B.3] states that the connected correlator is a sum of D^2-1 pure exponentials $C(X, Y; n) = \sum_{j\geq 2}^{D^2} c_{XY}(j)\lambda_j^n$, which always exists for a normal MPS via the transfer-map spectral decomposition.

Proof. ✓ Proof Given the expansion hypothesis, the equality is immediate. The source [CPGSV21, Section II.B.3] derives this expansion from the spectral decomposition of $\mathcal{E}_A^n$ restricted to the complement of the fixed-point eigenspace: the leading eigenvalue $\lambda_1 = 1$ contributes $\langle X_0 \rangle \langle Y_0 \rangle$, which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues. □

Theorem 17.5 (Exponential decay bound (interface)). Lean ✓ Stmt *If a constant $C_{XY} \in \mathbb{R}$ and $\lambda_2 \in \mathbb{C}$ satisfy, for all $n \geq 0$,*

$$|C(X, Y; n)| \leq C_{XY} \, |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [CPGSV21, Section II.B.3] obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and Sec. 4 provides the transfer-map gap condition $|\lambda_j| < 1$ for the subleading eigenvalues.

Proof. ✓ Proof Given the bound hypothesis, the estimate is immediate. The source [CPGSV21, Section II.B.3] obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the transfer-map gap condition $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues. □

Remark 17.6 (Transfer-map spectral hypothesis). The hypothesis "$|\lambda_2| < 1$" in Theorem 17.5 is a transfer-map gap condition: all non-leading eigenvalues of $\mathcal{E}_A$ lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of Chapter 16. The complementary transfer-map gap $\rho(\mathcal{E}_A - P) < 1$ is established for primitive channels (Theorem 4.51, Chapter 4). The mixed-transfer gap needed for block separation is supplied for irreducible trace-preserving tensors (Theorem 7.30), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in Theorem 17.11. Thus the analytical input needed to make Theorems 17.4 and 17.5 unconditional is already present; what remains is extracting the spectral expansion coefficients c_j, λ_j from the eigendecomposition of $\mathcal{E}_A$ (Issue #1447).

Definition 17.7 (Correlation length). Lean ✓ Stmt The correlation length associated with a subleading eigenvalue λ_2 of the transfer map ($|\lambda_2| < 1$) is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

Lemma 17.8 (Correlation length is positive). Lean ✓ Stmt *If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.*

Proof. ✓ Proof Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. □

17.3 Quantitative transfer-map gap bounds

The transfer-map gap theorems in Chapter 4 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 17.9 (Exponential convergence of injective primitive channels). Lean ✓ Stmt *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C\,(1-\delta)^n\,\|X\|.$$

The transfer-map gap δ exists by primitivity (Theorem 4.51, the complementary transfer-map gap).

Proof. ✓ Proof Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive transfer-map gap theorem for the complementary map. The Gelfand formula yields a geometric bound $\|\,(\mathcal{E}_A - P)^n\| \leq C\,(1-\delta)^n$ for suitable constants, giving the claimed estimate. □

Theorem 17.10 (Correlation length bound). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\mathrm{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \leq C\,e^{-n/\xi}\|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log\rho(\mathcal{E}_A - P)$.

Proof. ✓ Proof For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C\,(1-\delta)^n\,\|X\|$. Rewriting $(1-\delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1-\delta)$ yields the claimed estimate. □

Theorem 17.11 (Transfer-map gap from injectivity). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. ✓ Proof Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. □

17.4 Relation to parent Hamiltonians

Remark 17.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 16.17) whose ground state is the matrix product vector (Lemma 16.21), the exponential decay bound (Theorem 17.5) shows that ground-state correlations decay with a finite correlation length ξ.

17.5 Renormalization fixed points and zero correlation length

Definition 17.13 (Renormalization fixed point). Lean ✓ Stmt An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV16, Definition 3.2].

Theorem 17.14 (Kraus-isometry criterion for RFP). Lean ✓ Stmt *Suppose there exists an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1,i_2),j}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1,i_2),j} A^j$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV16, Theorem 3.1].

Proof. ✓ Proof Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2). Substituting the decomposition above and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. □

Definition 17.15 (Local orthogonality). Lean ✓ Stmt A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 17.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV16, Definition 3.5].

Theorem 17.16 (RFP normal tensor is injective). Lean ✓ Stmt *If A is a nonzero normal RFP tensor, then A is injective.*

Proof. ✓ Proof The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\operatorname{span}\{A^w : |w| \le n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. □

Theorem 17.17 (Left-canonical normal RFP tensor is injective). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV16, Appendix B].

Proof. ✓ Proof The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 17.16 applies. □

Theorem 17.18 (Unitary diagonal fixed point for a left-canonical normal RFP). Lean ✓ Stmt *Assume $D > 0$. Let A be a normal renormalization fixed point such that*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \text{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV16, Appendix B].

Proof. ✓ Proof By Theorem 17.17, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 9.20 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. □

Theorem 17.19 (Rank-one classification for RFP transfer maps). Lean ✓ Stmt *Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\rho$ is the rank-one fixed-point projection.*

Proof. ✓ Proof The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1-\delta)^n\|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \geq 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1-\delta)^{n+1}\|X\|$ for all n. Taking $n \to \infty$ gives $\mathcal{E}_A = P_\rho$. □

Theorem 17.20 (Full structural form for RFP tensors). Lean ✓ Stmt *A normal tensor A in canonical form II that is a renormalization fixed point admits a decomposition $A^i = X\Lambda U^i X^{-1}$, where Λ is diagonal positive and the family $U = (U^i)$ is an isometry in the physical indices.*

Proof. ✓ Proof Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry U, and transports the decomposition back through the conjugation. □

Theorem 17.21 (Canonical-form blocks are injective). Lean ✓ Stmt *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

Proof. ✓ Proof Immediate from the `block_injective` field of `IsCanonicalForm`. □

Theorem 17.22 (RG flow convergence for canonical-form tensors). Lean ✓ Stmt *For a tensor in canonical form with blocks (A_k) and weights (μ_k), the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \to \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \to \mathcal{E}_\infty(\rho)$ for all density matrices ρ. This is [CPGSV16, Appendix B].*

Proof. ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0. The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1-\delta)^n\|X\|$ (Theorem 17.9) then gives pointwise convergence $\mathcal{E}^n(X) \to P(X)$, and composing with the subsequence $2^n \to \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. □

Remark 17.23. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 17.24 (Correlations independent of distance). Lean ✓ Stmt For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 17.25 (Zero correlation length). Lean ✓ Stmt An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [CPGSV16, Definition 3.6].

Theorem 17.26 (CID implies RFP). Lean ✓ Stmt *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV16, Theorem 3.8].*

Proof. ✓ Proof Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. □

Theorem 17.27 (Zero correlation length iff idempotent transfer). Lean ✓ Stmt *An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV16, Theorem 3.8].*

Proof. ✓ Proof The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. □

Theorem 17.28 (Canonical-form blocks satisfy RFP iff ZCL). Lean ✓ Stmt *For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP–ZCL part of [CPGSV16, Theorem 3.10].*

Proof. ✓ Proof Apply Theorem 17.27 to the chosen canonical-form block. □

Chapter 18

Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension d, bond dimension D, and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

18.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 18.1 (GHZ tensor). Lean ✓ Stmt Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \mathrm{diag}(1, 0), \qquad A^1 = |1\rangle\langle 1| = \mathrm{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\mathrm{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 18.2 (GHZ transfer map is idempotent). Lean ✓ Stmt *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. ✓ Proof The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is idempotent. □

Theorem 18.3 (GHZ tensor is an RFP). Lean ✓ Stmt *The GHZ tensor is a renormalization fixed point.*

Proof. ✓ Proof Immediate from Theorem 18.2. □

Theorem 18.4 (GHZ tensor is not injective). Lean ✓ Stmt *The GHZ tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ *is the two-dimensional space of diagonal matrices in $M_2(\mathbb{C})$, which does not equal $M_2(\mathbb{C})$.*

Proof. ✓ Proof Every element of $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. □

Theorem 18.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). Lean ✓ Stmt *Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli-X matrix σ_x. The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\widetilde{A}_g$ defines the same MPV family as A, i.e. $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$.*

Proof. ✓ Proof The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\widetilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. □

Definition 18.6 (Virtual $\mathbb{Z}_2$ representation of the GHZ tensor). Lean ✓ Stmt The virtual $\mathbb{Z}_2$ representation on the bond space is the homomorphism $\mathbb{Z}_2 \to M_2(\mathbb{C})$ sending the identity to I and the generator to $\sigma_z = \operatorname{diag}(1, -1)$.

Theorem 18.7 (Virtual $\mathbb{Z}_2$ symmetry of the GHZ tensor). Lean ✓ Stmt *The matrix σ_z commutes with every GHZ matrix A^i. A non-scalar matrix commuting with all A^i is the hallmark of a non-injective ($\mathbb{Z}_2$-injective) tensor: the on-site symmetry of Theorem 18.5 swaps the two one-dimensional injective blocks $A^0 \leftrightarrow A^1$, while σ_z acts on each block by a phase.*

Proof. ✓ Proof Both A^0 and A^1 are diagonal, so each commutes with the diagonal matrix σ_z. □

Theorem 18.8 (GHZ order parameter is a transfer-map fixed point). Lean ✓ Stmt *The order-parameter operator σ_z is a fixed point of the GHZ transfer map, $\mathcal{E}_A(\sigma_z) = \sigma_z$.*

Proof. ✓ Proof Since $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ and both A^i are diagonal, the claim is the entrywise computation $\mathcal{E}_A(\sigma_z) = \sigma_z$. □

Remark 18.9. A primitive (injective) tensor has a one-dimensional transfer-map fixed-point space, spanned by a positive matrix that equals the identity only in a normalized gauge. That the GHZ transfer map fixes the additional non-scalar operator σ_z is the transfer-matrix signature of the extra non-decaying mode behind its long-range ferromagnetic correlations.

18.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 18.10 (AKLT tensor). Lean ✓ Stmt Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \qquad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \qquad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}}\sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$ for $\sigma_+ = \sqrt{2}|0\rangle\langle 1|$ and $\sigma_- = \sqrt{2}|1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin-$\frac{1}{2}$ to spin-$\frac{1}{2}$.

Theorem 18.11 (AKLT tensor is not injective). Lean ✓ Stmt *The AKLT tensor is not injective at a single site:* $\operatorname{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ *is the three-dimensional space of traceless matrices* $\mathfrak{sl}(2, \mathbb{C})$*, which is a proper subspace of* $M_2(\mathbb{C})$*.*

Theorem 18.12 (AKLT tensor is normal). Lean ✓ Stmt *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.*

Theorem 18.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the AKLT tensor). Lean ✓ Stmt *The AKLT tensor is on-site symmetric under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ subgroup of* $\mathrm{SO}(3)$ *generated by two commuting π-rotations about orthogonal axes, acting on the spin-1 physical index.*

Theorem 18.14 (Anticommuting virtual gauges of the AKLT symmetry). Lean ✓ Stmt *The two virtual matrices that implement the symmetry of Theorem 18.13, namely $i\sigma_y$ and σ_z, anticommute.*

Definition 18.15 (AKLT factor system). Lean ✓ Stmt The AKLT factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g,h) = (-1)^{(g_1+g_2)h_1}$ (the value -1 when $g_1 + g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges $i\sigma_y$ and σ_z; the diagonal term $g_1 h_1$ relative to the cluster case reflects $(i\sigma_y)^2 = -I$.

Lemma 18.16 (AKLT commutator phase). Lean ✓ Stmt *The commutator phase of the AKLT factor system on the two generators is -1.*

Theorem 18.17 (The AKLT factor system is a 2-cocycle). Lean ✓ Stmt *The AKLT factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1))$.*

Theorem 18.18 (The AKLT $\mathbb{Z}_2 \times \mathbb{Z}_2$ phase is non-trivial). Lean ✓ Stmt *The AKLT factor system (Theorem 18.17), built from the anticommuting gauges $i\sigma_y$ and σ_z (Theorem 18.14), is not a coboundary: its commutator phase on the two generators is -1. Its class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$ is therefore non-trivial, so the AKLT chain realizes a non-trivial SPT phase rather than a trivial product state.*

Remark 18.19. This $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the smallest subgroup of $\mathrm{SO}(3)$ that still carries a non-trivial 2-cocycle. That minimality, and the full continuous classification over $\mathrm{SO}(3)$, are not formalized (Remark 18.23).

Definition 18.20 (Cartesian AKLT tensor). Lean ✓ Stmt The Cartesian presentation of the AKLT tensor has $d = 3$, $D = 2$, and $A^x = \sigma_x$, $A^y = \sigma_y$, $A^z = \sigma_z$, the three Pauli matrices. This is the rotationally natural form used in [CPGSV21] (around line 1159); it is the AKLT state in the Cartesian basis of the spin-1 site, the counterpart of the $|m\rangle$-basis form of Definition 18.10.

Theorem 18.21 (The spin-$\frac{1}{2}$ cover is surjective onto $\mathrm{SO}(3)$). Lean ✓ Stmt *Every rotation $R \in \mathrm{SO}(3)$ is realized by conjugating the Pauli vector by some $U \in \mathrm{SU}(2)$: there is $U \in \mathrm{SU}(2)$ with $\left(\frac{1}{2}\operatorname{tr}(\sigma_i U \sigma_j U^{-1})\right)_{ij} = R$. Equivalently, the adjoint double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is surjective. The proof factors $R = R_z(\alpha)\, R_x(\beta)\, R_z(\gamma)$ by its ZXZ Euler decomposition and maps each coordinate rotation to an explicit $\mathrm{SU}(2)$ generator: $\mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto R_z(\theta)$ and $\left(\begin{smallmatrix} \cos\frac{\beta}{2} & -i\sin\frac{\beta}{2} \\ -i\sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{smallmatrix}\right) \mapsto R_x(\beta)$.*

Theorem 18.22 ($\mathrm{SO}(3)$ on-site symmetry of the AKLT tensor). Lean ✓ Stmt *The Cartesian AKLT tensor is on-site symmetric under the spin-1 (defining) representation of $\mathrm{SO}(3)$ on the physical index.*

Remark 18.23. The bond index carries the projective spin-$\frac{1}{2}$ representation: the gauge for a rotation is either of its two $\mathrm{SU}(2)$ preimages, so the assignment is a representation only up to sign and factors through the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$. Its class is the non-trivial element of $H^2(\mathrm{SO}(3), \mathrm{U}(1)) \cong \mathbb{Z}_2$, witnessing the SPT phase.

18.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 18.24 (Cluster-state tensor). Lean ✓ Stmt Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. The review [CPGSV21] writes the reflected convention $A^0 = |0\rangle\langle +|$, $A^1 = |1\rangle\langle -|$; the transpose taken here is the same state read in the opposite direction and carries the same SPT order.

Theorem 18.25 (Cluster-state tensor is not injective). Lean ✓ Stmt *The cluster-state tensor is not injective:* $\mathrm{span}_{\mathbb{C}}\{A^0, A^1\} = \mathrm{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ *has dimension* 2, *not* $\dim M_2(\mathbb{C}) = 4$.

Proof. ✓ Proof Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. □

Theorem 18.26 (Cluster-state tensor is normal). Lean ✓ Stmt *Although the cluster-state tensor is not injective at a single site, its length-2 blocking is injective: the four length-2 products span $M_2(\mathbb{C})$. In particular the cluster-state tensor is normal.*

Remark 18.27. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \tfrac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \tfrac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\tfrac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{\langle 0|, \langle 1|\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 18.28 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). Lean ✓ Stmt *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$.*

Theorem 18.29 (Anticommuting virtual gauges of the cluster-state symmetry). Lean ✓ Stmt *The virtual matrices implementing the two generators of Theorem 18.28, namely $V_1 = \sigma_z$ and $V_2 = \sigma_x$, anticommute ($V_1 V_2 = -V_2 V_1$).*

Definition 18.30 (Cluster-state factor system). Lean ✓ Stmt The cluster-state factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g, h) = (-1)^{g_2 h_1}$ (the value -1 when $g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges σ_z and σ_x, both squaring to the identity, so there is no diagonal term.

Lemma 18.31 (Cluster-state commutator phase). Lean ✓ Stmt *The commutator phase of the cluster-state factor system on the two generators is -1.*

Theorem 18.32 (The cluster-state factor system is a 2-cocycle). Lean ✓ Stmt *The cluster-state factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1))$.*

Theorem 18.33 (The cluster-state $\mathbb{Z}_2 \times \mathbb{Z}_2$ phase is non-trivial)**.** Lean ✓ Stmt *The cluster-state factor system (Theorem 18.32), built from the anticommuting gauges σ_z and σ_x (Theorem 18.29), is not a coboundary: its commutator phase on the two generators is -1. Its class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \cong \mathbb{Z}_2$ is therefore non-trivial, so the cluster state realizes a non-trivial SPT phase.*

Theorem 18.34 (The cluster state has string order under its $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry)**.** Lean ✓ Stmt *For every element g of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the length-2 blocked cluster-state tensor has string order in the sense of Definition 12.38 for the twist by g, with the maximally mixed boundary state $\Lambda = \frac{1}{2}\mathbb{1}$ as its stationary boundary state.*

Proof. ✓ Proof The blocked tensor is injective (Remark 18.27) and on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Theorem 18.28), and each group element acts by a real symmetric involutive permutation matrix, hence unitarily. The maximally mixed state $\Lambda = \frac{1}{2}\mathbb{1}$ is positive definite and has trace 1. The four blocked matrices satisfy $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the transfer map is unital and Λ is a fixed point of the adjoint transfer map. These are exactly the hypotheses of Theorem 12.61, which then yields string order for every g. The cluster state is thus the first explicit witness of that criterion. □

Chapter 19

Channel Representations and Normal Forms

This chapter collects the representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: the Choi–Jamiołkowski isomorphism, Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

19.1 Representations

The following definitions support the Choi–Jamiolkowski isomorphism (Theorem 19.6) and the representation theorems of [Wol12, Chapter 2].

Definition 19.1 (Partial traces). Lean Lean ✓ Stmt Let $X \in M_{d \cdot d'}(\mathbb{C})$ be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace* $\mathrm{tr}_A(X)$ and *right partial trace* $\mathrm{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\mathrm{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \qquad (\mathrm{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 19.2 (Maximally entangled state). Lean ✓ Stmt The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d}\delta_{i_1 j_1}\delta_{i_2 j_2}$ and $\mathrm{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 19.3 (Tensor product of a map with identity). Lean ✓ Stmt Given a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *tensor extension* $T \otimes \mathrm{id}$ acts on bipartite matrices $X \in M_{D\times D}(\mathbb{C})$ by applying T to each "slice":

$$(T \otimes \mathrm{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X^{(i_2,j_2)}_{ab} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

19.2 Choi–Jamiołkowski isomorphism

Definition 19.4 (Choi matrix). Lean ✓ Stmt The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is

$$\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \in M_{D\times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1,i_2),(j_1,j_2)} = \frac{1}{D}\,(T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiolkowski convention of [Wol12, Proposition 2.1].

Theorem 19.5 (Choi matrix of the identity map). Lean ✓ Stmt *The identity channel has Choi matrix* $|\Omega\rangle\langle\Omega|$.

Proof. ✓ Proof Apply the identity map in the definition of the Choi matrix. □

Theorem 19.6 (Complete positivity iff the Choi matrix is positive semidefinite). Lean ✓ Stmt *A linear map* $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ *is completely positive (in the Kraus sense) if and only if its Choi matrix* $\tau \geq 0$.

Proof. ✓ Proof ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c\,K_i(a,b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a,b) = v_m(a,b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. □

Theorem 19.7 (Trace preservation implies the partial-trace condition). Lean ✓ Stmt *If* T *is trace-preserving, then* $\mathrm{tr}_A(\tau) = \frac{1}{D}\mathbb{1}_D$.

Proof. ✓ Proof $(\mathrm{tr}_A(\tau))_{ij} = \mathrm{tr}(T(\Omega^{(ij)})) = \mathrm{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D}E_{ij}$ is the (i,j)-slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D}\delta_{ij}$. □

Theorem 19.8 (Hermiticity preservation and the Choi matrix). Lean ✓ Stmt *The Choi matrix* τ *is Hermitian if and only if* $T(B^\dagger) = T(B)^\dagger$ *for all* $B \in M_D(\mathbb{C})$.

Proof. ✓ Proof The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D}(T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B. □

Theorem 19.9 (Trace of the Choi matrix of a trace-preserving map). Lean ✓ Stmt *If* T *is trace-preserving, then* $\mathrm{tr}(\tau) = 1$.

Proof. ✓ Proof $\mathrm{tr}(\tau) = \mathrm{tr}(\mathrm{tr}_A(\tau)) = \mathrm{tr}(\frac{1}{D}\mathbb{1}_D) = 1$. □

19.3 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

Theorem 19.10 (Polarization of sesquilinear sandwiches). Lean ✓ Stmt *For any* $A, B, X \in M_D(\mathbb{C})$,

$$4\,AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i\,(A+iB)X(A+iB)^\dagger - i\,(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof Working entrywise, reduces to the scalar polarization identity $4\,\alpha\,\overline{\delta} = \sum_{k=0}^{3} i^k(\alpha + i^k\beta)\overline{(\gamma + i^k\delta)}$ in $\mathbb{C}$, which holds after substituting $i^2 = -1$. □

Theorem 19.11 (CP decomposition of sandwich sums). Lean ✓ Stmt *For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4\,T = T_1 - T_2 + i\,T_3 - i\,T_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families.*

Proof. ✓ Proof Apply the polarization identity summand-by-summand. □

Theorem 19.12 (No information without disturbance). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \mathrm{id}$. In particular, a quantum channel preserving every pure state equals the identity.*

Proof. ✓ Proof Rank-one outer products $|u\rangle\langle v|$ polarize into rank-one self-outer-products:

$$4\,|u\rangle\langle v| = \sum_{k=0}^{3} i^k |u + i^k v\rangle\langle u + i^k v|,$$

so T preserves every $|u\rangle\langle v|$. Since the matrix units $E_{ij} = |e_i\rangle\langle e_j|$ span $M_D(\mathbb{C})$, T equals the identity on a spanning set, hence everywhere. □

Definition 19.13 (Pure-state ensemble density). Lean ✓ Stmt Given a finite family $\{\psi_i\}_{i\in\iota}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i}\psi_i$.

Theorem 19.14 (Isometric mixing preserves the density). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ are related by an isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 19.15.*

Proof. ✓ Proof Expanding and applying the orthogonality relation $\sum_i V_{ij}\overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$:

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij}\overline{V_{ij'}}\right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

□

Theorem 19.15 (Hughston–Jozsa–Wootters converse). Lean ✓ Stmt *If two pure-state ensembles $\{\psi_i\}_{i\in\iota_1}$ and $\{\phi_j\}_{j\in\iota_2}$ induce the same pure-ensemble density operator and $|\iota_2| \leq |\iota_1|$, then there exists a tall isometric mixing matrix $V \in \mathbb{C}^{\iota_1\times\iota_2}$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij}\phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|\iota_1| \leq |\iota_2|$ follows by swapping the roles of the ensembles.*

Proof. ✓ Proof Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these K_i, $L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00}\cdot\rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. Theorem 19.26 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij}\,L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij}\phi_j$. □

Theorem 19.16 (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt *Under the cardinality hypothesis $|\iota_2| \leq |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.*

Proof. ✓ Proof Combine the two directions above. □

19.4 Kraus representation theorem

Theorem 19.17 (Trace preservation implies Kraus normalization). Lean ✓ Stmt *If* $T(X) = \sum_i K_i X K_i^\dagger$ *is trace-preserving, then* $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Proof. ✓ Proof For any N, $\mathrm{tr}((\sum_i K_i^\dagger K_i)\, N) = \sum_i \mathrm{tr}(K_i^\dagger K_i N) = \sum_i \mathrm{tr}(K_i N K_i^\dagger) = \mathrm{tr}(T(N)) = \mathrm{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. □

Theorem 19.18 (Kraus normalization implies trace preservation). Lean ✓ Stmt *If* $\sum_i K_i^\dagger K_i = \mathbb{1}$*, then* $T(X) = \sum_i K_i X K_i^\dagger$ *is trace-preserving.*

Proof. ✓ Proof $\mathrm{tr}(T(X)) = \sum_i \mathrm{tr}(K_i X K_i^\dagger) = \sum_i \mathrm{tr}(K_i^\dagger K_i X) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\, X) = \mathrm{tr}(X)$. □

Theorem 19.19 (Unitality implies Kraus unital normalization). Lean ✓ Stmt *If* $T(X) = \sum_i K_i X K_i^\dagger$ *satisfies* $T(\mathbb{1}) = \mathbb{1}$*, then*

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. ✓ Proof Evaluate the Kraus formula at the identity matrix. □

Theorem 19.20 (Unitary freedom in Kraus operators). Lean ✓ Stmt *Let* U *be a unitary* $r \times r$ *matrix* ($U^\dagger U = \mathbb{1}$) *and suppose* $K_j = \sum_\ell U_{j\ell} \widetilde{K}_\ell$. *Then* $\{K_j\}$ *and* $\{\widetilde{K}_\ell\}$ *define the same Kraus map: for every* $X \in M_D(\mathbb{C})$,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

Proof. ✓ Proof Substitute the combination formula, expand the triple sum over j, ℓ, ℓ', swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. □

Theorem 19.21 (Isometric mixing preserves the Kraus map). Lean ✓ Stmt *Let* W *be an isometry between two Kraus index spaces, so* $W^\dagger W = \mathbb{1}$*. If*

$$K_j = \sum_\ell W_{j\ell} \widetilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ *and* $\{\widetilde{K}_\ell\}$ *define the same completely positive map.*

Proof. ✓ Proof Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. □

Theorem 19.22 (The transition matrix of orthonormal Kraus families is unitary). Lean ✓ Stmt *Let* $\{K_j\}_{j=0}^{r-1}$ *and* $\{K_j'\}_{j=0}^{r-1}$ *be two Hilbert–Schmidt orthonormal Kraus families, and suppose*

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell}\, K_\ell'.$$

Then the transition matrix U *is unitary:* $U^\dagger U = \mathbb{1}_r$.

Proof. ✓ Proof Expand $\mathrm{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $UU^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. □

Theorem 19.23 (Dual map equality from primal map equality). Lean ✓ Stmt *If two Kraus families satisfy $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X, then $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$ for all Y.*

Proof. ✓ Proof Use the trace pairing: for all X, $\mathrm{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \mathrm{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \mathrm{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \mathrm{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$. Nondegeneracy of the trace pairing gives the result. □

Theorem 19.24 (Equal Stinespring Gramians). Lean ✓ Stmt *If two Kraus families define the same CPM, then $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$.*

Proof. ✓ Proof Specialise Theorem 19.23 at $Y = \mathbb{1}$. □

Theorem 19.25 (Rectangular Kraus freedom). Lean ✓ Stmt *If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \le r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.*

Proof. ✓ Proof Pad the smaller family to size r_1, establish Gram matrix equality $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^\dagger V = \mathbb{1}_{r_2}$. □

Theorem 19.26 (Rectangular Kraus freedom with general finite index types). Lean ✓ Stmt *Variant of Theorem 19.25 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2.*

Proof. ✓ Proof Reindex by choosing enumerations of those finite index sets and apply Theorem 19.25. □

Theorem 19.27 (Isometric freedom of Kraus representations). Lean ✓ Stmt *Let $\{B_\alpha\}_{\alpha\in\iota_1}$ and $\{A_j\}_{j\in\iota_2}$ be finite Kraus families with $|\iota_2| \le |\iota_1|$. Then the following are equivalent:*

1. *$\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.*
2. *There exists a matrix $V = (V_{\alpha j})$ with $V^\dagger V = \mathbb{1}$ such that $B_\alpha = \sum_j V_{\alpha j} A_j$ for every α.*

Proof. ✓ Proof The implication (1) ⇒ (2) is Theorem 19.26. Conversely, expand $\sum_\alpha B_\alpha X B_\alpha^\dagger$ using $B_\alpha = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^\dagger V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. □

Theorem 19.28 (Unitary freedom of Kraus representations). Lean ✓ Stmt *Let $\{B_\alpha\}_{\alpha\in\iota}$ and $\{A_j\}_{j\in\iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:*

1. *$\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all $X \in M_D(\mathbb{C})$.*
2. *There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_\alpha = \sum_j U_{\alpha j} A_j$ for every α.*

Proof. ✓ Proof Apply Theorem 19.27 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. □

Remark 19.29 (Choi rank and minimal Kraus cardinality). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt If a completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently,

$$\mathrm{rank}(\tau_E) \le r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\mathrm{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. ✓ Proof For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r. For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. □

19.5 Stinespring dilation

Definition 19.30 (Stinespring isometry). Lean ✓ Stmt Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 19.31 (Entry formula for the Stinespring isometry). Lean ✓ Stmt *For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,*

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 19.32 (Stinespring Gram identity). Lean ✓ Stmt *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 19.33 (Stinespring dual representation). Lean ✓ Stmt *For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,*

$$V^\dagger\,(A \otimes \mathbb{1}_r)\,V = \sum_j K_j^\dagger A\,K_j = T^*(A).$$

This is the Stinespring representation of the dual *(Heisenberg picture) map T^*.*

Proof. ✓ Proof Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i, j); the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l, giving $\sum_j K_j^\dagger A K_j$. □

Theorem 19.34 (The Stinespring isometry condition iff trace preservation). Lean ✓ Stmt $V^\dagger V = \mathbb{1}_D$ *if and only if* $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. *In particular, V is an isometry precisely when the Kraus map is trace-preserving.*

Proof. ✓ Proof $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}}\, V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}}\, (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$. □

Theorem 19.35 (Stinespring Schrödinger representation). Lean ✓ Stmt *The Kraus map* $T(\rho) = \sum_j K_j \rho K_j^\dagger$ *equals the partial trace over the dilation space:*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. ✓ Proof Expand $(V\rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. □

Theorem 19.36 (Existence of a Stinespring dilation). Lean ✓ Stmt *Every completely positive map admits an ancilla dimension r, Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that*

$$E(A) = V^\dagger \pi(A) V.$$

Proof. ✓ Proof Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. □

Theorem 19.37 (CPTP maps admit isometric Stinespring dilations). Lean ✓ Stmt *Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.*

Proof. ✓ Proof Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. □

19.6 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 19.38 (CP partial order). Lean ✓ Stmt For linear maps $S, T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, write $T \le S$ (in the CP order) when $S - T$ is completely positive.

Definition 19.39 (Block-top contraction). Lean ✓ Stmt For natural numbers r, s, let $C = C_{r,s} \in \mathbb{C}^{r\times(r+s)}$ be the rectangular $r \times (r+s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 19.40 (C is a co-isometry). Lean ✓ Stmt *The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.*

Proof. ✓ Proof Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij}\overline{C_{i'j}}$ collapses to $\delta_{ii'}$. □

Lemma 19.41 (C is a contraction). Lean ✓ Stmt *The block-top matrix satisfies $C^\dagger C \le \mathbb{1}_{r+s}$.*

Proof. ✓ Proof The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s, so $\mathbb{1} - C^\dagger C$ is positive semidefinite. □

Lemma 19.42 (Intertwining of Stinespring isometries). Lean ✓ Stmt *Let $K : \{0, \dots, r-1\} \to M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \to M_D(\mathbb{C})$ be two Kraus families. Write $K \mathbin{+\!\!+} L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then*

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K +\!\!+ L}.$$

Proof. ✓ Proof Entrywise expansion of both sides: the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of $V_{K+\!\!+L}$ on those coordinates, which is precisely V_K. □

Theorem 19.43 (Ordered CP maps via Stinespring contractions). Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be CP maps with $T_1 \le T_2$. Then there exist an ancilla dimension m, Heisenberg-form Stinespring matrices V_1, V_2 realizing T_1, T_2 via $T_i(A) = V_i^\dagger(A \otimes \mathbb{1})V_i$, and a contraction $\widetilde{C}$ on the dilation space with $\widetilde{C}^\dagger\widetilde{C} \le \mathbb{1}$ such that*

$$V_1 = (\mathbb{1}_D \otimes \widetilde{C})\, V_2.$$

This is [Wol12, Theorem 2.3].

Proof. ✓ Proof Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 +\!\!+ L^\dagger$. Then $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$, and the latter equals $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$ which is the Heisenberg form for K_2, yielding $T_2(A) = V_{K_2}^\dagger(A \otimes \mathbb{1})V_{K_2}$. Taking $\widetilde{C} = C_{r_1,s}$ the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \widetilde{C})V_{K_2}$ is Lemma 19.42, and $\widetilde{C}^\dagger\widetilde{C} \le \mathbb{1}$ is Lemma 19.41. ☐

Definition 19.44 (Block-diagonal projectors on the dilation space). Lean Lean ✓ Stmt For r, s, let $P_{\mathrm{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\mathrm{bot}} = \mathbb{1} - P_{\mathrm{top}}$. Both are PSD (for P_{bot}, by Lemma 19.41) and $P_{\mathrm{top}} + P_{\mathrm{bot}} = \mathbb{1}$.

Theorem 19.45 (Radon–Nikodym theorem for CP maps). Lean ✓ Stmt *Let $T_1, T_2 : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m, a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \to \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for $i = 1, 2$ and every $A \in M_D(\mathbb{C})$,*

$$T_i(A) = V^\dagger(A \otimes P_i)V.$$

This is [Wol12, Theorem 2.4].

Proof. ✓ Proof Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2, and form $K = K_1 +\!\!+ L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\mathrm{top}}$, $P_2 = P_{\mathrm{bot}}$ as in Definition 19.44. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger(A \otimes \mathbb{1})(\mathbb{1} \otimes C)$ rewrites $V^\dagger(A \otimes P_{\mathrm{top}})V$ as $((\mathbb{1} \otimes C)V)^\dagger(A \otimes \mathbb{1})((\mathbb{1} \otimes C)V)$, which equals $V_{K_1}^\dagger(A \otimes \mathbb{1})V_{K_1} = T_1(A)$ by the intertwining Lemma 19.42. For the complementary block, $A \otimes P_{\mathrm{bot}} = A \otimes \mathbb{1} - A \otimes P_{\mathrm{top}}$ and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. ☐

Theorem 19.46 (Open-system representation of quantum channels). Lean ✓ Stmt *Every quantum channel $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that*

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V\rho V^\dagger)_{(i,k),(j,k)} = \mathrm{tr}_r(V\rho V^\dagger).$$

Proof. ✓ Proof Apply the existence of an isometric Stinespring dilation (Theorem 19.37) and the Schrödinger-picture identity (Theorem 19.35). ☐

Theorem 19.47 (Open-system representation, partial-trace form). Lean ✓ Stmt *Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \mathrm{tr}_E(V\rho V^\dagger)$.*

Proof. ✓ Proof Rewrite the componentwise identity of Theorem 19.46 as a partial trace over the ancilla factor. ☐

Remark 19.48 (Unitary form). The fully unitary open-system form $T(\rho) = \mathrm{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$ of [Wol12, Theorem 2.5] follows from the isometric form by extending the Stinespring isometry V to a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$; this uses orthonormal basis extension and is left to a later theorem.

19.7 POVMs and Naimark dilation

Definition 19.49 (Positive operator-valued measure). Lean ✓ Stmt A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D.$$

Definition 19.50 (Naimark Kraus square roots). Lean Lean ✓ Stmt For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots.*

Theorem 19.51 (Naimark square roots satisfy the Kraus normalization). *Lean* *✓ Stmt* *For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has*

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. ✓ Proof Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. □

Definition 19.52 (Naimark isometry). Lean ✓ Stmt The *Naimark isometry* $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \qquad V_{(a,i),\,k} = (M_i)_{a,k}.$$

Definition 19.53 (Naimark projective measurement). Lean Lean ✓ Stmt The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \qquad (P_i)_{(a,b),(c,d)} = \delta_{a,c}\delta_{b,i}\delta_{d,i}.$$

Theorem 19.54 (Naimark isometry condition). *Lean* *✓ Stmt* *The Naimark isometry satisfies* $V^\dagger V = \mathbb{1}_D$.

Proof. ✓ Proof The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. □

Theorem 19.55 (Naimark projectors form a projective measurement). *Lean* *Lean* *Lean* *Lean* *✓ Stmt* *The Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_iP_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.*

Proof. ✓ Proof Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. □

Theorem 19.56 (Naimark dilation). *Lean* *✓ Stmt* *Every POVM $\{E_i\}_{i=0}^{n-1}$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,*

$$E_i = V^\dagger P_i V.$$

Proof. ✓ Proof Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}}\,(M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. □

Theorem 19.57 (Existential Naimark dilation). Lean ✓ Stmt *For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r, an isometry $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.*

Proof. ✓ Proof Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. □

Definition 19.58 (Naimark dilation as a structure). Lean ✓ Stmt A *Naimark dilation* of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i.

Theorem 19.59 (Canonical dilation satisfies the Naimark dilation axioms). Lean ✓ Stmt *The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.*

Proof. ✓ Proof Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. □

Theorem 19.60 (Concrete uniqueness for canonical projectors). Lean ✓ Stmt *Let $V : \mathbb{C}^D \to \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = WV_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.*

Proof. ✓ Proof For each outcome i, the i-th block of V has Gram matrix E_i. Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = WV_0$. □

Definition 19.61 (POVM from a PSD resolution of identity on a dilation). Lean ✓ Stmt Given an isometry $V : \mathbb{C}^D \to \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=0}^{n-1}$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 19.62 (Quantum instrument). Lean ✓ Stmt A *quantum instrument* with n outcomes is a family $\{\Phi_i\}_{i=0}^{n-1}$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 19.63 (Instrument operations). Lean Lean Lean Lean ✓ Stmt Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i, the outcome probability $p_i(\rho) = \operatorname{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 19.64 (Instrument total map is a channel). Lean ✓ Stmt *The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.*

Proof. ✓ Proof Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. □

Theorem 19.65 (Conservation of probability for instruments). Lean ✓ Stmt *For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \operatorname{tr}(\Phi_i(\rho))$ sum to $\operatorname{tr}(\rho)$.*

Proof. ✓ Proof Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \operatorname{tr}(\Phi_i(\rho)) = \operatorname{tr}((\sum_i \Phi_i)(\rho)) = \operatorname{tr}(\rho)$. □

Theorem 19.66 (Non-negativity of instrument probabilities). Lean ✓ Stmt *If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \operatorname{tr}(\Phi_i(\rho))$ is a non-negative real number.*

Proof. ✓ Proof Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. □

19.8 Trace-pairing expansion in transfer-matrix form

Definition 19.67 (Trace-self-dual basis). Lean ✓ Stmt A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \operatorname{tr}(\sigma_i X)\sigma_i.$$

Definition 19.68 (Trace-pairing coefficients). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \operatorname{tr}(\sigma_i T(\sigma_j)).$$

Theorem 19.69 (Trace-pairing expansion of a linear map). Lean ✓ Stmt *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij} \operatorname{tr}(\sigma_j \rho)\sigma_i, \qquad t_{ij} = \operatorname{tr}(\sigma_i T(\sigma_j)).$$

Proof. ✓ Proof Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. □

19.9 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by $SL(2,\mathbb{C})$ filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 19.81) requires the compactness and minimisation argument for $SL(2,\mathbb{C})$ filterings in [Wol12, Propositions 2.9 and 2.11]; see Section 19.10.

Theorem 19.70 (SVD for positive semidefinite matrices). Lean ✓ Stmt *Every positive semidefinite matrix $M \in M_D(\mathbb{C})$ admits a decomposition*

$$M = U \operatorname{diag}(\sigma) U^\dagger,$$

where $U \in \mathcal{U}(D)$ is unitary and $\sigma : \{0, \dots, D-1\} \to \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. ✓ Proof The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. □

Theorem 19.71 (SVD existence for invertible matrices). Lean ✓ Stmt *Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition*

$$M = U \operatorname{diag}(\sigma) V^\dagger,$$

with $U, V \in \mathcal{U}(D)$ unitary and $\sigma_i > 0$ for all i.

Proof. ✓ Proof Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V \operatorname{diag}(\lambda) V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \operatorname{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := MV\Sigma^{-1}$. Then $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \operatorname{diag}(\lambda) \Sigma^{-1} = \mathbb{1}$, so U is unitary, and $U\Sigma V^\dagger = MV(\Sigma^{-1}\Sigma)V^\dagger = MVV^\dagger = M$. □

Theorem 19.72 (SVD representation of a transfer matrix). Lean ✓ Stmt *Every invertible transfer matrix $\widehat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ admits a singular value decomposition $\widehat{T} = U \operatorname{diag}(\sigma) V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.*

Proof. ✓ Proof Direct specialisation of Theorem 19.71 to the index type $\{0, \ldots, D-1\} \times \{0, \ldots, D-1\}$ used by the transfer-matrix representation. □

Remark 19.73 (Sorted / unique singular values). Theorem 19.71 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M. Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

19.10 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.3].

Definition 19.74 (SL-filtering operation). An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$ where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible and have Kraus rank 1.

Definition 19.75 (Doubly-stochastic map). A linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *doubly-stochastic* if $T(\mathbb{1}) \propto \mathbb{1}$ and the reduced density matrix $\operatorname{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

Lemma 19.76 (Infimum attainment for SL-filterings). *For a positive-definite Choi matrix τ, the infimum of $\operatorname{tr}[(S_2 \otimes_k S_1)\tau(S_2 \otimes_k S_1)^\dagger]$ over $S_1, S_2 \in M_D(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained.* ***Not yet proved****—requires compactness of bounded subsets of $\mathrm{SL}(D, \mathbb{C})$ and the extreme-value theorem.*

Theorem 19.77 (Generic normal form for CP maps). *Let $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic.* ***Not yet proved****—proof blocked on Lemma 19.76.*

Definition 19.78 (Diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ for its matrix in the normalized Pauli basis $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$. The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of $\widehat{T'}$ is zero.

Definition 19.79 (Non-diagonal Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some $x \in [0,1]$,

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

Definition 19.80 (Singular Lorentz normal form)**.** For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, write $\widehat{T'}$ in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to $(1+\sigma_3)/2$. Trace preservation supplies the first row of the displayed matrix.

Theorem 19.81 (Lorentz normal form for qubit channels)**.** *For every qubit channel $T : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, there exist $\mathrm{SL}(2,\mathbb{C})$-filterings Φ_1, Φ_2 such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular.* ***Not yet proved****—proof blocked on Lemma 19.76 and the Lorentz group classification of $\mathrm{SL}(2,\mathbb{C})$ orbits.*

19.11 Determinant of a quantum channel

Definition 19.82 (Channel determinant). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, Section 6.1].

Remark 19.83 (Channel determinant as an operator determinant). Lean ✓ Stmt The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$.

Definition 19.84 (Unitary channel)**.** Lean Lean ✓ Stmt For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 19.85 (Unitary channels have determinant one)**.** *Lean* *Lean* *✓ Stmt* *If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.*

Proof. ✓ Proof Vectorization identifies T with a Kronecker product $\overline{U} \otimes U$, whose determinant is $\overline{\det U}^D (\det U)^D = 1$. ☐

Theorem 19.86 (Determinant bound for positive trace-preserving maps)**.** *Lean* *Lean* *✓ Stmt* *For any positive trace-preserving map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, $|\det T| \le 1$. This is [Wol12, Theorem 6.1(1)].*

Proof. ✓ Proof Every eigenvalue μ of T satisfies $|\mu| \le 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \le 1$ follows by induction. □

Theorem 19.87 (Determinant one iff the channel is unitary)**.** Lean Lean ✓ Stmt *For a CPTP map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$,*

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

Proof. ✓ Proof Reverse direction is Theorem 19.85. Forward direction: $|\det T| = 1$ together with $|\mu| \le 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$-automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \ge T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y. Hence T^*, and therefore T, preserves products. By the Skolem–Noether theorem, every $*$-automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U. Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. □

Chapter 20

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 20.1 establishes definitions and the basic exponential form (Proposition 20.14). Section 20.2 gives the Duhamel formula and perturbation bound. Section 20.3 develops the generator theory and proves the GKSL theorem.

20.1 Dynamical semigroups and the exponential form

Definition 20.1 (Dynamical semigroup). Lean ✓ Stmt A family of linear maps $T : \mathbb{R} \to (M_D(\mathbb{C}) \to_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \mathrm{id}$.

This is [Wol12, Equation (7.1)].

Definition 20.2 (Continuous dynamical semigroup). Lean ✓ Stmt A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 20.3 (Exponential semigroup). Lean Lean ✓ Stmt Given a generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 20.4 (Semigroup law for the exponential). Lean Lean ✓ Stmt *For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.*

Proof. ✓ Proof Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. □

Theorem 20.5 (Exponential semigroup at time 0). Lean ✓ Stmt *The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof The exponential of the zero endomorphism is the identity. □

Theorem 20.6 (Continuity of the exponential semigroup). Lean Lean ✓ Stmt *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. ✓ Proof The exponential is analytic (convergence radius $= \infty$), hence continuous. □

Theorem 20.7 (Derivative of the exponential semigroup). Lean ✓ Stmt *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt} e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Equation (7.2)].

Proof. ✓ Proof Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \mathrm{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \mathrm{End}(M_D(\mathbb{C}))$. □

Theorem 20.8 (Derivative at the origin). Lean ✓ Stmt *One has*

$$\left. \frac{d}{dt} e^{tL} \right|_{t=0} = L.$$

Proof. ✓ Proof This is Theorem 20.7 at $t = 0$ together with Theorem 20.5. □

Theorem 20.9 (Generator uniqueness). Lean ✓ Stmt *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. ✓ Proof Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. □

Theorem 20.10 (CLM and linear-map forms coincide). Lean ✓ Stmt *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. ✓ Proof This is a direct unpacking of the definition of the linear-map exponential semigroup. □

Theorem 20.11 (Pointwise derivative formula). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. ✓ Proof Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 20.10. □

Theorem 20.12 (Linear-map exponential semigroup at time 0). Lean ✓ Stmt *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. ✓ Proof Transport Theorem 20.5 through Theorem 20.10. □

Theorem 20.13 (Exponential semigroup as a dynamical semigroup). Lean ✓ Stmt *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. ✓ Proof Transport the continuous-linear-map semigroup law from Theorem 20.4 through Theorem 20.10; the initial condition is Theorem 20.12. □

Theorem 20.14 (Every continuous semigroup is exponential). Lean ✓ Stmt *Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form*

$$T_t = e^{tL}$$

for some generator $L \in \mathrm{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [Wol12, Proposition 7.1].

Proof. ✓ Proof Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s\, ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. □

20.2 Perturbation theory

Theorem 20.15 (Duhamel formula). Lean ✓ Stmt *Let $\Delta = L' - L$. For $t \geq 0$,*

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta\, e^{sL'}\, ds.$$

This is [Wol12, Lemma 7.1].

Proof. ✓ Proof Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta\, e^{sL'}$ (Theorem 20.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s)\, ds$. □

Theorem 20.16 (Perturbation bound). Lean ✓ Stmt *For $t \geq 0$,*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|\Delta\| \cdot \sup_{s \in [0,t]} \|e^{sL}\| \cdot \sup_{s \in [0,t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [Wol12, Corollary 7.1].

Proof. ✓ Proof Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 20.15). □

Corollary 20.17 (Perturbation bound under unit-norm control). Lean ✓ Stmt *For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then*

$$\|e^{tL'} - e^{tL}\| \leq t\, \|L' - L\|.$$

Proof. ✓ Proof Apply Theorem 20.16 and bound both suprema by 1. □

Lemma 20.18 (Dyson-series summability from factorial majorant). Lean ✓ Stmt *Assume a factorial majorant of Dyson–Phillips iterates:*

$$\|D_n(t)\| \leq M\, \frac{(t\, \|L' - L\|\, M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. ✓ Proof This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. □

Theorem 20.19 (Factorial norm bound for Dyson iterates). Lean ✓ Stmt *For $s \in [0, t]$ and $M = \sup_{u \in [0,t]} \|e^{uL}\|$,*

$$\|\widetilde{T}_s^{(n)}\| \leq M\, \frac{(s\, \|\Delta\|\, M)^n}{n!}.$$

Proof. ✓ Proof Induction on n. The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n\,du = s^{n+1}/(n+1)$, recovering the factorial. □

Corollary 20.20 (Dyson series summability). Lean ✓ Stmt *The Dyson–Phillips series* $\sum_n \widetilde{T}_t^{(n)}$ *converges in operator norm for every* $t \geq 0$.

Proof. ✓ Proof Compose the factorial norm bound (Theorem 20.19) with the Weierstrass M-test step (Lemma 20.18). □

Corollary 20.21 (Factorial norm bound at the final time). Lean ✓ Stmt *For* $t \geq 0$*, setting* $s = t$ *in Theorem 20.19 gives*

$$\|\widetilde{T}_t^{(n)}\| \leq M\,\frac{(t\,\|\Delta\|\,M)^n}{n!},$$

where $M = \sup_{u\in[0,t]} \|e^{uL}\|$.

Proof. ✓ Proof This is the special case $s = t$ of Theorem 20.19. □

Theorem 20.22 (Continuity of Dyson iterates). Lean ✓ Stmt *Each Dyson iterate* $t \mapsto \widetilde{T}_t^{(n)}$ *is continuous in the time parameter.*

Proof. ✓ Proof By induction on n. The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L}\Delta\,\widetilde{T}_s^{(n)}\,ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\widetilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. □

Theorem 20.23 (Factorial bound on the Dyson remainder). Lean ✓ Stmt *For* $s \in [0,t]$, $M = \sup_{u\in[0,t]} \|e^{uL}\|$*, and* $M' = \sup_{u\in[0,t]} \|e^{uL'}\|$*:*

$$\|T'_s - \textstyle\sum_{n<N} \widetilde{T}_s^{(n)}\| \leq M'\,\frac{(s\,\|\Delta\|\,M)^N}{N!}.$$

Proof. ✓ Proof Induction on N. The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L}\Delta\,R_N(u)\,du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. □

Theorem 20.24 (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt *For* $t \geq 0$*, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}.$$

Proof. ✓ Proof The partial-sum remainder $\|e^{tL'} - \sum_{n<N} \widetilde{T}_t^{(n)}\| \leq M'\,(t\|\Delta\|M)^N/N!$ tends to zero as $N \to \infty$, since $c^N/N! \to 0$ for any constant c. Together with the summability of the series (Corollary 20.20), this identifies the sum via uniqueness of limits. □

Corollary 20.25 (Tsum form of the Dyson series). Lean ✓ Stmt $\sum_{n=0}^{\infty} \widetilde{T}_t^{(n)} = e^{tL'}$ *(tsum form).*

Proof. ✓ Proof Immediate from Theorem 20.24. □

20.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 20.54), which shows that such generators have the standard Lindblad form.

20.3.1 Generator decomposition and conditional complete positivity

Definition 20.26 (Generator decomposition). Lean ✓ Stmt A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

Definition 20.27 (Conditional complete positivity). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Proposition 7.2].

Definition 20.28 (Trace-annihilating map). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is *trace-annihilating* if $\mathrm{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 20.29 (Trace constraint). Lean ✓ Stmt A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

Theorem 20.30 (Trace constraint implies trace-annihilating). Lean ✓ Stmt *If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.*

Proof. ✓ Proof Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\mathrm{tr}(L(\rho)) = \mathrm{tr}((\sum_i K_i^\dagger K_i)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = \mathrm{tr}((\kappa + \kappa^\dagger)\rho) - \mathrm{tr}(\kappa\rho) - \mathrm{tr}(\kappa^\dagger\rho) = 0$. □

20.3.2 The Lindblad form

Definition 20.31 (Lindblad form). Lean ✓ Stmt A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \Big(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \Big),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Equation (7.21)].

Theorem 20.32 (Lindblad form is trace-annihilating). Lean ✓ Stmt *Every Lindblad form defines a trace-annihilating linear map.*

Proof. ✓ Proof The Hamiltonian part satisfies $\mathrm{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\mathrm{tr}(L_j \rho L_j^\dagger) = \mathrm{tr}(L_j^\dagger L_j \rho) = \mathrm{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. □

Theorem 20.33 (Lindblad form equals generator decomposition). Lean ✓ Stmt *A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$. This is [Wol12, Equation (7.24)].*

Proof. ✓ Proof Compute $\kappa^\dagger = -iH + \frac{1}{2}\sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. □

Theorem 20.34 (Lindblad form is CCP). Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

Proof. ✓ Proof By Theorem 20.33, the Lindblad form equals a generator decomposition with ϕ CP. □

Definition 20.35 (Commutator form of Lindblad equation). Lean ✓ Stmt The *commutator form* of the Lindblad equation writes the generator as

$$L(\rho) = i[\rho, H] + \frac{1}{2}\sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho,\, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Equation (7.22)].

Theorem 20.36 (Commutator form equals standard Lindblad form). Lean ✓ Stmt *The commutator form (Definition 20.35) defines the same linear map as the standard Lindblad form (Definition 20.31).*

Proof. ✓ Proof Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j\rho, L_j^\dagger]$ yields the standard dissipator terms. □

20.3.3 Characterization of conditional complete positivity

Theorem 20.37 (Conditional complete positivity implies projected Choi positivity). Lean ✓ Stmt *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Proposition 7.2].

Proof. ✓ Proof Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P, so the projected Choi matrix reduces to the projected Choi matrix of ϕ, which is positive. □

Theorem 20.38 (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If L is Hermiticity-preserving and $P((L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Proposition 7.2].*

Proof. ✓ Proof Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiolkowski isomorphism applied to Q gives the CP map ϕ, and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ. □

20.3.4 Completely positive semigroups and conditionally completely positive generators

Theorem 20.39 (CP semigroups have CCP generators). Lean ✓ Stmt *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof From $(e^{tL} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 20.38 gives CCP. □

Theorem 20.40 (CCP generators yield CP semigroups). Lean ✓ Stmt *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \qquad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL}, and the cone of CP maps is closed. □

Theorem 20.41 (CP semigroups iff CCP generators). Lean ✓ Stmt *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. ✓ Proof Combine Theorems 20.39 and 20.40. □

20.3.5 Freedom in generator representation

Theorem 20.42 (Traceless Kraus operators exist). Lean ✓ Stmt *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K_j' = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Proposition 7.4].*

Proof. ✓ Proof Set $c_j = -\operatorname{tr}(K_j)/d$. □

Theorem 20.43 (Shift invariance of generators). Lean ✓ Stmt *If $K_j' = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Equation (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. ✓ Proof Direct algebraic expansion. □

20.3.6 Uniqueness of the traceless Lindblad form

Definition 20.44 (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if $\operatorname{tr}(L_j) = 0$ for every j.

Theorem 20.45 (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Proposition 7.4(2)].*

Proof. ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

Theorem 20.46 (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Proposition 7.4(2)].*

Proof. ✓ Proof After identifying $\phi = \phi'$ (Theorem 20.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ. Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ. The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 20.47 (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. ✓ Proof Combine Theorems 20.45 and 20.46. □

20.3.7 Trace-annihilation and trace preservation

Theorem 20.48 (Trace-annihilating generators yield trace-preserving semigroups). Lean ✓ Stmt *If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.*

Proof. ✓ Proof For fixed ρ, the function $f(t) = \mathrm{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \mathrm{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\mathrm{tr}(e^{tL}(\rho)) = f(t) = f(0) = \mathrm{tr}(\rho)$. □

Theorem 20.49 (Trace-preserving semigroups have trace-annihilating generators). Lean ✓ Stmt *If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.*

Proof. ✓ Proof Differentiate the identity $\mathrm{tr}(e^{tL}(\rho)) = \mathrm{tr}(\rho)$ at $t = 0$ from the right. □

20.3.8 The GKSL/Lindblad theorem

Definition 20.50 (GKSL generator). Lean ✓ Stmt A linear map $L : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 20.51 (A trace-constrained generator decomposition gives a GKSL generator). Lean ✓ Stmt *If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

Proof. ✓ Proof CP follows from Theorem 20.40. The trace constraint implies trace-annihilation by Theorem 20.30, so Theorem 20.48 gives trace preservation. □

Theorem 20.52 (GKSL generators admit a trace-constrained generator decomposition). Lean ✓ Stmt *If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. ✓ Proof Apply Theorem 20.39 to obtain a CCP decomposition. Then apply Theorem 20.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. □

Theorem 20.53 (GKSL iff conditional complete positivity and trace annihilation). Lean ✓ Stmt *L is a GKSL generator if and only if L is CCP and trace-annihilating.*

Proof. ✓ Proof Forward: apply Theorem 20.39 to the CP part and Theorem 20.49 to the TP part. Reverse: combine Theorem 20.40 with Theorem 20.48. □

Theorem 20.54 (GKSL iff Lindblad form). Lean ✓ Stmt *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Theorem 7.1].

Proof. ✓ Proof Forward: apply Theorem 20.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 20.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 20.33 gives the Lindblad formula. Reverse: Theorems 20.34 and 20.32 show that every Lindblad form satisfies the right-hand side of Theorem 20.53. □

20.3.9 Kossakowski matrix form

Definition 20.55 (Kossakowski form). Lean ✓ Stmt A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \tfrac{1}{2} \sum_{k,l} C_{lk}([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 20.56 (Kossakowski form iff Lindblad form). Lean ✓ Stmt *A linear map admits a Kossakowski form iff it admits a Lindblad form.*

Proof. ✓ Proof Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. □

20.4 Primitivity and irreducibility of QDS

20.4.1 Auxiliary spectral and semigroup lemmas

Definition 20.57 (Quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 20.58 (Semigroup power identity). Lean ✓ Stmt *For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.*

Theorem 20.59 (Eigenvector propagation under powers). Lean ✓ Stmt *If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.*

Lemma 20.60 (Density matrices are nonzero). Lean ✓ Stmt *Every density matrix is nonzero.*

Theorem 20.61 (Compression invariance is stable under powers). Lean ✓ Stmt *If a corner compression is invariant under E, it remains invariant under every power E^n.*

Theorem 20.62 (Generator eigenvectors along the exponential semigroup). Lean ✓ Stmt *If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.*

Theorem 20.63 (Spectral mapping for semigroup generators). Lean ✓ Stmt *If μ is in the spectrum of L, then $e^{t\mu}$ is in the spectrum of e^{tL}.*

Theorem 20.64 (Root-of-unity rigidity for phases). Lean ✓ Stmt *If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.*

Theorem 20.65 (Peripheral generator eigenvalues are purely imaginary). Lean ✓ Stmt *Peripheral spectral points of the generator have vanishing real part.*

Theorem 20.66 (Peripheral cardinality bound). Lean ✓ Stmt *The number of peripheral eigenvalues is bounded by* $\dim(M_D(\mathbb{C}))$.

Theorem 20.67 (Bounded order under peripheral power-closure). Lean ✓ Stmt *If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding* $\dim(M_D(\mathbb{C}))$ *equals* 1.

Theorem 20.68 (Peripheral powers stay peripheral under irreducibility). Lean ✓ Stmt *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 20.69 (Continuous-linear-map power evaluation). Lean ✓ Stmt *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 20.70 (Spectral-radius criterion from eigenvalue bounds). Lean ✓ Stmt *If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1.*

Theorem 20.71 (Primitive powers annihilate trace-zero matrices). Lean ✓ Stmt *If E is an irreducible primitive channel with faithful fixed density σ, then $E^n(X) \to 0$ for every matrix X with $\mathrm{tr}(X) = 0$.*

Proof. ✓ Proof Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1, so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. □

Theorem 20.72 (Fixed points of an irreducible channel are trace multiples). Lean ✓ Stmt *If E is an irreducible channel with faithful fixed density σ, then every nonzero fixed point V of E satisfies $V = \mathrm{tr}(V)\sigma$.*

Proof. ✓ Proof Subtract the trace multiple $\mathrm{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. □

Corollary 20.73 (Nonzero fixed points have nonzero trace). Lean ✓ Stmt *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. ✓ Proof Otherwise the preceding theorem would force the fixed point to vanish. □

Theorem 20.74 (Peripheral eigenvectors become periodic fixed points). Lean ✓ Stmt *For every positive time t, a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. ✓ Proof Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt}. □

Theorem 20.75 (Peripheral eigenvectors with nonzero trace). Lean ✓ Stmt *For every positive time t, under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. ✓ Proof Apply Theorem 20.74 and then use the previous corollary at the irreducible time pt. □

Theorem 20.76 (Trace-preserving eigenvectors force eigenvalue 1). Lean ✓ Stmt *If E is trace-preserving, $E(V) = \mu V$, and $\mathrm{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. ✓ Proof Take traces in $E(V) = \mu V$ and use trace preservation. □

Theorem 20.77 (Peripheral collapse under irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. ✓ Proof Combine Theorem 20.75 with Theorem 20.76. □

Theorem 20.78 (Primitivity from irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. ✓ Proof The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 20.77 collapses the peripheral set to $\{1\}$. □

Remark 20.79 (Semigroup irreducible/primitive equivalence). ✓ Stmt The semigroup analog of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 20.95 and 20.96).

Remark 20.80 (Reduction to a smaller time step for semigroups). ✓ Stmt Starting from one irreducible time t_0, we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u, and deduce primitivity of T_u.

Theorem 20.81 (The stationary density is fixed for all times). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. ✓ Proof By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ. □

Theorem 20.82 (Fixed points at the irreducible time are trace multiples). Lean ✓ Stmt *If T_{t_0} is irreducible with faithful fixed density σ, then every fixed point V of T_{t_0} satisfies $V = \operatorname{tr}(V)\sigma$.*

Proof. ✓ Proof This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0}. □

Definition 20.83 (Residual slice index). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 20.84 (Residual slice time). Lean ✓ Stmt For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s\,\mathrm{fract}\left(\frac{nu}{s}\right).$$

Theorem 20.85 (Residual slice decomposition). Lean Lean ✓ Stmt *For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and*

$$nu = m_n s + r_n.$$

Proof. ✓ Proof This is the floor-plus-fraction decomposition of nu/s, multiplied by s. □

Theorem 20.86 (A residual slice can vanish). Lean ✓ Stmt *Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \to 0$ as $n \to \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.*

Proof. ✓ Proof The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. □

Theorem 20.87 (Reduced-time eigenvector has nonzero trace). Lean ✓ Stmt *Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u, then T_u admits a μ-eigenvector with nonzero trace.*

Theorem 20.88 (Peripheral collapse at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.*

Theorem 20.89 (Primitivity at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, T_u is primitive.*

Theorem 20.90 (Irreducibility descends to a reduced time step). Lean ✓ Stmt *If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.*

Proof. ✓ Proof Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0}; irreducibility of T_{t_0} rules this out. □

Theorem 20.91 (Existence of an irreducible reduced time step). Lean ✓ Stmt *If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that*

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. ✓ Proof Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0}, channelity comes from the semigroup hypotheses, and Theorem 20.90 gives irreducibility of T_u. □

Theorem 20.92 (Fixed points of the primitive reduced time step are trace-scalars). Lean ✓ Stmt *If T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \mathrm{tr}(\tau)\sigma$.*

Theorem 20.93 (Existence of a primitive reduced time step). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.*

Theorem 20.94 (Irreducibility propagates to all positive times). Lean ✓ Stmt *If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.*

Proof. ✓ Proof Let σ be the unique faithful fixed density of T_{t_0}. By Theorem 20.81, it is fixed for all times, and Theorem 20.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 20.93; then the trace-scalar description of fixed points from Theorem 20.92 gives the uniqueness criterion for irreducibility at every positive time. □

Theorem 20.95 (An irreducible semigroup is primitive). Lean ✓ Stmt *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Proposition 7.5].*

Proof. ✓ Proof First apply Theorem 20.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 20.78 to conclude primitivity of every positive-time slice. □

Theorem 20.96 (Irreducibility iff primitivity for a quantum dynamical semigroup). Lean ✓ Stmt *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists\, t_0 > 0,\ T_{t_0} \text{ irreducible}) \iff (\forall\, t > 0,\ T_t \text{ primitive and irreducible}).$$

This is [Wol12, Proposition 7.5].

Proof. ✓ Proof Forward: apply Theorem 20.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. □

20.5 Kernel of the adjoint Liouvillian

Definition 20.97 (Faithful stationary state). Lean ✓ Stmt A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 20.98 (Adjoint dissipator). Lean ✓ Stmt For a Lindblad operator L_j, the adjoint dissipator on observables is

$$\mathcal{D}^*_{L_j}(A) = L_j^\dagger A L_j - \tfrac{1}{2} L_j^\dagger L_j A - \tfrac{1}{2} A L_j^\dagger L_j.$$

Definition 20.99 (Adjoint generator). Lean ✓ Stmt The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}^*_{L_j}(A).$$

Theorem 20.100 (Trace-pairing characterization of the adjoint generator). Lean ✓ Stmt *For every density matrix ρ and every observable A,*

$$\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A).$$

Proof. ✓ Proof Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. □

Definition 20.101 (Commutant of the Lindblad data). Lean ✓ Stmt The commutant of a Lindblad form is the set of matrices commuting with H, with every Lindblad operator L_j, and with every adjoint $L_j^\dagger$.

Definition 20.102 (Adjoint kernel). Lean ✓ Stmt The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 20.103 (Commutant lies in the adjoint kernel). Lean Lean ✓ Stmt *If an observable commutes with H, with every L_j, and with every $L_j^\dagger$, then it lies in* $\ker(L^*)$.

Proof. ✓ Proof The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}^*_{L_j}(A)$ is zero under the commutation hypotheses. □

Theorem 20.104 (Adjoint kernel lies in the commutant). Lean ✓ Stmt *Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\operatorname{tr}(\rho\, L^*(A)) = \operatorname{tr}(L(\rho)\, A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A:*

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \ \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Theorem 7.2].

Proof. ✓ Proof From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\operatorname{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j. Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. □

Theorem 20.105 (Adjoint kernel equals the commutant). Lean ✓ Stmt *Under the same hypotheses as Theorem 20.104,*

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Theorem 7.2].

Proof. ✓ Proof The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 20.104. □

20.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

Definition 20.106 (Nontrivial projection). Lean ✓ Stmt A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 20.107 (Rank-deficient fixed density). Lean ✓ Stmt Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 20.108 (Rank-deficient kernel element). Lean ✓ Stmt Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 20.109 (Invariant compression). Lean ✓ Stmt Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 20.110 (Block-upper-triangular Lindblad form). Lean ✓ Stmt Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ for all j, where $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$.

Definition 20.111 (Generator-preserved compression). Lean ✓ Stmt For a projection P, the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P\,L(PXP)\,P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 20.112 (Reducible quantum dynamical semigroup). Lean ✓ Stmt A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 20.113 (Semigroup invariance implies generator invariance). Lean ✓ Stmt *If the semigroup preserves $PM_d(\mathbb{C})P$ for all non-negative times, then the generator preserves the same compression.*

Proof. ✓ Proof Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. ☐

Theorem 20.114 (Generator invariance implies semigroup invariance). Lean ✓ Stmt *If the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. ✓ Proof Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. ☐

Theorem 20.115 (Reducibility as a generator-level criterion). Lean ✓ Stmt *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. ✓ Proof Combine Theorems 20.113 and 20.114. ☐

Theorem 20.116 (Rank-deficient fixed densities and kernel elements). Lean ✓ Stmt *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in* $\ker L$. *This is [Wol12, Proposition 7.6, (1)⇔(2)].*

Proof. ✓ Proof The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator–semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. ☐

Theorem 20.117 (A rank-deficient fixed density yields an invariant compression). Lean ✓ Stmt *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1)⇒(3)].*

Proof. ✓ Proof Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0. Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. □

Theorem 20.118 (Invariant compression yields block-upper-triangular Lindblad data). Lean ✓ Stmt *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. ✓ Proof First pass from semigroup invariance to generator invariance by Theorem 20.113. The generator-level vanishing condition forces $(1-P)L_jP = 0$ and $(1-P)\kappa P = 0$ through the sum-of-squares identity. □

Theorem 20.119 (Block-upper-triangular Lindblad data yield invariant compression). Lean ✓ Stmt *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. ✓ Proof The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 20.114. □

Theorem 20.120 (Invariant compression and triangular Lindblad data). Lean Lean ✓ Stmt *For a GKSL generator L, the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3)⇔(4)].*

Proof. ✓ Proof Combine Theorems 20.119 and 20.118. □

Theorem 20.121 (Triangular Lindblad data give deficient kernels). Lean ✓ Stmt *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P, then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Proposition 7.6, (4)⇒(2)].*

Proof. ✓ Proof The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m. Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. □

Theorem 20.122 (Equivalent formulations of reducibility). Lean ✓ Stmt *For a GKSL generator $L : M_D(\mathbb{C}) \to M_D(\mathbb{C})$, the four conditions*

1. *rank-deficient fixed density,*
2. *rank-deficient kernel element,*
3. *invariant compression, and*
4. *block-upper-triangular Lindblad form*

are equivalent. This is [Wol12, Proposition 7.6].

Proof. ✓ Proof The cycle (1)↔(2), (1)→(3), (3)↔(4), and (4)→(2) closes the equivalence chain. □

20.6.1 Sufficient conditions for non-reducibility

Definition 20.123 (Lindblad span). Lean ✓ Stmt The $\mathbb{C}$-linear span of the Lindblad operators $\{L_j\}$.

Definition 20.124 (Hermitian-closed Lindblad span). Lean ✓ Stmt The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$-subspace of $M_d(\mathbb{C})$.

Definition 20.125 (Trivial Lindblad commutant). Lean ✓ Stmt The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 20.126 (Kossakowski rank). Lean ✓ Stmt The minimal number of Lindblad operators across all GKSL representations of L, which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 20.127 (κ block vanishing from generator compression). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2}\sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1-P)L_jP = 0$ for every j, then $(1-P)\kappa P = 0$.*

Proof. ✓ Proof From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1-P)L(P) = 0$: the $(1-P)\varphi(P)$ term vanishes because each $(1-P)L_jP = 0$, and the $(1-P)(P\kappa^\dagger)$ term vanishes because $(1-P)P = 0$. What remains is $(1-P)\kappa P = 0$. ☐

Theorem 20.128 (Full algebra generation forbids block-upper-triangular form). Lean ✓ Stmt *Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. By Theorem 20.127, $(1-P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1-P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1-P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. ☐

Theorem 20.129 (Hermitian span with trivial commutant forbids triangular form). Lean ✓ Stmt *If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.*

Proof. ✓ Proof Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P. The block vanishing $(1-P)L_jP = 0$ and Hermitian closure give $PL_j(1-P) = 0$ as well, so $[P, L_j] = 0$ for all j. But then P lies in the commutant of the Lindblad span, contradicting triviality. ☐

Theorem 20.130 (Large Kossakowski rank forbids block-upper-triangular form). Lean ✓ Stmt *If the Kossakowski rank satisfies*

$$\mathrm{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. ✓ Proof A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. ☐

Theorem 20.131 (No block-upper-triangular Lindblad form implies non-reducibility). Lean ✓ Stmt *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. ✓ Proof A reducible semigroup would have an invariant compression, and Theorem 20.118 would then produce a block-upper-triangular Lindblad form. □

Theorem 20.132 (Full algebra generation implies non-reducibility). Lean ✓ Stmt *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Corollary 7.2(1)].*

Proof. ✓ Proof By Theorem 20.128, no block-upper-triangular decomposition exists. Then apply Theorem 20.131. □

Theorem 20.133 (Hermitian span and trivial commutant imply non-reducibility). Lean ✓ Stmt *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2(2)].*

Proof. ✓ Proof By Theorem 20.129, no block-upper-triangular decomposition exists. Then apply Theorem 20.131. □

Theorem 20.134 (Large Kossakowski rank implies non-reducibility). Lean ✓ Stmt *If* $\mathrm{rank}(C) > d^2 - d$*, then the QDS is not reducible. This is [Wol12, Corollary 7.2(3)].*

Proof. ✓ Proof The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 20.130. Then apply Theorem 20.131. □

Chapter 21

Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorems 5.12 and 5.13]; Hansen–Pedersen's Jensen inequality [HP82]; and Lieb's concavity theorem [Lie73]). They are provided here as cited inputs, since the corresponding matrix-analysis proofs for real powers $x \mapsto x^p$ on the Loewner order and for the matrix logarithm are not developed elsewhere in this document.

21.1 Diagonal Jensen inequality

Lemma 21.1 (Diagonal Jensen inequality). Lean ✓ Stmt *Let $f : \mathbb{R} \to \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix on a finite-dimensional space indexed by n, and let $v \in \mathbb{C}^n$ satisfy $\langle v, v\rangle = 1$. Then*

$$f(\Re\langle v, Av\rangle) \le \Re\langle v, f(A)\, v\rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof Write $A = U\,\mathrm{diag}(\mu)\,U^*$ by the spectral theorem and set $w = U^* v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n. The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\Re\langle v, Av\rangle = \sum_i p_i \mu_i$ and $\Re\langle v, f(A)v\rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See [Bha97, Chapter V]. □

21.2 Trace concavity and convexity of matrix powers

Theorem 21.2 (Trace concavity of real powers). Lean Lean ✓ Stmt *For $p \in [0, 1]$, PSD matrices A_1, A_2, and $t \in [0, 1]$:*

$$t\,\Re\,\mathrm{tr}(A_1^p) + (1-t)\Re\,\mathrm{tr}(A_2^p) \le \Re\,\mathrm{tr}((tA_1 + (1-t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \geq 0$ be an orthonormal eigenbasis of A. A direct computation gives $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t\, a_j + (1-t)\, b_j$ where $a_j = \Re\langle\psi_j, A_1\psi_j\rangle$, $b_j = \Re\langle\psi_j, A_2\psi_j\rangle$. Scalar concavity of $x \mapsto x^p$ on $[0,\infty)$ yields $\mu_j^p \geq t\, a_j^p + (1-t)\, b_j^p$, and the diagonal Jensen inequality (Lemma 21.1, concave form) gives $a_j^p \leq \Re\langle\psi_j, A_1^p\psi_j\rangle$ (and similarly for b_j). Summing over j and using $\sum_j \Re\langle\psi_j, M\psi_j\rangle = \Re \operatorname{tr} M$ yields the conclusion. See [Bha97, Chapter V]. □

Theorem 21.3 (Trace convexity of real powers). Lean Lean ✓ Stmt *For $p \in [1,2]$, PSD matrices A_1, A_2, and $t \in [0,1]$:*

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t\, \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Identical eigenbasis reduction as in Theorem 21.2, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0,\infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. □

21.3 Operator Jensen inequality for positive maps

Definition 21.4 (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices $\{C_i\}_{i\in\iota}$ and a defect block S, one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S. One also forms the scalar block-diagonal matrix whose ι-blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 21.5 (Compression auxiliary for Jensen). Lean ✓ Stmt *Let Y be positive definite and let W be an isometry. Then*

$$(W^*YW)^{-1} \leq W^*Y^{-1}W.$$

Proof. ✓ Proof Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^* & W^*YW \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives the stated inequality. □

Lemma 21.6 (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt *If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i\in\iota} C_i C_i^*,$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i\in\iota} w_i\, C_i C_i^* + t\, SS^*,$$

the defect relation rewrites as

$$\sum_{i\in\iota} C_i C_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with $t > 0$ implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i\in\iota} w_i^{-1}\, C_i C_i^* + t^{-1}\, SS^*.$$

Proof. ✓ Proof Expand the block matrix multiplication entrywise. The defect identity gives the isometry relation after summing the ι-blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry. □

Lemma 21.7 (Finite-POVM resolvent inequality). Lean ✓ Stmt *Let $w_i \geq 0$ and $t > 0$. If the defect block satisfies*

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then

$$\left(\sum_{i \in \iota} w_i \, C_i C_i^* + t \, \mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} \, C_i C_i^* + t^{-1} \, SS^*.$$

Proof. ✓ Proof Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries $w_i + t$ and defect entry t. The compression identities rewrite the compressed matrix as $\sum_i w_i C_i C_i^* + t\mathbb{1}$ and its inverse compression as the displayed upper bound. □

Theorem 21.8 (Jensen for concave real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0, 1]$:*

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.13] for concave $f(x) = x^p$.

Theorem 21.9 (Map form of Jensen for concave real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [0, 1]$, the inequality $T(A^p) \leq (TA)^p$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 21.10 (Jensen for convex real powers). Lean Not ready *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1, 2]$:*

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.11] for the subunital operator-convex case $f(x) = x^p$.

Theorem 21.11 (Map form of Jensen for convex real powers). Lean ✓ Stmt *For a positive subunital map T, a positive semidefinite matrix A, and $p \in [1, 2]$, the inequality $(TA)^p \leq T(A^p)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

Theorem 21.12 (Jensen for concave log). Lean Not ready *For a positive* unital *map T and positive-definite A:*

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.13] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitality.

Theorem 21.13 (Map form of Jensen for the concave logarithm). Lean ✓ Stmt *For a positive unital map T and a positive-definite matrix A, the inequality $T(\log A) \leq \log(TA)$ holds.*

Proof. ✓ Proof Direct application of the axiom. □

21.4 Lieb concavity theorem

Theorem 21.14 (Lieb concavity). Lean Not ready *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A,B) \mapsto \operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:*

$$t\,\mathfrak{R}\operatorname{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t)\mathfrak{R}\operatorname{tr}(K^\dagger A_2^s K B_2^{1-s}) \le \mathfrak{R}\operatorname{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}).$$

See [Lie73, And79].

Theorem 21.15 (Map form of Lieb concavity). Lean ✓ Stmt *For $s \in [0,1]$, any matrix K, and positive-definite matrices A_1, A_2, B_1, B_2, the function $(A,B) \mapsto \mathfrak{R}\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, with the same inequality as Theorem 21.14.*

Proof. ✓ Proof Direct application of the axiom. □

Corollary 21.16 (Lieb concavity, $K = \mathbb{1}$ case). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, and positive-definite matrices A_1, A_2, B_1, B_2, the map $(A,B) \mapsto \mathfrak{R}\operatorname{tr}(A^s B^{1-s})$ is jointly concave:*

$$t\,\mathfrak{R}\operatorname{tr}(A_1^s B_1^{1-s}) + (1-t)\mathfrak{R}\operatorname{tr}(A_2^s B_2^{1-s}) \le \mathfrak{R}\operatorname{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 21.15 by taking $K = \mathbb{1}$.

Proof. ✓ Proof Specialize Theorem 21.15 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. □

Corollary 21.17 (Concavity of Lieb's map in the first argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A_1, A_2, B, the map $A \mapsto \mathfrak{R}\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\mathfrak{R}\operatorname{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t)\mathfrak{R}\operatorname{tr}(K^\dagger A_2^s K B^{1-s}) \le \mathfrak{R}\operatorname{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 21.15 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1-t)B = B$. □

Corollary 21.18 (Concavity of Lieb's map in the second argument). Lean ✓ Stmt *For $s \in [0,1]$, $t \in [0,1]$, matrix K, and positive-definite matrices A, B_1, B_2, the map $B \mapsto \mathfrak{R}\operatorname{tr}(K^\dagger A^s K B^{1-s})$ is concave:*

$$t\,\mathfrak{R}\operatorname{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t)\mathfrak{R}\operatorname{tr}(K^\dagger A^s K B_2^{1-s}) \le \mathfrak{R}\operatorname{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

Proof. ✓ Proof Specialize Theorem 21.15 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1-t)A = A$. □

Chapter 22

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

22.1 Von Neumann entropy

Definition 22.1 (Von Neumann entropy). Lean ✓ Stmt For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = -\sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 22.2 (Entropy is non-negative). Lean ✓ Stmt *For any density matrix ρ, $S(\rho) \geq 0$.*

Proof. ✓ Proof Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. □

Lemma 22.3 (Eigenvalue sum). Lean ✓ Stmt *The eigenvalues of a density matrix sum to 1.*

Proof. ✓ Proof Follows from $\operatorname{tr}(\rho) = \sum_i \lambda_i = 1$. □

Theorem 22.4 (Eigenvalue bound). Lean ✓ Stmt *Each eigenvalue of a density matrix lies in $[0, 1]$.*

Proof. ✓ Proof Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

Theorem 22.5 (Entropy upper bound). Lean ✓ Stmt *For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,*

$$S(\rho) \leq \log D.$$

Proof. ✓ Proof By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. □

22.2 Tripartite partial traces

Definition 22.6 (Partial trace over A). Lean ✓ Stmt For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A \, \rho_{ABC})_{(b_1,c_1)(b_2,c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a,b_1,c_1)(a,b_2,c_2)}.$$

Definition 22.7 (Partial trace over C). Lean ✓ Stmt The partial trace over C:

$$(\mathrm{tr}_C \, \rho_{ABC})_{(a_1,b_1)(a_2,b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1,b_1,c)(a_2,b_2,c)}.$$

Definition 22.8 (Partial trace over AC). Lean ✓ Stmt The partial trace over A and C:

$$(\mathrm{tr}_{AC} \, \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a,b_1,c)(a,b_2,c)}.$$

Lemma 22.9 (Partial trace preserves Hermiticity). *Lean* *✓ Stmt* *If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.*

Proof. ✓ Proof Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. ☐

22.3 Strong subadditivity

Theorem 22.10 (Strong subadditivity). *Lean* *Not ready* *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 22.11 (SSA equality). Lean ✓ Stmt A tripartite density matrix ρ_{ABC} satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 22.12 (Quantum Markov decomposition). Lean ✓ Stmt A *Hayashi Markov decomposition* of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B, a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \, \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi's presentation of quantum Markov structure [Hay06]; the block decomposition used by the MPDO argument is the structure theorem of [HJPW04].

Theorem 22.13 (Hayashi SSA-equality characterization). Lean Not ready *For a tripartite density matrix ρ_{ABC},*

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of arXiv:1606.00608. The equality conditions are reviewed in [Hay06, Rus02]; the structural block-decomposition formulation used for MPDOs appears in [HJPW04].

22.4 Mutual information

Definition 22.14 (Mutual information). Lean ✓ Stmt The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 22.15 (Mutual information is non-negative). Lean ✓ Stmt *For any bipartite density matrix ρ_{AB}, $I(A{:}B) \geq 0$.*

Proof. ✓ Proof Apply strong subadditivity (Theorem 22.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A{:}C) \geq 0$. □

22.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 22.16 (Entropy formulation of von Neumann entropy). Lean ✓ Stmt This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 22.1.

Theorem 22.17 (Entropy formulation of strong subadditivity). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 22.10. The underlying axiom's proof in [LR73] is deferred to the umbrella entropy issue.

Definition 22.18 (Entropy formulation of quantum Markov decomposition). Lean ✓ Stmt This is the entropy formulation of the Hayashi Markov decomposition from Definition 22.12.

Theorem 22.19 (SSA equality iff quantum Markov decomposition). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC}, equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B.*

This formulation introduces no new axiom: it is the same equality criterion as Theorem 22.13.

Definition 22.20 (Entropy formulation of mutual information). Lean ✓ Stmt This formulation has the same value $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 22.14.

Theorem 22.21 (Subadditivity from trivial-middle SSA). Lean ✓ Stmt *For a tripartite density matrix ρ_{ABC} with $\dim B = 1$,*

$$S(\rho_{ABC}) \le S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. ✓ Proof Apply Theorem 22.17 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 22.27, hence its entropy vanishes by Theorem 22.28. □

22.6 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity $I_L \le I_{L+1}$ (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound $I_L \le 4 \log D$ (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (taken as an axiom) and the single-system $S(\rho) \le \log D$ bound; neither introduces a new axiom.

Theorem 22.22 (Entropy nonnegativity for finite index sets). Lean ✓ Stmt *For any PSD Hermitian matrix ρ with $\mathrm{tr}(\rho) = 1$ on an arbitrary finite index set, $S(\rho) \ge 0$. This is the analog of Theorem 22.2 for arbitrary finite index sets: it does not require the index set to be $\{0, \dots, D-1\}$, so it applies to bipartite density matrices on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.*

Proof. ✓ Proof The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply $x \log x \le 0$ on $[0, 1]$ pointwise and sum. □

Theorem 22.23 (Mutual information is monotone under enlargement). Lean ✓ Stmt *For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,*

$$I(A{:}B)_{\mathrm{tr}_C \rho_{ABC}} \le I(A{:}BC)_{\rho_{ABC}},$$

where the left-hand side is the bipartite mutual information of the reduced state $\rho_{AB} = \mathrm{tr}_C(\rho_{ABC})$ and the right-hand side is evaluated by expanding $I(A{:}BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$.

Proof. ✓ Proof Expanding both sides in entropy form and cancelling the common $S(\rho_A)$ term, the inequality reduces to strong subadditivity $S(\rho_{ABC}) + S(\rho_B) \le S(\rho_{AB}) + S(\rho_{BC})$. The bipartite B-reduced state of ρ_{AB} matches the tripartite B-reduced state $\mathrm{tr}_{AC}(\rho_{ABC})$, ensuring the $S(\rho_B)$ term is the one supplied by Theorem 22.17. □

Theorem 22.24 (Elementary area-law bound on mutual information). Lean ✓ Stmt *For a bipartite density matrix ρ_{AB} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d_A, d_B \ge 1$ whose single-system reduced states are obtained by partial trace,*

$$I(A{:}B) \le \log d_A + \log d_B.$$

Proof. ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine $S(\rho_A) \le \log d_A$ and $S(\rho_B) \le \log d_B$ (Theorem 22.5) with $S(\rho_{AB}) \ge 0$ (Theorem 22.22). □

22.7 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 22. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

Theorem 22.25 (Partial trace over A preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_A(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\operatorname{tr}(\operatorname{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_{a} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 22.26 (Partial trace over C preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_C(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions,

$$\operatorname{tr}(\operatorname{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_{c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 22.27 (Partial trace over AC preserves the full trace). Lean ✓ Stmt *For any tripartite matrix ρ_{ABC},*

$$\operatorname{tr}(\rho_{ABC}) = \operatorname{tr}(\operatorname{tr}_{AC}(\rho_{ABC})).$$

Proof. ✓ Proof Unfolding the definitions and interchanging the outer sums,

$$\operatorname{tr}(\operatorname{tr}_{AC}(\rho_{ABC})) = \sum_{b} \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \operatorname{tr}(\rho_{ABC}).$$

□

Theorem 22.28 (Entropy vanishes in dimension 1). Lean ✓ Stmt *If $\rho \in M_1(\mathbb{C})$ is Hermitian and $\operatorname{tr}(\rho) = 1$, then*

$$S(\rho) = 0.$$

Proof. ✓ Proof The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$:

$$\lambda_0 = \operatorname{tr}(\rho) = 1, \qquad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

Chapter 23

Matrix Product Operators and Density Operators

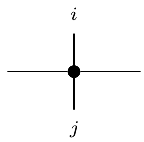
Fix a physical dimension d and a bond dimension D. An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

23.1 MPO tensors

Definition 23.1 (MPO tensor). Lean ✓ Stmt An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \qquad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j. Diagrammatically,



The black node denotes the tensor M, the horizontal legs are virtual, the upper leg is the ket index i, and the lower leg is the bra index j.

Definition 23.2 (Doubled-index MPS view). Lean ✓ Stmt By combining the pair (i, j) into a single physical index in $\{0, \ldots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D:

$$M^{\mathrm{MPS}}_{(i,j)} = M^{ij},$$

identifying $\{0, \ldots, d-1\} \times \{0, \ldots, d-1\}$ with $\{0, \ldots, d^2-1\}$ via the product encoding.

Definition 23.3 (Word evaluation). Lean ✓ Stmt For words $u = (i_1, \ldots, i_N)$ and $v = (j_1, \ldots, j_N)$ in $\{0, \ldots, d-1\}^N$, the *word evaluation* is

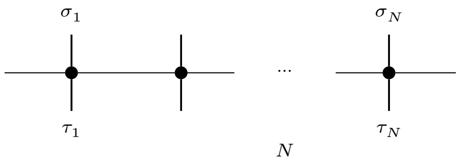
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \cdots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 23.4 (MPO operator family). Lean ✓ Stmt The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma,\tau} = \mathrm{tr}(M^{\sigma,\tau}).$$

Diagrammatically,



each black node denotes the same local tensor M, and the matrix entry $\rho^{(N)}(M)_{\sigma,\tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 23.5 (Hermitian MPO tensor). Lean ✓ Stmt An MPO tensor is *Hermitian* if, for all $i, j \in \{0, \dots, d-1\}$,

$$M^{ij} = (M^{ji})^\dagger$$

23.2 Transfer maps

Definition 23.6 (MPO transfer map). Lean ✓ Stmt The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 23.7 (Identification with the doubled-index MPS transfer map). *Lean* *✓ Stmt* *The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.*

Proof. ✓ Proof After identifying the single physical index of the doubled tensor with a pair (i, j), the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. □

Theorem 23.8 (Positivity of the MPO transfer map). *Lean* *✓ Stmt* *If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.*

Proof. ✓ Proof By Lemma 23.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. □

23.3 Zero correlation length

Definition 23.9 (Zero correlation length for MPO tensors). Lean ✓ Stmt An MPO tensor M has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analog of the renormalization fixed-point condition for MPS tensors; see [CPGSV16, definition of MPDO zero correlation length, lines 736–741] and [CPGSV21, Section II.E.2, lines 937–939].

Theorem 23.10 (MPO ZCL iff doubled-index MPS RFP). *Lean* *✓ Stmt* *An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.*

Proof. ✓ Proof By Lemma 23.7, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. □

23.4 MPDOs and LPDOs

Definition 23.11 (MPDO). Lean ✓ Stmt An MPO tensor is an *MPDO* if, for every $N \in \mathbb{N}$,

$$\rho^{(N)}(M) \geq 0$$

This is the global positivity condition of [CPGSV16, Section 4].

Definition 23.12 (LPDO). Lean ✓ Stmt An MPO tensor is an *LPDO* if there exist a Kraus dimension d_K, an inner bond dimension D', a bond-space identification $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that, for all physical indices i, j,

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e back to $\{0, \dots, D-1\}$. See [CPGSV16, Section 4.3].

Theorem 23.13 (LPDOs are MPDOs). Lean ✓ Stmt *Every LPDO tensor is an MPDO.*

Proof. ✓ Proof By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N-site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e:

$$M^{\sigma_1\tau_1} \cdots M^{\sigma_N\tau_N} = \left(\sum_{\kappa} P_\kappa \otimes \overline{Q_\kappa}\right)\Big|_e,$$

where $P_\kappa = A^{(\sigma_1,\kappa_1)} \cdots A^{(\sigma_N,\kappa_N)}$ and $Q_\kappa = A^{(\tau_1,\kappa_1)} \cdots A^{(\tau_N,\kappa_N)}$. Since the trace is invariant under the reindexing by e, we apply $\mathrm{tr}(X \otimes Y) = \mathrm{tr}(X)\,\mathrm{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma,\tau} = \sum_{\kappa} \mathrm{tr}(P_\kappa)\overline{\mathrm{tr}(Q_\kappa)} = \sum_{\kappa} \psi_\kappa(\sigma)\overline{\psi_\kappa(\tau)},$$

where $\psi_\kappa(\sigma) = \mathrm{tr}(P_\kappa)$. Hence $\rho^{(N)}(M) = \sum_\kappa |\psi_\kappa\rangle\langle\psi_\kappa| \geq 0$. ◻

23.5 Purification RFP

Definition 23.14 (Purifying MPS tensor). Lean ✓ Stmt Given a purifying family $A^{(i,k)} \in M_{D'}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $k \in \{0, \dots, d_K-1\}$, the *purifying MPS tensor* is

$$A^{\mathrm{pur}}_{(i,k)} = A^{(i,k)},$$

an MPS tensor with physical dimension $d \cdot d_K$ and bond dimension D'.

Definition 23.15 (Purification RFP). Lean ✓ Stmt An MPO tensor M has a *purification RFP* if it admits an LPDO witness whose purifying MPS tensor is a pure-state renormalization fixed point. This is [CPGSV16, definition of purification RFP, lines 756–759].

Remark 23.16 (PRFP does not imply MPO ZCL). Lean ✓ Stmt Purification RFP is strictly weaker than MPO zero correlation length: there exists an MPO tensor with purification RFP whose transfer map is not idempotent. Equivalently, purification RFP does not imply MPO zero correlation length.

Theorem 23.17 (Purification RFP implies LPDO). Lean ✓ Stmt *A purification-RFP tensor is, in particular, an LPDO.*

Proof. ✓ Proof The PRFP condition provides an LPDO witness by construction. ◻

23.6 Pure-state recovery inside the MPO formalism

Definition 23.18 (MPDO renormalization fixed point). Lean ✓ Stmt An MPO tensor M is an *MPDO renormalization fixed point* if its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

Theorem 23.19 (MPDO RFP iff MPO ZCL). Lean ✓ Stmt *MPDO renormalization fixed points are exactly MPO tensors with zero correlation length.*

Proof. ✓ Proof Both conditions assert $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$; they are definitionally identical. □

Theorem 23.20 (MPDO RFP iff doubled-index MPS RFP). Lean ✓ Stmt *The MPDO RFP condition is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

Proof. ✓ Proof Unfold the MPDO RFP condition to its definitional equivalent, MPO ZCL, then apply Theorem 23.10. □

Definition 23.21 (Diagonal pure-state embedding as an MPO). Lean ✓ Stmt An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 23.22 (Transfer map of the diagonal pure-state embedding). Lean ✓ Stmt *The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\mathrm{MPO}}} = \mathcal{E}_A.$$

Proof. ✓ Proof In the double sum defining $\mathcal{E}_{A^{\mathrm{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X (A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. □

Theorem 23.23 (Pure-state recovery of the RFP condition). Lean ✓ Stmt *For an MPS tensor A, its diagonal MPO embedding is MPDO-RFP if and only if A is an RFP:*

$$A^{\mathrm{MPO}} \textit{ is MPDO-RFP} \iff A \textit{ is RFP.}$$

Proof. ✓ Proof Both conditions assert idempotence of the corresponding transfer map. Theorem 23.22 identifies those transfer maps, so the two predicates are equivalent. □

Theorem 23.24 (Pure-state recovery of zero correlation length). Lean ✓ Stmt *For an MPS tensor A, the diagonal MPO embedding is MPDO-RFP if and only if A has zero correlation length.*

Proof. ✓ Proof Combine Theorem 23.23 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 17.27). □

23.7 Simple local structure from SAL and ZCL

For the simple MPDO case in Appendix C.2 of [CPGSV16], the strong-area-law hypothesis is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

Definition 23.25 (Local η-structure for a three-site reduced state). Lean ✓ Stmt For a three-site density operator ρ_{ABC}, the local η-structure is the quantum Markov decomposition on the middle subsystem B produced by equality in strong subadditivity. Concretely, after a unitary change of basis on B, one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$$

such that

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

Theorem 23.26 (SAL implies local η-structure). Lean ✓ Stmt *If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy form of the simple-MPDO strong area law — then it admits the local η-structure of Definition 23.25.*

Proof. ✓ Proof This is exactly the forward implication of the Hayashi equality characterization: equality in strong subadditivity yields a quantum Markov decomposition on the middle subsystem. □

Definition 23.27 (Injective inverse-map layer for a simple MPO tensor). Lean Lean Lean Lean ✓ Stmt A simple MPO tensor K is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor K^{-1} together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

Theorem 23.28 (Inverse-map identities for an injective simple MPO tensor). Lean Lean Lean Lean ✓ Stmt *For an injective simple MPO tensor, contracting K^{-1} with K recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.*

Definition 23.29 (Explicit neighboring η-operator data). Lean Lean ✓ Stmt Fix a Hayashi decomposition witness h_η. An explicit neighboring η-family is a dependent family $(\eta_{k,h})$ indexed by sector pairs, where $\eta_{k,h}$ acts on the neighboring bond space $B_k^R \otimes B_h^L$. Requiring positivity for each pair gives the corresponding positive η-data.

Definition 23.30 (Trace matrices of explicit η-data). Lean Lean Lean Lean ✓ Stmt Every explicit neighboring η-family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix T used in the rank-one step.

Theorem 23.31 (Positivity of bipartite partial traces). Lean Lean ✓ Stmt *If ρ is positive semidefinite on a bipartite space $A \otimes B$, then both reduced operators $\mathrm{tr}_A\,\rho$ and $\mathrm{tr}_B\,\rho$ are positive semidefinite.*

Proof. ✓ Proof Write each partial trace as the finite sum, over the traced index, of a principal submatrix of ρ. Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite. □

Definition 23.32 (Hayashi–Markov extraction of explicit η-operators). Lean ✓ Stmt Every Hayashi decomposition witness h_η canonically produces an explicit neighboring η-family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \mathrm{tr}_C(\rho^{(k)}_{b_2 C}) \otimes \mathrm{tr}_A(\rho^{(h)}_{Ab_1}).$$

Positive semidefiniteness of each $\eta_{k,h}$ follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction layer; the remaining connection to Appendix C.2 is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

Theorem 23.33 (Trace matrix of the Hayashi–Markov η-operators). Lean Lean ✓ Stmt *The trace matrix induced by the extracted family is entrywise $T_{k,h} = 1$ in both its complex-valued form and its real-valued form, matching the normalization $T_{k,h} = a_k b_h$ used in $a_k = b_h = 1$.*

Proof. ✓ Proof Each $\rho^{(k)}_{b_2 C}$ is a density matrix, hence has unit trace; the partial trace tr_C preserves the trace. The analogous statement holds for $\mathrm{tr}_A\, \rho^{(h)}_{Ab_1}$. Multiplicativity of the trace under Kronecker products gives $T_{k,h} = 1 \cdot 1 = 1$, and taking real parts gives the real-valued statement. □

Theorem 23.34 (Entrywise nonnegativity of the real trace matrix). Lean ✓ Stmt *Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of T is a separate structural condition.*

Proof. ✓ Proof The trace of a positive semidefinite neighboring operator is non-negative in the complex order. Taking real parts gives the stated entrywise nonnegativity. □

Definition 23.35 (Auxiliary matrix predicates for the rank-one step). Lean Lean Lean ✓ Stmt The auxiliary predicates state three matrix-theoretic ingredients used in the rank-one step of Appendix C.2, Lemma C.4: constant traces of all positive powers, existence of a rank-one factorization $T = ab^\top$, and the conditional Perron–Frobenius implication from primitivity together with constant trace powers to such a factorization.

This conditional implication is not a globally provable theorem. As a universal statement over primitive non-negative matrices it is false; a concrete 3×3 counterexample appears in the archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

Theorem 23.36 (Trace square of a Hermitian matrix). Lean ✓ Stmt *Let T be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues λ_i. Then*

$$\mathrm{tr}(T^2) = \sum_i \lambda_i^2.$$

Proof. ✓ Proof The spectral theorem writes T as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix. □

Theorem 23.37 (PSD trace-power rank-one criterion). Lean Lean ✓ Stmt *Let T be a finite real square matrix. If T is positive semidefinite, normalized by $\mathrm{tr}(T) = 1$, and has constant traces on all positive powers, then T has a rank-one factorization. Consequently, under the same positive-semidefinite and trace-normalization hypotheses, the conditional rank-one criterion used in Lemma C.4 follows.*

Proof. ✓ Proof The spectral theorem diagonalizes the positive semidefinite matrix with non-negative real eigenvalues. The trace gives that their sum is 1, while the second trace moment gives that the sum of their squares is also 1. Hence exactly one eigenvalue is equal to 1 and all others vanish, so the diagonal form is rank one. Unitary conjugation preserves the existence of an outer-product factorization. □

Theorem 23.38 (SAL and ZCL imply rank-one T (conditional form))**.** Lean ✓ Stmt *Let T be a finite real square matrix. Assume that T is primitive, that* $\mathrm{tr}(T) = 1$, *and that every positive power of T has the same trace as T. If one further supplies the Perron–Frobenius hypothesis that this particular primitive non-negative matrix with constant trace powers is rank one, then*

$$T = ab^\top$$

for some vectors a and b, with normalization

$$a \cdot b = 1.$$

Proof. ✓ Proof The Perron–Frobenius hypothesis gives the factorization $T = ab^\top$. Taking traces and using $\mathrm{tr}(ab^\top) = a \cdot b$ converts the normalization $\mathrm{tr}(T) = 1$ into $a \cdot b = 1$. □

Theorem 23.39 (PSD-corrected rank-one T criterion)**.** Lean ✓ Stmt *Let T be a finite real square matrix. Assume that T is primitive, positive semidefinite, normalized by* $\mathrm{tr}(T) = 1$, *and has constant traces on all positive powers. Then*

$$T = ab^\top$$

for some vectors a and b, with $a \cdot b = 1$.

Proof. ✓ Proof The positive-semidefinite matrix criterion diagonalizes T, uses the first two trace moments to show that exactly one eigenvalue is nonzero, and hence supplies the auxiliary rank-one criterion. The conditional Lemma C.4 statement then gives the factorization and trace normalization. □

23.8 Horizontal and Vertical Canonical Forms

Definition 23.40 (Horizontal canonical form for an MPO)**.** Lean Not ready An MPO tensor is in *horizontal canonical form* if its doubled-index MPS tensor generates the same MPV family as a block-diagonal tensor $\bigoplus_k \mu_k A_k$ whose blocks are injective, left-canonical, and have nonzero weights μ_k.

Definition 23.41 (Vertical canonical form for an MPO)**.** Lean Not ready An MPO tensor is in *vertical canonical form* if its diagonal MPS tensor $\{M^{ii}\}_{i=0}^{d-1}$ generates the same MPV family as a block-diagonal tensor obtained from a basis of normal tensors by repeating each block according to a multiplicity and weighting each copy by a positive scalar. This is the flattened form of the decomposition $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ from [CPGSV16, Proposition 4.13].

Remark 23.42 (Finite witnesses for block-injective canonical form). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let (A_k) be a family of block tensors, and fix a length L. Write A_k^w for the length-L word evaluation of A_k along a word w. The finite-length spanning hypothesis says that the tuple-valued evaluations

$$w \longmapsto (A_k^w)_k$$

span the full product algebra of block matrices. The corresponding finite-length separation property asserts that there exists a length L such that if, for every word w of length L,

$$\sum_k \operatorname{tr}(\Delta_k A_k^w) = 0$$

then these pairings vanish on a spanning set for the product algebra, so the induced linear functional on the product algebra is zero; by nondegeneracy of the product trace pairing, every Δ_k vanishes. In other words, a single finite blocking length already separates the blocks by their word evaluations. Combined with injectivity, left-canonicality, and nonzero block weights, this finite-length separation yields the horizontal canonical-form data of the block-diagonal tensor.

The entrywise scalar family of block-matrix coefficients encodes the same finite-length data. Linear independence of that scalar family implies the product-algebra spanning statement, hence a block-injective canonical-form witness. The same criterion therefore gives horizontal canonical-form data directly. The duplicate-scalar counterexample shows that this finite-length witness is strictly stronger than blockwise injectivity, left-canonicality, and nonzero weights alone: two identical scalar blocks satisfy those hypotheses, but the block-injective canonical-form separation condition fails and no blocking length makes the entrywise family linearly independent.

The block-injectivity proposition in [CPGSV16, lines 340–345] gives a concrete sufficient criterion for the spanning condition: after a common blocking length each block is block-injective, and after a further finite blocking length one can form linear combinations of word evaluations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with this finite block-separation data produces the product-algebra spanning condition, hence the horizontal canonical-form data.

Theorem 23.43 (Finite block-separation criterion). Lean ✓ Stmt *If a block family admits the finite-length data described in the block-injectivity proposition of [CPGSV16, lines 340–345], then it is in block-injective canonical form.*

Proof. ✓ Proof This is the direct implication from the finite-length data to the finite-length trace-separation statement. What remains open in the full block-injectivity proposition of [CPGSV16, lines 340–345] is the derivation of this finite witness from separated canonical-form / BNT data. □

Lemma 23.44 (Blockwise equality from agreement on the MPV family). Lean ✓ Stmt *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length-L block words is faithful across all blocks simultaneously. If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block A_k.*

Proof. ✓ Proof This is Lemma L of [CPGSV16, Appendix A], using the finite criterion from Theorem 23.43. Expanding the hypothesis over a word of length $L+1$ beginning with the fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the Y-Z difference, a tuple of block matrices whose trace pairings against every length-L block word vanish. The block-injective canonical-form hypothesis then forces each block difference to be zero, and dividing by the nonzero weight μ_k^{L+1} concludes the blockwise equality of the two first-site insertions. □

Theorem 23.45 (Horizontal canonical form implies vertical canonical form). Not ready *Every MPDO in horizontal canonical form is also in vertical canonical form.*

Proof. Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 23.44 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition. □

23.9 Fusion isometries

Following [CPGSV16, Section 4.5], the next definitions describe the transfer-map content of the fusion-isometry picture. For each blocked size $n \geq 1$, the doubled-index blocked tensor of an MPO has a blocked transfer map E_n acting on bond-space matrices, and the support algebra is modeled by a subspace through which E_n factors as a retract.

Definition 23.46 (Transfer-map fusion-isometry datum). Lean ✓ Stmt A *transfer-map fusion-isometry datum* at blocked size n for an MPO tensor consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$T_n : M_D(\mathbb{C}) \to \mathcal{A}_n, \qquad S_n : \mathcal{A}_n \to M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \mathrm{id}_{\mathcal{A}_n}, \qquad S_n \circ T_n = E_n,$$

where E_n denotes the blocked transfer map.

Definition 23.47 (Fusion-isometry RFP predicate). Lean ✓ Stmt An MPO tensor M satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size $n \geq 1$ there exists transfer-map fusion-isometry data for the blocked transfer map E_n.

Theorem 23.48 (Fusion-isometry data imply the MPDO RFP condition). Lean Lean ✓ Stmt *A one-site fusion-isometry datum implies the MPDO RFP condition. Hence the fusion-isometry formulation, which supplies such data at every positive blocked size, implies the same condition.*

Proof. ✓ Proof Apply the definition at blocked size $n = 1$. If $E_1 = S_1 \circ T_1$ and $T_1 \circ S_1 = \mathrm{id}$, then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since E_1 is the original transfer map, this is exactly the MPDO RFP condition. □

Theorem 23.49 (One-site fusion retract criterion). Lean ✓ Stmt *A one-site transfer-map fusion-isometry datum exists if and only if the MPO transfer map is idempotent.*

Proof. ✓ Proof The forward implication is the retract calculation $E_1^2 = S_1 T_1 S_1 T_1 = S_1 T_1$. Conversely, if E_1 is idempotent, then it factors through its range, giving the required one-site datum. □

Theorem 23.50 (Transfer-map fusion criterion for MPDO RFP). Lean ✓ Stmt *The fusion-isometry formulation is equivalent to the MPDO RFP condition.*

Proof. ✓ Proof The forward implication is Theorem 23.48. Conversely, if the transfer map E is idempotent, then every positive blocked transfer map equals E and therefore factors through its range. This range-factorization gives the required fusion-isometry data. □

Corollary 23.51 (Pure-state recovery). Lean ✓ Stmt *For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.*

Proof. ✓ Proof Combine Theorem 23.50 with the diagonal pure-state recovery theorem for the MPO predicate. □

Theorem 23.52 (All-blocked fusion reduces to one-site fusion). Lean ✓ Stmt *The fusion-isometry formulation for all positive blocked sizes is equivalent to the existence of a one-site transfer-map fusion-isometry datum.*

Proof. ✓ Proof Combine the one-site criterion with the equivalence between the all-blocked fusion formulation and transfer-map idempotence. □

23.10 Algebra structure

Definition 23.53 (Algebra-structure data for an MPDO). Lean ✓ Stmt A family of support $*$-algebras $\{\mathcal{A}_n\}_{n\geq 0}$, one at every blocking size, equipped with blocking maps $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \to \mathcal{A}_{2n}$ and inclusion maps $\iota_n : \mathcal{A}_n \to \mathcal{A}_{n+1}$ such that, after viewing all elements inside the ambient bond-space matrix algebra, m_n is ordinary matrix multiplication and ι_n is the ambient inclusion.

Definition 23.54 (Algebra-structure RFP predicate). Lean ✓ Stmt An MPO tensor M satisfies the *algebra-structure formulation* used here when there exist algebra-structure data whose support algebra at every positive blocking size coincides with the fixed-point algebra of the adjoint blocked transfer map. This is a nontrivial algebraic condition, but it is still weaker than the full coefficient formulation of [CPGSV16, Theorem 4.14(ii)]; that formulation also includes the coefficient family $c^{(L)}_{\alpha,\beta,\gamma}$.

Theorem 23.55 (RFP yields a stationary algebra tower under a faithful fixed point). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer map is idempotent, then there exists a stationary family of support algebras, all equal to the fixed-point algebra of the adjoint transfer map, with multiplication given by matrix multiplication and inclusions given by the identity.*

Proof. ✓ Proof Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives a $*$-subalgebra of adjoint fixed points. Idempotence identifies every positive blocked transfer map with the original one, so the same fixed-point algebra works at every blocking size. □

Theorem 23.56 (Fusion yields a stationary algebra tower under a faithful fixed point). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer-map fusion criterion holds, then the MPO satisfies the algebra-structure formulation of the renormalization fixed-point condition.*

Proof. ✓ Proof Combine Theorem 23.48 with Theorem 23.55. □

Theorem 23.57 (Stabilized blocked adjoint fixed points yield algebra data). Lean ✓ Stmt *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If, for every positive blocked size n, the adjoint fixed-point space of the blocked transfer map agrees with the adjoint fixed-point space of the original transfer map, then the algebra-structure formulation holds.*

Proof. ✓ Proof Use the adjoint fixed-point algebra of the original transfer map as a constant support-algebra tower. The stabilization hypothesis identifies that same algebra with the adjoint fixed-point space of every positive blocked transfer map, so the stationary family satisfies the definition. □

This does not give the full converse direction of [CPGSV16, Theorem 4.14]: the compatibility condition above only sees stabilized adjoint fixed-point algebras, not the coefficient/BNT data. In particular, Theorem IV.13(ii) also requires the special diagonal matrices $\chi_{\alpha,\beta,\gamma}$. The coefficient data attached to the algebra tower are: a chosen basis of each support algebra, coordinate maps into a finite scalar family, and the corresponding multiplication / inclusion coefficient arrays.

Definition 23.58 (Blocked coefficients for the algebra tower). Lean ✓ Stmt For each blocked size n, once a basis of $\mathcal{A}_n$ is chosen, every element of $\mathcal{A}_n$ is identified with a finite coefficient family on that basis.

Theorem 23.59 (Reconstruction from blocked coefficients). Lean ✓ Stmt *Reconstructing a coefficient family gives the corresponding basis expansion $\sum_i a_i b_i$ in $\mathcal{A}_n$.*

Proof. ✓ Proof This is exactly the finite-basis reconstruction formula for the chosen basis of $\mathcal{A}_n$. □

Definition 23.60 (Blocked multiplication coefficients). Lean ✓ Stmt The blocking multiplication map m_n determines a coefficient family for the product of any two chosen basis elements of $\mathcal{A}_n$, now viewed in the basis of $\mathcal{A}_{2n}$.

Theorem 23.61 (Blocked multiplication reconstructs from its coefficients). Lean ✓ Stmt *In the ambient bond-space matrix algebra, the product of two chosen basis elements of $\mathcal{A}_n$ is the sum of the basis elements of $\mathcal{A}_{2n}$ weighted by these extracted coefficients.*

Proof. ✓ Proof Apply the reconstruction formula of Theorem 23.59 to the element $m_n(b_i, b_j) \in \mathcal{A}_{2n}$. □

Theorem 23.62 (Compatible adjoint fixed points reconstruct from blocked coefficients). Lean ✓ Stmt *If the algebra tower is compatible with an MPO tensor M, then every adjoint blocked fixed point of the blocked transfer map E_n belongs to $\mathcal{A}_n$ and is recovered exactly from its extracted coefficient family.*

Proof. ✓ Proof Compatibility identifies $\mathcal{A}_n$ with the adjoint fixed-point algebra of E_n, so the fixed point first becomes an element of $\mathcal{A}_n$ and then Theorem 23.59 reconstructs it. □

Theorem 23.63 (Adjoint fixed points descend through the algebra tower). Lean ✓ Stmt *If an MPO tensor satisfies the algebra-structure formulation, then every adjoint fixed point of the blocked transfer map E_n at a positive blocking size n is already an adjoint fixed point of the unblocked transfer map E_1.*

Proof. ✓ Proof Compatibility at size n places X in $\mathcal{A}_n$. The inclusion map ι_n lifts X into $\mathcal{A}_{n+1}$, and since ι_n is just an inclusion it does not change the underlying operator, so we continue to denote the lifted element by X. Thus compatibility at size $n+1$ yields $E_{n+1}^\dagger X = X$. Rewriting $E_{n+1} = E_n \circ E_1$ and applying $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$ together with the hypothesis $E_n^\dagger X = X$ extracts $E_1^\dagger X = X$. □

Lemma 23.64 (Reverse inclusion of adjoint fixed points). Lean ✓ Stmt *Every adjoint fixed point of the unblocked transfer map E_1 is an adjoint fixed point of every blocked transfer map E_n.*

Proof. ✓ Proof Induction on n using $E_{n+1} = E_n \circ E_1$ and $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$; the base case is $E_0 = \mathrm{id}$. This step does not require the algebra-structure data. □

Lemma 23.65 (Blocked powers of adjoint eigenvectors). Lean ✓ Stmt *If $E = E_1$ and $E^\dagger X = \lambda X$, then $E_n^\dagger X = \lambda^n X$ for every n.*

Proof. ✓ Proof The case $n = 0$ is $E_0 = \mathrm{id}$. For the induction step, $E_{n+1} = E_n \circ E_1$, hence

$$E_{n+1}^\dagger X = E_1^\dagger E_n^\dagger X = E_1^\dagger(\lambda^n X) = \lambda^n E_1^\dagger X = \lambda^{n+1} X.$$

□

Theorem 23.66 (Finite-order adjoint eigenvalues for algebra towers). Lean ✓ Stmt *Assume the algebra-structure formulation holds. If $X \neq 0$, $E_1^\dagger X = \lambda X$, and $\lambda^n = 1$ for some $n > 0$, then $\lambda = 1$.*

Proof. ✓ Proof By Lemma 23.65, $E_n^\dagger X = \lambda^n X = X$. The descent theorem gives $E_1^\dagger X = X$. Thus $\lambda X = X$, so $(\lambda - 1)X = 0$. Since $X \neq 0$, one obtains $\lambda = 1$. □

Theorem 23.67 (Adjoint fixed-point equality for algebra towers). Lean ✓ Stmt *If an MPO tensor satisfies the algebra-structure formulation, then for every positive blocking size n the adjoint fixed points of E_n are exactly the adjoint fixed points of E_1.*

Proof. ✓ Proof Combine Theorem 23.63 with Lemma 23.64. □

Theorem 23.68 (Algebra data give the stationary fixed-point tower). Lean ✓ Stmt *Under the trace-preserving normalization and a positive-definite fixed point, any algebra-structure witness forces the stationary adjoint fixed-point tower itself to be compatible with the MPO tensor.*

Proof. ✓ Proof Apply the adjoint fixed-point equality obtained from the algebra-structure predicate to the stationary tower constructed from the faithful fixed point. □

Remark 23.69 (Why algebra data do not imply fusion data). Write $E = E_1$ for the one-site transfer map. The present algebra predicate gives, for every $n > 0$,

$$\mathrm{Fix}((E^n)^\dagger) = \mathrm{Fix}(E^\dagger).$$

Hence a vector X with $E^\dagger X = \lambda X$ and $\lambda^n = 1$ lies in $\mathrm{Fix}(E^\dagger)$, so $(\lambda - 1)X = 0$ and $\lambda = 1$ by Theorem 23.66. It does not imply $E^2 = E$. For instance, if Π_{diag} is the projection onto diagonal matrices and $0 < \varepsilon < 1$, set

$$E(X) = \varepsilon\, X + (1 - \varepsilon)\, \Pi_{\mathrm{diag}}(X),$$

so that

$$E^n(X) = \Pi_{\mathrm{diag}}(X) + \varepsilon^n (X - \Pi_{\mathrm{diag}}(X)), \qquad E^2 - E = (\varepsilon^2 - \varepsilon)(\mathrm{id} - \Pi_{\mathrm{diag}}).$$

The fixed-point algebra is diagonal for every $n > 0$, but $E^2 \neq E$.

The missing source step is the coefficient argument in [CPGSV16, Appendix C.4]. One needs the same-length product law

$$O_L(M_\alpha) O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) O_L(M_\gamma)$$

and the length-one idempotent trace identity

$$m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

The proof of [CPGSV16, Theorem 4.14] uses these identities, positivity, and the simple-matrix lemma [CPGSV16, lines 1155–1163] to give

$$\{\nu_{\gamma,k}\}_k = \{\mu_{\alpha,k_1} \mu_{\beta,k_2} \chi_{\alpha,\beta,\gamma,k_3}\}_{\alpha,\beta,k_1,k_2,k_3}, \qquad \mathrm{tr}(\nu_\gamma) = m_\gamma.$$

The reconstruction maps in [CPGSV16, lines 2065–2085] then use

$$\widetilde{R}_2\left(\bigoplus_{\gamma} X_\gamma\right) = \bigoplus_{\gamma} \frac{\nu_\gamma}{m_\gamma} \otimes X_\gamma, \qquad \widetilde{R}_1\left(\bigoplus_{\gamma} X_\gamma\right) = \bigoplus_{\gamma} \frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma.$$

Thus $\mathrm{tr}(\nu_\gamma) = m_\gamma$ and the defining equality $m_\gamma = \mathrm{tr}(\mu_\gamma)$ normalize the factors ν_γ/m_γ and μ_γ/m_γ, respectively, in $T = \widetilde{R}_2\widetilde{T}R_1$ and $S = \widetilde{R}_1\widetilde{S}R_2$. The scalar Newton–Girard power-sum identity needed to obtain the trace-power form is already available as Theorem 10.19; what remains is the MPDO construction of the BNT-label witness carrying these two displayed equations.

23.10.1 Diagonal χ-matrices and the trace-power formula

The following statements isolate the form of [CPGSV16, Theorem 4.14(ii)]: the structure coefficients of the length-L operator algebra are trace-powers of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. They state the coefficient identity and the elementary trace-power identity for diagonal matrices.

Definition 23.70 (Diagonal χ family)**.** Lean ✓ Stmt A family of diagonal complex matrices $\chi_{\alpha,\beta,\gamma}$ indexed by ordered triples drawn from a common index type, each of a specified size. This is the matrix $\chi_{\alpha,\beta,\gamma}$ of [CPGSV16, Theorem 4.14(ii)].

Theorem 23.71 (Trace of χ^L equals the sum of L-th powers of entries)**.** Lean ✓ Stmt *For every triple (α, β, γ), if $\chi_{\alpha,\beta,\gamma}$ has size $r_{\alpha,\beta,\gamma}$, then for every $L \geq 0$,*

$$\mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi^L_{\alpha,\beta,\gamma,k},$$

where $\chi_{\alpha,\beta,\gamma,k}$ are the diagonal entries.

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 23.72 (Trace-power form of a structure-coefficient family)**.** Lean ✓ Stmt A binary compatibility predicate between an abstract structure-coefficient family $c^{(L)}_{\alpha,\beta,\gamma}$ and a diagonal χ family: the pair (c, χ) is said to be in *trace-power form* when

$$c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi^L_{\alpha,\beta,\gamma,k}$$

for every $L \geq 0$ and every triple (α, β, γ), where $r_{\alpha,\beta,\gamma}$ is the size of $\chi_{\alpha,\beta,\gamma}$. This is the formal analog of the target identity of [CPGSV16, Theorem 4.14(ii)]; the existential version—"there exists χ for which (c, χ) has trace-power form"—is obtained by quantifying over χ.

Definition 23.73 (BNT-label coefficient family)**.** Lean Lean Lean Lean ✓ Stmt The same-length coefficient system $c^{(L)}_{\alpha,\beta,\gamma}$ from Theorem IV.13(ii), indexed by fixed BNT labels α, β, γ. The coefficient is attached to the length-L product $O_L(M_\alpha)O_L(M_\beta)$, not to the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \to \mathcal{A}_{2n}$. When a diagonal χ-family has already been constructed, it also canonically determines the coefficient family $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) = \sum_r \chi^L_{\alpha,\beta,\gamma,r}$, as in [CPGSV16, Appendix C.4].

Definition 23.74 (BNT-label operator family). Lean ✓ Stmt A BNT-label operator family records, for each chain length L, the operators $O_L(M_\alpha)$ indexed by the fixed BNT labels α. The ambient algebra at length L is left abstract here; the later comparison step must relate it to the chosen blocked support algebras.

Definition 23.75 (Same-length BNT product form). Lean ✓ Stmt A BNT-label operator family and a BNT-label coefficient system have same-length product form when, for every positive length L,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma).$$

This is the product identity appearing in Theorem IV.13(ii), stated independently of the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \to \mathcal{A}_{2n}$.

Theorem 23.76 (Same-length BNT product expansion). *Lean* *✓ Stmt* *Under same-length BNT product form, the product of two length-L BNT-label operators is the corresponding finite linear combination of length-L BNT-label operators.*

Proof. ✓ Proof This is exactly the defining equality of same-length product form. □

Definition 23.77 (BNT-label trace-scalar family). Lean ✓ Stmt A BNT-label trace-scalar family records the scalars $m_\alpha = \operatorname{tr}(\mu_\alpha)$ indexed by the fixed BNT labels α, as in the idempotent condition of Theorem IV.13(ii).

Definition 23.78 (BNT idempotent coefficient form). Lean ✓ Stmt A BNT-label trace-scalar family and a BNT-label coefficient system have idempotent coefficient form when, for every BNT label γ,

$$m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta.$$

This is the idempotent condition in Theorem IV.13(ii), stated separately from the trace-power formula.

Theorem 23.79 (BNT idempotent coefficient expansion). *Lean* *✓ Stmt* *Under BNT idempotent coefficient form, each m_γ is the corresponding finite linear combination of products $m_\alpha m_\beta$ with the length-one coefficients $c^{(1)}_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This is exactly the defining equality of BNT idempotent coefficient form. □

Definition 23.80 (Blocked-basis BNT-label assignment). Lean Lean ✓ Stmt The BNT-label assignment consists, for every positive blocked length n, of a source label map σ_n from the chosen basis of $\mathcal{A}_n$ and a target label map τ_n from the chosen basis of $\mathcal{A}_{2n}$ to the fixed BNT labels. It also gives the combined label map $\sigma_n \sqcup \tau_n$ on the disjoint union of these two blocked bases.

Definition 23.81 (BNT comparison with blocked coefficients). Lean Lean Lean ✓ Stmt A comparison between the BNT-label coefficients and the chosen blocked-basis coefficients consists of a BNT-label assignment together with the equality

$$c^{(n)}_{i,j,k} = c^{(n)}_{\sigma_n(i),\sigma_n(j),\tau_n(k)}.$$

This is a blocked-basis comparison: the left hand side is expanded in the chosen basis of $\mathcal{A}_{2n}$, so the statement is not itself the same-length BNT product law. This records the coefficient-comparison statement separately from the construction of the label maps from the MPDO tensor.

Theorem 23.82 (Blocked coefficients are pulled back from BNT labels). Lean ✓ Stmt *Under a BNT comparison with blocked coefficients, each chosen blocked-basis multiplication coefficient is the corresponding BNT-label coefficient.*

Proof. ✓ Proof This is exactly the defining equality of the comparison. □

Definition 23.83 (Positive-length BNT-label trace-power form). Lean ✓ Stmt A BNT-label coefficient system and a diagonal χ-family are compatible when, for every positive length L,

$$c^{(L)}_{\alpha,\beta,\gamma} = \operatorname{tr}(\chi^{L}_{\alpha,\beta,\gamma}).$$

The same matrix $\chi_{\alpha,\beta,\gamma}$ is used for all positive L.

Theorem 23.84 (BNT-label coefficient as a positive-length trace power). Lean ✓ Stmt *Under positive-length BNT-label trace-power form, for every $L > 0$, $c^{(L)}_{\alpha,\beta,\gamma}$ is the trace of $\chi^{L}_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof Combine the defining positive-length trace-power identity with the trace formula for powers of a diagonal matrix. □

Theorem 23.85 (Canonical BNT coefficients from a χ-family). Lean ✓ Stmt *The coefficient family canonically determined by a diagonal χ-family satisfies the positive-length trace-power condition with respect to the same χ-family.*

Proof. ✓ Proof This is immediate from the definition of the canonical coefficient family. □

Definition 23.86 (Positive BNT-label χ trace-power witness). Lean Lean ✓ Stmt A positive BNT-label χ witness consists of a length-independent diagonal $\chi_{\alpha,\beta,\gamma}$-family, positivity of every diagonal entry, and the positive-length trace-power identity for the BNT-label coefficients. This is the coefficient statement of Theorem IV.13(ii); constructing the witness from an MPDO tensor and comparing it to blocked bases remain separate obligations. If the coefficient family is chosen canonically from the same χ-family, positivity of the diagonal entries alone gives such a witness.

Theorem 23.87 (Positive BNT-label witness gives trace powers). Lean ✓ Stmt *A positive BNT-label χ witness gives, for every $L > 0$, the formula $c^{(L)}_{\alpha,\beta,\gamma} = \operatorname{tr}(\chi^{L}_{\alpha,\beta,\gamma})$.*

Proof. ✓ Proof Apply the trace reformulation of the positive-length trace-power identity from the witness. □

Theorem 23.88 (Same-length BNT product with chi-trace coefficients). Lean ✓ Stmt *If the same-length BNT product identity holds and the BNT-label coefficients come from a positive length-independent χ-witness, then for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \operatorname{tr}(\chi^{L}_{\alpha,\beta,\gamma})O_L(M_\gamma).$$

Proof. ✓ Proof Substitute the positive-length trace-power formula into the same-length BNT product expansion. □

Theorem 23.89 (Same-length BNT product for canonical chi coefficients). Lean ✓ Stmt *If the same-length BNT product identity holds with the coefficient family canonically determined by a diagonal χ-family, then for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \operatorname{tr}(\chi^{L}_{\alpha,\beta,\gamma})O_L(M_\gamma).$$

Proof. ✓ Proof Substitute the canonical trace-power formula for the coefficients into the same-length BNT product expansion. □

Theorem 23.90 (BNT idempotent coefficients as chi traces). Lean ✓ Stmt *If the BNT trace scalars satisfy the idempotent coefficient condition and the BNT-label coefficients come from a positive length-independent χ-witness, then*

$$m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

Proof. ✓ Proof Substitute the length-one trace-power formula into the idempotent coefficient identity. □

Theorem 23.91 (BNT idempotent coefficients for canonical chi coefficients). Lean ✓ Stmt *If the BNT trace scalars satisfy the idempotent coefficient condition with the coefficient family canonically determined by a diagonal χ-family, then*

$$m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

Proof. ✓ Proof Substitute the length-one canonical trace formula into the idempotent coefficient identity. □

Theorem 23.92 (Blocked coefficients inherit BNT trace powers). Lean Lean Lean ✓ Stmt *If the chosen blocked-basis coefficients are compared with the fixed BNT-label coefficient system, and that BNT-label system has a positive length-independent χ-witness, then each positive-length blocked coefficient is the trace power of the corresponding BNT-label χ-matrix:*

$$c^{(n)}_{i,j,k} = \mathrm{tr}(\chi^n_{\sigma_n(i),\sigma_n(j),\tau_n(k)}).$$

Proof. ✓ Proof Use the comparison equality to replace the blocked-basis coefficient by the corresponding BNT-label coefficient, and then apply the BNT-label trace-power formula. In the special case where the compared BNT-label coefficient family is already the canonical family determined by χ, the same conclusion follows directly from the canonical trace formula. □

Definition 23.93 (Blocked χ family for multiplication coefficients). Lean ✓ Stmt A dependent diagonal matrix family indexed by the blocked multiplication triples (n, i, j, k), where i and j label basis elements of $\mathcal{A}_n$ and k labels a basis element of $\mathcal{A}_{2n}$. This is a blocked-basis analog of the uniform BNT-label family in [CPGSV16, Theorem 4.14(ii)]; the remaining uniformity gap is recorded in `docs/paper-gaps/cpgsv17_blocked_chi_uniformity.tex`.

Theorem 23.94 (Trace of a blocked χ power). Lean ✓ Stmt *For each blocked multiplication triple (n, i, j, k) and every exponent L, the trace of the L-th power of its diagonal matrix is the sum of the L-th powers of the corresponding diagonal entries.*

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 23.95 (Trace-power form for blocked multiplication coefficients). Lean ✓ Stmt The blocked multiplication coefficients are in trace-power form when, for every positive blocked size n and every multiplication triple (i, j, k), the coefficient $c^{(n)}_{i,j,k}$ equals

$$\sum_r \chi^n_{n,i,j,k,r}.$$

Equivalently, by Theorem 23.94, it is the trace of the n-th power of the corresponding diagonal matrix. The exponent n is the source blocking length of the two factors in $\mathcal{A}_n$; the product is expanded in the basis of $\mathcal{A}_{2n}$.

Theorem 23.96 (Blocked coefficient as a trace power). Lean ✓ Stmt *Under trace-power form for the blocked multiplication coefficients, for every positive blocked size n, the coefficient $c^{(n)}_{i,j,k}$ is the trace of the n-th power of the diagonal matrix attached to (n, i, j, k).*

Proof. ✓ Proof Combine the defining trace-power identity with the trace formula for powers of a diagonal matrix. □

Definition 23.97 (Positive blocked χ trace-power witness). Lean ✓ Stmt A positive blocked χ trace-power witness consists of a blocked χ family, positivity of all its diagonal entries, and the positive-length trace-power identity for the blocked multiplication coefficients. This is the blocked-basis analog of the positive diagonal matrices in [CPGSV16, Theorem 4.14(ii)]; the uniform BNT-label gap is recorded in `docs/paper-gaps/cpgsv17_blocked_chi_uniformity.tex`.

Theorem 23.98 (BNT witnesses give blocked χ witnesses). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Suppose that the chosen blocked-basis coefficients are compared with a fixed BNT-label coefficient system, and that this BNT-label system has a positive length-independent χ-witness. Then the chosen blocked bases carry a positive blocked χ trace-power witness. The construction is obtained by pulling back the BNT-label $\chi_{\alpha,\beta,\gamma}$-matrices along the source and target label maps.*

Proof. ✓ Proof Reindex the positive BNT-label χ-family along the blocked-basis label map. Positivity is preserved by reindexing, and the trace-power identity follows by substituting the blocked-basis comparison into the BNT-label trace-power formula. □

Definition 23.99 (BNT-label theorem data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The BNT-label theorem data consist of the fixed BNT-label coefficient system, the same-length BNT operator family and product identity, the trace scalars and idempotent condition, the positive length-independent χ-witness, and the comparison with the chosen blocked bases. They also determine the pulled-back blocked-basis χ-family. These data are the source-side hypotheses from which the blocked-basis consequences below are derived. In the source-side special case where the coefficient family is canonically determined by the same χ-family, theorem data are built from that χ-family, its positivity, and the product, idempotent, and blocked-comparison statements for the resulting canonical coefficients. The same product and idempotent predicates may also be rephrased using the canonical coefficient family determined by the χ-matrices carried by the data, with the corresponding displayed equations.

Theorem 23.100 (BNT theorem data give the same-length product law). Lean ✓ Stmt *BNT-label theorem data give the same-length product equation*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma)$$

for every positive length L.

Proof. ✓ Proof This is the same-length product law carried by the theorem data. □

Theorem 23.101 (BNT theorem data give the idempotent scalar law). Lean ✓ Stmt *BNT-label theorem data give the idempotent scalar equation*

$$m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta.$$

Proof. ✓ Proof This is the idempotent coefficient law carried by the theorem data. □

Theorem 23.102 (BNT theorem data give the blocked coefficient comparison). Lean Lean Lean ✓ Stmt *BNT-label theorem data give the comparison between the blocked-basis coefficients and the BNT-label coefficients through the source and target label maps. Since the positive-length coefficients agree with the canonical coefficient family determined by the same χ-matrices, the comparison can also be written with those canonical coefficients.*

Proof. ✓ Proof This is the blocked-basis coefficient comparison carried by the theorem data. □

Theorem 23.103 (BNT theorem data give coefficient trace powers). Lean Lean ✓ Stmt *BNT-label theorem data give the coefficient identity $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$ for every positive length L. Equivalently, at every positive length, the coefficient family agrees with the canonical coefficient family determined by the same χ-matrices.*

Proof. ✓ Proof This is the positive-length χ-trace identity belonging to the positive BNT-label χ-witness in the theorem data. □

Theorem 23.104 (BNT theorem data give positive chi entries). Lean ✓ Stmt *BNT-label theorem data give positivity of every diagonal entry of $\chi_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This is the positivity condition carried by the positive BNT-label χ-witness in the theorem data. □

Theorem 23.105 (BNT theorem data give the chi-trace product law). Lean ✓ Stmt *BNT-label theorem data give the same-length product expansion with coefficients written as traces of powers of the length-independent $\chi_{\alpha,\beta,\gamma}$-matrices.*

Proof. ✓ Proof Apply the chi-trace product law to the product identity and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 23.106 (BNT theorem data give the chi-trace idempotent law). Lean ✓ Stmt *BNT-label theorem data give the idempotent scalar identity with the length-one coefficients written as traces of the corresponding χ-matrices.*

Proof. ✓ Proof Apply the chi-trace idempotent law to the idempotent condition and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 23.107 (BNT theorem data give blocked coefficient trace powers). Lean ✓ Stmt *BNT-label theorem data give the blocked-basis coefficient trace-power formula obtained by pulling the BNT-label $\chi_{\alpha,\beta,\gamma}$-matrices back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the blocked-basis coefficient trace-power theorem to the comparison and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 23.108 (BNT theorem data give a positive blocked χ witness). Lean Lean Lean Lean Lean ✓ Stmt *BNT-label theorem data give a positive blocked-basis χ trace-power witness by pulling the length-independent BNT-label χ-family back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the BNT-to-blocked witness construction to the comparison and the positive BNT-label χ-witness belonging to the theorem data. □

Theorem 23.109 (BNT theorem data give blocked trace-power form). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from BNT-label theorem data satisfies the blocked trace-power predicate.*

Proof. ✓ Proof This is the trace-power part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 23.110 (BNT theorem data give positive blocked chi families). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from BNT-label theorem data has positive diagonal entries.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 23.111 (BNT theorem data give positive pulled-back blocked entries). Lean ✓ Stmt *The pulled-back blocked-basis χ-entries obtained from BNT-label theorem data are positive.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the theorem data. □

Theorem 23.112 (BNT theorem data give pulled-back blocked trace powers). Lean ✓ Stmt *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis χ-matrices obtained from BNT-label theorem data.*

Proof. ✓ Proof Apply the trace-power identity of the positive blocked-basis χ-witness obtained from the theorem data. □

Definition 23.113 (Existential BNT-label theorem witness). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt An existential BNT-label theorem witness consists of the finite BNT-label type, the same-length operator spaces, their algebraic structure, and the corresponding BNT-label theorem data, including the pulled-back blocked-basis χ-family. The construction of such a witness from an MPDO tensor is the outstanding step toward Theorem IV.13(ii). The proposition-level form is the nonemptiness of this witness type. In the source-side special case where the coefficients are canonically determined by a χ-family, the witness is obtained from that same χ-family together with the remaining product, idempotent, and blocked-comparison statements. For any such witness, the product and idempotent predicates can be rephrased with the canonical coefficient family determined by its χ-matrices, with the corresponding displayed equations.

Theorem 23.114 (BNT theorem witnesses give source equations). Lean Lean Lean Lean Lean Lean ✓ Stmt *A proposition-level BNT-label theorem witness contains the source-side same-length product law, the idempotent scalar law, the positive-length trace-power law, and positivity of the diagonal $\chi_{\alpha,\beta,\gamma}$-entries. After unpacking the witness, the same-length product and idempotent equations can first be written with the BNT-label coefficients $c^{(L)}_{\alpha,\beta,\gamma}$, and then with the coefficients expressed as $\operatorname{tr}(\chi^L_{\alpha,\beta,\gamma})$ and $\operatorname{tr}(\chi_{\alpha,\beta,\gamma})$, respectively. At every positive length, the BNT-label coefficient family also agrees with the canonical coefficient family determined by these χ-matrices. Consequently, the product and idempotent predicates may be stated directly using that canonical coefficient family, and the same displayed source equations hold with those canonical coefficients.*

Proof. ✓ Proof Unpack the existential witness and apply the corresponding witness-level equations. □

Theorem 23.115 (BNT theorem witnesses give coefficient trace powers). Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives the formula $c^{(L)}_{\alpha,\beta,\gamma} = \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma})$ for every positive length L. Equivalently, at positive lengths, its coefficient family agrees with the canonical coefficient family determined by the same χ-matrices.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.116 (BNT theorem witnesses give positive chi entries). Lean ✓ Stmt *An existential BNT-label theorem witness gives positivity of every diagonal entry of $\chi_{\alpha,\beta,\gamma}$.*

Proof. ✓ Proof This follows from the positivity condition in the positive BNT-label χ-witness. □

Theorem 23.117 (BNT theorem witnesses give the same-length product law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the product identity $O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma c^{(L)}_{\alpha,\beta,\gamma} O_L(M_\gamma)$ for every positive length L.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.118 (BNT theorem witnesses give the idempotent scalar law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the idempotent scalar identity $m_\gamma = \sum_{\alpha,\beta} c^{(1)}_{\alpha,\beta,\gamma} m_\alpha m_\beta$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.119 (BNT theorem witnesses give the blocked coefficient comparison). Lean Lean Lean Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives the comparison between the blocked-basis coefficients and the BNT-label coefficients through the source and target label maps. The same comparison is also available with the canonical coefficient family determined by the witness's χ-matrices. The proposition-level nonempty form of the witness gives the same comparisons after choosing a witness.*

Proof. ✓ Proof Apply the corresponding theorem-data comparison carried by the witness. □

Theorem 23.120 (BNT theorem witnesses give the chi-trace product law). Lean ✓ Stmt *An existential BNT-label theorem witness gives, for every positive length L, the identity $O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \mathrm{tr}(\chi^L_{\alpha,\beta,\gamma}) O_L(M_\gamma)$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.121 (BNT theorem witnesses give the chi-trace idempotent law). Lean ✓ Stmt *An existential BNT-label theorem witness gives the idempotent scalar identity $m_\gamma = \sum_{\alpha,\beta} \mathrm{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta$.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.122 (BNT theorem witnesses give blocked coefficient trace powers). Lean ✓ Stmt *An existential BNT-label theorem witness gives the blocked-basis coefficient trace-power formula obtained by pulling the BNT-label χ-matrices back along the blocked-basis comparison maps.*

Proof. ✓ Proof Apply the corresponding theorem-data identity carried by the witness. □

Theorem 23.123 (BNT theorem witnesses give positive blocked χ witnesses). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *An existential BNT-label theorem witness gives a positive blocked-basis χ trace-power witness for the chosen algebra-structure data. The blocked-basis χ-family is the pullback of the BNT-label χ-family along the source and target comparison maps. The proposition-level nonempty form of the witness gives the corresponding existential blocked-basis trace-power family, both in coefficient form and in matrix-trace form.*

Proof. ✓ Proof Apply the corresponding theorem-data consequence carried by the witness. ☐

Theorem 23.124 (BNT theorem witnesses give blocked trace-power form). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from an existential BNT-label theorem witness satisfies the blocked trace-power predicate.*

Proof. ✓ Proof This is the trace-power part of the positive blocked-basis χ-witness obtained from the existential witness. ☐

Theorem 23.125 (BNT theorem witnesses give positive blocked chi families). Lean ✓ Stmt *The pulled-back blocked-basis χ-family obtained from an existential BNT-label theorem witness has positive diagonal entries.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the existential witness. ☐

Theorem 23.126 (BNT theorem witnesses give positive pulled-back blocked entries). Lean ✓ Stmt *The pulled-back blocked-basis χ-entries obtained from an existential BNT-label theorem witness are positive.*

Proof. ✓ Proof This is the positivity part of the positive blocked-basis χ-witness obtained from the existential witness. ☐

Theorem 23.127 (BNT theorem witnesses give pulled-back blocked trace powers). Lean ✓ Stmt *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis χ-matrices obtained from an existential BNT-label theorem witness.*

Proof. ✓ Proof Apply the trace-power identity of the positive blocked-basis χ-witness obtained from the existential witness. ☐

Theorem 23.128 (Positive blocked χ witness gives trace powers). Lean ✓ Stmt *A positive blocked χ trace-power witness gives, for every positive blocked size n, the formula $c^{(n)}_{i,j,k} = \mathrm{tr}(\chi^n_{n,i,j,k})$.*

Proof. ✓ Proof Apply the trace reformulation of the trace-power identity from the witness. ☐

23.11 Commuting form and GSNNCH

Definition 23.129 (Commuting-form data for an MPDO). Lean ✓ Stmt For a fixed chain length $N \geq 2$, this consists of a positive semidefinite two-site operator B such that its translated copies on the periodic chain commute pairwise.

Definition 23.130 (Commuting-form property). Lean ✓ Stmt An MPO tensor has the commuting-form property if for every $N \geq 2$ its N-site operator is a positive scalar multiple of the product of the translated copies of some commuting two-site operator B.

Definition 23.131 (GSNNCH). Lean ✓ Stmt An MPO tensor is a Gibbs state of a nearest-neighbor commuting Hamiltonian if every finite-chain operator admits the equivalent positive-operator presentation $\rho^{(N)} = c \prod_i B_{i,i+1}$ with $c > 0$ and commuting nearest-neighbor factors. The projector-limit convention in [CPGSV16] identifies this with the exponential Gibbs form.

Theorem 23.132 (GSNNCH iff commuting form). Lean ✓ Stmt *For an MPO tensor, the GSNNCH condition is equivalent to the existence of commuting-form data on every finite chain.*

Proof. ✓ Proof A positive normalization constant converts commuting-form data into a GSNNCH witness, and every GSNNCH witness remembers its underlying commuting-form data. ☐

Definition 23.133 (GSNNCH with zero correlation length). Lean ✓ Stmt This is the conjunction of the GSNNCH condition with MPO zero correlation length, matching item (iii) of the simple-MPDO equivalence in [CPGSV16].

Theorem 23.134 (GSNNCH–ZCL iff commuting form with ZCL). Lean ✓ Stmt *The GSNNCH–ZCL branch is equivalent to the conjunction of the commuting-form property with MPO zero correlation length.*

Proof. ✓ Proof Unfold the definition of GSNNCH with zero correlation length and apply Theorem 23.132. ☐

Definition 23.135 (Translation-invariant two-site bond). Lean ✓ Stmt This records a single positive two-site bond together with its translated copies on every finite periodic chain, requiring those translated copies to commute pairwise.

Definition 23.136 (η-local commuting-form data). Lean ✓ Stmt This defines the eta-local form of Appendix C.2: the translation-invariant positive two-site bond data together with the requirement that the corresponding product realizes the MPO at every finite length.

Definition 23.137 (Finite-chain data from a translation-invariant bond). Lean ✓ Stmt A translation-invariant positive two-site bond determines finite-chain commuting-form data at every length $N \geq 2$.

Theorem 23.138 (Component identities for finite-chain bond data). Lean Lean Lean ✓ Stmt *The finite-chain data associated to a translation-invariant bond retains the original local bond and the supplied lower bound $2 \leq N$, and the factor at site i is the translated copy of that local two-site bond.*

Proof. ✓ Proof These are the component identities for the finite-chain construction. ☐

Theorem 23.139 (η-local structure carries commuting form). Lean Lean Lean Lean Lean ✓ Stmt *The η-local structure itself determines a commuting-form witness for every chain length $N \geq 2$. In particular, the translated nearest-neighbor bonds commute and the MPDO at length N is a positive scalar multiple of their product, as in [CPGSV16, Proposition C.8].*

Proof. ✓ Proof At length N, apply the commuting two-site bond and realization identity provided by the η-local structure. ☐

Theorem 23.140 (η-local structure gives GSNNCH). Lean ✓ Stmt *The same η-local structure gives a GSNNCH witness on every finite periodic chain.*

Proof. ✓ Proof Evaluate the local structure at each chain length $N \geq 2$. ☐

Theorem 23.141 (η-local structure with ZCL gives GSNNCH–ZCL). Lean Lean ✓ Stmt *Once the η-local structure gives the commuting nearest-neighbor product form, adding zero correlation length gives the GSNNCH–ZCL case of the simple-MPDO equivalence.*

Proof. ✓ Proof Use the commuting-form witness carried by the η-local structure and combine it with the ZCL hypothesis. □

Theorem 23.142 (η-local structure implies commuting form). Lean ✓ Stmt *Once the explicit neighboring operators $\eta_{k,h}$ have been assembled into an η-local structure, the global commuting-form property follows.*

Proof. ✓ Proof This is the commuting-form consequence already carried by the η-local structure. □

Theorem 23.143 (η-local structure implies GSNNCH). Lean ✓ Stmt *Once the explicit neighboring operators $\eta_{k,h}$ have been assembled into the η-local structure, the GSNNCH condition follows.*

Proof. ✓ Proof This is the GSNNCH witness already carried by the η-local structure. □

Remark 23.144 (Entropy-side construction). The remaining entropy-side step is the construction of a single η-local product form from SAL(K). The local $\eta_{k,h}$ are already supplied by Definition 23.32. What remains is the tensor-coordinate assembly

$$B = \sum_{k,h} \mathrm{Id}_{H_L^{(k)}} \otimes \eta_{k,h} \otimes \mathrm{Id}_{H_R^{(h)}}, \qquad B \geq 0,$$

and, for every $N \geq 2$,

$$\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}, \qquad c_N > 0, \qquad [B_{i,i+1}, B_{j,j+1}] = 0.$$

These equations are exactly the η-local product hypothesis; after that, Theorem 23.142 gives the commuting-form conclusion. Zero correlation length enters only in the subsequent RFP and GSNNCH–ZCL conclusions.

23.12 Blocked-RFP construction for simple MPDOs

Definition 23.145 (Local simple-MPDO structure). Lean ✓ Stmt This structure records the local Appendix C.2 ingredients: a normalized three-site reduced state with equality in strong subadditivity, the resulting local η-structure, and the rank-one conclusion for the auxiliary primitive matrix T, supplied here through the conditional rank-one criterion of Theorem 23.38.

Theorem 23.146 (Local simple-MPDO structure from SAL–ZCL hypotheses). Lean Lean ✓ Stmt *The hypotheses of Lemmas C.3–C.4 determine the local simple-MPDO structure. There are two forms. The conditional form assumes the rank-one implication for T. The corrected form assumes instead that $T \geq 0$ as a matrix, with $\mathrm{tr}(T) = 1$ and $\mathrm{tr}(T^m) = 1$ for every $m > 0$, and then applies Theorem 23.39.*

Definition 23.147 (Simple-MPDO blocked-RFP structure). Lean ✓ Stmt For an MPO tensor K, this records the local Appendix C.2 ingredients, the commuting-form property, and zero correlation length. The local component consists of the hypotheses used in the entropy-side argument for commuting form.

Theorem 23.148 (Blocked-RFP structure from SAL–ZCL and commuting form). Lean Lean ✓ Stmt *The SAL–ZCL hypotheses together with a commuting-form witness determine the simple-MPDO blocked-RFP structure. In the PSD-corrected form, the matrix input is* $T \geq 0$, $\mathrm{tr}(T) = 1$, *and* $\mathrm{tr}(T^m) = 1$ *for all* $m > 0$, *rather than an abstract Perron–Frobenius rank-one assumption for* T.

Theorem 23.149 (Blocked-RFP structure from η-local structure). Lean ✓ Stmt *Given the local Appendix C.2 data, ZCL, and an* η*-local product* $\sigma^{(N)}(K) = c_N \prod_n B_{n,n+1}$ *with* $c_N > 0$, $B_{n,n+1} \geq 0$, *and* $[B_{i,i+1}, B_{j,j+1}] = 0$, *one obtains the simple-MPDO blocked-RFP structure for* K.

Proof. ✓ Proof The η-local product gives the commuting-form property for K; the local Appendix C.2 hypotheses and ZCL fill the other two fields. □

Theorem 23.150 (Blocked-RFP structure from SAL–ZCL and η-local structure). Lean Lean ✓ Stmt *Suppose the local hypotheses of Lemmas C.3–C.4 give* ρ_{ABC}, $I(A:C|B)_\rho = 0$, *a primitive matrix* T *with* $\mathrm{tr}(T) = 1$ *and* $\mathrm{tr}(T^m)$ *independent of* m, *together with the conditional rank-one criterion for* T. *Suppose also that* K *has zero correlation length and that the assembled* η*-local data give* $\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}$ *with* $c_N > 0$, $B_{n,n+1} \geq 0$, *and* $[B_{i,i+1}, B_{j,j+1}] = 0$ *for all* i, j. *Then these data determine the simple-MPDO blocked-RFP structure for* K. *In the PSD-corrected form, the conditional rank-one criterion is replaced by* $T \geq 0$, $\mathrm{tr}(T) = 1$, *and* $\mathrm{tr}(T^m) = 1$ *for all* $m > 0$.

Proof. ✓ Proof First form the local structure from the SAL–ZCL local hypotheses and the conditional rank-one criterion for T. The η-local product formula supplies the commuting-form component, and ZCL supplies the remaining component of the blocked-RFP structure. □

Theorem 23.151 (Zero correlation length gives a blocked fusion-isometry witness). Lean ✓ Stmt *If an MPO tensor* K *has zero correlation length, then it has a blocked fusion-isometry witness at size* 2 *in the sense of Definition 23.46.*

Proof. ✓ Proof Zero correlation length is the MPDO RFP condition. Applying the converse direction of Theorem 23.50 to that RFP witness yields fusion-isometry data at every positive blocked size; evaluating it at size 2 produces the blocked witness. □

Theorem 23.152 (Blocked-RFP data give a blocked fusion-isometry witness). Lean ✓ Stmt *Given the blocked-RFP data for an MPO tensor* K, *one obtains a blocked fusion-isometry witness at size* 2.

Proof. ✓ Proof Apply the zero-correlation-length theorem to the ZCL component of the blocked-RFP data. □

Lemma 23.153 (Blocked-RFP data give the fusion formulation). Lean ✓ Stmt *Given the blocked-RFP data for an MPO tensor* K, *one obtains the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The implication used here is

$$\mathrm{ZCL}(K) \implies \mathrm{RFP}(K) \implies \mathrm{RFP}_{\mathrm{fusion}}(K).$$

□

Theorem 23.154 (Simple-MPDO RFP chain from blocked-RFP data). Lean ✓ Stmt *Given the blocked-RFP data for an MPO tensor* K, *one obtains the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size* 2, *and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The commuting-form and zero-correlation-length assumptions give the GSNNCH-with-ZCL case. Zero correlation length also gives the blocked fusion-isometry witness and the transfer-map fusion formulation. □

Lemma 23.155 (Size-2 fusion witness from η-local structure). Lean ✓ Stmt *If the η-local structure has been constructed for an MPO tensor K, then zero correlation length gives the blocked fusion-isometry witness at size 2.*

Proof. ✓ Proof The implication used here is

$$\mathrm{ZCL}(K) \Longrightarrow \mathrm{RFP}(K) \Longrightarrow \mathrm{FusionWitness}_2(K).$$

□

Lemma 23.156 (Fusion formulation from η-local structure). Lean ✓ Stmt *If the η-local structure has been constructed for an MPO tensor K, then zero correlation length gives the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof This is the implication

$$\mathrm{ZCL}(K) \Longrightarrow \mathrm{RFP}(K) \Longrightarrow \mathrm{RFP}_{\mathrm{fusion}}(K).$$

□

Theorem 23.157 (Simple-MPDO RFP chain from η-local structure). Lean Lean ✓ Stmt *If the η-local structure has been constructed for an MPO tensor K, then zero correlation length gives the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The η-local structure supplies the commuting nearest-neighbor product form. The remaining two implications are

$$\mathrm{ZCL}(K) \Longrightarrow \mathrm{RFP}(K) \Longrightarrow \mathrm{RFP}_{\mathrm{fusion}}(K) \quad \text{and} \quad \mathrm{ZCL}(K) \Longrightarrow \mathrm{RFP}(K) \Longrightarrow \mathrm{FusionWitness}_2(K).$$

□

Theorem 23.158 (Simple-MPDO RFP chain from SAL–ZCL and η-local structure). Lean Lean ✓ Stmt *Assume the local SAL–ZCL hypotheses of Lemmas C.3–C.4 and the assembled η-local product form, with the conditional rank-one criterion for T, $\sigma^{(N)}(K) = c_N \prod_{n=1}^{N} B_{n,n+1}$ with $c_N > 0$, $B_{n,n+1} \geq 0$, and $[B_{i,i+1}, B_{j,j+1}] = 0$. Then K is GSNNCH with ZCL, admits a blocked fusion-isometry witness at size 2, and satisfies the transfer-map fusion formulation of MPDO RFP. The PSD-corrected form uses $T \geq 0$ together with $\mathrm{tr}(T^m) = 1$ for all $m > 0$ in place of the conditional matrix rank-one hypothesis.*

Proof. ✓ Proof Combine the local SAL–ZCL hypotheses and the η-local product form into the blocked-RFP structure, then apply Theorem 23.154. □

Theorem 23.159 (Simple-MPDO RFP chain from commuting form and ZCL). Lean ✓ Stmt *From the commuting-form property and zero correlation length one simultaneously recovers the GSNNCH-with-ZCL criterion, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.*

Proof. ✓ Proof The commuting-form and zero-correlation-length hypotheses give the GSNNCH–ZCL branch via Theorem 23.134. The blocked witness is Theorem 23.151. The full transfer-map fusion formulation follows by applying the converse direction of Theorem 23.50 to zero correlation length, which is the MPDO RFP condition. □

Remark 23.160 (SAL–ZCL implication for the GSNNCH–ZCL criterion). Not ready The implication from SAL and ZCL alone to the conclusion of Appendix C.2 requires a derivation of the commuting-form property from the local simple-MPDO structure of Lemmas C.3–C.4. Once that derivation is available, Theorem 23.159 gives the GSNNCH–ZCL branch and the blocked fusion-isometry witness.

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